REASONING GP · CLINIC GUIDES

Lab Results Handbook

> Abnormal blood results in UK primary care, one double-sided sheet per result

Each guide gives you what needs action today, how to check the result is real, result bands with the action and timing for each, causes and tests to send, treatment of the common causes, when to refer, follow-up, pitfalls, and a plain-English phrase for the patient.

43	89	8	Oct 2026
two-page guides	A4 pages	sections	checked against UK guidance

ACT TODAY
The values and combinations that need 999, same-day or urgent action, with thresholds.

RESULT BANDS
What each band means, the action and the timing: the heart of every guide.

MANAGE AND REFER
First-line treatment of the common causes, and when and where to refer.

FOLLOW-UP
When to recheck, the target, and when to stop, plus the pitfalls.

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Version 1.1 · October 2026 · next review April 2027
gpreasoning.uk · Not a substitute for clinical judgement

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Decision support, not a substitute for clinical judgement. Use your laboratory's reference ranges, check doses in the SPC and your local formulary, and follow your local pathways. Each guide names its national sources; the full interactive pathway for every topic is on gpreasoning.uk.

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Raised creatinine or falling eGFR

CLINIC GUIDE

A raised serum creatinine, or a low or falling eGFR, in adults 18+. Not for children, pregnancy, or people on dialysis or with a kidney transplant (renal team).

NICE **NG148** · KDIGO 2012 · **NG203**
UKKA 2023 · TA1075 · NG12 · MHRA
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/raised-creatinine.html

PAGE 1 · INTERPRET

is it dangerous? → acute or chronic? → how bad? → what is the cause?

! ACT TODAY — POTASSIUM, OVERLOAD, OBSTRUCTION, SEPSIS

NG148 · UKKA 2023 · NG12

TIER	RESULT OR COMBINATION	ACTION
999	K^+ **≥6.5 mmol/L**, or hyperkalaemic ECG change at any K^+ · pulmonary oedema · uraemia (drowsy, asterixis, pericardial rub) · anuria with deterioration · sepsis with high-risk features · fever, loin pain, obstruction (pyonephrosis)	Emergency admission; never give fluid to an overloaded patient
Same day	**AKI stage 2–3** · K^+ **6.0–6.4** if acutely unwell or with AKI (if well: see 11) · AKI with **blood + protein on dipstick** (vasculitis, glomerulonephritis) · retention with AKI · AKI with no clear cause or unable to drink · myeloma with high calcium or suspected cord compression · a new eGFR **<15**	Same-day hospital; nephritic urine → same-day nephrology; new eGFR <15 → urgent renal discussion
Suspected cancer pathway	Electrophoresis or free light chains suggesting **myeloma** · unexplained visible haematuria aged **45+**	NG12 referral: haematology or urology

1 IS IT REAL — AND IS IT NEW?

NG148 · NG203

- **Find the baseline**: a recent stable pre-illness creatinine, or a representative value from the previous 7–365 days. Old results are the most useful test.
- **Stage on creatinine, not eGFR**: eGFR assumes a steady state and lags in AKI.
- **New low eGFR, well, no baseline**: repeat U&E **within 2 weeks** to exclude AKI; CKD only if it persists for **>3 months**.
- **Falsely raised**: meat meal before the test, creatine supplements, high muscle mass, heavy exercise; trimethoprim or cimetidine block secretion (creatinine rises, GFR does not fall).
- **Falsely reassuring**: low muscle mass in frail, amputee or wasted patients. Consider **cystatin C** when muscle mass makes eGFR unreliable (NG203).
- **Step change on known CKD** = AKI on CKD until proven otherwise.

2 RESULT BANDS AND ACTION

NG148 · KDIGO · NG203

RESULT (VS BASELINE)	MEANS	ACTION · TIMING
Rise **≥26 µmol/L in 48 h**, or **1.5–1.9×** in 7 days	AKI stage 1	Home only if well, reversible cause, drinking: U&E **24–48 h**
2.0–2.9×	AKI stage 2	**Same-day** hospital
≥3×, or **≥354** with an acute rise	AKI stage 3	Same day; nephrology within 24 h
New low eGFR, well	AKI or CKD?	Repeat **within 2 weeks**
eGFR <60 or ACR >3 for **>3 months**	CKD	Stage G and A (6); CKD bundle (8)
ACEi/ARB start: eGFR fall **≥25%** or creatinine rise **≥30%**	Beyond the expected dip	Look for volume loss, NSAIDs; smaller rises: continue

AKI also: urine output <0.5 mL/kg/h for >6 h. A stable 354 in CKD is not stage 3.

3 CLINICAL CLUES

history → where is the problem?

Pre-renal	Vomiting, diarrhoea, poor intake, sepsis, bleeding, heart failure; postural drop; urea disproportionately high
Drugs	NSAIDs (including OTC), ACEi/ARB, diuretics, SGLT2i, lithium, trimethoprim, PPIs, contrast; the "triple whammy"
Post-renal	Poor stream, retention, constipation, pelvic cancer, single kidney; palpable bladder
Intrinsic	Rash, joint pain, haemoptysis, sinus symptoms; frothy urine, oedema; new PPI or antibiotic with rash; bloody diarrhoea

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Dry, on "sick-day" drugs	**Pre-renal, drug**	Obs, volume; drug review
LUTS, palpable bladder	**Obstruction**	Bladder scan, USS
Blood + protein on dipstick	**Glomerulonephritis**, vasculitis	Same-day nephrology
60+, bone pain, anaemia, high Ca^{2+}	**Myeloma**	Electrophoresis, light chains
Long lie, myalgia, dark urine	**Rhabdomyolysis**	CK
Bloody diarrhoea, low platelets	**HUS**	FBC, film; same day
New PPI or antibiotic, rash	**Interstitial nephritis**	Dipstick; stop the drug
Diabetes, BP, slow decline	**CKD**	ACR; old results

Causes often combine: a dehydrating illness on an ACEi, a diuretic and an NSAID is the classic community AKI.

5 WORK-UP CHECKLIST

NG148 · NG203 · NG12

- ☐ **U&E + K^+**; previous results
- ☐ **Urine dipstick** before any catheter
- ☐ **Urine ACR**
- ☐ FBC, CRP, bicarbonate
- ☐ Calcium, LFTs, glucose
- ☐ **ECG** if K^+ ≥6.0
- ☐ Myeloma screen if 60+ with clues
- ☐ CK if long lie
- ☐ **Renal USS** (indications below)
- ☐ **KFRE** in CKD G3–G5

USS: AKI with possible obstruction or no clear cause within 24 h (6 h if pyonephrosis). CKD: accelerated progression, haematuria, obstructive symptoms, family history of polycystic kidneys (over 20), eGFR <30. Never use the KFRE during AKI.

6 STAGE CKD: G AND A

NG203 · KDIGO

EGFR	G	ACR MG/MMOL	A
≥90	G1	<3	A1
60–89	G2	3–30	A2
45–59 · 30–44	G3a · G3b	>30	A3
15–29 · <15	G4 · G5	G1–G2: CKD only with another marker	

Confirm an ACR of 3–70 with an early-morning sample; ≥70 needs no repeat. Risk rises with both G and A.

7 READ THE URINE

dipstick before any catheter

Blood + protein	No UTI or catheter trauma: glomerulonephritis or vasculitis → same-day nephrology with AKI
Leucocytes, nitrites	With fever or loin pain and a rising creatinine: think infected obstruction
Negative	Typical of pre-renal and drug causes; does not exclude myeloma (dipstick misses light chains)

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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8 MANAGEMENT OF THE COMMONEST CAUSES

NG148 · NG203 · UKKA · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Any AKI · sick-day drug review	Stop NSAIDs. Hold metformin. Review ACEi/ARB, diuretics and SGLT2i individually — stopping a diuretic in heart failure can cause overload. Review renally cleared drugs (lithium, DOACs, gabapentinoids, opioids)	Write the restart plan: usually 24–48 h after eating and drinking normally, with a repeat U&E. Restart renoprotective drugs once recovered
Pre-renal, stage 1, at home	Oral fluids if dry and not overloaded; treat the cause; antibiotics dosed to renal function	U&E within **24–48 h** and until back to baseline
Retention, obstruction	Catheterise (record residual volume); treat constipation; stop anticholinergics	Same-day review; post-obstructive diuresis usually needs hospital
K^+ 5.5–5.9, well	Repeat on a fresh, promptly processed sample; review ACEi/ARB, spironolactone, trimethoprim, supplements, salt substitutes	See the hyperkalaemia guide
Confirmed CKD	**ACEi or ARB**, titrated to the maximum tolerated dose, if ACR ≥70 (any BP), ACR >30 with hypertension, or ACR >3 with diabetes; not normally started if K^+ >5.0. **Dapagliflozin 10 mg** per TA1075. **Atorvastatin 20 mg**	U&E and K^+ **1–2 weeks** after each start or dose change. BP: <140/90 if ACR <70; <130/80 if ACR ≥70. Never combine an ACEi with an ARB
Suspected myeloma, stable	FBC, calcium, ESR or plasma viscosity, serum protein electrophoresis, serum free light chains	Results suggest myeloma → suspected cancer pathway referral; avoid NSAIDs

Dapagliflozin (TA1075): added to an optimised ACEi/ARB when eGFR is 20 to <45, or 45–90 with ACR ≥22.6 mg/mmol or type 2 diabetes. Expect a small early eGFR dip; do not stop for it.

9 DOSE BY THE RIGHT NUMBER

MHRA · SPC

CREATININE CLEARANCE
Cockcroft–Gault for DOACs, and for aged 75+, BMI <18 or >40, and narrow-margin renally cleared drugs (digoxin, sotalol) — not eGFR.

METFORMIN
eGFR 45–59: maximum 2 g a day. 30–44: maximum 1 g a day, review. **<30: stop.** Hold in AKI or dehydration.

NITROFURANTOIN
Avoid if eGFR <45; a short course only with caution at 30–44. Trimethoprim raises K^+ and creatinine.

OTHERS
Gabapentinoids, opioids, lithium, aciclovir, LMWH: renal dosing per the SPC. Check eGFR before contrast.

10 FOLLOW-UP AND MONITORING CHECKLIST

NG148 · NG203

- ☐ AKI at home: U&E **24–48 h**, then until back to baseline
- ☐ New low eGFR: repeat **within 2 weeks**, then confirm at 3 months
- ☐ Held drugs restarted with a U&E; renoprotective drugs not left off
- ☐ After AKI: U&E, ACR and BP at about 3 months; watch for CKD for at least 3 years
- ☐ ACEi/ARB start or increase: U&E and K^+ at **1–2 weeks**
- ☐ CKD: eGFR and ACR at least yearly, more often by G and A category
- ☐ **KFRE** in G3–G5; plot the eGFR trend, not the snapshot
- ☐ Code the AKI or CKD; written sick-day advice; NSAID warning on the record
- ☐ Chase USS, myeloma screen, ACR — named clinician responsible
- ☐ Annual: BP to target, statin, NSAID and dose review, vaccinations

11 WHEN AND WHERE TO REFER

NG148 · NG203 · NG12

999	K^+ ≥6.5 or ECG change; pulmonary oedema; uraemia; anuria with deterioration; sepsis; pyonephrosis
Same day	AKI stage 2–3; K^+ 6.0–6.4 if acutely unwell or with AKI (if well, repeat within 1 day after discussing with the renal/medical team); nephritic urine; retention with AKI; no clear cause
Suspected cancer	Myeloma on electrophoresis or light chains; visible haematuria 45+
Nephrology (NG203)	5-year KFRE >5% · ACR ≥70 (unless diabetes, appropriately treated) · ACR ≥30 with haematuria · sustained eGFR fall ≥25% with a category change, or ≥15 mL/min/1.73m² within 12 months · BP poorly controlled on 4+ drugs · rare or genetic cause · suspected renal artery stenosis
Local practice	eGFR <30 not known to nephrology: commonly used locally, not an NG203 criterion
Urology	Obstruction needing specialist care; chronic retention

12 SPECIAL GROUPS

NG203 · NG148 · SPC

- **Heart failure**: review diuretics with care — stopping can cause overload; restart the ACEi/ARB early.
- **Frail or low muscle mass**: a "normal" creatinine may hide poor function; consider cystatin C.
- **Could become pregnant**: ACE inhibitors and ARBs are teratogenic — discuss contraception and plans before starting.
- **Lithium**: AKI → hold and check an urgent level; psychiatry plan before restarting.
- **Diabetes**: hold metformin and the SGLT2i when unwell; restart when eating and drinking.

13 PITFALLS AND NEVER-EVENTS

NG148 · NG203

- **Never** label CKD on one eGFR, or use the KFRE during an AKI.
- **Never** stop an ACEi/ARB for the expected rise (<30% creatinine).
- **Never** give fluid to an overloaded patient.
- **Never** combine an ACEi with an ARB.
- A held ACEi/ARB never restarted is a common, silent harm.
- An AKI alert is acted on the same day, not filed.

14 SAFETY-NET — SAY IT, THEN DOCUMENT IT

NG148

Call 999	Collapse, chest pain, severe breathlessness, a racing or irregular heart, new confusion
Same day	Passing much less urine; cannot keep fluids down; very drowsy; new swelling or breathlessness
Sick days	Vomiting, diarrhoea or fever with poor intake: pause the named tablets; restart when eating and drinking normally for 24–48 h

15 SAY IT LIKE THIS

plain English

"Your kidney blood test has changed. Often this is from being dehydrated or from a medicine, and it usually recovers. Stop the ibuprofen and pause [tablets] until we recheck on [date]; we'll tell you when to restart them."

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Hyponatraemia: the low sodium CLINIC GUIDE

Interpreting a serum sodium below 135 mmol/L in adults 18+. Not for pregnancy or the peripartum period (obstetric guidance, NICE NG235), children, or hospital correction with hypertonic saline.

SfE emergency guidance 2022 · NICE CKS
European guideline 2014 · NICE **NG243** · NG12
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/hyponatraemia.html

PAGE 1 · INTERPRET — is it dangerous? → is it real? → how low, how fast → what is the cause?

! ACT TODAY — SYMPTOMS AND SPEED OUTRANK THE NUMBER

SfE 2022 · NICE CKS · NG243 · NG12

TIER	RESULT OR FEATURE	ACTION
999	Any symptom attributable to low sodium, at **any level**: seizure, GCS ≤14, drowsiness, confusion, vomiting, respiratory distress; also headache or nausea without vomiting · any neurology with a fall over **<48 h** (postoperative, exercise, MDMA, bowel preparation, water intoxication) · shock (HR **>120**, SBP **<90**) · **suspected adrenal crisis**: low Na^+ with hypotension, vomiting, collapse or raised K^+	Emergency admission. Adrenal crisis: IM or IV hydrocortisone now if trained and authorised; never wait for cortisol
Same day	Na^+ **<125 mmol/L** even if well · known onset within 48 h · hypervolaemic (heart failure, cirrhosis, nephrotic) · with hypokalaemia, malnutrition, alcohol dependence or liver disease · cannot keep fluids down	Same-day hospital or acute medical assessment
Suspected cancer pathway	Chest X-ray suggesting lung cancer · aged 40+ with unexplained haemoptysis. Urgent chest X-ray if 40+ with ≥2 of cough, fatigue, breathlessness, chest pain, weight or appetite loss (≥1 if ever smoked)	NG12 referral. Low sodium alone is not an NG12 criterion, but lowers the bar for imaging

1 CONFIRM IT IS REAL

repeat urgently if unexpected

- **Repeat U&E the same day** if the result is new, unexpected, below 130 mmol/L or there are symptoms. Send the urine tests at the same time.
- **Sample from a drip arm**: hypotonic or dextrose fluid dilutes the sample. Resample from the other arm.
- **Pseudohyponatraemia**: very high triglycerides or a paraprotein (myeloma) with indirect electrode methods; **measured osmolality is normal**. No treatment needed.
- **Hyperglycaemia** (or mannitol): water shifts out of cells, so osmolality is **high**. Treat the glucose; do not apply a single fixed correction factor.
- Only **hypotonic** hyponatraemia (osmolality **<275 mOsm/kg**) is true water excess. Normal serum osmolality is **275–295 mOsm/kg**.
- **Duration unknown? Treat it as chronic**: the safer correction rate applies.

2 SEVERITY BANDS — THE ASYMPTOMATIC PATIENT

SfE 2022 · NICE CKS

NA^+ MMOL/L	MEANS	ACTION AND TIMING
130–134 mild	Often a drug, mild illness or chronic water excess	Primary care if well, stable and the cause is clear and reversible: stop the drug, recheck promptly
125–129 moderate	Real risk of falls, confusion and further fall	**Same-day** face-to-face review, bloods and urine before treatment; discuss admission or referral with the specialist; **early repeat** if managed outside hospital
<125 profound	High risk of deterioration and of harm from correction	**Same-day hospital assessment**, even if apparently well
Any level + symptoms or a fall over <48 h	Possible cerebral oedema	**999** — see Act today

Remote or out-of-hours result: volume status cannot be judged by phone, so Na^+ below 130 mmol/L needs a face-to-face examination.

3 CLASSIFY IN FOUR STEPS

European guideline 2014 · SfE 2022

1 · SERUM OSMOLALITY
<275 true hypotonic · 275–295 pseudo · >295 glucose or mannitol.

2 · URINE OSMOLALITY
≤100: water excess or low solute. >100: vasopressin is active, go to step 3.

3 · URINE SODIUM
≤30: low effective volume (losses, heart failure, cirrhosis). >30: SIAD, diuretic, adrenal, renal salt loss.

4 · EXAMINE AND REVIEW DRUGS
Volume status refines, not replaces, the urine tests. Unsure → say so and ask.

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Thiazide, SSRI, carbamazepine, PPI	**Drug-induced**	Drug review; urine Na^+
Urine osm **≤100 mOsm/kg**	**Water excess**, low solute (beer potomania)	Fluid and alcohol history
D&V, postural drop >20 mmHg; urine Na^+ **≤30**	**Extra-renal loss**	U&E, urea
Oedema, ascites; urine Na^+ ≤30	**Heart failure, cirrhosis, nephrotic**	LFTs, albumin; specialist
Pigmentation, low BP, raised K^+	**Adrenal insufficiency**	8–9 am cortisol (do not delay)
Euvolaemic, urine osm >100, urine Na^+ **>30**	**SIAD** (a diagnosis of exclusion)	TFTs, cortisol, drug review, CXR
40+, smoker, weight loss	**Lung cancer** (small cell)	Chest X-ray
High glucose, high serum osm	**Translocational**	Glucose

Urine Na^+ is unreliable on diuretics, in CKD or after saline. Never label SIAD without TFTs, cortisol, urine tests and a drug review.

5 WORK-UP CHECKLIST

bloods and urine together

- ☐ **U&E**: K^+, urea, creatinine
- ☐ **Spot urine osmolality**: first discriminator
- ☐ Glucose: excludes hyperglycaemia
- ☐ **8–9 am cortisol** if adrenal cause possible
- ☐ Lipids, protein electrophoresis if pseudo
- ☐ Lying and standing BP, JVP, oedema
- ☐ **Measured serum osmolality**
- ☐ **Spot urine Na^+**: before any IV fluid
- ☐ **TFTs**: hypothyroidism mimics SIAD
- ☐ LFTs, albumin: liver, nephrotic
- ☐ **Every medicine**, alcohol and fluid intake
- ☐ **Chest X-ray** if SIAD has no clear cause

Cortisol: interpret with the local assay and NG243; do not delay treatment for it. ADH (vasopressin) levels are not useful in primary care.

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6 MANAGEMENT OF THE COMMONEST CAUSES

SfE 2022 · European guideline 2014 · NICE CKS · NG243

CAUSE	FIRST LINE	CHECK AND CONTINUE
Drug-induced	Stop the thiazide, SSRI, carbamazepine or other likely drug if safe; review unnecessary PPIs	Recheck Na^+ promptly after stopping. Persists → investigate as SIAD. Never restart a thiazide without sodium monitoring
Hypovolaemic (D&V, diuretic)	Oral rehydration if stable and drinking; stop the thiazide. Shocked → 999. **Never fluid-restrict**	Once volume is restored, a brisk water diuresis can raise Na^+ fast: recheck closely
SIAD	Remove the cause. **Individualised fluid restriction** counting every drink (tea, soup, juice); no generic fixed volume without specialist input. Keep normal dietary salt and solute	Review the sodium and adherence early. Urine:plasma osmolality ratio >1 → restriction alone unlikely to work: specialist advice
Hypothyroidism	Treat overt hypothyroidism on the thyroid pathway (check TSH and free T4)	Recheck Na^+. Mild hypothyroidism rarely explains significant hyponatraemia: keep looking
Adrenal insufficiency	Crisis → hydrocortisone and 999. Suspected, not in crisis → same-day endocrinology (NG243)	Sick-day rules, steroid card; recheck U&E as advised
Heart failure, cirrhosis	Treat the condition with the specialist team; **no reflex IV saline**	Diuretic plan per specialist; daily weights
Low solute, beer	Cut alcohol, restore nutrition	High risk of over-rapid correction
Acute: exercise, MDMA, postoperative	Any symptom → 999; no hypotonic fluids while waiting	Hospital correction; sodium checked on recovery

Correction limits (hospital, maximum not target): rise no more than **10 mmol/L in the first 24 h**, then **8 mmol/L in each further 24 h** (SfE 2022); slower still with alcohol dependence, malnutrition, liver disease or hypokalaemia. Over-correction risks irreversible osmotic demyelination. **Specialist or hospital only:** hypertonic saline, oral urea, tolvaptan; demeclocycline is not advised (European guideline).

7 FOLLOW-UP AND MONITORING CHECKLIST

NICE CKS · SfE 2022

- ☐ **Recheck Na^+ promptly** after stopping a drug, changing fluids or starting treatment
- ☐ Na^+ 125–129 not admitted: **early repeat** agreed with the specialist
- ☐ Fluid restriction reviewed early: sodium, adherence, thirst
- ☐ After discharge: recheck at the interval named in the discharge plan; ask for one if missing
- ☐ **Target**: stable Na^+ ≥130 with the cause found; normalising is not always needed in chronic mild cases
- ☐ Widen intervals once Na^+ is stable above 130; review the cause yearly in chronic SIAD
- ☐ Re-refer if Na^+ falls below 125 again or new neurology appears
- ☐ Remote result below 130 → face-to-face examination, not a phone plan
- ☐ Cause and culprit drug recorded; future prescribers warned

8 WHEN AND WHERE TO REFER

SfE 2022 · NICE CKS · NG12

999	Symptoms at any level; acute fall with neurology; shock; adrenal crisis
Same day	Na^+ <125; onset <48 h; hypervolaemic; alcohol, malnutrition, liver disease or low K^+; worsening despite stopping the drug
Same-day advice	Na^+ 125–129: discuss admission or referral; suspected adrenal insufficiency
Suspected cancer	CXR suggests lung cancer; 40+ with unexplained haemoptysis
Endocrinology	New SIAD without cause (urgent); confirmed SIAD stable at 130–134; intermediate cortisol
Other	Nephrology (CKD, refractory) · hepatology (cirrhosis) · falls service if Na^+ <130 with falls

9 SPECIAL SITUATIONS

SfE 2022 · NICE CKS · NG243

- **Older people who fall**: check sodium; Na^+ <130 with falls → falls service as well as the cause.
- **MDMA or endurance exercise**: acute, can be severe in the young; any symptom → 999.
- **On long-term steroids**: never stop abruptly; low Na^+ with vomiting or low BP is adrenal crisis until proven otherwise.
- **Heart failure on ACE inhibitor or ARB**: low sodium is a marker of severity; seek specialist advice before changing diuretics.

10 PITFALLS AND NEVER-EVENTS

SfE 2022 · European guideline

- **Never** reassure because a patient seems well below 125 mmol/L, or call a number "mild" when there are symptoms or a fast fall.
- **Never** fluid-restrict a dry patient, or give saline for SIAD or to an oedematous patient.
- Low Na^+ plus raised K^+, each looking mild: **think adrenal crisis**.
- **Never** chase normal sodium quickly: a fast rise is more dangerous than the low level.
- Mild chronic hyponatraemia still causes falls, unsteadiness and poor attention.
- **Never** start urea, tolvaptan or demeclocycline in primary care, or give hypertonic saline outside hospital.
- Never accept "SIADH" on a discharge letter without the cause and the plan; ask for them.

11 SAFETY-NET AND SELF-CARE — SAY IT, THEN DOCUMENT IT

SfE 2022 · NICE CKS

CALL 999
A fit, collapse or fainting, unusual drowsiness, new or worse confusion, severe headache, vomiting or breathing difficulty.

GP OR 111 TODAY
Feeling sick, dizzy on standing, unsteady or a new fall, cramps, very thirsty with little urine, cannot keep up with fluids.

FLUID LIMIT
Every drink counts, including tea, soup and fruit. Use a measured jug and write it down. Attend the repeat blood test even if well.

PREVENT THE NEXT ONE
Endurance exercise: drink to thirst, not to a schedule. Tell every new prescriber about the low sodium.

12 SAY IT LIKE THIS

plain English

"The sodium, or salt, level in your blood is lower than it should be. This is usually not about how much salt you eat: more often a medicine or your body holding on to too much water is diluting it. We'll find the cause, fix it gently rather than quickly, and recheck your blood. If you become confused, very drowsy, have a fit or keep vomiting, call 999."

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Hypernatraemia: the high sodium CLINIC GUIDE

Interpreting a serum sodium above 145 mmol/L in adults 18+. Not for infants and children (hypernatraemic dehydration → 999 or urgent paediatric admission), or hospital fluid and desmopressin treatment.

NICE CG174 (IV fluids) · **CG185** (lithium)
JBDS-IP HHS 2022 · SfE cranial DI 2018
CQC regulation 14 (hydration)
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/hypernatraemia.html

PAGE 1 · INTERPRET — is it dangerous? → is it real? → how high, how fast → why can't this person get or keep water?

! ACT TODAY — SYMPTOMS, TEMPO AND ACCESS TO WATER OUTRANK THE NUMBER

CG174 · JBDS · SfE

TIER	RESULT OR FEATURE	ACTION
999	Raised sodium with **confusion, drowsiness, seizure, shock or severe dehydration** · high sodium and **unable to drink** or deteriorating · polyuria after head injury, neurosurgery, pituitary surgery, meningitis or encephalitis (acute central DI) · suspected **HHS** (high glucose, dehydration, drowsy) · focal neurology (intracranial cause until proven otherwise)	Emergency transfer; correction is a hospital task
Same day	Na^+ **150–154 mmol/L** · 146–149 if new, rising, unexplained, polyuric, frail, confused, or cannot ensure intake · polyuria with dilute urine (possible DI) · on **lithium** and dehydrated · dehydration with AKI	Same-day acute medical assessment; DI → urgent endocrinology
Admit	Na^+ **≥155 mmol/L** (CG174), or sooner if symptomatic, acutely rising, hypovolaemic, unable to take oral fluids or clinically unstable · **≥160** = severe	Urgent admission; ≥160: **immediate senior/specialist input**
Safeguarding	Dependent adult with poor intake, recurrent, or with weight loss, pressure ulcers or poor care · suspected salt poisoning	Same-day safeguarding concern (Care Act 2014)

1 CONFIRM IT IS REAL

repeat urgently, examine in person

- **Repeat U&E urgently** if unexpected, with glucose and paired serum and urine osmolality. A spurious high sodium is uncommon: **believe it if the patient is dry, drowsy or not drinking**.
- **Sample error**: drawn above a saline or sodium bicarbonate drip; evaporation from a delayed or uncapped sample. Resample from the other arm.
- **Sodium gain**: saline, bicarbonate, hypertonic feeds, salt enemas, sodium-containing effervescent or soluble medicines.
- **Hyperglycaemia** dilutes sodium: in **HHS** it may look normal, then rise as glucose falls. Check capillary glucose and ketones in every dehydrated patient.
- **Duration unknown? Treat it as chronic**: most community cases have built up over days.
- **Not by phone**: hydration, cognition and access to drinks cannot be judged remotely.

2 SEVERITY BANDS

use local laboratory ranges

NA^+ MMOL/L	MEANS	ACTION AND TIMING
146–149, well	Mild; often fever, heat or poor intake	Home only if well, cognitively intact, cause clear and a **named person gives measured drinks**; urgent repeat U&E
146–149, at risk	Frail, confused, polyuric, unclear or rising	**Same-day** medical review
150–154	Significant water deficit	**Same-day acute medical assessment** in most; immediate if symptomatic, not drinking, or DI/HHS possible
155–159	Oral correction often impractical	**Urgent admission** (CG174); 999 if drowsy, confused or not drinking
≥160	Severe	**Admit**; immediate senior/specialist input

Unknown duration = chronic. In hospital, sodium should fall no faster than about **0.5 mmol/L per hour** and **10–12 mmol/L in 24 h**: too fast a fall causes cerebral oedema.

3 READ THE URINE — THE KEY DISCRIMINATOR

paired with serum osmolality

CONCENTRATED URINE
For example **>600 mOsm/kg**: the kidney is saving water appropriately. Think intake, insensible or GI loss.

INAPPROPRIATELY DILUTE
Dilute urine despite a high sodium: **DI**, central or nephrogenic. Same-day assessment; urgent endocrinology discussion.

LARGE VOLUME, NOT DILUTE
Osmotic diuresis: glucose (HHS) or urea. Check glucose and ketones first.

INTERPRET WITH CARE
Suggestive, not diagnostic: allow for diuretics, recent intake, renal function and hyperglycaemia.

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Frail, dementia, dysphagia, care home	**Inadequate intake** (commonest)	Urine osm: concentrated (>600 mOsm/kg)
Fever, sweating, heat, burns	**Insensible loss**	Temperature, CRP; source
Diarrhoea, lactulose, vomiting	**GI water loss**	U&E, urea
Lithium, polyuria, thirst	**Nephrogenic DI**	Lithium level, calcium, urine osm
Head injury, pituitary surgery	**Central DI**	Urine osm (dilute); endocrinology
High glucose, drowsy	**HHS**	Glucose, ketones → 999
Loop diuretic, laxatives, osmotic agents	**Drug-induced water loss**	Drug review; U&E
Saline, feeds, effervescent tablets	**Sodium gain**	Drug and feed review

Polyuria: more than about **50 mL/kg/day** (about 3 L/day). Dilute urine with high sodium → DI or osmotic diuresis, not "poor intake".

5 WORK-UP CHECKLIST

bloods, urine and the drinks

- ☐ **U&E**, creatinine: AKI?
- ☐ **Serum osmolality**
- ☐ **Calcium**: hypercalcaemia → NDI
- ☐ **12-hour lithium level** if on lithium
- ☐ GCS, orientation, focal neurology
- ☐ **Swallow**; who gives the drinks?
- ☐ **Glucose**, HbA1c, ketones
- ☐ **Urine osmolality** and urine Na^+
- ☐ Potassium: low K^+ → NDI
- ☐ Urine volume if polyuric
- ☐ Lying and standing BP, weight, temperature
- ☐ **Drugs**: diuretics, lactulose

RESULT	MEANS
Rising creatinine	Dehydration with AKI: same day
High calcium, low K^+	Reversible causes of nephrogenic DI
High glucose, ketones	HHS or DKA: 999
Lithium level high, or symptoms	Toxicity: same-day advice; toxicity can occur in range

Not in primary care: water-deprivation or copeptin testing, IV fluids, starting desmopressin.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 SAFE TO MANAGE AT HOME ONLY IF — EVERY BOX TICKED AND DOCUMENTED

CG174 · CQC reg 14

- ☐ Na^+ 146–149, not rising; no Act-today feature
- ☐ Clinically well, cognitively intact, not polyuric
- ☐ Can drink; a named person gives measured drinks
- ☐ Glucose, calcium and U&E checked; drugs reviewed
- ☐ Urgent repeat booked; safety-net given
- ☐ Safeguarding considered for any dependent adult

7 MANAGEMENT OF THE COMMONEST CAUSES

CG174 · CG185 · JBDS · SfE · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Poor intake, mild (146–149)	**Plain water** by mouth, not sports drinks or saline; measured drinks given by a named person. **Never fluid-restrict**	Urgent repeat U&E. Nobody to give drinks → same-day assessment
Fever, heat	Extra plain water; treat the fever and its cause	Care home: check the fluid target and who is giving drinks
Diarrhoea, vomiting	Oral rehydration solution plus free water; review diuretics and lactulose	Not keeping fluids down → same-day assessment
Sodium gain	Stop sodium-containing effervescent or soluble medicines where possible; review feeds	Recheck after the change
Lithium nephrogenic DI	Keep hydrated; **do not stop lithium abruptly** without urgent specialist advice unless toxic. Amiloride, thiazide or dose change only with psychiatry, endocrinology or nephrology	Thiazides raise lithium levels. If amiloride is started: U&E and K^+ after starting and each dose change
Central DI	Desmopressin to the written endocrinology plan only; no community start for a new case	Do not over-drink on desmopressin (hyponatraemia). Vomiting or no water access is an emergency
Hypercalcaemia	Treat the calcium (hypercalcaemia pathway)	DI usually settles as calcium normalises
HHS	999. Hospital JBDS pathway: 0.9% sodium chloride first, despite the high sodium	Sodium may rise as glucose falls: follow osmolality

8 FOLLOW-UP AND MONITORING CHECKLIST

CG185 · SfE · CG174

- ☐ **Urgent repeat U&E** for mild cases at home; booked before the patient leaves
- ☐ **Target**: sodium back in range, falling slowly, with the cause removed
- ☐ Named person, measured drinks and fluid chart in place
- ☐ **Lithium**: level every **3 months** in year 1, then every **6 months** (3-monthly if higher risk); U&E/eGFR, calcium and TFTs at least every **6 months** (CG185)
- ☐ Lithium with polyuria: urine volume and osmolality, then specialist advice
- ☐ **Desmopressin**: sodium after every dose change or illness, per the endocrinology plan
- ☐ Diuretics and laxatives reviewed against intake
- ☐ Safeguarding question answered and recorded
- ☐ Cause recorded in the problem list

9 WHEN AND WHERE TO REFER

CG174 · JBDS · SfE 2018

999	Confusion, drowsiness, seizure, shock, severe dehydration, unable to drink; acute central DI; HHS
Same day	Na^+ 150–154; 146–149 at risk; dehydration with AKI; lithium and dehydrated
Admit	Urgent hospital assessment/admission: Na^+ ≥155 (CG174), or sooner if symptomatic, acutely rising, hypovolaemic, unable to take oral fluids or clinically unstable. ≥160 is severe → immediate senior/specialist input
Endocrinology	Suspected DI (urgent discussion); pituitary or hypothalamic cause; recurrent or chronic DI
Nephrology	Nephrogenic DI from CKD; lithium DI with renal impairment
Safeguarding	Neglect in a dependent adult; salt poisoning: same day

10 SPECIAL GROUPS

CQC reg 14 · CG185 · SfE

- **Care home residents**: a documented daily fluid target, monitored intake and thickened fluids where needed for dysphagia; hydration is a legal standard (CQC regulation 14).
- **Central DI**: medical alert and written sick-day plan; unconscious patients given only isotonic fluid can become severely hypernatraemic.
- **After pituitary surgery**: DI, then a low-sodium phase, then possibly permanent DI; sodium in either direction needs the specialist team.

11 PITFALLS AND NEVER-EVENTS

JBDS · SfE · CG174

- "Not thirsty": thirst fails in older people and with reduced consciousness.
- "Encourage fluids" when nobody can bring, hold or help with a drink.
- "Poor intake" when the urine is **dilute**: think DI or osmotic diuresis.
- A normal-looking sodium with high glucose: **HHS**.
- **Never** fluid-restrict, never drop sodium fast, never arrange water-deprivation testing from primary care.
- A sodium that settles in a dependent adult: the safeguarding question remains.
- **Never** stop lithium abruptly, or start amiloride or a thiazide, without specialist advice.

12 SAFETY-NET AND SELF-CARE — SAY IT TO THE PATIENT AND CARER, THEN DOCUMENT IT

CG174 · SfE

CALL 999
New confusion, drowsiness, a fit, collapse, or being unable to drink.

PRACTICE OR 111 TODAY
Vomiting, drinking less, passing much less urine, or large volumes of urine with thirst.

DRINKS
Plain water little and often, offered by a named person; keep a simple chart if a carer helps.

ON DESMOPRESSIN
Follow the written plan; do not over-drink; vomiting or no water access → urgent help.

13 SAY IT LIKE THIS

plain English

"Your blood is too salty, which almost always means your body is short of water rather than that you've eaten too much salt. We'll put the water back slowly, because doing it too fast can be harmful, and find out why it happened. If you become confused, very sleepy or can't drink, call 999."

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Hyperkalaemia: the high potassium CLINIC GUIDE

Interpreting a raised potassium result in adults 18+. Not for children, DKA (hospital protocol), patients on dialysis (renal unit) or hospital IV treatment.

UKKA hyperkalaemia guideline 2023 (upd. 2026)
NICE **NG203** (CKD) · NG106 · NG136
NICE TA1148 (2026) · TA623 · MHRA
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/hyperkalaemia.html

PAGE 1 · INTERPRET — is it dangerous? → is it real? → how high, how fast → why?

! ACT TODAY — TRANSFER FIRST, CONFIRM IN PARALLEL

UKKA 2023

TIER	RESULT OR FEATURE	ACTION
999	K^+ **≥6.5 mmol/L** on any sample · **any ECG change** that could be hyperkalaemia (peaked T, flat or absent P, long PR, broad QRS, bradycardia, sine wave) · symptoms at any level: weakness, paralysis, palpitations, syncope · **AKI**, oliguria or anuria · missed dialysis, rhabdomyolysis, tumour lysis · rapidly rising K^+ · suspected digoxin toxicity · high K^+ with low Na^+ and hypotension (adrenal crisis)	Emergency transfer; **do not wait for a repeat**. ECG only if it does not delay transfer. No IV calcium in routine primary care. Withhold contributing drugs and hand over what was withheld
Same day	K^+ **6.0–6.4** if acutely unwell or AKI · K^+ 5.5–5.9 with rising creatinine, acute illness, symptoms, a rapid rise, or no ECG or reliable follow-up possible	Urgent 12-lead ECG and face-to-face assessment; admit on ECG, trend, renal function and clinical state

A normal ECG does **not** exclude severe hyperkalaemia. A critical result must never sit in an inbox or a routine repeat queue overnight: name who acts out of hours.

1 CONFIRM IT IS REAL — ONLY IF NO RED FLAG

UKKA 2023

- **Haemolysis**: check the lab comment; difficult draw, small needle.
- **Delayed processing** or a refrigerated sample: K^+ leaks from red cells. Common with late collections and rural transport.
- **Fist clenching, prolonged tourniquet**: falsely high.
- **EDTA contamination** (wrong tube order): very high K^+ with a very low calcium.
- **High platelets or white cells**: ask for a **plasma** (lithium heparin) K^+.
- **Repeat** with free-flow venepuncture, no fist clenching, prompt processing, marked "urgent".

Never presume pseudohyperkalaemia at ≥6.5, or with symptoms, ECG changes, AKI or a rising trend. A normal haemolysed repeat after a genuine high result does not reassure.

2 SEVERITY BANDS — THE WELL PATIENT

UKKA 2023 · NG203

K^+ MMOL/L	MEANS	ACTION AND TIMING
5.1–5.4	Above the reference range; not hyperkalaemia	On RAAS drugs or in CKD: remove contributors, recheck in **1–4 weeks**
5.5–5.9 mild	Usually drugs plus CKD	Review the **same working day**; repeat within **3 days**; ECG if cardiac disease, rapid rise, renal impairment or symptoms
6.0–6.4 moderate	Risk of arrhythmia	**ECG**; unwell or AKI → **same-day hospital**; well → hold contributors, discuss with renal/medical team, repeat within **1 day**
≥6.5 severe	Cardiac arrest risk at any moment	**999**; confirm in parallel

Remote or out-of-hours: ≥6.0 cannot be managed by telephone. No ECG today or cannot attend → emergency assessment.

3 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
ACE inhibitor, ARB, MRA, amiloride	**RAAS drugs** (commonest)	Drug list; eGFR trend
Trimethoprim, NSAID, heparin, ciclosporin, tacrolimus	**Other drugs**	Drug list, OTC ibuprofen
Creatinine up, oliguria, sepsis	**AKI**	U&E vs baseline; bladder
CKD 4–5	**Reduced excretion**	eGFR, bicarbonate
Low Na^+, low BP, pigmentation	**Adrenal insufficiency**	Cortisol only if no delay
Diabetic CKD, low bicarbonate	**Type 4 RTA**	Renal assessment
Salt substitute, supplements	**Potassium load**	Ask about LoSalt
High glucose, unwell	**DKA**, insulin lack	Glucose, ketones
Muscle injury, chemotherapy	**Cell breakdown**	CK, urate via acute team

4 WORK-UP CHECKLIST

conditional, not a panel

- ☐ **Repeat U&E**, clean sample
- ☐ **12-lead ECG** at ≥6.0
- ☐ **FBC**: platelets, white cells
- ☐ Postural BP, volume status
- ☐ Sodium: low → adrenal?
- ☐ **eGFR trend**: AKI?
- ☐ Bicarbonate, glucose
- ☐ **Full drug list**, OTC and salt
- ☐ Bladder: retention?
- ☐ Digoxin level if toxicity possible

Only with a trigger: CK, urate and phosphate (muscle injury, tumour lysis) via acute services; aldosterone and renin are specialist tests. ECG: compare with old tracings; a broad QRS may be bundle branch block or pacing, but do not let that reassure.

5 READING THE ECG

UKKA 2023

Early	Tall, narrow, peaked T waves (V2–V4)
Later	Long PR, flat or absent P, broadening QRS
Pre-arrest	Sine wave, bradycardia, junctional rhythm
Any of these	Emergency transfer, whatever the number

6 IF YOU TRANSFER — HAND OVER

UKKA 2023

- The K^+, its **trend and timing**, and whether the sample was haemolysed.
- eGFR and creatinine against baseline; urine output; dialysis status.
- The ECG if done; **every drug** and which ones you withheld.
- Suspected cause: AKI, muscle breakdown, tumour lysis, adrenal, digoxin.

7 SAFE TO MANAGE IN PRIMARY CARE ONLY IF — EVERY BOX TICKED AND DOCUMENTED

UKKA 2023

- ☐ K^+ 5.5–5.9, well, no AKI, no Act-today feature
- ☐ Contributors stopped; cause acted on
- ☐ ECG normal where indicated; sample quality checked
- ☐ Repeat within 3 days booked; named reviewer
- ☐ Sodium, bicarbonate and glucose reviewed
- ☐ Written escalation threshold, safety-net, sick-day plan

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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8 MANAGEMENT OF THE COMMONEST CAUSES — MILD, WELL, NO AKI

UKKA 2023 · NG203 · NG106 · NG136 · TA1148 · TA623

CAUSE	FIRST LINE	CHECK AND CONTINUE
Supplements, salt substitutes	Stop potassium supplements and LoSalt-type salt substitutes	Ask again at every review: patients restart them
Trimethoprim, NSAIDs, heparin	Stop or switch where possible, including OTC ibuprofen	Record NSAID avoidance in the notes
ACE inhibitor, ARB	Remove other contributors first, then consider dose reduction or advice. CKD: **stop at K^+ ≥6.0** once other potassium-raising drugs are stopped (NG203)	K^+ and eGFR **1–2 weeks** after any start or dose change. Heart failure or albuminuric CKD: plan reintroduction, not indefinite stop
MRA (spironolactone, eplerenone)	Reduce or withhold on the clinical picture; seek advice before stopping in heart failure	MHRA: potentially fatal hyperkalaemia with an ACE inhibitor or ARB. Avoid starting if K^+ >5.0 unless specialist-directed
Diet, dehydration	Individualised swaps, not blanket bans on milk, pulses, nuts and vegetables; correct dehydration. **Never diet alone at ≥6.0**	Advanced CKD → renal dietitian sets any target
Persistent, limiting a needed RAAS drug	Binder: CKD 3b–5 or heart failure, not on dialysis, RAAS not optimised because of K^+; confirmed K^+ **≥5.5** for sodium zirconium cyclosilicate (TA1148; can start in primary care, UKKA 2026), **≥6.0** for patiromer (TA623)	Patiromer binds magnesium: monitor it. Separate binders from other oral drugs as the SPC states. Never a substitute for emergency transfer

9 SICK-DAY PLAN — PERSONALISED, IN WRITING

UKKA · NG203

Name which medicines to pause (ACE inhibitor, ARB, MRA, diuretics, metformin, SGLT2 inhibitor) during significant vomiting, diarrhoea or not drinking, **and when to restart**. Check U&E after the illness before restarting after AKI or high K^+. Avoid NSAIDs altogether rather than pausing them.

10 FOLLOW-UP AND MONITORING CHECKLIST

UKKA 2023 · NG203 · MHRA

- ☐ 5.5–5.9: **repeat within 3 days**, named reviewer, written escalation threshold
- ☐ 6.0–6.4, well: renal/medical team discussion, **repeat within 1 day**; still ≥6.0 → admit
- ☐ 5.1–5.4 on RAAS: recheck in **1–4 weeks**
- ☐ Any RAAS dose change: K^+ and eGFR at **1–2 weeks** (NG203)
- ☐ New trimethoprim, NSAID or second RAAS drug: K^+ at **1 week**
- ☐ After discharge: K^+ and eGFR within days, at the interval in the plan; check what was stopped and plan reintroduction
- ☐ **Target**: K^+ back in the reference range while keeping the prognostic drugs where possible
- ☐ Binder: monitoring as the initiating service sets; magnesium on patiromer
- ☐ Monitoring register and recalls for overdue bloods

11 WHEN AND WHERE TO REFER

UKKA 2023 · TA1148 · TA623

999	≥6.5; any ECG change; symptoms; AKI or oliguria; rapid rise; digoxin toxicity; adrenal crisis
Same day	6.0–6.4 if unwell or AKI; 5.5–5.9 with AKI, illness, symptoms or no ECG or follow-up
Nephrology	Persistent despite removing contributors; falling eGFR; type 4 RTA; binder decision
Cardiology	Heart failure where K^+ is limiting prognostic drugs
Endocrinology	Stable suspected Addison's (crisis is 999)

12 WHAT THE HOSPITAL WILL DO — HOSPITAL ONLY, SO YOU CAN EXPLAIN IT AND CHECK THE DISCHARGE PLAN

UKKA 2023

IV CALCIUM
Protects the heart for a while; does **not** lower potassium. Not for routine primary care.

INSULIN–GLUCOSE, SALBUTAMOL
Move potassium into cells. Glucose is monitored for hours afterwards because of late hypoglycaemia.

REBOUND
These shift potassium rather than remove it: it rises again unless the cause is fixed or the potassium is removed.

REMOVAL
Dialysis via the renal team for refractory cases. Calcium resonium is no longer part of acute treatment.

13 PITFALLS AND NEVER-EVENTS

UKKA 2023

- **Never** repeat a K^+ ≥6.5 instead of transferring, or call it "probably haemolysis" without asking the lab.
- **Never** reassure because the patient feels well or the ECG is normal.
- "Chronic and stable" without comparing the trend.
- Blaming diet with a normal eGFR and no drugs, or with a low sodium or bicarbonate beside it.
- **Never** stop a heart failure drug for good without a reintroduction plan.
- **Never** start a binder instead of emergency transfer.
- **Never** leave a critical result in an inbox or a routine repeat queue overnight.

14 SAFETY-NET — SAY IT, THEN DOCUMENT IT

UKKA 2023

CALL 999
Collapse or blackout; chest pain; a racing or very slow heartbeat with feeling faint; legs or arms going weak.

PRACTICE OR 111 TODAY
Palpitations, feeling faint, new numbness or weakness, cannot keep fluids down, passing much less urine.

AVOID
Salt substitutes (LoSalt and similar); ibuprofen and other anti-inflammatories without checking first.

KEEP THE APPOINTMENT
Come for the repeat blood test even if you feel well; bring a list of every medicine.

15 SAY IT LIKE THIS

plain English

"Your potassium is higher than it should be. Potassium helps your heart beat in rhythm, so we take a high level seriously even when you feel well. One of your medicines and your kidneys are the likely reasons. We'll adjust things and recheck within a few days. If you feel faint, your heart races or your legs go weak, call 999."

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Hypokalaemia: the low potassium CLINIC GUIDE

Interpreting a potassium below 3.5 mmol/L in adults 18+. Not for children, DKA, hospital IV replacement, or specialist eating-disorder care.

NHS SPS treating acute hypokalaemia 2024
NICE **NG136** · NG69 · NG12 · CG32
RCPsych CR233 (MEED) · Sando-K SPC
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/hypokalaemia.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → how low → renal loss, gut loss or shift?

! ACT TODAY — NEVER MANAGE K⁺ BELOW 2.5 IN THE COMMUNITY

SPS 2024

TIER	RESULT OR FEATURE	ACTION
999	Any low K⁺ with **arrhythmia**, chest pain, syncope or collapse · **ascending weakness**, unable to walk, breathing difficulty · K⁺ <2.5 and unstable · on **digoxin** with toxicity features (bradycardia, heart block, vomiting, confusion, visual change) and unstable · rhabdomyolysis (muscle pain, dark urine, AKI) and unstable	Emergency transfer. **Never IV potassium in primary care**; do not delay for outpatient tests
Same day	K⁺ **<2.5 mmol/L** even if well (usually admission) · significant ECG change: frequent ectopics, marked ST/T change, prominent U waves · K⁺ 2.5–2.9 with symptoms, vomiting, dehydration, digoxin, heart disease, renal impairment or a falling level · cannot take oral treatment	Hospital assessment for monitored replacement

A normal ECG does not exclude important hypokalaemia. Severe alkalosis, rapid-onset hypertension, proximal weakness and hyperglycaemia in a smoker: think **ectopic ACTH** → same-day endocrine advice.

1 CONFIRM IT IS REAL

repeat with magnesium

- **Very high white cell count** (leukaemia): cells take up K⁺ in a delayed or warm sample → falsely low. Ask for prompt processing.
- **Drip arm**: sample above dextrose or insulin infusion. Resample from the other arm.
- **Recent salbutamol nebs or insulin**: a real but transient shift into cells.
- **Repeat U&E with magnesium**, bicarbonate and glucose when the result is unexpected or does not fit; do not delay action at <2.5 or with ECG change.

2 SEVERITY BANDS

SPS 2024

K⁺ MMOL/L	MEANS	ACTION AND TIMING
3.0–3.4 mild	Usually no symptoms; often a diuretic	Primary care: treat the cause, review the diuretic, oral potassium if needed; recheck soon
2.5–2.9 moderate	Weakness, cramps, constipation; U waves	**ECG and assessment the same day**. Oral only if well, absorbing, normal ECG, reliable follow-up; early recheck
<2.5 severe	Arrhythmia, paralysis, rhabdomyolysis	**Hospital** for IV replacement and monitoring; 999 if unstable

3 CLASSIFY THE MECHANISM

bicarbonate, BP and drugs first

RENAL LOSS
Diuretics (commonest). With hypertension and alkalosis: primary aldosteronism, Cushing's. Acidosis: renal tubular acidosis. Young, normotensive, low Mg: Bartter or Gitelman.

GUT LOSS
Vomiting (alkalosis, kidney loses K⁺), diarrhoea, laxatives, high-output stoma. Treat the cause.

SHIFT INTO CELLS
Insulin, salbutamol, alkalosis, refeeding, thyrotoxic periodic paralysis. Replace cautiously: rebound high K⁺.

LOW INTAKE
Rarely alone: "tea and toast" with a diuretic, alcohol, eating disorders. Low magnesium keeps K⁺ low.

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Thiazide or loop diuretic	**Diuretic** (commonest)	Mg, bicarbonate
Hypertension, high bicarbonate	**Primary aldosteronism**	Aldosterone:renin ratio
Normal BP, alkalosis, no diuretic, dental erosion	**Purging**, laxatives	Ask directly; MEED risk
D&V, stoma, IBD	**GI loss**	U&E, postural BP
Alcohol, PPI, diarrhoea	**Low magnesium**	Magnesium
Low bicarbonate, normal anion gap	**Renal tubular acidosis**	Venous gas; nephrology
Moon face, striae, proximal weakness	**Cushing's**	Endocrine-led tests
Episodic paralysis, tremor	**Thyrotoxic periodic paralysis**	TFTs
Liquorice, steroids, salbutamol	**Drug or diet**	Drug and diet history

5 WORK-UP CHECKLIST

proportionate to the picture

- ☐ **Repeat U&E**, eGFR
- ☐ **Bicarbonate**; gas if unwell
- ☐ **ECG**: symptoms, moderate, digoxin
- ☐ Power, reflexes, bowel sounds
- ☐ TFTs if periodic weakness
- ☐ **ARR** if hypertensive
- ☐ **Magnesium**: every time
- ☐ Glucose
- ☐ **BP**, lying and standing
- ☐ **Full drug list**, laxatives, liquorice
- ☐ CK if muscle pain
- ☐ Urine K⁺ only if cause unclear

Well, stable 3.0–3.4 with an obvious diuretic cause: repeat U&E with bicarbonate and magnesium and review the diuretic before wider tests. Aldosterone:renin ratio: correct the potassium first and follow the laboratory's advice on interfering drugs. Never use a 9 am cortisol to screen for Cushing's.

6 READING THE ECG

SPS 2024

Early	U waves (V2–V4), flattened T waves
Later	ST depression, long PR; U wave merging with T (apparent long QT)
Severe	Ventricular ectopics, broad QRS, VT: emergency

7 SAFE IN PRIMARY CARE ONLY IF — EVERY BOX

SPS 2024

- ☐ K⁺ 3.0–3.4, or 2.5–2.9 meeting the oral criteria; no Act-today feature
- ☐ ECG done where indicated and normal
- ☐ Magnesium, bicarbonate, BP checked; cause acted on
- ☐ Recheck booked; safety-net given and recorded

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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8 MANAGEMENT OF THE COMMONEST CAUSES

SPS 2024 · Sando-K SPC · NG136 · NG69

CAUSE	FIRST LINE	CHECK AND CONTINUE
Replacement regimen oral potassium chloride	**Sando-K** effervescent: **12 mmol K⁺ per tablet**, dissolved in water, with or after food. **Mild (3.0–3.4): 24 mmol** (2 tablets) two or three times a day. **Moderate (2.5–2.9), well: 24–36 mmol** three or four times a day (SPS 2024). Lower doses in renal impairment	Recheck U&E and magnesium; not rising → check magnesium, adherence and losses before increasing. Warn about nausea and diarrhoea. Count potassium in other medicines
Diuretic	Lowest effective dose; review the need. Potassium-sparing strategy only after checking renal function and interacting drugs, with specialist or formulary advice	Thiazide for hypertension: review the drug choice (NG136). An ACE inhibitor or ARB raises K⁺
Low magnesium	Correct at the same time; preparation and dose per the formulary and SPC. Hospital advice if severe, symptomatic or renal impairment	K⁺ often stays low until magnesium is corrected; oral magnesium can cause diarrhoea
GI loss	Treat the cause; stop laxatives; oral rehydration	Cannot keep oral treatment down → same-day assessment
Primary aldosteronism suspected	Endocrine advice **before** starting spironolactone: it interferes with testing	Adenoma may be curable by surgery
Purging, eating disorder	Ask without judgement; assess physical risk with MEED; refer the eating-disorder service (NG69)	Replacing K⁺ without addressing purging repeats the result. Watch for refeeding (CG32)

Modified-release potassium chloride tablets are less suitable: more GI irritation (SPS). **Hospital only:** IV potassium, premixed and glucose-free, by infusion pump, never as a bolus. Potassium-containing salt substitutes are not a treatment for hypokalaemia.

9 SAFETY-NET AND SELF-CARE — SAY IT, THEN DOCUMENT IT

SPS 2024

CALL 999
Chest pain, fainting, palpitations with feeling faint or breathless, cannot walk, breathing difficulty, dark urine with severe muscle pain.

PRACTICE TODAY
Cramps or weakness worse despite the tablets, palpitations, vomiting or diarrhoea stopping you taking them.

TAKING SANDO-K
Dissolve fully in water; take with or after food; do not chew. Tell us about side effects.

AVOID
Liquorice in quantity; laxatives you do not need. Normal balanced diet with fruit and vegetables.

10 FOLLOW-UP AND MONITORING CHECKLIST

SPS 2024 · NG136

- ☐ **Recheck U&E and magnesium** soon after starting replacement or changing the diuretic; earlier for moderate
- ☐ **Target**: K⁺ back within the reference range
- ☐ **Stop** potassium once in range and the cause is fixed; continue only for ongoing losses
- ☐ Recheck after stopping or after any new ACE inhibitor, ARB or potassium-sparing drug
- ☐ Over-replacement risk: CKD, ACE inhibitor, ARB, MRA, amiloride
- ☐ Long-term diuretic: U&E after each dose change and in illness, then at individualised intervals
- ☐ Recurrent without a cause, or with hypertension → investigate
- ☐ Cause recorded in the problem list

11 WHEN AND WHERE TO REFER

SPS 2024 · NG69 · NG12

999	Arrhythmia, chest pain, syncope, collapse, paralysis, breathing difficulty; unstable
Same day	<2.5; significant ECG change; 2.5–2.9 with risk features; vomiting; dehydration
Endocrinology	Hypertension with low K⁺ and raised ARR; Cushingoid; thyrotoxic paralysis; persistent low K⁺ off diuretics
Nephrology	RTA; Bartter or Gitelman; refractory despite K⁺, Mg and diuretic change
Eating disorder	Suspected purging (NG69)
Cancer pathway	Ectopic ACTH: 40+ with NG12 symptoms → urgent chest X-ray

12 SPECIAL SITUATIONS

SPS 2024 · NG69 · CG32

- **On digoxin or anti-arrhythmics**: low K⁺ raises the arrhythmia risk even at a usual digoxin level; ECG, and a lower threshold for same-day assessment.
- **Heart failure on a loop diuretic**: balance fluid against potassium; adding or raising an ACE inhibitor, ARB or MRA changes the potassium, so recheck after every change.
- **Refeeding** (very low weight, eating disorder, alcohol, prolonged poor intake): potassium, phosphate and magnesium can fall fast once feeding restarts (CG32).
- **Thyrotoxic or familial periodic paralysis**: the low K⁺ is a shift, so replace cautiously and watch for rebound hyperkalaemia; treat the thyroid.

13 PITFALLS AND NEVER-EVENTS

SPS 2024 · NG136

- **Never** give IV potassium in primary care.
- **Never** reassure because the patient feels well or the ECG is normal.
- "Refractory" before the magnesium is checked and corrected.
- "It's the diuretic" without the BP, bicarbonate and magnesium.
- A diuretic explanation in a young normotensive patient: ask about purging.
- Leaving potassium running indefinitely; adding an ACE inhibitor on top.

14 SAY IT LIKE THIS

plain English

"Your potassium is a little low. Potassium keeps your muscles and heart working properly. Your water tablet is the most likely reason. These dissolving tablets will top it up, and we'll recheck the blood soon. If you get palpitations with feeling faint, chest pain or your legs go weak, call 999."

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Hypomagnesaemia: the low magnesium CLINIC GUIDE

Serum Mg below 0.7 mmol/L, or a low K^+ or Ca^{2+} that will not correct, in adults 18+. Not for children, hospital IV treatment, obstetric magnesium sulfate or refeeding.

NHS SPS treating acute hypomagnesaemia 2024
MHRA Drug Safety Update, PPIs (2012) · NICE CG184 · CG32
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/magnesium-deficiency.html

PAGE 1 · INTERPRET — is it dangerous? → is it real? → how low → gut, kidney, drug or shift?

! ACT TODAY — THE ECG AND SYMPTOMS OUTRANK THE NUMBER

SPS 2024 · resuscitation guidance

TIER	RESULT OR FEATURE	ACTION
999	Low Mg with **torsades de pointes**, ventricular or haemodynamically significant arrhythmia, or **significant QT prolongation** · syncope or collapse with a suspected electrolyte arrhythmia · **seizure, tetany** or severe neuromuscular symptoms · weakness with dysphagia or breathing difficulty	Emergency transfer, **whatever the exact level**. IV magnesium is hospital-only
Same day	Mg **<0.5 mmol/L** (severe), or any symptomatic low Mg · low K^+ or Ca^{2+} that will not correct · AKI, eGFR <30, dialysis or oliguria (magnesium accumulates) · cannot take or absorb oral treatment · suspected **refeeding** (CG32)	Same-day hospital assessment; ECG and K^+, Ca^{2+} with it. Refeeding: not an isolated magnesium plan

On cisplatin, amphotericin or aminoglycosides: replacement is directed by the prescribing specialist team. Pregnancy: manage with the obstetric team.

1 CONFIRM IT IS REAL

repeat if unexpected or discordant

- **Serum Mg is about 1% of body magnesium**: a normal level does not exclude depletion, and a low level often means a large deficit.
- **EDTA contamination** (wrong tube order): falsely low Mg and Ca with a very high K^+.
- **Haemolysis** releases magnesium from red cells and can mask a low level.
- **Low albumin** lowers total Mg; interpret with the albumin.
- **Repeat with U&E, adjusted Ca^{2+}, phosphate** if unexpected; do not delay action when the ECG or symptoms are concerning.

2 SEVERITY BANDS

SPS 2024 · use local ranges

MG MMOL/L	MEANS	ACTION AND TIMING
0.5–0.7 mild	Often no symptoms; cramps, tremor	If stable, eGFR ≥30 and no red flag: **oral replacement** in primary care and treat the cause
<0.5 severe	Arrhythmia, seizures, tetany; low K^+ and Ca^{2+}	**Same-day** hospital assessment; IV in hospital if symptomatic or oral unsuitable
Any level + arrhythmia, QT, seizure	Life-threatening	**999**

Remote: a level below 0.5, or refractory low K^+ or Ca^{2+}, needs same-day assessment, not a message.

3 WHY MAGNESIUM MATTERS TO THE OTHER RESULTS

SPS 2024

POTASSIUM
Low Mg makes the kidney waste K^+: potassium stays low until magnesium is replaced.

CALCIUM
Low Mg blocks PTH release and action: calcium alone will not correct a low Ca^{2+}.

THE HEART
Long QT, torsades; digoxin toxicity at a "therapeutic" level. Check digoxin and Mg together.

PHOSPHATE
Low with refeeding and alcohol: check it with every low magnesium.

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Long-term PPI	**PPI**: gut absorption (MHRA)	PPI indication, dose, duration
Loop or thiazide diuretic	**Renal loss**	K^+, drug review
Alcohol, poor diet	**Intake + renal loss**	K^+, phosphate, Ca^{2+}, LFTs
Chronic diarrhoea, IBD, stoma, bariatric surgery	**GI loss**	Coeliac serology if malabsorption
Cisplatin, ciclosporin, tacrolimus, aminoglycoside	**Drug renal wasting**	Specialist team
Young, low K^+, alkalosis, low urine Ca	**Gitelman** (or Bartter)	Nephrology
Poorly controlled diabetes	**Glycosuria**	Glucose, HbA1c
Starvation, feeding restarted	**Refeeding**	Same day (CG32)

5 WORK-UP CHECKLIST

SPS 2024 · SPCs

- ☐ **U&E, eGFR**: before any replacement
- ☐ **Adjusted Ca^{2+}**, albumin
- ☐ **ECG**: symptoms, low K^+, QT drugs, heart disease
- ☐ **Every medicine**: PPI, diuretic, laxatives
- ☐ Coeliac serology if malabsorption
- ☐ **K^+**: often low too
- ☐ **Phosphate**
- ☐ Glucose or HbA1c if relevant
- ☐ Alcohol, diarrhoea, nutrition
- ☐ Digoxin level if on digoxin and unwell

Chvostek's sign is non-specific; Trousseau's is more specific. Urine magnesium or fractional excretion helps separate renal from gut loss but is a specialist-level test for persistent, unexplained cases.

6 READING THE ECG

SPS 2024

Look for	Long QT, long PR, broad QRS, flat T waves
Danger	Torsades de pointes, ventricular ectopics, VT: 999
Context	QT drugs, low K^+, digoxin, heart disease

7 SAFE IN PRIMARY CARE ONLY IF — EVERY BOX

SPS 2024

- ☐ Mg 0.5–0.7, stable; no Act-today feature
- ☐ eGFR ≥30; ECG done where indicated and normal
- ☐ K^+, Ca^{2+}, phosphate checked; cause acted on
- ☐ Repeat booked; safety-net given and recorded

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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8 MANAGEMENT OF THE COMMONEST CAUSES

SPS 2024 · SPCs · MHRA 2012 · CG184

CAUSE	FIRST LINE	CHECK AND CONTINUE
Replacement regimen mild, stable, eGFR ≥30	**Magnesium aspartate** 10 mmol sachet: **1–2 sachets daily** (10–20 mmol; SPC), or **magnesium glycerophosphate** 4 mmol chewable: **1–2 tablets three times a day** (12–24 mmol; SPC). SPS: **10–24 mmol a day** in divided doses. No meaningful difference between salts: choose by formulary and tolerance	Start low, with food; **diarrhoea limits the dose** and can worsen losses. Glycerophosphate is contraindicated if GFR <30 or phosphate is high. Check the SPC for interaction intervals
PPI	Review the indication; lowest effective dose; step down or stop where the indication allows (an H2-receptor antagonist may suit some)	Barrett's, severe oesophagitis or high-risk gastroprotection: continue with oral Mg and periodic levels
Loop or thiazide diuretic	Lowest effective dose; do not switch loop to thiazide just for magnesium	Amiloride only with specialist advice: hyperkalaemia risk
Alcohol	Treat the alcohol use; replace documented low Mg; thiamine per protocol (Pabrinex contains no magnesium)	Check K^+, phosphate, Ca^{2+} and nutrition; watch for refeeding
GI loss	Treat the cause; investigate malabsorption	Oral Mg may worsen diarrhoea: lower, divided doses
Low K^+ or Ca^{2+} with it	Replace magnesium first or alongside	Refractory despite replacement → same-day hospital

Hospital only: IV magnesium sulfate with ECG, BP, renal and electrolyte monitoring; IM magnesium is painful and not for routine general practice. **Do not** prescribe magnesium for non-specific cramps, diabetes, or bone health without a documented low level.

9 SAFETY-NET AND SELF-CARE — SAY IT, THEN DOCUMENT IT

SPS 2024

CALL 999
Palpitations with dizziness, fainting or collapse; a fit; painful muscle spasms; new weakness with trouble swallowing or breathing.

PRACTICE TODAY
Cannot take or keep down the magnesium; worsening cramps, tremor or palpitations.

TAKING IT
With food, in divided doses. Diarrhoea → tell us; a lower dose often helps. Come for the repeat blood test.

DIET
Nuts, seeds, wholegrains, pulses and green vegetables help, but do not replace treatment of a low level.

10 FOLLOW-UP AND MONITORING CHECKLIST

SPS 2024 · SPCs · MHRA

- ☐ **Repeat Mg** with K^+, Ca^{2+} and renal function after about **7 days** of oral treatment (SPS)
- ☐ **Target**: Mg within the laboratory range, with K^+ and Ca^{2+} corrected
- ☐ **Stop**: transient cause → 1–2 days after the level normalises (SPS); continue if the cause persists
- ☐ Long-term supplement: Mg every **3–6 months** (SPC), more often in CKD
- ☐ Long-term PPI: consider Mg before and periodically, especially with digoxin or diuretics (MHRA)
- ☐ Annual PPI review with a step-down trial (CG184)
- ☐ Low despite tolerated oral Mg → adherence, diarrhoea, renal loss, all drugs
- ☐ Cause and PPI indication recorded

11 WHEN AND WHERE TO REFER

SPS 2024 · CG32

999	Torsades, arrhythmia, significant QT, syncope, seizure, tetany, breathing or swallowing weakness
Same day	<0.5; symptomatic; refractory low K^+ or Ca^{2+}; AKI or eGFR <30; cannot take oral; refeeding
Nephrology	Unexplained, persistent, recurrent or familial; low K^+ with alkalosis (Gitelman, Bartter); renal advice before replacement in CKD 4–5, dialysis or transplant
Gastro	Uninvestigated malabsorption; post-bariatric; PPI cannot stop and Mg stays low
Endocrine	Primary aldosteronism

12 SPECIAL SITUATIONS

SPS 2024 · MHRA · CG32 · SPCs

- **On digoxin**: low magnesium (and low K^+) increases digoxin toxicity at a usual level; check both if nausea, visual change, bradycardia or confusion.
- **QT-prolonging drugs** (some antipsychotics, antimicrobials, methadone, anti-arrhythmics): correct magnesium and potassium first; ECG.
- **CKD**: magnesium is excreted by the kidney and accumulates; seek renal advice before replacement in CKD 4–5, dialysis or transplant.
- **Alcohol dependence, eating disorders, very poor intake**: think refeeding before restarting nutrition; same-day assessment if high risk (CG32).

13 PITFALLS AND NEVER-EVENTS

SPS 2024 · MHRA

- Calling K^+ or Ca^{2+} "refractory" before magnesium is checked and replaced.
- **Never** start oral magnesium without an eGFR.
- **Never** give IV or IM magnesium in general practice.
- Trusting a "therapeutic" digoxin level when Mg is low.
- Adding a QT-prolonging drug before low Mg is corrected.
- Assuming a long-term PPI still has an indication.

14 SAY IT LIKE THIS

plain English

"Your magnesium is low. It's a mineral your muscles, nerves and heart need, and it also helps keep your potassium and calcium right. Your indigestion tablet may be stopping you absorbing it, so we'll look at that and give you a supplement. If it upsets your stomach, tell us and we'll lower the dose."

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Hypercalcaemia: the raised calcium

CLINIC GUIDE

Albumin-adjusted calcium above 2.6 mmol/L in adults 18+. Not for children (paediatric or endocrine advice), pregnancy (specialist MDT) or the acute hospital treatment itself.

NICE **NG132** (2019) · NICE **NG12** (myeloma)
Society for Endocrinology emergency guidance (2016)
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/hypercalcaemia.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → PTH-mediated or not?

! ACT TODAY — THE LEVEL AND THE PATIENT

Society for Endocrinology · NG12

TIER	RESULT OR COMBINATION	ACTION
Emergency	Adjusted Ca **>3.5 mmol/L**, however well (dysrhythmia and coma risk) · **any level** with confusion, reduced GCS, seizure, arrhythmia, severe dehydration, AKI or falling urine output, persistent vomiting or unable to keep fluids down · on **digoxin** with arrhythmia or toxicity concern · back pain with leg weakness (cord compression)	Emergency transfer (999 if confusion, collapse or arrhythmia); hand over the calcium trend, eGFR, drugs, observations
Same day	Adjusted Ca **3.0–3.5 mmol/L** · symptomatic at **any level below 3.0** · substantial rise on previous results · suppressed PTH with concerning symptoms or malignancy features · postural hypotension and hyponatraemia (adrenal insufficiency)	Same-day assessment or specialist discussion; hospital if symptomatic, rising, dehydrated, AKI, malignancy, abnormal ECG or vomiting
Suspected cancer pathway	Hypercalcaemia with bone or back pain, anaemia, renal impairment or raised ESR/plasma viscosity → FBC, calcium, ESR/PV, **protein electrophoresis + serum free light chains**	Results suggest myeloma → suspected cancer pathway referral (NG12)

Severity is never read off the number alone: a modest calcium in a sick patient is still an emergency. Never delay urgent assessment for a repeat sample.

1 CONFIRM IT IS REAL

NG132 · use local laboratory ranges

- Use the **laboratory-reported albumin-adjusted calcium**, not a hand-applied formula: methods differ between laboratories. Normal is about **2.2–2.6 mmol/L**.
- **Repeat at least once** if ≥2.6 mmol/L, or ≥2.5 with features of PHPT. Timing follows the level and symptoms; act on the first result if high or symptomatic.
- Spurious or transient rise: prolonged tourniquet or venous stasis, dehydration, a sample taken during acute illness. Compare with previous results.
- **Do not** request ionised calcium routinely for PHPT (NG132); discuss with the laboratory only if adjustment is unreliable (critical illness, severe low albumin).
- Measure **PTH** once the raised calcium is confirmed: random, non-fasting, with a calcium on the same sample. Do not defer it for months.

2 SEVERITY BANDS

Society for Endocrinology · NG132

ADJUSTED CA	MEANS	ACTION AND TIMING
2.6–2.99	Mild; often asymptomatic	Repeat, then PTH; outpatient work-up if well. Symptomatic → same day
≥2.85 in PHPT	NG132 surgical referral feature	Refer to a parathyroid surgeon
3.0–3.5	Moderate; tolerated if the rise was slow, often needs treatment	**Same-day** assessment or specialist discussion
>3.5	Severe; urgent correction needed	**Emergency** hospital assessment, even if well

Symptoms, rate of rise, AKI and dehydration move a patient up a band. Below 3.0 usually needs no urgent correction.

3 READ THE PTH AGAINST THE CALCIUM

NG132

Raised or non-suppressed	PTH-mediated: PHPT (commonest outpatient cause), FHH, tertiary (CKD), lithium. Specialist interpretation
Above the midpoint, PHPT suspected	Seek endocrine advice
Below the midpoint, Ca ≥2.6	Seek specialist advice
Suppressed (below lab range)	Non-PTH-mediated: malignancy, myeloma, vitamin D excess, granulomatous disease. Urgent parallel work-up

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Raised PTH, low phosphate, stones	**Primary HPT**	25-OH vitamin D, eGFR; NG132 assessment
Lifelong mild rise, family history	**FHH**	Urine calcium excretion (ratio <0.01 is a clue only)
Longstanding CKD	**Tertiary HPT**	Renal team
Thiazide, calcium or vitamin D, antacids	**Drugs, milk-alkali**	Drug review; recheck after stopping
On lithium	**Lithium-associated HPT**	PTH; prescriber and endocrine advice
Weight loss, sweats, nodes, known cancer	**Malignancy**	FBC, LFTs, CXR; NG12 criteria
Back pain, anaemia, renal impairment	**Myeloma**	ESR/PV, electrophoresis, free light chains
Hilar nodes, erythema nodosum	**Sarcoidosis**	CXR; respiratory assessment
Weight loss, tremor, tachycardia	**Thyrotoxicosis**	TSH
Postural hypotension, low sodium	**Adrenal insufficiency**	Same-day assessment
Bed rest, spinal injury, Paget's	**Immobility**	ALP; a contributor, not a diagnosis

Malignancy is the commonest cause in hospital; PHPT is the commonest in outpatients. A suppressed PTH means urgent evaluation, not an automatic cancer referral.

5 WORK-UP CHECKLIST

NG132 · NG12

- ☐ **Adjusted Ca + albumin**: confirm
- ☐ **U&E, eGFR**: AKI, CKD
- ☐ Magnesium, ALP, LFTs
- ☐ **25-OH vitamin D**: deficiency masks PHPT
- ☐ **Electrophoresis + free light chains** if myeloma fits
- ☐ CXR if cancer or sarcoid suspected
- ☐ **Intact PTH** + same-sample Ca
- ☐ **Phosphate**: low in PHPT
- ☐ **FBC**, ESR or plasma viscosity
- ☐ TSH
- ☐ **ECG** if ≥3.0, rising or symptomatic
- ☐ Urine calcium if FHH considered

Low phosphate with a non-suppressed PTH supports PTH-mediated disease; a raised phosphate or ALP points elsewhere but is not diagnostic.
Not routine in primary care: PTHrP, 1,25-dihydroxyvitamin D, ACE, ionised calcium, sestamibi or neck imaging (specialist-directed). ECG: short QT or arrhythmia supports severity; no fixed QTc cut-off.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

NG132 · Society for Endocrinology · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Acute or severe (any cause)	Arrange transfer and hand over. Hospital starts **monitored IV saline**; bisphosphonate, denosumab, calcitonin, steroids and dialysis are specialist choices (doses not given here)	**No IV fluids in the community** without monitoring and a protocol. A loop diuretic is **not** calcium-lowering treatment
Mild PHPT, no surgical feature	Conservative surveillance in primary care. **Normal dietary calcium** (restriction raises PTH). Adequate hydration within cardiac and renal limits	Annual adjusted Ca + eGFR; DXA at diagnosis and every **2–3 years**; renal ultrasound at diagnosis. Review surgical criteria yearly
PHPT + low vitamin D	Correct the deficiency, monitoring calcium and renal function	Seek specialist advice first if calcium is markedly raised
PHPT, surgery declined or unsuitable	Cinacalcet, **endocrinology-initiated**: Ca ≥2.85 with symptoms, or ≥3.0 with or without (NG132)	Specialist monitoring. It does not improve bone density or prevent stones
Thiazide, supplements	Stop the thiazide or vitamin D where appropriate; **review** calcium supplements needed for osteoporosis	Recheck adjusted Ca. Still raised → investigate for unmasked PHPT
Lithium	**Never stop it unilaterally**: contact the prescriber or psychiatry team	Endocrine assessment if hypercalcaemia persists
Malignancy, sarcoid	Oncology- or respiratory-led treatment; steroids are a specialist decision	Monitor calcium and eGFR as their plan directs

7 CONFIRMED PHPT — THE REFERRAL DECISION

NG132

REFER TO SURGEON
Symptoms of hypercalcaemia, renal stones, fragility fracture or osteoporosis, or adjusted Ca ≥2.85 mmol/L.

NONE OF THESE
Still consider surgical referral; age under 50 is a specialist shared-decision point, not an NG132 criterion.

SURVEILLANCE
Asymptomatic and stable: annual Ca + eGFR, DXA every 2–3 years, revisit referral each year.

NO SURGERY
Declined, unsuitable or failed → endocrinology; cinacalcet only if Ca thresholds are met.

8 SELF-CARE IN STABLE MILD DISEASE

NG132

- **Hydration**: drink normally and avoid dehydration, within cardiac and renal limits; illness with vomiting → seek advice early.
- **Keep normal dietary calcium**: cutting out dairy raises PTH. No unprescribed calcium or vitamin D.
- Stay active; avoid immobility and excess alcohol. Weight-bearing exercise for bone health.
- Stone-former: fluids within limits, less salt, moderate animal protein; recurrent stones → urology.

"Your calcium is a little high. The commonest reason is a small, non-cancerous overactive gland in the neck, but we need a second blood test to check the hormone that controls calcium. Drink normally, do not cut out dairy, and call 999 if you become confused or drowsy, or get seen the same day if very thirsty or being sick."

9 FOLLOW-UP AND MONITORING

NG132

- ☐ **Mild PHPT**: adjusted Ca + eGFR **annually**; decide each year: survey or refer
- ☐ **DXA** (spine, hip, distal radius) at diagnosis and every **2–3 years**
- ☐ **Renal ultrasound** at diagnosis, then only if a stone is suspected
- ☐ Drug stopped → recheck Ca; not normal → investigate for PHPT
- ☐ After discharge: Ca, U&E, drugs reconciled, specialist follow-up confirmed
- ☐ After parathyroidectomy: surgical team's plan for low calcium ("hungry bone") recorded
- ☐ **Same day** if a routine Ca returns **≥3.0**, or symptoms at any level
- ☐ Not NG132 recalls: routine annual 24-hour urine Ca, routine 2–3-yearly renal scans

10 WHEN AND WHERE TO REFER

NG132 · NG12

Emergency	>3.5 · any level with confusion, arrhythmia, AKI, severe dehydration or vomiting · digoxin toxicity concern
Same day	3.0–3.5 · symptomatic below 3.0 · suppressed PTH with worrying features
Suspected cancer	Tests suggest myeloma, or other NG12 criteria are met; not for a low PTH alone
Parathyroid surgeon	Confirmed PHPT with symptoms, renal stones, fragility fracture or osteoporosis, or Ca **≥2.85**; consider even without these
Endocrinology	PHPT not meeting surgical criteria, possible FHH, lithium or tertiary HPT, unclear PTH
Respiratory	Suspected sarcoidosis or other granulomatous disease

11 PITFALLS AND NEVER-EVENTS

NG132

- **Never** stop lithium unilaterally · never start community IV fluids unmonitored · never use a loop diuretic to lower calcium.
- **Never** refer every suppressed PTH straight to a cancer pathway, or miss myeloma in hypercalcaemia with back pain or anaemia.
- Do not use **bisphosphonates** to lower calcium long term in PHPT (NG132) · do not restrict dietary calcium · no blanket "drink 2–3 L a day" in heart failure, CKD or frailty.
- Low vitamin D can mask PHPT: recheck calcium after correcting it.
- A single low urine calcium ratio does not diagnose FHH or rule out surgery · do not order ACE or PTHrP to "screen".

12 SAFETY-NET AND SPECIAL GROUPS

Society for Endocrinology · NG132

- **999**: confusion, drowsiness or collapse · seizure · palpitations or chest pain · vomiting and unable to keep fluids down · back pain with leg weakness. **Same day**: worsening weakness or lethargy, unable to eat or drink for more than a day, severe bone pain, much less urine.
- **Pregnancy**: specialist (endocrine and obstetric) advice; do not start cinacalcet or bisphosphonates. Offer parathyroid surgery to women with PHPT who are considering pregnancy (NG132). **Children**: paediatric or endocrine advice; adult thresholds do not apply.
- **Digoxin**: hypercalcaemia potentiates toxicity. **CKD**: tertiary HPT is usually on the renal pathway. **Known cancer**: calcium and any antiresorptive follow the oncology plan.

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Hypocalcaemia: the low calcium

CLINIC GUIDE

Albumin-adjusted calcium below the reference range in adults 18+. Not for children or neonates, pregnancy (specialist advice), dialysis (renal team) or hospital IV treatment.

Society for Endocrinology emergency (2016)
MHRA · SPC (denosumab) · NICE **NG203** · NG12
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/hypocalcaemia.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → what does the PTH say?

! ACT TODAY — SYMPTOMS BEAT THE NUMBER

Society for Endocrinology

TIER	RESULT OR COMBINATION	ACTION
999	**Tetany** or carpopedal spasm · **laryngospasm or stridor** · **seizure** · palpitations with syncope, marked QT prolongation or ventricular arrhythmia · stridor and neck swelling after neck surgery (haematoma: airway emergency)	Emergency admission for **IV calcium gluconate** with ECG monitoring. Never attempt IV calcium in the surgery
Emergency	Adjusted Ca **<1.90 mmol/L**, even if asymptomatic · suspected **acute pancreatitis** at any calcium	Same-day hospital assessment; oral calcium alone is too slow. Never leave it in an inbox overnight
Same day	Adjusted Ca **1.90–2.09** newly found · perioral tingling, fingertip paraesthesia or cramps at any level · **after thyroid or parathyroid surgery** with any tingling or cramps · hypoparathyroidism with vomiting or diarrhoea	Face-to-face assessment with an **ECG** (not by telephone); contact the operating or endocrine team

Symptoms usually start below about 1.9 mmol/L but depend on how fast calcium falls. Correct low magnesium and potassium too: they add to the arrhythmia risk.

1 CONFIRM IT IS REAL

use local laboratory ranges

- **Low albumin** lowers total calcium while ionised calcium is normal: read the **laboratory's adjusted calcium**, never the total alone.
- **EDTA contamination** (wrong tube order or decanting) gives a spuriously very low calcium with a very high potassium: repeat before acting if the patient is well.
- Hyperventilation (alkalosis) lowers ionised calcium and causes tingling with a normal total.
- **Ionised calcium** (blood gas) if unwell, critically ill, or the result does not fit the picture.
- **When to repeat**: symptoms or <1.90 → do not wait for a repeat; 1.90–2.09 → same-day repeat with assessment; mild and well → repeat with the work-up bloods. Never prescribe calcium on one total calcium.

2 SEVERITY BANDS

<1.90 severe: Society for Endocrinology · local lab range

ADJUSTED CA	MEANS	ACTION AND TIMING
2.10 to lower limit	Mild; usually asymptomatic	Work-up below; treat the cause (often vitamin D deficiency)
1.90–2.09	Mild to moderate	**Same day**: face-to-face, ECG, repeat
<1.90	Severe	**Emergency** admission for IV calcium
Any level + symptoms	Treat as severe	Tetany, seizure, stridor → **999**

Normal range is laboratory-specific, commonly about 2.20–2.60 mmol/L. In acute pancreatitis a calcium <2.0 marks severe disease (Glasgow–Imrie).

3 CLINICAL CLUES AT THE DESK

examination

Early	Perioral tingling, fingertip paraesthesia, cramps
Severe	Carpopedal spasm, laryngospasm, seizures, confusion
Signs	Chvostek (facial twitch on tapping) and Trousseau (spasm with BP cuff inflated): more often positive in acute than chronic
ECG	Prolonged QT from a long ST segment; read the QTc against local thresholds
Neck scar	Thyroid or parathyroid surgery, even years ago: think hypoparathyroidism
Chronic	Cataracts, dry skin, brittle nails, basal ganglia calcification

4 CAUSES — READ THE PTH

clue → think → first test

CLUE	THINK	FIRST TEST
High PTH, low phosphate, raised ALP, risk group	**Vitamin D deficiency** (commonest)	25-OH vitamin D
High PTH, high phosphate, low eGFR	**CKD**	U&E, phosphate trend
High PTH, diarrhoea, weight loss, bariatric surgery	**Malabsorption**	Coeliac serology; vitamin D
Low or normal PTH, recent neck surgery	**Post-surgical hypoparathyroidism** (commonest low-PTH cause)	Urgent Ca, PTH; operating team
Low or normal PTH, no surgery	**Autoimmune or familial hypopara**	Mg; endocrinology
Alcohol, PPI, diuretic, cisplatin, diarrhoea	**Low magnesium**	Magnesium; recheck PTH after
Recent denosumab or bisphosphonate	**Drug-induced**	Ca, eGFR, vitamin D
After parathyroidectomy	**Hungry bone**	Surgical team's plan
Abdominal pain, vomiting	**Acute pancreatitis**	Same-day admission
Bone pain, raised ALP, high phosphate, weight loss	**Osteoblastic metastases** (prostate, breast)	PSA or breast exam; NG12 criteria
Anticonvulsants, rifampicin	**Vitamin D breakdown**	25-OH vitamin D

High PTH = the parathyroids are responding (secondary). Low or inappropriately normal PTH = the parathyroids are failing (hypoparathyroidism): specialist territory.

5 WORK-UP CHECKLIST

send together

- ☐ **Adjusted Ca + albumin**: confirm
- ☐ **Magnesium**: low Mg blocks PTH
- ☐ **25-OH vitamin D**
- ☐ ALP: raised in vitamin D deficiency
- ☐ Coeliac serology if malabsorption
- ☐ **Intact PTH**: the branch point
- ☐ **Phosphate**: high in hypopara, CKD
- ☐ **U&E, eGFR**: CKD
- ☐ **ECG** if symptomatic, acute or severe
- ☐ Drug history: denosumab, PPI

25-OH vitamin D: <25 nmol/L deficient · 25–50 may be inadequate in some · >50 sufficient. 1,25-dihydroxyvitamin D is specialist-directed.

Vitamin D risk groups: darker skin, housebound or care home, pregnancy and breastfeeding, malabsorption, anticonvulsants or rifampicin.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

Society for Endocrinology · ROS · NG203 · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Severe or symptomatic	999 or immediate transfer. Hospital gives IV calcium gluconate with ECG monitoring, then an infusion	Hospital also corrects magnesium. No IV calcium in primary care
Vitamin D deficiency (high PTH, mild, well)	**Replacement**: colecalciferol loading, about **300,000 IU** over 6–10 weeks (e.g. 50,000 IU weekly for 6 weeks), then **maintenance** 800–2,000 IU daily. Calcium if dietary intake is poor	Recheck adjusted Ca, PTH and phosphate at **2–4 weeks**. First exclude hypercalcaemia, PHPT, granulomatous disease, stones
Low magnesium	**Correct magnesium first**: calcium will not hold until it is replaced. Review PPI, diuretic, alcohol	Recheck Mg and PTH after. Severe, symptomatic or renal impairment → hospital
Mild, stable, cause known	Oral calcium per local formulary: **carbonate with meals**; citrate if on a PPI or achlorhydria. Start vitamin D at the same time	Review at **2–4 weeks**; target adjusted Ca ≥2.1 mmol/L. Not responding → endocrinology
CKD	Correct vitamin D deficiency with colecalciferol or ergocalciferol (NG203)	Alfacalcidol or calcitriol only renal-led, especially G4–G5 or dialysis
Hypoparathyroidism	**Endocrinology-initiated** calcitriol or alfacalcidol plus calcium; cholecalciferol alone does not work	Low-normal calcium target and urine calcium per specialist plan; overtreatment harms kidneys

7 SELF-CARE AND THE CONVERSATION

NHS · ROS

Diet: 700 mg calcium a day for adults (milk, cheese, yoghurt, fortified plant milks, tinned fish with bones, calcium-set tofu). **Vitamin D**: everyone should consider 10 micrograms (400 IU) daily in autumn and winter; all year if rarely outdoors. Sources: oily fish, egg yolk, fortified foods.

"Your calcium is a bit low, most likely because you are short of vitamin D, which helps you absorb calcium. The vitamin D course should fix it, and we will recheck your blood in a few weeks. If you get tingling round your mouth, cramps or twitching hands, get seen the same day."

8 DRUGS THAT LOWER CALCIUM — AND THE FIX

MHRA · SPC

DRUG	HOW	WHAT TO DO
Denosumab	Blocks bone resorption; highest risk with low vitamin D or CrCl <30	Correct calcium and vitamin D before starting; co-prescribe calcium and vitamin D
Bisphosphonates	Reduce bone resorption	Correct hypocalcaemia before giving; check renal function
PPIs (long term)	Low magnesium; less calcium absorbed	Review need; consider an H2 blocker if magnesium stays low
Anticonvulsants, rifampicin	Faster vitamin D breakdown	Check 25-OH vitamin D; supplement
Loop diuretics, cisplatin, cinacalcet, foscarnet	Calcium or magnesium loss; PTH suppression	Check Mg; discuss with the prescriber or specialist

9 FOLLOW-UP AND MONITORING

SPC · MHRA · specialist plan

- [ ] **Mild**: adjusted Ca, PTH, phosphate at **2–4 weeks**; target ≥2.1; stop calcium when the cause is fixed and intake is adequate
- [ ] After loading: 25-OH vitamin D only if adherence, malabsorption or severe deficiency is a concern
- [ ] **Denosumab**: Ca, eGFR, vitamin D before starting; Ca **before each dose**; extra check **within 2 weeks** of the first dose if predisposed (CrCl <30, dialysis)
- [ ] Prescribe calcium and vitamin D alongside denosumab
- [ ] **Hypoparathyroidism**: Ca, eGFR, urine calcium per specialist plan; dose changes specialist-led
- [ ] **After neck surgery**: follow the operating team's plan

10 WHEN AND WHERE TO REFER

Society for Endocrinology · NG12

999	Tetany, stridor, seizure, collapse with arrhythmia
Emergency	Adjusted Ca <1.90 · suspected pancreatitis · symptomatic after neck surgery
Endocrinology (urgent)	Low or inappropriately normal PTH · pseudohypoparathyroidism · not responding to calcium and vitamin D at 6 weeks
Renal	CKD with refractory hypocalcaemia or secondary HPT needing active vitamin D
Gastroenterology	Malabsorption; positive coeliac serology (keep gluten in until seen)
Suspected cancer	Bone pain, raised ALP, weight loss → NG12 criteria; aged 60+ with bone pain → myeloma tests

11 SAFETY-NET AND SPECIAL GROUPS

Society for Endocrinology · BTA

- **999**: hands or feet go into spasm · noisy or difficult breathing · fit · palpitations with fainting. **Same day**: new tingling round the mouth or fingertips, cramps, or twitching.
- **After thyroidectomy**: tingling in the first days is low calcium until proven otherwise; seizure in the first 24–48 hours is hypocalcaemia until proven otherwise. **Hypoparathyroidism**: if calcium or calcitriol cannot be kept down (vomiting, diarrhoea), contact the endocrine team or attend hospital urgently; no self-increase of calcitriol; carry a medical alert card.

12 PITFALLS AND NEVER-EVENTS

MHRA · SPC

- **Never** give oral calcium alone for Ca <1.90 or symptoms · never assess tingling or a Ca <2.10 by telephone only · never miss magnesium in "resistant" hypocalcaemia.
- **Never** start denosumab with uncorrected hypocalcaemia or vitamin D deficiency (MHRA).
- Do not treat a low total calcium caused by low albumin · do not start alfacalcidol automatically in CKD G3b.
- Calcium carbonate taken with a PPI, fasting, is poorly absorbed · take a bisphosphonate on an empty stomach, apart from calcium (SPC).

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Abnormal bone profile CLINIC GUIDE

Calcium, phosphate, ALP and PTH read together, in adults 18+. Not for children (ALP is physiologically high while growing), pregnancy (placental ALP) or dialysis (renal team).

NICE **NG132** (2019) · **NG12** · NG234
Society for Endocrinology (2016) · ROS (2020)
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/abnormal-bone-profile.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → which pattern?

! ACT TODAY

Society for Endocrinology · NG12 · NG234

TIER	RESULT OR COMBINATION	ACTION
Emergency	Adjusted Ca **>3.5 mmol/L** however well, or **≥3.0** with significant symptoms · high Ca with confusion, vomiting, dehydration, AKI or ECG change · adjusted Ca **<1.90**, or low Ca with tetany, seizure, laryngospasm, QT prolongation or arrhythmia	Same-day hospital; **999** if unstable. Stop calcium and vitamin D supplements
Same day	Adjusted Ca **3.0–3.5**, stable and asymptomatic · back pain with new neurological signs in known or suspected cancer (cord compression)	Face-to-face or specialist discussion · cord compression: acute oncology team now
Suspected cancer pathway	Raised ALP with bone pain, pathological fracture or weight loss (breast, prostate, lung, kidney, thyroid) · aged **60+** with persistent bone or back pain or unexplained fracture → FBC, Ca, ESR/PV, **protein electrophoresis + free light chains**	Suspected cancer pathway referral for the likely primary, or if tests suggest myeloma (NG12)

Persistent hypercalcaemia with a **suppressed PTH**: urgent parallel work-up for malignancy and myeloma with specialist advice; same day if unwell.

1 CONFIRM IT IS REAL

NG132 · local laboratory

- Use the laboratory's **adjusted calcium**; do not re-apply a formula to a value already adjusted. Typical range 2.20–2.60 mmol/L; phosphate 0.8–1.5 mmol/L.
- **EDTA contamination**: very low Ca with very high K (and often low ALP) → repeat on a fresh sample.
- **Haemolysis or delayed separation** can falsely raise phosphate and potassium.
- **Biotin** supplements interfere with some PTH assays: ask, and tell the laboratory.
- **Raised ALP without disease**: healing fracture, pregnancy, recent surgery. Repeat and trend; check GGT.
- **Repeat Ca at least once** if ≥2.6 (or ≥2.5 with features of PHPT). PTH once confirmed on two occasions, with a same-sample calcium. Never delay transfer of an unwell patient to repeat.

2 SEVERITY BANDS

Society for Endocrinology · NG132

RESULT	MEANS	ACTION AND TIMING
Ca **>3.5**	Severe high	**Emergency**, however well
Ca **3.0–3.5**	Moderate high	**Same day**; emergency if symptomatic
Ca **2.6–2.99**	Mild high	Repeat, then PTH; routine work-up
Ca **≥2.85** + PHPT	Surgical threshold	Parathyroid surgeon (NG132)
Ca **1.90** to lower limit	Mild to moderate low	PTH, Mg, vitamin D; same day if symptoms
Ca **<1.90**	Severe low	**Emergency** IV calcium
Isolated ALP, normal GGT	Bone source	Repeat and trend; bone work-up

3 HISTORY AND EXAMINATION

what changes the plan

Drugs	Thiazide, lithium, calcium, vitamin D or A, antacids, biotin, bisphosphonate, denosumab
Bones, stones, groans	Bone pain, renal stones, constipation, low mood: hypercalcaemia
Proximal weakness	Stand from a chair without arms: osteomalacia
Deformity	Enlarged skull, bowed legs, deafness: Paget's
Nodes, breast, prostate	Lymphoma, sarcoid, metastases

4 CAUSES — READ THE PATTERN

clue → think → first test

PATTERN	THINK	FIRST TEST
High Ca, raised or non-suppressed PTH	**Primary HPT** (commonest outpatient cause)	eGFR, 25-OH D, urine Ca; DXA
Mildly high Ca, normal PTH, low urine Ca, family history	**FHH**	Ca:Cr clearance ratio (<0.01 supports, not diagnostic)
High Ca, **suppressed PTH**	**Malignancy, myeloma**, sarcoid, vitamin D excess, thyrotoxicosis	FBC, LFTs, SPE + SFLC, CXR, TSH, 25-OH D
Low Ca, high PTH, low phosphate	**Vitamin D deficiency**, malabsorption	25-OH D; coeliac serology
Low Ca, low phosphate, high ALP, bone pain, proximal weakness, waddling gait	**Osteomalacia**	25-OH D; X-ray (Looser zones)
Low Ca, high phosphate, high PTH, low eGFR	**CKD-MBD**	Trend Ca, PO_4, PTH, ALP, eGFR
Low Ca, low or normal PTH	**Hypoparathyroidism**, low Mg	Mg; ask about neck surgery
Isolated high ALP, normal GGT, Ca, PO_4; older adult	**Paget's disease**	X-ray painful bones; not diagnosed on ALP alone
High ALP, bone pain, weight loss	**Bone metastases**	Examine breast, prostate, lung; NG12
High PTH, normal Ca	**Secondary HPT** (or normocalcaemic PHPT by exclusion)	25-OH D, eGFR, Ca intake; repeat

GGT is the quickest bone-or-liver test: raised GGT → liver ALP; normal GGT and LFTs → bone ALP. Bone-specific ALP if the pattern is unclear.

5 WORK-UP CHECKLIST

NG132 · NG12

- ☐ **Adjusted Ca**, albumin, phosphate
- ☐ **PTH** with same-sample Ca
- ☐ U&E, eGFR; magnesium
- ☐ **SPE + free light chains** if myeloma possible
- ☐ CXR if lymphoma or sarcoid suspected
- ☐ **ALP + GGT**, LFTs
- ☐ **25-OH vitamin D**
- ☐ FBC, ESR or PV
- ☐ X-ray of painful bones
- ☐ Urine Ca if FHH possible

Specialist-directed only: PTHrP, 1,25-dihydroxyvitamin D, parathyroid imaging. ACE is not a sarcoid screen; PSA only via the symptom-led prostate pathway.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

ROS · NG132 · Paget's guideline 2019 · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Vitamin D deficiency (25-OH D <25 nmol/L), osteomalacia	**Replacement** when rapid correction is needed: loading about **300,000 IU** in total, e.g. colecalciferol **50,000 IU weekly × 6 weeks** or **4,000 IU daily × 10 weeks**; then **maintenance 800–2,000 IU daily**. Calcium only if dietary intake is poor or osteomalacia	Check calcium after loading (can unmask PHPT). 25-OH D recheck only if symptomatic, malabsorption or poor adherence. PTH should settle within 3–6 months
Primary HPT	Refer if NG132 features; otherwise surveillance agreed with endocrinology. **Normal dietary calcium**; adequate hydration; review thiazides	Annual adjusted Ca + eGFR; DXA every 2–3 years
Paget's disease (specialist-led)	For **bone pain** or significant active disease, not to normalise ALP: zoledronic acid **5 mg IV** single infusion (or risedronate 30 mg daily for 2 months)	Check renal function, Ca and vitamin D first; dental review. Flu-like reaction after IV is common
CKD-MBD	Correct vitamin D deficiency; phosphate binders and active vitamin D are **renal-led**	Trend Ca, phosphate, PTH, ALP, eGFR together
Hypoparathyroidism	Alfacalcidol + calcium, **specialist-led** doses	Monitoring frequency set by endocrinology
On denosumab	Ca checked and corrected **before each dose**; adequate calcium and vitamin D	**Never stop or delay** without an alternative plan (rebound vertebral fractures)

7 ISOLATED RAISED ALP — STEP BY STEP

bone or liver?

1 · REPEAT AND TREND
Ask about recent fracture or surgery, pregnancy, bone pain, drugs, weight loss. Check GGT and LFTs.

2 · GGT RAISED
Liver source: follow the abnormal LFT pathway first.

3 · GGT NORMAL
Bone source: Ca, phosphate, 25-OH D, PTH, eGFR; bone-specific ALP if unclear. X-ray painful bones.

4 · STILL UNEXPLAINED
Persistent or marked, or bone pain → metabolic bone services. Pain with weight loss → NG12.

8 SELF-CARE AND THE CONVERSATION

NHS · ROS

Calcium: 700 mg a day from food for adults. **Vitamin D**: 10 micrograms (400 IU) daily in autumn and winter; all year if rarely outdoors. Weight-bearing exercise, no smoking, alcohol within 14 units a week.

"Your bone blood test is a bit out of balance. It is most often a simple vitamin D shortage or a small overactive gland, but I want to check a couple more things, including a hormone level, so we treat the right cause."

9 FOLLOW-UP AND MONITORING

NG132 · ROS · specialist plan

- [ ] **PHPT surveillance**: adjusted Ca + eGFR at least **annually**; DXA every **2–3 years**; renal ultrasound if a stone is suspected
- [ ] Refer when Ca reaches ≥2.85, stones, osteoporosis, fracture or symptoms
- [ ] **After vitamin D loading**: calcium; PTH still high once replete → consider coexisting PHPT
- [ ] **Paget's**: ALP and symptom response as the specialist advises; hearing loss → audiology
- [ ] **Raised ALP**: repeat and trend; still raised and non-hepatic → bone work-up
- [ ] **Denosumab**: dose never overdue; Ca before each dose

10 WHEN AND WHERE TO REFER

NG132 · NG12

Emergency	Ca >3.5, or ≥3.0 with symptoms · Ca <1.90 or tetany, seizure, stridor
Suspected cancer	Myeloma tests abnormal · raised ALP with bone pain and weight loss
Parathyroid surgeon	Confirmed PHPT with symptoms, stones, osteoporosis or fragility fracture, or Ca **≥2.85**; consider even without
Endocrinology	PHPT below surgical criteria · possible FHH · new hypoparathyroidism · PTH high after vitamin D repletion
Metabolic bone	Persistent non-hepatic ALP with bone pain; suspected Paget's
Renal	Rising PTH, persistent high phosphate or complex CKD-MBD
Respiratory	High Ca with bilateral hilar nodes (sarcoid)

11 PITFALLS AND NEVER-EVENTS

NG132 · SPC

- **Never** diagnose Paget's on ALP alone, or treat it to normalise ALP · never diagnose normocalcaemic PHPT from one raised PTH.
- **Never** use a loop diuretic to lower calcium, or restrict dietary calcium in PHPT · do not order PTHrP or ACE to screen.
- Check renal function, calcium and vitamin D before any bisphosphonate or denosumab; alendronic acid is not recommended if CrCl <35 mL/min (SPC).
- Age alone does not justify GP-only PHPT surveillance: agree it with endocrinology.
- A low urine Ca ratio does not prove FHH · parathyroid imaging localises, it does not diagnose.

12 SAFETY-NET AND SPECIAL GROUPS

tell the patient

999: confusion or drowsiness, a fit or collapse, chest pain or irregular heartbeat, tight throat or noisy breathing with tingling or cramps, back pain with new leg weakness, numbness or loss of bladder or bowel control. **Same day**: vomiting or unable to keep fluids down, severe thirst or passing much more urine, new tingling round the mouth or fingers, cramps, new bone pain or a fracture. **On a bisphosphonate or denosumab**: report thigh, groin or hip pain and jaw or mouth symptoms. If calcium is high, take no calcium or vitamin D supplements unless agreed.

- **Pregnancy**: placental ALP rises normally; do not chase it. **Young adults still growing**: bone ALP can be physiologically high.
- **On oncology bone treatment** (zoledronic acid or denosumab 120 mg monthly for metastases, distinct from 60 mg six-monthly for osteoporosis): oncology-led; dental review, calcium and vitamin D, renal function.
- **CKD G4–G5 or dialysis**: interpret calcium, phosphate and PTH trends with the renal team; no universal PTH target outside dialysis.

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Vitamin D deficiency CLINIC GUIDE

A low 25-hydroxyvitamin D in adults 18+. Not for children or rickets, treatment in pregnancy, CKD G4–G5 or dialysis, or after bariatric surgery: use those specialist pathways.

ROS vitamin D (2020) · **SACN** (2016)
NICE NG203 · NG247 · CKS
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/vitamin-d-deficiency.html

PAGE 1 · INTERPRET

is it dangerous? → should it have been tested? → how low, and why?

! ACT TODAY — CHECK THE CALCIUM FIRST

Society for Endocrinology · ROS

TIER	RESULT OR COMBINATION	ACTION
999	Low vitamin D with **tetany**, seizure, laryngospasm, stridor or arrhythmia, or adjusted Ca **<1.90 mmol/L**	Emergency admission for IV calcium; do not wait to start vitamin D. Check magnesium
Same day	**Hypercalcaemia on high-dose vitamin D** with nausea, vomiting, thirst, polyuria, confusion or renal impairment (toxicity is a calcium diagnosis, not a 25-OH D number)	Stop vitamin D and calcium; Ca, phosphate, eGFR, ECG if severe. Ca **>3.5** or confusion → emergency
Urgent	25-OH D **<25 nmol/L** with bone pain, proximal weakness, waddling gait, raised ALP, low phosphate (**osteomalacia**) · child with bowed legs or wrist widening (**rickets**)	Osteomalacia work-up + specialist metabolic bone advice (not loading alone) · urgent paediatrics

Low vitamin D with a **high calcium**: do not simply replace it. Think PHPT, malignancy, granulomatous disease (sarcoid, TB, lymphoma) and seek advice first.

1 CONFIRM IT IS REAL — AND NEEDED

ROS · SACN · NG247

- **Test only when the result changes management**: suspected osteomalacia, bone pain or proximal myopathy, malabsorption, persistent deficiency on treatment, complex osteoporosis, before IV zoledronate or denosumab per protocol.
- **Do not test** well adults just because they are dark-skinned, housebound, older, pregnant or on an enzyme inducer: give 10 micrograms (400 IU) daily without a test.
- Levels are **seasonal** (lowest after winter) and assays differ between laboratories: compare like with like.
- **1,25-dihydroxyvitamin D** is not a test of deficiency; it is specialist-directed.
- Unexpected result with a normal PTH and calcium → **repeat** before a loading course; check adherence and magnesium.

2 SEVERITY BANDS

SACN · ROS · CKS

25-OH D	MEANS	ACTION AND TIMING
<25 nmol/L	Deficient (SACN): osteomalacia risk	Check Ca, eGFR first. Treat: loading if rapid correction needed, then maintenance
25–50	May be inadequate in some	Maintenance if fracture, osteoporosis, antiresorptive planned, symptoms, raised PTH or malabsorption; else 400 IU advice
>50	Adequate for most	Reassure; public-health 10 micrograms daily
Any + high Ca	Not simple deficiency	Hold vitamin D; investigate calcium

No universal 50–75 nmol/L target; a higher target is a specialist decision (malabsorption, osteomalacia).

3 CLINICAL FEATURES THAT CHANGE THE PLAN

examination

Osteomalacia	Bone pain and tenderness, proximal weakness (trouble rising from a chair or climbing stairs), waddling gait, fractures
Low calcium	Tingling round the mouth or fingers, cramps, Chvostek or Trousseau sign
High calcium	Thirst, polyuria, nausea, confusion: suspect toxicity or another cause
Mimics	Myopathy is easily mistaken for deconditioning, PMR or fibromyalgia; CK is normal or only mildly raised
Response	Strength and pain often improve within 4–6 weeks of treatment, before the bloods normalise

4 CAUSES AND PATTERNS

clue → think → first test

CLUE	THINK	FIRST TEST
Dark skin, covered, housebound, care home, winter	**Low sunlight** (commonest)	None needed: supplement
Diarrhoea, weight loss, IBD, bariatric surgery	**Malabsorption**	Coeliac serology; specialist protocol
Carbamazepine, phenytoin, phenobarbital, rifampicin, antiretrovirals	**Enzyme induction**	Review supplement; bone-health risk
Low eGFR	**CKD**	Ca, phosphate, PTH (NG203)
Jaundice, cirrhosis	**Liver disease**	LFTs
BMI >30	**Obesity** (stored in fat)	Standard dose; recheck if persists
Low 25-OH D, high PTH, low-normal Ca, high ALP	**Secondary HPT**, osteomalacia	Phosphate, ALP; X-ray if pseudofractures
Low 25-OH D, low Ca, low or normal PTH	**Hypoparathyroidism** or low Mg	Magnesium; then endocrinology
Low 25-OH D, high Ca, suppressed PTH	**PTH-independent hypercalcaemia**	CXR, FBC, specialist advice
Low despite adequate treatment	**Adherence**, malabsorption	Ask; coeliac screen

5 WORK-UP CHECKLIST

when testing is justified

- ☐ **25-OH vitamin D**
- ☐ **Adjusted calcium**: before treating
- ☐ **Phosphate**: low in osteomalacia
- ☐ **ALP**: raised in osteomalacia
- ☐ **PTH**: secondary HPT
- ☐ **eGFR**: CKD
- ☐ LFTs: liver disease
- ☐ Magnesium if Ca low
- ☐ Coeliac serology if malabsorption
- ☐ X-ray if Looser zones suspected

Proximal myopathy: stand from a chair without using the arms, climb stairs, watch the gait.

6 BEFORE AN ANTIRESORPTIVE

MHRA · SPC

- **Denosumab**: correct hypocalcaemia and ensure adequate calcium and vitamin D first; monitor closely if eGFR <30.
- **IV zoledronate**: renal function, calcium and vitamin D before the infusion.
- **Oral bisphosphonate**: correct deficiency, but loading is not automatic if mild and treatment is urgent.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 WHICH REGIMEN?

ROS · SACN

CALCIUM HIGH
Hold vitamin D and calcium. Investigate the calcium (PHPT, granulomatous disease, malignancy) first.

LOAD, THEN MAINTAIN
<25 nmol/L with symptoms, osteomalacia, malabsorption or a potent antiresorptive imminent.

MAINTENANCE ONLY
<25 uncomplicated and not urgent, or 25–50 with a risk factor: 800–2,000 IU daily.

PUBLIC HEALTH DOSE
>50, or at-risk and untested: 10 micrograms (400 IU) daily.

8 REPLACEMENT REGIMENS — COLECALCIFEROL

ROS (Vitamin D and bone health) · SPS · SPC

LOADING OPTION 1
50,000 IU weekly × 6 weeks (300,000 IU)

LOADING OPTION 2
20,000 IU twice weekly × 7 weeks (280,000 IU)

LOADING OPTION 3
4,000 IU daily × 10 weeks (280,000 IU)

THEN MAINTENANCE
800–2,000 IU daily, long term while the risk persists

SITUATION	FIRST LINE	CHECK AND CONTINUE
<25 nmol/L, rapid correction needed	**Loading** (about 300,000 IU in total over 6–10 weeks), then maintenance. Not automatic for every level <25: weigh symptoms, planned antiresorptive, renal function, adherence	Record the total dose, duration and maintenance plan so high doses are not continued by accident (SPS)
Less urgent, or with an oral bisphosphonate	**Maintenance** 800–2,000 IU daily without loading	Continue while the risk factor persists
Public health	10 micrograms (400 IU) daily in autumn and winter; all year if rarely outdoors, covered or dark-skinned	Over the counter; no test
Calcium	Only if dietary intake is under 700 mg a day, hypocalcaemia, or an osteoporosis or osteomalacia plan	Routine high-dose calcium: constipation, hypercalcaemia, stones
CKD	Colecalciferol or ergocalciferol for nutritional deficiency (NG203)	Alfacalcidol or calcitriol only under renal guidance
Malabsorption, bariatric	Dose and route per gastroenterology or bariatric protocol	Recheck 25-OH D at about 3 months

9 FOLLOW-UP AND MONITORING

ROS · SPC

- [ ] **Adjusted calcium about 1 month** after loading ends: it may unmask PHPT
- [ ] **No routine repeat 25-OH D** in a well adult; consider at about 3 months if symptoms persist, malabsorption, osteomalacia, poor adherence
- [ ] Target: **>50 nmol/L** is adequate for most
- [ ] No blanket annual 25-OH D or PTH tests
- [ ] PTH may lag: still high once replete → check Ca, eGFR; endocrinology
- [ ] **Stop** high-dose treatment at the end of loading; continue maintenance while the risk persists, not automatically lifelong
- [ ] Before denosumab or IV zoledronate: Ca, eGFR and vitamin D corrected

10 WHEN AND WHERE TO REFER

ROS · NG203

999	Tetany, seizure, stridor, Ca <1.90
Same day	Hypercalcaemia on treatment with symptoms
Urgent paediatrics	Suspected rickets, deformity, fracture
Metabolic bone, endocrine	Suspected osteomalacia · resistant hypocalcaemia · PTH high despite repletion
Respiratory, specialist	Possible granulomatous disease: hold supplements
Gastroenterology	Persistently low despite adequate treatment
Renal	CKD with persistent secondary HPT after repletion
Obstetrics	Confirmed deficiency in pregnancy

11 SAFETY-NET AND SPECIAL GROUPS

SACN · NG247 · NHS

- **999**: a fit, severe muscle spasms, noisy or difficult breathing, or confusion. **Same day**: tingling round the mouth or fingers, or cramps; on high-dose vitamin D, nausea, vomiting, thirst or passing a lot of urine. **Review within 2 weeks** if bone pain or weakness persists or worsens.
- **Pregnancy and breastfeeding**: 10 micrograms (400 IU) daily; do not test routinely or for a risk factor alone; Healthy Start vitamins are free for those eligible. **Breastfed infants**: 8.5–10 micrograms daily from birth; formula-fed babies taking more than 500 mL a day do not need drops.
- **Care homes and housebound**: 10 micrograms daily all year, without testing. **Enzyme-inducing antiepileptics**: review supplementation and fracture risk; any drug change needs neurology input. **Bariatric surgery**: specialist protocol; calcium citrate is preferred.

12 PITFALLS AND NEVER-EVENTS

SPS · ROS

- **Never** load without checking calcium and renal function · never replace vitamin D in hypercalcaemia or suspected sarcoid without advice.
- **Never** leave a loading dose on repeat · never treat osteomalacia as "just a loading course".
- Do not use alfacalcidol or calcitriol for simple deficiency, or in CKD G3a–G5 without renal advice.
- Do not prescribe vitamin D just to prevent falls (NG249), infections or other non-bone conditions.
- Prescribe by dose in IU or micrograms, not brand.

13 SELF-CARE AND THE CONVERSATION

NHS · SACN

Few foods contain much vitamin D: oily fish, egg yolk, fortified spreads and cereals. Sunburn is never needed; supplements are safer. Tie a daily dose to an existing habit; set a reminder for weekly doses.

"Your vitamin D is low, which is very common in the UK because there isn't enough sunshine in winter. A short course of higher-dose tablets tops you up, then a small daily dose keeps you there. You don't need repeated blood tests unless you still feel unwell."

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Raised HbA1c or glucose CLINIC GUIDE

A raised HbA1c or plasma glucose in adults 18+. Not for children (same-day paediatric referral), pregnancy or gestational diabetes (antenatal pathway, NICE NG3), or review of known diabetes.

NICE **NG28** (Feb 2026) · NG17 · PH38
NG12 · JBDS DKA/HHS · MHRA · WHO 2011
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/hyperglycaemia.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → what does the number mean? → what type is it?

! ACT TODAY — KETONES, DEHYDRATION AND THE CANCER FLAG

JBDS · NG17 · NG12 · MHRA

TIER	RESULT OR FEATURE	ACTION
999	**DKA**: vomiting, abdominal pain, deep rapid breathing, drowsiness, or blood ketones **≥3.0 mmol/L** (glucose may be near-normal on an SGLT2 inhibitor) · **HHS**: very high glucose (often ≥30 mmol/L) with profound dehydration and confusion · foot infection or sepsis with systemic illness	Emergency admission. Check glucose and blood ketones now; never wait for a lab result
Same day	**Possible new type 1** at any age or BMI: weight loss, marked thirst and polyuria, onset over weeks, ketones (blood **1.6–2.9 mmol/L**) · hyperglycaemia in **pregnancy** · active foot disease (ulcer, hot swollen foot)	Same-day diabetes, antenatal diabetes or foot team. Never "metformin and review"
Urgent CT	Aged **60+** with **new-onset diabetes and weight loss** · jaundice aged 40+	NG12 urgent direct-access CT abdomen (ultrasound if CT unavailable); jaundice → suspected cancer pathway referral

1 CONFIRM IT IS REAL

NG28 · WHO 2011

- **Asymptomatic: repeat** to confirm, ideally within 2 weeks; diabetes needs a second result ≥48 mmol/mol. **Symptomatic** (thirst, polyuria, weight loss): one result is enough.
- **Random glucose ≥11.1 mmol/L** diagnoses diabetes only with symptoms. Capillary meter readings are never diagnostic: use venous plasma glucose.
- **HbA1c reflects 8–12 weeks**, so it misleads when glucose rose recently or red-cell life is altered.
- **HbA1c unreliable → use glucose**: symptoms under 2 months, suspected type 1, acute illness, steroids or antipsychotics, pancreatic damage or surgery; pregnancy; anaemia, haemolysis, recent transfusion or blood loss, haemoglobinopathy; advanced CKD, HIV treatment. Seek laboratory advice.
- **Stress hyperglycaemia**: raised glucose in acute illness with a normal HbA1c — recheck HbA1c 3 months after recovery.

2 RESULT BANDS AND ACTION

NG28 · PH38 · WHO 2011

RESULT	MEANS	ACTION · TIMING
HbA1c **<42**	Normal	Symptoms or unreliable HbA1c → fasting glucose
HbA1c **42–47** FPG **5.5–6.9**	Non-diabetic hyperglycaemia: high risk, not diabetes	NHS Diabetes Prevention Programme; HbA1c **annually**; CV risk
HbA1c **≥48** once, well	Possible diabetes	Repeat to confirm before labelling
HbA1c **≥48** ×2, or once + symptoms	Diabetes	Classify the type (3–4), baseline tests (5), start care
FPG **≥7.0** · random **≥11.1** + symptoms	Diabetes (venous plasma)	Asymptomatic FPG: confirm with a second result
Any level + ketones, vomiting or drowsiness	DKA, HHS or new type 1	**Act today**

HbA1c in mmol/mol; glucose in mmol/L. 42–47 is never a diabetes diagnosis on its own.

3 CLINICAL CLUES — IS IT TYPE 2?

NG28 · NG17

PACE AND SYMPTOMS
Thirst, polyuria, thrush. Onset over weeks or rapid weight loss → insulin deficiency.

TYPE 1 / LADA
Ketones, weight loss, autoimmune disease, poor response to tablets. Age or BMI never excludes it.

DRUGS AND HISTORY
Steroids, antipsychotics, tacrolimus; pancreatitis, alcohol; past gestational diabetes.

EXAMINE
Weight, BP, hydration; feet (pulses, monofilament); Cushingoid features; acanthosis.

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Gradual, overweight, family history, South Asian or African-Caribbean heritage	**Type 2** (commonest)	Repeat HbA1c; baseline (5)
Weight loss, ketones, rapid onset	**Type 1 / LADA**	Ketones today; GAD antibodies
60+ with weight loss; steatorrhoea, back pain	**Pancreatic cancer**, type 3c	Urgent CT (NG12); faecal elastase
On steroids or antipsychotics	**Drug-induced**	Afternoon glucose (HbA1c lags)
Young, not obese, dominant family history, antibody negative	**MODY**	Diabetes team: genetics
Arthropathy, raised ferritin, liver disease	**Haemochromatosis**	Ferritin, transferrin saturation
Striae, proximal weakness, bruising	**Cushing's** (or acromegaly)	Overnight dexamethasone test
Acute illness, normal HbA1c	**Stress hyperglycaemia**	HbA1c 3 months after recovery

Unsure of the type and unwell or ketotic → treat as possible type 1 the same day.

5 WORK-UP CHECKLIST

NG28 · NG203 · NG238

- ☐ **Ketones** and glucose today if symptomatic
- ☐ **U&E, eGFR**: gates metformin and SGLT2i
- ☐ Lipids, **QRISK3**, BP
- ☐ Urine dipstick: ketones, infection
- ☐ **GAD antibodies** if type 1 or LADA possible
- ☐ **Foot check**: monofilament, pulses
- ☐ **Repeat HbA1c** or venous plasma glucose
- ☐ **Urine ACR**: early-morning sample
- ☐ LFTs (fatty liver), FBC
- ☐ TSH, ECG if clinically indicated
- ☐ Ferritin, TSAT if haemochromatosis
- ☐ **Eye screening** referral at diagnosis

Never delay same-day diabetes-team assessment, or insulin, waiting for antibodies; C-peptide later if the type is still uncertain.

6 AGREE A TARGET

NG28

48 mmol/mol	Lifestyle, or drugs not causing hypoglycaemia (including metformin + SGLT2i)
53 mmol/mol	On a sulfonylurea or insulin; intensify when HbA1c reaches **58**
Frail	An individualised, relaxed target agreed with the patient and carer; avoid hypoglycaemia

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 MANAGEMENT OF THE COMMONEST CAUSES

NG28 (Feb 2026) · PH38 · JBDS · MHRA · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Type 2, well	Structured education and lifestyle from day one. **MR metformin** 500 mg once daily with food, increased weekly to 1–2 g daily (maximum 2 g); then add an **SGLT2 inhibitor** once metformin is tolerated: dapagliflozin 10 mg or empagliflozin 10 mg once daily	Metformin: eGFR **<45** review the dose, **<30** stop. SGLT2i: eGFR gate depends on the drug and licence (SPC). Not tolerated → SGLT2i alone. Frail, at risk from an SGLT2i → metformin alone; DPP-4 inhibitor if metformin not tolerated
Established CVD · heart failure	Triple first line: MR metformin + SGLT2i + subcutaneous semaglutide up to 1 mg weekly, usually stepwise. Heart failure → the SGLT2i is the priority	Already stable on metformin alone: no mass switch, but review for an SGLT2i
Very symptomatic, no ketones	Consider a sulfonylurea or insulin with diabetes-team advice, then rationalise	Any ketones or doubt about type → same-day diabetes team
Steroid-induced	Capillary glucose before the evening meal; morning gliclazide or insulin may be needed	Follow JBDS steroid guidance; reduce as the steroid falls
Non-diabetic hyperglycaemia	NHS Diabetes Prevention Programme, weight loss, activity; no routine medication	Annual HbA1c. Metformin only if HbA1c keeps rising despite intensive lifestyle change (PH38)

Alongside glucose: atorvastatin 20 mg if QRISK3 ≥10% (80 mg if established CVD); ACE inhibitor or ARB if CKD with ACR ≥3 mg/mmol; stop smoking; foot and eye screening.

8 SGLT2 INHIBITOR AND METFORMIN SAFETY

NG28 · MHRA · JBDS

BEFORE THE SGLT2I
Ask about previous DKA, dehydration and a very-low-carbohydrate or ketogenic diet. Do not start while unwell. Warn about genital thrush.

SICK-DAY RULES
Vomiting, diarrhoea or poor intake: pause the SGLT2i and metformin; never stop insulin. Restart when eating and drinking normally for 24–48 h.

KETONES, NOT GLUCOSE
Unwell on an SGLT2i: check blood ketones even if glucose is normal. ≥3.0 mmol/L or DKA symptoms → 999.

REMISSION
Weight loss improves HbA1c and can bring remission: NHS Type 2 Path to Remission where eligible. Screening continues.

9 FOLLOW-UP AND MONITORING CHECKLIST

NG28 · MHRA

- [ ] Asymptomatic first result ≥48: **repeat** before coding diabetes
- [ ] Metformin tolerability once titrated, before adding the SGLT2i
- [ ] **HbA1c at 3 months**; then 3–6-monthly until stable, then 6-monthly
- [ ] Above target: adherence, lifestyle, intercurrent illness, right target, right type — then intensify
- [ ] Annual care processes: HbA1c, BP, lipids, ACR, eGFR, BMI, smoking, feet, eye screening
- [ ] Non-diabetic hyperglycaemia: **annual HbA1c**; stress hyperglycaemia: HbA1c 3 months after recovery
- [ ] Metformin: check B12 if symptoms of deficiency; consider periodic monitoring if risk factors (e.g. higher dose, longer duration, low-normal baseline)
- [ ] **Rethink the type**: rapid tablet failure, ongoing weight loss or ketones at any point
- [ ] Chase antibodies, CT, ACR and the eye-screening invitation
- [ ] On a sulfonylurea or insulin: ask about hypoglycaemia and driving at every review

10 WHEN AND WHERE TO REFER

NG28 · NG17 · NG12

999	DKA, HHS; sepsis or foot infection with systemic illness
Same day	Possible type 1 or LADA with symptoms or ketones; pregnancy; active foot disease
Urgent CT	60+ with new diabetes and weight loss (NG12)
Diabetes team, urgent	Severe symptoms not settling; catabolic weight loss; possible LADA, well
Diabetes team, routine	Possible MODY or pancreatic diabetes; difficult steroid-induced diabetes; diagnostic doubt
Remission	NHS Type 2 Path to Remission where eligible; dietitian if complex needs
Endocrinology	Suspected Cushing's, acromegaly or phaeochromocytoma
Everyone	Structured education; eye screening at diagnosis; NHS Diabetes Prevention Programme for 42–47

11 SPECIAL GROUPS

NG28 · NG3 · DVLA

- **Pregnant or planning pregnancy**: same-day antenatal diabetes team if pregnant; pre-conception care before stopping contraception. HbA1c is not used to diagnose gestational diabetes.
- **Frail or limited life expectancy**: relaxed target agreed with patient and carer; avoid hypoglycaemia; caution with an SGLT2i.
- **Ketosis-prone type 2**, notably in people of African or African-Caribbean heritage: DKA can occur in "type 2" — check ketones when unwell.
- **Driving**: insulin → tell the DVLA; on tablets, car drivers usually need not unless DVLA hypoglycaemia triggers apply; Group 2 drivers on a sulfonylurea or insulin must tell the DVLA.

12 PITFALLS AND NEVER-EVENTS

NG28 · NG17 · NG12

- **Never** label diabetes on one asymptomatic HbA1c, or on a capillary meter reading.
- **Never** start metformin and "review in 3 months" in a 60+ with weight loss, or with ketones.
- A normal glucose does not exclude DKA on an SGLT2 inhibitor: check ketones.
- A normal HbA1c does not exclude diabetes after recent onset, in anaemia or haemoglobinopathy.
- An HbA1c of 48 in a frail care-home resident on a sulfonylurea is a harm marker.

13 SAFETY-NET — SAY IT, THEN DOCUMENT IT

NG28 · JBDS

Call 999	Vomiting, tummy pain, fast deep breathing, drowsiness or confusion — even if the sugar is not very high on a "flozin"
Same day	Very thirsty, passing lots of urine and losing weight; cannot keep fluids down; a new sore, redness or swelling on a foot
Hypos	On a sulfonylurea or insulin: shaky, sweaty or confused → fast sugar, then a snack; tell us about any severe or repeated hypo

14 SAY IT LIKE THIS

plain English

"Your blood test shows your sugar has been running high over the last two to three months. We'll recheck it to be sure, and look at your heart, kidneys, eyes and feet too. If you start vomiting, get tummy pain, fast deep breathing or feel very drowsy, call 999 — that can happen even when the sugar isn't very high."

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Hypoglycaemia: the low glucose CLINIC GUIDE

Low glucose in adults 18+, with or without diabetes. Not for children, pregnancy (maternity diabetes team), neonates or hospital inpatient protocols.

JBDS hypoglycaemia (2023) · NICE **NG17** · **NG28**
DVLA fitness to drive · NICE CKS · MHRA
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/hypoglycaemia.html

PAGE 1 · INTERPRET

treat first → is it real? → why did it happen?

! ACT TODAY — TREAT BEFORE YOU INVESTIGATE

JBDS · NG28

TIER	RESULT OR COMBINATION	ACTION
999	Unconscious, seizing or **unable to swallow** with a low glucose · glucose still **<4.0 mmol/L after 3 treatment cycles** · consciousness or focal signs not recovering once glucose is corrected · low glucose with Addisonian features (adrenal crisis)	**Nothing by mouth**; **glucagon 1 mg IM** (or 3 mg nasal); recovery position. Adrenal crisis: emergency hydrocortisone per protocol
Same day	**Sulfonylurea** hypo that was severe or recurrent, needed glucagon or IV glucose, or with a long-acting or MR agent, renal impairment, frailty or uncertain overdose · recurrent hypos in a frail care-home resident on insulin or a sulfonylurea	Hospital observation: recurrence possible for **24–36 hours** · same-day medication review
Urgent	**Recurrent spontaneous hypoglycaemia without diabetes**, no obvious cause · unexplained hypos with possible access to insulin or sulfonylureas	Urgent endocrinology (999 only if unwell); consider error, exposure, safeguarding

1 CONFIRM IT IS REAL

JBDS · specialist practice

- **In diabetes**: treat any capillary glucose **<4.0 mmol/L** with symptoms now; no lab confirmation needed.
- **Without diabetes**: never diagnose from a meter alone. Document **Whipple's triad**: symptoms, a low **laboratory** glucose at the time, and relief when it is corrected.
- **Pseudohypoglycaemia**: meters are least accurate at low values; cold or poorly perfused fingers; sugar on the skin; a venous sample left unseparated keeps using glucose (fluoride tube needed); CGM lags and can read falsely low when the sensor is lain on.
- **Mimics**: anxiety, arrhythmia, TIA or stroke, seizure.
- **When to repeat**: recheck capillary glucose **10–15 minutes** after each treatment. Non-diabetic: take the lab sample during the next episode, before treatment, only if safe.

2 SEVERITY BANDS

diabetes · JBDS · NG17

GLUCOSE	MEANS	ACTION AND TIMING
3.0–3.9 mmol/L	Level 1 (alert value)	15–20 g fast-acting glucose now; recheck at 10–15 min
<3.0	Level 2: clinically significant	Treat now; review the regimen and the cause
Any level, needs help	Level 3: severe	Unable to swallow → 999 + glucagon; diabetes team review
Non-diabetic	No universal cut-off	Whipple's triad, then endocrinology

3 HISTORY THAT DISCRIMINATES

JBDS

Timing	Fasting (insulinoma, Addison's, alcohol) · after meals (post-bariatric, reactive) · nocturnal (basal insulin)
Symptoms	Warning: sweating, tremor, palpitations, hunger. Brain: confusion, slurred speech, drowsiness, fits. Brain symptoms first = impaired awareness
Who treated it	Needed help = level 3, whatever the number
What changed	Dose, new drug, meals, exercise, alcohol, weight loss, renal function
Driving	Group 1 or 2 licence; any hypo at the wheel

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Insulin, missed meal, exercise, alcohol	**Insulin excess** (commonest)	Dose timing vs hypo timing; CGM
Gliclazide or glibenclamide, older, low eGFR	**Sulfonylurea**	eGFR; HbA1c
Night-time lows, waking headache	**Basal insulin excess**	CGM overnight
Late lows after exercise	**Increased insulin sensitivity**	Exercise plan review
Heavy drinking, overnight or morning	**Alcohol**	LFTs, GGT; alcohol history
Erratic glucose, lumpy sites	**Lipohypertrophy**	Examine sites
Pigmentation, low BP, low Na, high K	**Adrenal insufficiency**	Cortisol ± ACTH; 999 if crisis
Lows after meals, past bariatric surgery	**Post-bariatric hypoglycaemia**	Whipple's triad; endocrinology
Fasting lows, no diabetes	**Insulinoma** (rare)	Insulin, C-peptide in an episode
Low C-peptide with high insulin	**Exogenous insulin**	Drug screen; safeguarding
Sepsis, liver or renal failure	**Critical illness**	Hospital assessment

5 WORK-UP CHECKLIST

JBDS · NG17 · NG28

- ☐ **Capillary glucose**: treat first
- ☐ **U&E, eGFR**: drug clearance
- ☐ **Injection sites**: lipohypertrophy
- ☐ **Gold score** (awareness)
- ☐ **HbA1c**: is it too low?
- ☐ LFTs, GGT: liver, alcohol
- ☐ **CGM or diary**: timing pattern
- ☐ Cortisol ± ACTH if Addisonian

No diabetes, during a spontaneous episode and before treatment if safe: lab glucose, insulin, C-peptide, proinsulin, β-hydroxybutyrate, sulfonylurea screen. A 72-hour fast, mixed-meal test and imaging are specialist-only.

6 WHO IS AT HIGH RISK

NG17 · NG28

Impaired awareness	Gold score **≥4** (1 always aware to 7 never): high risk of severe hypos. Ask at least annually in type 1 (NG17)
Older, frail, dementia	On insulin or a sulfonylurea; living alone
Kidney or liver disease	Reduced clearance; eGFR <30 needs diabetes-team advice
Beta-blockers, neuropathy	Blunted warning symptoms
Alcohol, eating disorders	Low glycogen; erratic intake
HbA1c below target	On insulin or a sulfonylurea, especially older people: over-treatment
After bariatric surgery	Lows after meals; endocrine or metabolic service
Living alone	A severe hypo may go untreated: CGM alarms, check-ins

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 TREATING THE EPISODE

JBDS (2023) · SPC

1 · CONSCIOUS, CAN SWALLOW
15–20 g fast-acting glucose: glucose tablets (count by product), **150–200 mL fruit juice**, or a 60 mL glucose shot. Check drink labels: some have been reformulated with less sugar.

2 · RECHECK AT 10–15 MIN
Still **<4.0** → repeat, up to **3 cycles**. Can swallow but cannot eat: 40% glucose gel into the cheek.

3 · ONCE >4.0
Long-acting carbohydrate: a slice of bread, two biscuits, or the next meal if due. Not chocolate (fat slows absorption).

CANNOT SWALLOW
999, nothing by mouth. **Glucagon 1 mg IM** or **3 mg nasal** by a trained carer; IV glucose by a skilled professional. Glucagon is less reliable with sulfonylureas, liver disease, alcohol or starvation.

On **acarbose**: use glucose (tablets, gel, solution), not sucrose. Never omit the next insulin dose; review it.

8 PREVENTING THE NEXT ONE — BY CAUSE

NG17 · NG28 · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Insulin	Match hypo timing to the dose responsible (late-morning lows: breakfast insulin; night lows: basal). Adjust per the diabetes-team plan; no universal percentage	CGM or diary; structured education (e.g. DAFNE) in type 1; check injection sites
Sulfonylurea	**Lowest effective dose or stop** (de-intensify, do not swap to another sulfonylurea). Avoid glibenclamide in older people and renal impairment	Diabetes-team advice if eGFR <30, recurrent hypos or frailty
Older or frail, type 2	**Relax the HbA1c target** individually (NG28); de-intensify insulin or sulfonylurea	Review HbA1c and hypos at each review
Impaired awareness	Education, regimen review, avoid recurrent lows; CGM per NICE criteria	In type 1, do not relax targets as the main fix
No diabetes	Urgent endocrinology once Whipple's triad is documented	Imaging only after biochemical confirmation

9 SELF-CARE AND THE CONVERSATION

JBDS · NG17

Carry glucose at all times (car, bedside, work). **Type 1**: glucagon at home, carers trained, expiry checked. **Alcohol**: eat before and with it; check glucose overnight and before morning insulin. **Exercise**: check before, during and after; lows can come hours later.

"Your sugar dropped too low. Take something sugary straight away, check again after 15 minutes, then eat a snack like a sandwich or biscuits. Always carry glucose, and we will look at your tablets or insulin so this does not keep happening."

10 FOLLOW-UP, DRIVING AND MONITORING

DVLA · NG17

- [ ] **After a severe episode**: review cause, doses, glucagon training, CGM, driving
- [ ] **Annual**: count level 1–3 events; Gold score; glucagon in date; carries glucose
- [ ] **Driving, insulin**: check glucose within 2 hours before driving and every 2 hours on a journey; glucose in the car
- [ ] Do not drive during or straight after a hypo; wait the DVLA recovery period
- [ ] Group 2 (HGV, bus): full awareness, no severe hypo in 12 months; stricter checks
- [ ] Give written DVLA advice and record it
- [ ] **Sulfonylurea**: eGFR and HbA1c at least yearly; avoid glibenclamide in older people

11 WHEN AND WHERE TO REFER

JBDS · NG17 · NG28

999	Level 3; still <4.0 after 3 cycles; not recovering; adrenal crisis
Same day	High-risk sulfonylurea hypo · focal neurology · non-diabetic and unwell
Endocrinology (urgent)	Spontaneous hypos without diabetes · post-bariatric · adrenal insufficiency
Diabetes team	Any level 3 episode · Gold ≥4 · recurrent hypos on insulin · over-treated older people
CGM	All adults with type 1 (NG17); type 2 on multiple daily insulin with recurrent or severe hypos, impaired awareness, unable to finger-prick, or testing 8+ times a day (NG28)

12 PITFALLS AND NEVER-EVENTS

JBDS · MHRA

- **Never** give oral glucose to someone who cannot swallow · never send home a high-risk sulfonylurea hypo after one sugary drink.
- **Never** stop basal insulin in type 1 on sick days · SGLT2 inhibitor: stop in acute illness; check ketones if DKA suspected, even with normal glucose (MHRA).
- Do not diagnose non-diabetic hypoglycaemia on a meter reading, or image before biochemistry.
- Slow recovery after correction is not "hypo brain damage" until stroke, seizure, sepsis, intoxication and other causes are excluded.

13 SAFETY-NET AND SICK DAYS

JBDS · MHRA · TREND-UK

- **999** (tell carers): cannot swallow, unconscious or fitting; still low after 3 treatments; not waking up properly once glucose is normal. **Same day**: a hypo on a sulfonylurea that needed help or keeps coming back; new confusion or weakness.
- **Sick days**: type 1 never stops basal insulin (urgent advice if vomiting, not eating or ketones) · insulin changes in type 2 are individualised · **sulfonylurea**: withhold or seek advice when not eating · pumps: never stop without a basal-insulin plan · test glucose more often.
- **Older or frail on insulin or a sulfonylurea**: more lows, fewer warnings; review at least yearly and de-intensify early.

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Abnormal lipid profile CLINIC GUIDE

Interpreting an abnormal cholesterol, HDL or triglyceride result in adults 18+. Not for children (paediatric FH pathways) or acute coronary syndrome. Very high triglycerides: the High triglycerides guide.

NICE **NG238** (2023) · **CG71** FH (updated 2019)
NICE **QS100** · QRISK3 · HEART UK
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/abnormal-lipid-profile.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → which band? → clues → why? → work it up → is it FH?

! ACT TODAY

NG238 · CG71

TIER	RESULT OR PICTURE	ACTION
999	Acute, ongoing or recent **chest pain**, whatever the cholesterol	999 as possible ACS
Same day	Triglycerides (TG) at **any level** + abdominal pain or vomiting: **pancreatitis**?	Hospital, 999 if unwell; no repeat first
Same day	Raised TG + thirst, polyuria, weight loss or vomiting: **DKA or HHS**?	Glucose, ketones; hospital if raised or unwell
Urgent	TG **>20 mmol/L** not explained by alcohol or poor glycaemic control · LDL **>13 mmol/L** in an adult (possible homozygous FH)	Urgent lipid specialist · FH centre without delay
Urgent	Heavy proteinuria, oedema and low albumin (**nephrotic syndrome**)	Urgent medical assessment

1 IS IT REAL? CONFIRM BEFORE YOU LABEL

NG238 · CG71

FASTING OR NOT
A **non-fasting** full profile is enough for risk assessment. Fasting matters only for TG: repeat fasting if TG is **≥4.5 mmol/L** or the result is unexpected.

REPEAT FIRST
Never label on one result. FH needs at least **2 LDL measurements** after secondary causes are excluded.

WRONG MOMENT
Acute illness, infection, surgery or a recent MI lowers cholesterol for a time; pregnancy raises it. Recheck when well.

LDL UNRELIABLE
Calculated LDL fails when TG is above **4.5 mmol/L**: use **non-HDL** (total minus HDL) or a direct LDL.

2 WHICH BAND, AND WHAT TO DO

mmol/L · use your laboratory's range

Below the FH bands, overall CVD risk (box 7) decides, not the number.

RESULT	MEANS	ACTION
TC <7.5, no FH features	Polygenic or normal	CVD risk: QRISK3 where valid (box 7). Routine
TC >7.5 or **LDL >4.9** (adult)	Possible FH	FH assessment (box 6), especially with CHD before 60 in patient or first-degree relative
TC >9.0 or **non-HDL >7.5**	Severe hypercholesterolaemia	Specialist assessment, even without a family history (NG238)
LDL >13 (adult)	Possible homozygous FH	Specialist FH centre without delay (CG71)
HDL <1.0 men, **<1.2** women	Risk marker, not a target	Lifestyle; no specific drug; counted in QRISK3 via TC:HDL
TG 4.5–9.9	QRISK3 may underestimate	Fasting repeat; secondary causes; non-HDL >7.5 → specialist advice
TG 10–20	Pancreatitis risk	Fasting repeat after 5 days but within 2 weeks; still >10 → specialist advice
TG >20	Chylomicronaemia	Urgent specialist review unless alcohol or poor glycaemic control explains it

3 CLINICAL CLUES

examination that changes the plan

Tendon xanthomata	Achilles, hand extensors: FH (definite on Simon Broome with TC >7.5)
Corneal arcus, xanthelasma	Significant in a young patient: assess for FH
Eruptive xanthomata	Buttocks, elbows: very high TG
Central obesity, acanthosis	Insulin resistance; low HDL with high TG
Oedema, frothy urine	Nephrotic syndrome
Itch, jaundice	Cholestasis: very high cholesterol
Pulses, bruits	Undiagnosed PAD or carotid disease → secondary prevention

4 WHY? SECONDARY CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Tired, cold, weight gain	**Hypothyroidism**	TSH
Thirst, weight loss, diabetes	**Diabetes**	HbA1c
Oedema, raised creatinine	**CKD, nephrotic**	eGFR, ACR
Itch, raised ALP or GGT	**Cholestasis**	LFTs
Regular or heavy drinking	**Alcohol**	AUDIT-C, GGT
Raised BMI, waist, BP	**Metabolic syndrome**	HbA1c
Antipsychotics, steroids, retinoids, antiretrovirals, ciclosporin, amiodarone, thiazides, beta-blockers, oestrogens	**Drugs**	Drug review

5 WORK-UP CHECKLIST

NG238 · QS100

- ☐ **Lipids**: TC, HDL, non-HDL, TG
- ☐ **HbA1c**
- ☐ **LFTs** (ALT, GGT)
- ☐ Alcohol (AUDIT-C)
- ☐ Family history of premature CHD
- ☐ **TSH**
- ☐ **U&E, eGFR** + urine ACR
- ☐ BMI, waist, BP, smoking
- ☐ Drug review
- ☐ QRISK3, only where valid

6 IS THIS FH?

CG71 · NG238

Suspect: TC >7.5 or LDL >4.9 (adult) and/or CHD <60 in patient or first-degree relative. No QRISK3.

SIMON BROOME	TC >7.5 OR LDL >4.9, PLUS
Definite	Tendon xanthomata (patient or 1st/2nd-degree relative) or DNA-confirmed
Possible	MI in a 1st-degree relative <60 or 2nd-degree <50, or TC >7.5 in an adult relative

Possible/definite or **DLCN >5** → FH service: DNA, **cascade testing**. Search records: <30, TC >7.5; 30+, TC >9.0.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 FRAME THE PATIENT: WHO GETS A STATIN

NG238 · CG71 · doses per SPC and local formulary

PATIENT	RISK TOOL	STATIN
Established CVD (MI, angina, stroke/TIA, PAD, revascularisation)	None: secondary prevention	**Atorvastatin 80 mg**; lower if interactions, adverse-effect risk or preference
Suspected or confirmed FH	None: QRISK3 underestimates	High-intensity statin; lipid clinic (box 6)
CKD (eGFR <60 and/or albuminuria)	Not QRISK3	**Atorvastatin 20 mg**, primary or secondary prevention
Type 1 diabetes	Not QRISK3	Offer atorvastatin 20 mg if over 40, diabetes >10 years, nephropathy or other CVD risk factors
Aged 25–84, none of the above	**QRISK3 ≥10%**	**Offer atorvastatin 20 mg** after an informed discussion
	QRISK3 <10%	Lifestyle; do not rule out a statin if preferred or risk is underestimated
Aged 85+	QRISK3 not appropriate	Consider a statin after individual assessment

QRISK3 may underestimate risk with severe mental illness, HIV, autoimmune disease, antipsychotics, corticosteroids and TG 4.5–9.9 mmol/L.

8 TARGETS AND NEXT STEPS

NG238 · CG71

GROUP	TARGET AT 2–3 MONTHS
Primary prevention	>40% fall in non-HDL from baseline
Secondary prevention	LDL ≤2.0 or non-HDL ≤2.6 mmol/L
FH	≥50% fall in LDL from the untreated baseline

Not at target: adherence → lifestyle → secondary cause → step up the statin → ezetimibe (full ladder in the High cholesterol protocol).

Lifestyle, alongside a statin: Mediterranean-style diet; unsaturated for saturated fat; no plant sterols or stanols for CVD prevention; 150 min/week activity; alcohol ≤14 units/week; stop smoking.

Low HDL: no specific drug; address smoking, inactivity, weight, insulin resistance and TG. A high HDL does not cancel a raised LDL or non-HDL.

Mixed pattern (raised LDL or non-HDL and TG): exclude diabetes, hypothyroidism, alcohol and drugs; use non-HDL, not calculated LDL; premature CVD in the family → think familial combined hyperlipidaemia → lipid clinic.

Raised TG: manage on the High triglycerides guide.

9 FOLLOW-UP AND MONITORING

NG238 · NHS England

- [ ] **Lipids** 2–3 months after starting or changing treatment
- [ ] **Transaminases**: baseline, within 3 months and at 12 months; then only if indicated
- [ ] ALT raised but <3× ULN: not a reason to stop or withhold; stop only if it stays >3× ULN
- [ ] **CK** only with muscle symptoms. Baseline >5× ULN → repeat after 7 days; still >5× → do not start
- [ ] Severe muscle pain, weakness or dark urine → stop the statin; same-day CK and renal function
- [ ] On treatment: CK 4–10× ULN or intolerable → stop 4–6 weeks, rechallenge; >10× ULN → urgent; <4× → look for other causes
- [ ] Baseline CK raised but <5× ULN → start at a lower dose
- [ ] Annual medicines review ± non-fasting non-HDL
- [ ] Not on a statin: reassess QRISK3 and lipids periodically

10 WHEN AND WHERE TO REFER

NG238 · CG71

999 / same day	Chest pain (999); TG with abdominal pain or vomiting; DKA or HHS
Urgent	TG >20 not due to alcohol or glycaemic control; LDL >13 (adult); nephrotic syndrome
FH service	Possible or definite Simon Broome, or DLCN >5: DNA and cascade testing
Lipid specialist	TC >9.0 or non-HDL >7.5; TG still >10 on repeat; TG 4.5–9.9 with non-HDL >7.5; suspected familial combined hyperlipidaemia
Lipid clinic	Not at target on maximum tolerated treatment; statin intolerance (local pathway)
Primary care	Polygenic, secondary causes, QRISK3 ≥10%, low HDL

11 SPECIAL GROUPS

CG71 · NG238

- **Pregnancy**: lipids rise normally, so do not interpret them; statins are not used in pregnancy — stop them before conception.
- **FH family**: relatives, including children of a parent with FH, are tested through the specialist FH service.
- **HIV**: QRISK3 underestimates risk; check antiretroviral interactions before choosing a statin.
- **Severe mental illness, antipsychotics**: risk underestimated; lipids often worsen on treatment.

12 PITFALLS AND NEVER-EVENTS

NG238 · CG71

- **Never** run QRISK3 in FH, established CVD, CKD or type 1 diabetes.
- **Never** trust calculated LDL when TG is above 4.5 mmol/L.
- **Never** stop a statin for a mild ALT rise, or check CK routinely.
- **Never** label on one result taken during acute illness.
- Do not miss hypothyroidism, nephrotic syndrome or cholestasis behind a high cholesterol.

13 SAY IT LIKE THIS

plain English

"Your cholesterol is one part of your risk of a heart attack or stroke, not a disease on its own. We'll check it isn't caused by something else, like your thyroid, and work out your overall risk before deciding together whether a tablet would help."

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Raised triglycerides CLINIC GUIDE

Interpreting a raised triglyceride (TG) in adults 18+. Not for children, pregnancy (obstetric and lipid co-management) or acute pancreatitis care (hospital). Cholesterol-led lipid management: the Lipids pathway.

NICE **NG238** (2023) · **NG104** pancreatitis
NICE **TA805** (2022) · **HST13** · MHRA · SPCs
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/high-triglycerides.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → how high? → why? → work it up

! ACT TODAY

NG238 · NG104

TIER	RESULT OR PICTURE	ACTION
999 / same day	**Any TG level** + severe epigastric pain radiating to the back, nausea or vomiting: possible **acute pancreatitis** (can occur below 10 mmol/L)	Hospital now; do not wait for a fasting repeat. A normal amylase or lipase does not exclude it in lipaemic blood
Same day	Raised TG with thirst, polyuria, weight loss, vomiting or acute illness: possible **DKA or HHS**	Capillary glucose and blood ketones now; hospital if ketones raised or unwell
Urgent specialist	TG **>20 mmol/L** not explained by excess alcohol or poor glycaemic control · eruptive xanthomata, lipaemia retinalis, or recurrent pancreatitis	Urgent lipid specialist review (NG238); same day if any pain or illness
Urgent	Raised TG with heavy proteinuria, oedema and low albumin (nephrotic syndrome)	Urgent nephrology discussion at any TG level

1 IS IT REAL? CONFIRM BEFORE YOU ACT

NG238

FASTING OR NOT
Non-fasting lipids are fine for routine CVD risk. Repeat **fasting** when TG is **≥4.5 mmol/L**, the result is unexpected, or chylomicronaemia is suspected.

10–20 MMOL/L
Repeat fasting after at least **5 days but within 2 weeks**, and review secondary causes meanwhile.

RECENT TRIGGERS
A binge, a heavy fatty meal, new steroids or a diabetes flare can raise a single result. Ask before you label.

LDL UNRELIABLE
Calculated LDL is unreliable when TG is above 4.5 mmol/L: use **non-HDL cholesterol** (total minus HDL) or a direct LDL. Milky plasma can disrupt other tests.

2 HOW HIGH, AND WHAT TO DO

NICE NG238 thresholds · use your laboratory's range

TRIGLYCERIDE	MEANS	ACTION	TIMING
<4.5 mmol/L	CVD risk marker	Assess overall CVD risk (QRISK3); manage LDL/non-HDL, usually with a statin where indicated	Routine
4.5–9.9 mmol/L	Raised; risk tools may **underestimate** CVD risk	Confirm fasting; treat secondary causes; optimise other risk factors. Non-HDL **>7.5 mmol/L** → specialist advice	Routine
10–20 mmol/L	Pancreatitis risk	Fasting repeat in 5 days–2 weeks; urgent review of causes; safety-net. Still **>10** → specialist advice	Within 2 weeks
>20 mmol/L	Chylomicronaemia; high pancreatitis risk	Urgent specialist review unless explained by alcohol or poor glycaemic control; stop alcohol	Urgent
Any level + pain or vomiting	Possible pancreatitis	Emergency assessment	Now
Fasting ≥1.7, CVD, on a statin, LDL >1.04 to ≤2.60	Residual risk	Consider icosapent ethyl (TA805)	Routine

3 EXAMINATION CLUES

targeted

Eruptive xanthomata	Small yellow papules on buttocks, elbows: very high TG
Lipaemia retinalis	Creamy retinal vessels: chylomicronaemia
Tendon xanthomata	Achilles, knuckles: think familial hypercholesterolaemia
Acanthosis, central fat	Insulin resistance, metabolic syndrome
Epigastric tenderness	Pancreatitis until proven otherwise

4 WHY? FIND THE CAUSE

clue → think → first test

CLUE	THINK	FIRST TEST
Thirst, weight loss, known diabetes	**Poor glycaemic control**	HbA1c, glucose
Regular or heavy drinking	**Alcohol**	AUDIT-C, GGT
Central obesity, high BP	**Metabolic syndrome**	Waist, HbA1c
Tired, cold, weight gain	**Hypothyroidism**	TSH
Oedema, frothy urine, CKD	**Renal disease**	eGFR, urine ACR
Steroids, thiazides, beta-blockers, isotretinoin, oestrogens, antiretrovirals	**Drugs**	Drug review
Very high TG, pancreatitis, family history	**Genetic chylomicronaemia**	Specialist
High TG and LDL, premature CVD in family	**Familial combined**	Lipids in relatives

5 WORK-UP CHECKLIST

NG238

- ☐ **Fasting lipids** + non-HDL
- ☐ **TSH**
- ☐ **U&E/eGFR**
- ☐ BMI, waist, BP
- ☐ Alcohol units (AUDIT-C)
- ☐ **HbA1c**, glucose
- ☐ **LFTs + GGT**
- ☐ Urine dipstick or ACR
- ☐ QRISK3 (may underestimate)
- ☐ Drug review

Look for eruptive xanthomata (buttocks, elbows) and, if TG is very high, lipaemia retinalis. Ketones only if unwell or hyperglycaemic, not because of a TG number.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY CAUSE AND LEVEL

NG238 · TA805 · HST13 · MHRA · SPCs

SITUATION	FIRST LINE	CHECK AND CONTINUE
Any raised TG	Treat the secondary cause first: diabetes control, alcohol, thyroid, renal disease, drugs. **Stop sugary drinks and fruit juice**, cut refined carbohydrate, lose weight, regular activity	Repeat lipids once the cause is addressed; TG often falls quickly
TG >10 mmol/L	Stop alcohol; **specialist-directed low-fat diet** (dietitian), not a generic Mediterranean diet	Specialist advice if still >10 on repeat
Raised CVD risk	**Statin first** for overall atherogenic risk (NG238). Do not routinely offer a fibrate or omega-3 for CVD prevention	Target: secondary prevention LDL ≤2.0 or non-HDL ≤2.6 mmol/L; primary prevention >40% non-HDL reduction
Persistent severe TG	**Fenofibrate on specialist advice**, mainly for pancreatitis risk; formulation-specific renal dosing (SPC). Never combine gemfibrozil with a statin (MHRA)	Baseline renal function and LFTs; CK only if muscle symptoms
CVD on a statin (TA805)	**Icosapent ethyl 2 g twice daily** if fasting TG ≥1.7 and LDL >1.04 to ≤2.60 mmol/L	Watch for AF and bleeding (anticoagulants, antiplatelets)
Confirmed FCS	Volanesorsen: specialist centre only (HST13)	Platelet monitoring per SPC

7 LIFESTYLE THAT LOWERS TRIGLYCERIDES

NG238 · UK CMO

SUGAR FIRST
Sugary drinks, fruit juice and refined carbohydrate drive TG as much as fat. Swap to water or unsweetened drinks.

ALCOHOL
Reduce substantially; stop completely while TG is severely raised or after pancreatitis. Record units every review.

WEIGHT
Waist loss matters most; effect varies with diabetes control and genetics. Offer weight-management services.

ACTIVITY
About 150 minutes a week of moderate aerobic activity, spread through the week; the effect fades within days of stopping.

8 FOLLOW-UP AND MONITORING CHECKLIST

NG238 · SPCs

- ☐ **10–20 mmol/L**: fasting repeat 5 days–2 weeks later; still >10 → specialist
- ☐ Repeat lipids after the cause is treated (fasting if TG markedly raised)
- ☐ No universal TG target: in severe TG the aim is less pancreatitis risk; use specialist-agreed targets
- ☐ Manage LDL and non-HDL by CVD-risk category
- ☐ **Fenofibrate**: renal function and LFTs per SPC; reassess if eGFR falls
- ☐ **Icosapent ethyl**: AF symptoms, bleeding risk
- ☐ **After TG pancreatitis**: alcohol abstinence; specialist target
- ☐ Planning pregnancy: stop statins when pregnancy is possible
- ☐ Alcohol units recorded at every review

9 WHEN AND WHERE TO REFER

NG238 · NG104

999 / same day	Possible pancreatitis at any TG; DKA or HHS; >20 with pain or illness
Urgent lipid clinic	TG >20 not due to alcohol or poor glycaemic control; suspected chylomicronaemia; recurrent pancreatitis
Specialist advice	Still >10 on fasting repeat; 4.5–9.9 with non-HDL >7.5
Nephrology	Nephrotic syndrome (urgent)
Dietitian	TG >10, pancreatitis, chylomicronaemia, pregnancy
Lipid clinic, routine	Suspected familial combined hyperlipidaemia with premature CVD

10 SPECIAL GROUPS

NG238 · NG104 · HST13

- **Pregnancy**: TG rises normally; with pre-existing high TG it can reach pancreatitis levels → obstetric and lipid co-management. Avoid fibrates unless a specialist advises; stop statins when pregnancy is possible and do not restart until breastfeeding ends.
- **Diabetes**: a very high TG may be the first sign of new or uncontrolled diabetes; control glucose before adding lipid drugs.
- **Chylomicronaemia**: very-low-fat diets only under a specialist dietitian (risk of essential fatty-acid and fat-soluble vitamin deficiency).
- **After pancreatitis**: steatorrhoea → think exocrine insufficiency; check faecal elastase and involve gastroenterology.

11 PITFALLS AND NEVER-EVENTS

NG238 · NG104 · MHRA

- **Never** wait for a fasting repeat when there is abdominal pain or vomiting.
- **Never** trust calculated LDL when TG is above 4.5 mmol/L.
- **Never** start a fibrate just to lower CVD risk, or combine gemfibrozil with a statin.
- **Never** treat over-the-counter fish oil as icosapent ethyl.
- Do not miss new diabetes, hypothyroidism or nephrotic syndrome behind a high TG.

12 SAY IT LIKE THIS

plain English

"Your triglycerides — a type of fat in the blood — are very high, and at this level they can inflame the pancreas. Cutting out alcohol and sugary drinks brings them down quickly. If you get severe tummy pain going through to your back, or vomiting, go straight to A&E."

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Abnormal thyroid function tests

CLINIC GUIDE

Reading TSH, FT4 and FT3 in adults 18+. Not for pregnancy (trimester ranges, RCOG GTG76), children, thyroid cancer follow-up or known pituitary disease. Treatment detail: Hypothyroidism and Hyperthyroidism Clinic Guides.

NICE **NG145** (2019, updated 2023) · **NG12**
RCOG **GTG76** (2025) · MHRA · SPS
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/abnormal-tfts.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → read the pattern → find the cause

! ACT TODAY — THESE DO NOT WAIT FOR A REPEAT

NG145 · NG12 · NG196

TIER	RESULT OR PICTURE	ACTION
999	**Thyroid storm**: thyrotoxic + fever + marked tachycardia or AF + confusion, heart failure, vomiting or jaundice · **Myxoedema coma**: hypothyroid + drowsy or confused + hypothermia · new AF with low TSH and shock, syncope, chest pain or heart failure	Clinical diagnoses: do not wait for TFTs
Same day	On **carbimazole or PTU** with sore throat, mouth ulcers or fever · eye disease with reduced acuity, colour change or corneal exposure · low FT4, non-raised TSH plus hypotension, vomiting, low Na^+ or hypoglycaemia	Stop drug, same-day FBC · ophthalmology · emergency care
Urgent	**Low FT4 with low, normal or mildly raised TSH** (possible central hypothyroidism) · new overt hyperthyroidism in pregnancy, AF or eye disease	9 am cortisol **before** any levothyroxine where feasible; urgent endocrinology
Suspected cancer pathway	**Unexplained thyroid lump, whatever the TSH**; stridor → 999	Consider suspected cancer pathway referral (NG12)

1 IS IT REAL? CONFIRM BEFORE YOU LABEL IT

NG145 · MHRA 2023 · SPS

BIOTIN
Hair, skin and nail supplements can make TSH and FT4 falsely high or low, depending on the assay. Ask every time; stop, tell the lab, repeat.

ACUTE ILLNESS, DRUGS
Non-thyroidal illness, steroids and dopamine lower TSH; recovery can raise it. Avoid testing when acutely ill; repeat once well.

TIMING, ADHERENCE
Levothyroxine just before the test raises FT4; tablets taken only before blood tests give high TSH with normal FT4. TSH lags a dose change by 6–8 weeks.

DISCORDANT PATTERN
Results that do not fit the patient (heterophile antibodies, assay error) → ask the lab to recheck on another platform before acting.

Subclinical results always need a repeat: TSH with FT4 about **3 months** later (many mild rises settle). Pregnancy: only the trimester-specific range applies.

2 READ THE PATTERN — WHAT IT MEANS AND WHAT TO DO

NG145 · use your laboratory's reference ranges

TSH	FT4 / FT3	MEANS	ACTION AND TIMING
Low (<0.1)	Either raised	**Overt hyperthyroidism**	TRAb; refer; consider a beta-blocker while waiting
Low	FT3 only raised	**T3 toxicosis**	Manage as overt hyperthyroidism
<0.1	Both normal	Subclinical hyperthyroidism	Repeat; <0.1 twice ≥3 months apart + thyroid disease or symptoms → specialist advice
0.1 to below range	Both normal	Mild subclinical	Repeat; if untreated, TSH every 6 months
Raised	FT4 low	**Overt primary hypothyroidism**	Offer levothyroxine now (page 2)
≥10 mU/L	FT4 normal	Subclinical hypothyroidism	Repeat at 3 months; ≥10 twice → consider levothyroxine
Above range, <10	FT4 normal	Subclinical, mild	Repeat at 3 months; under 65 + symptoms → 6-month trial
Low, normal, mildly ↑	FT4 low	**Central** or non-thyroidal illness	9 am cortisol; urgent endocrinology
Normal or raised	FT4 high	Interference or recent dose; rarely TSHoma, resistance	Biotin, dose timing; discuss with lab, endocrinology
On levothyroxine	TSH ↑ · TSH suppressed	Under-replaced · over-replaced	Adherence, then increase · reduce

Testing (NG145): TSH alone first in most adults; the lab adds FT4 if TSH is raised, and FT4 + FT3 if TSH is low. Request TSH + FT4 together if pituitary disease is suspected.

3 FIND THE CAUSE

clue → think → first test

CLUE	THINK	FIRST TEST
Diffuse goitre, bruit, eye signs	**Graves'**	TRAb
Older, nodular, TRAb negative	**Toxic nodular**	Specialist scan
Tender gland after a virus	**Subacute thyroiditis**	ESR/CRP; repeat TFTs
Delivery in the past year	**Postpartum thyroiditis**	Repeat TFTs (hypo phase)
Firm painless goitre	**Hashimoto's**	TPO antibodies once
Amiodarone, lithium, immunotherapy	**Drug-induced**	Previous TFTs
Dizzy, pigmented, tired	**Addison's, pituitary**	9 am cortisol

4 WORK-UP CHECKLIST

NG145 · SPS

- ☐ **Repeat TFTs** if subclinical
- ☐ **TRAb** in hyperthyroidism
- ☐ Biotin and drug history
- ☐ **ECG** if palpitations or AF
- ☐ **9 am cortisol** if central or Addison's
- ☐ **FT3** whenever TSH is low
- ☐ **TPO antibodies** once, raised TSH
- ☐ Pregnancy status and plans
- ☐ **FBC, LFTs** before any antithyroid drug
- ☐ Ultrasound only for a lump

TRAb negative → think thyroiditis or toxic nodular disease. Do not repeat TPO antibodies or scan a normal neck. On amiodarone, read with previous TFTs: a raised FT4 alone can be the drug effect.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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5 MANAGEMENT BY THE COMMONEST PATTERNS

NG145 · MHRA · SPC

CONFIRMED PATTERN	FIRST LINE	CHECK AND CONTINUE
Overt primary hypothyroidism *replacement, usually lifelong*	**Levothyroxine**: under 65 without CVD about **1.6 micrograms/kg/day**; aged 65+ or CVD start **25–50 micrograms** daily and titrate slowly. Empty stomach, same routine; separate from calcium and iron by 4 hours	TSH every 3 months until stable, then yearly; adjust in 25 microgram steps; aim TSH in range, never suppressed
Subclinical hypothyroidism	TSH ≥10 on 2 tests 3 months apart → consider levothyroxine. Above range but <10, under 65 with symptoms → consider a **6-month trial**	Trial: symptoms no better with TSH in range → consider stopping. Untreated → monitor (box 6)
Overt hyperthyroidism	Refer endocrinology; consider a beta-blocker while waiting. **Carbimazole** is started by or on the advice of the specialist; doses are specialist-directed	First prescription: sore-throat rule, verbally and in writing; effective contraception (MHRA)
Thyroiditis	Supportive only (beta-blocker); **no antithyroid drug**	Recheck TFTs for a hypothyroid phase
Subclinical hyperthyroidism	Exclude drugs, recent illness, levothyroxine excess; specialist advice if TSH <0.1 twice with disease or symptoms	Weigh AF and fracture risk; treatment is not automatic in older adults

6 FOLLOW-UP AND MONITORING CHECKLIST

NG145 · SPS

- ☐ **Subclinical result**: repeat TSH + FT4 at about 3 months
- ☐ **Levothyroxine**: TSH every 3 months until 2 similar results in range, then yearly
- ☐ **Untreated subclinical hypo**: TSH + FT4 yearly if TPO positive or previous surgery or radioiodine; otherwise every 2–3 years
- ☐ **Untreated subclinical hyper**: TSH every 6 months
- ☐ **Antithyroid drug**: TSH, FT4, FT3 every 6 weeks until TSH in range, then TSH every 3 months
- ☐ **After stopping (Graves')**: TSH within 8 weeks, then every 3 months for a year, then yearly
- ☐ **After radioiodine**: TSH, FT4, FT3 every 6 weeks for 6 months; hypothyroidism is expected
- ☐ **Amiodarone, lithium**: TFTs 6-monthly (SPS)
- ☐ **Thyroiditis**: recheck TFTs to catch the hypothyroid phase

7 WHEN AND WHERE TO REFER

NG145 · NG12 · RCOG GTG76

999	Thyroid storm, myxoedema coma, unstable AF
Same day	Possible agranulocytosis; sight-threatening eye disease; adrenal crisis
Suspected cancer	Unexplained thyroid lump (NG12)
Urgent endocrinology	Low FT4 with non-raised TSH; overt hyperthyroidism with AF, eye disease or severe symptoms; amiodarone thyrotoxicosis
Antenatal endocrine	Any abnormal TFT in pregnancy, or Graves' now or before
Routine or advice	New Graves' or toxic nodular disease; subclinical hyperthyroidism meeting NG145 criteria; high TSH despite adherence; persistent symptoms on optimal levothyroxine (liothyronine is a specialist decision)

8 SPECIAL SITUATIONS THAT CHANGE THE READING

RCOG GTG76 · NG145 · SPS

SITUATION	WHAT THE RESULT CAN MEAN	WHAT TO DO
Pregnancy	TSH normally falls in the first trimester (hCG); non-pregnant ranges mislead	Trimester-specific range only. On levothyroxine: increase the dose as soon as the test is positive, urgent TFTs, joint obstetric–endocrine care. Graves' now or before → TRAb
After delivery	Postpartum thyroiditis: often a hyper phase, then a hypo phase	Supportive treatment; repeat TFTs; decide later whether replacement is lifelong
Amiodarone	Raised FT4 with normal TSH can be the drug effect; hypo- and hyperthyroidism both occur	Compare with baseline. Low TSH with raised FT4/FT3 → urgent endocrinology. Do not stop amiodarone without cardiology
Lithium · immunotherapy	Lithium: hypothyroidism common. Checkpoint inhibitors: thyroiditis, often hyper then hypo	Treat hypothyroidism and continue lithium · follow the oncology protocol
Older adults	A mildly raised TSH is common; a suppressed TSH carries AF and fracture risk	Treatment rarely helps mild subclinical disease; do not file a suppressed TSH as "near normal"

9 PITFALLS AND NEVER-EVENTS

NG145 · MHRA · SPC

- **Never** diagnose subclinical disease on one result, or test during acute illness unless thyroid disease is suspected.
- **Never** titrate central hypothyroidism on TSH, or give levothyroxine before cortisol when Addison's is possible.
- **Never** give carbimazole for thyroiditis, or skip the written sore-throat warning.
- **Never** act on a discordant result before asking about biotin and timing.
- **Never** raise levothyroxine to suppress TSH for persistent tiredness: check anaemia, B12, coeliac disease, sleep and mood.
- Do not stop amiodarone without cardiology; a suppressed TSH after thyroid cancer may be the oncology target.

10 REASSURE AND SAY IT LIKE THIS

NG145 · plain English

- A mildly abnormal TSH in acute illness that settles on recovery needs no treatment.
- Positive TPO antibodies with a normal TSH are not a reason to treat.
- Untreated mild subclinical disease: most people stay well; the plan is monitoring, not tablets.

"Your thyroid test is a little outside the normal range, but one result can be thrown off by a recent illness, some supplements or timing. Before we label anything, we'll repeat it in about three months. If it's still off, we'll decide together whether treatment would actually help you."

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Raised prolactin CLINIC GUIDE

Interpreting a raised prolactin in adults 18+. Not for pregnancy or breastfeeding (physiological rise), children, or known pituitary tumours already under endocrinology.

Society for Endocrinology · Pituitary Society **2023**
SfE/BSE/BHVS echo statement **2019** · NICE **NG12**
Cabergoline SPC · MHRA · Reviewed **Oct 2026** · next **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/hyperprolactinaemia.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → how high? → why?

! ACT TODAY

SfE apoplexy guidance · NICE NG12

TIER	RESULT OR PICTURE	ACTION
999	**Pituitary apoplexy**: sudden severe headache with visual loss, double vision, vomiting, collapse or confusion, with or without a known pituitary tumour	Transfer; **hydrocortisone 100 mg** IM/IV if adrenal insufficiency suspected or unstable — do not delay transfer
Same day	Raised prolactin with a **new visual-field defect** (classically bitemporal) or new cranial-nerve palsy	Same-day specialist or emergency assessment: MRI, formal fields
Urgent	Prolactin **>5000 mIU/L** without a drug cause · raised with headache, hypogonadism or amenorrhoea · persists after drug review	Urgent endocrinology ± pituitary MRI
Suspected cancer pathway	Aged **50+** with discharge, retraction or other change in **one nipple only**, whatever the prolactin	Breast suspected cancer pathway referral (NG12). A raised prolactin alone is never a cancer referral

1 IS IT REAL? MOST MILD RISES ARE NOT

SfE · local laboratory protocol

STRESS OF THE SAMPLE
Venepuncture, exercise, sex, sleep, a breast examination or nipple stimulation all raise prolactin. Repeat rested; a cannulated sample if needle stress is suspected.

MACROPROLACTIN
Bound, inactive prolactin reads as high. Ask for PEG precipitation when symptoms are absent or out of proportion; act on the **monomeric** result.

PREGNANCY, DRUGS, THYROID
Pregnancy test, drug list, TSH before anything else. Dopamine blockers, SSRIs, opioids and oestrogens raise it.

HOOK EFFECT
A large pituitary mass with only a modest prolactin → ask the lab to repeat on dilution. Check units (mIU/L vs µg/L) and sex-specific range.

2 HOW HIGH, AND WHAT IT USUALLY MEANS

Endocrine Society · Pituitary Society 2023 · use your laboratory's range

RESULT	USUALLY MEANS	ACTION	TIMING
Mildly raised, single result, no symptoms	Stress artefact or macroprolactin	Repeat rested with macroprolactin screen	Routine
Macroprolactin positive, monomeric normal	Macroprolactinaemia: inactive, harmless	Reassure; no MRI, no treatment	Discharge
Raised on a dopamine antagonist	Drug-induced (antipsychotics, metoclopramide, domperidone)	Review with the prescriber; repeat if safely withheld	Agreed interval
Raised with high TSH	Primary hypothyroidism (TRH drive)	Treat the thyroid; recheck prolactin	Once euthyroid
Persistent, unexplained monomeric rise	Microprolactinoma, stalk effect	Endocrinology ± pituitary MRI	Urgent if symptoms
Usually <2000 mIU/L with a pituitary mass	Stalk compression by a non-prolactin tumour	Specialist; dopamine agonist will not shrink it	Urgent
>5000 mIU/L	Prolactinoma strongly suggested, often macro (≥10 mm)	Urgent endocrinology; check visual fields	Urgent
Large mass, modest prolactin	Hook effect hiding a macroprolactinoma	Ask the lab to repeat on dilution	Urgent
Any level + visual or neuro signs	Mass effect	Same-day assessment	Today

Prolactin level does not reliably predict symptoms. A symptomatic patient may need imaging at a much lower level; local endocrine pathways set MRI thresholds.

3 SYMPTOMS THAT MAKE A RAISED RESULT MATTER

SfE

Women	Scanty or absent periods, galactorrhoea, infertility, low libido
Men	Low libido, erectile dysfunction, infertility; often present late with headache or field loss
Anyone	Headache, visual change, features of low pituitary function; long-standing hypogonadism → bone loss

4 FIND THE CAUSE

clue → think → first test

CLUE	THINK	FIRST TEST
No symptoms, high result	**Macroprolactin**	PEG screen
Antipsychotic, antiemetic	**Drug-induced**	Drug review
Tired, cold, weight gain	**Hypothyroidism**	TSH, FT4
Missed period, breast change	**Pregnancy**	hCG
Amenorrhoea, galactorrhoea	**Prolactinoma**	Repeat; endocrinology
Irregular cycles, hirsutism	**PCOS** (mild rise)	Testosterone, SHBG
Renal or liver disease	**Reduced clearance**	U&E/eGFR, LFTs
Headache, field loss, low libido	**Pituitary mass**	Fields; specialist MRI
Chest wall scar, piercing, zoster	**Neurogenic**	Exclude others first

5 WORK-UP CHECKLIST

SfE · Endocrine Society

- ☐ **Repeat prolactin**, rested
- ☐ **TSH** ± FT4
- ☐ U&E/eGFR, LFTs
- ☐ **Visual fields**: confrontation; formal if any doubt
- ☐ Ask: periods, libido, erections, discharge, headache
- ☐ **Macroprolactin** (PEG)
- ☐ **Pregnancy test**
- ☐ Full drug and supplement list
- ☐ LH, FSH, oestradiol or testosterone if hypogonadal
- ☐ MRI via endocrinology, not first line

Galactorrhoea from prolactin is usually bilateral and milky; one-sided or bloody discharge is a breast question. Avoid repeated breast examination before sampling.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

SfE · Pituitary Society 2023 · cabergoline SPC · MHRA

CAUSE	FIRST LINE	CHECK AND CONTINUE
Macroprolactinaemia	Reassure and explain; no treatment, MRI or repeat testing if monomeric prolactin is normal and there are no symptoms	Reassess only if symptoms develop
Drug-induced	Discuss stopping, reducing or switching with the prescriber. **Never stop an antipsychotic** without the mental-health team; a switch or aripiprazole add-on is their decision. Do not start a dopamine agonist empirically	Repeat prolactin if the drug is safely withheld; still raised → endocrine advice ± MRI
Hypothyroidism	Levothyroxine (see Abnormal TFTs guide)	Recheck prolactin once euthyroid
Microprolactinoma *specialist-initiated*	**Cabergoline** 0.5 mg a week (one dose or two), increased by 0.5 mg a week at monthly intervals; usual 1 mg a week, range 0.25–2 mg; above 1 mg a week give in divided doses (SPC)	Baseline echocardiogram before long-term use; prolactin schedule set by endocrinology
Macroprolactinoma	Cabergoline, specialist-led; surgery if resistant, intolerant or apoplexy	MRI and formal fields arranged by endocrinology
Stalk compression	Neurosurgery and endocrinology; **no dopamine agonist**	Pituitary function and fields by the specialist

7 LIVING WITH A PROLACTINOMA — WHAT TO COVER

DVLA · CoSRH · cabergoline SPC

IT IS BENIGN
Not a cancer; most are controlled with tablets and never need surgery. The Pituitary Foundation has good patient information.

TAKING CABERGOLINE
Same days each week, with food to ease nausea. Do not stop or change the dose without the specialist: prolactin can rise again within weeks.

DRIVING
A visual-field defect must be reported to the DVLA; no driving until fields are confirmed safe. Sleepiness on cabergoline → do not drive.

CONTRACEPTION, BONES
Microprolactinoma: combined hormonal contraception is generally fine if otherwise eligible (CoSRH); macro → ask endocrinology. Weight-bearing exercise, calcium, vitamin D.

8 FOLLOW-UP AND MONITORING

SPC · MHRA · SfE 2019

- [ ] **Repeat after drug review**: normal → stop investigating
- [ ] **Cabergoline**: echocardiogram at baseline, then at 5 years on usual doses; sooner at higher doses or with new murmur
- [ ] Ask about **impulse-control** problems (gambling, spending, hypersexuality, binge eating)
- [ ] **Sleepiness** or sudden sleep onset → do not drive
- [ ] Mood change, especially with past psychosis or mania
- [ ] **Pregnancy plans**: agree with endocrinology before trying
- [ ] **Bone health** if long hypogonadism or amenorrhoea
- [ ] Dose changes and withdrawal: specialist decisions
- [ ] Macroprolactinaemia: discharge, note it on the record

9 WHEN AND WHERE TO REFER

SfE · NG12

999	Suspected pituitary apoplexy
Same day	New visual-field defect or cranial-nerve palsy
Suspected cancer	One-nipple change aged 50+ (breast, NG12)
Urgent endocrinology	>5000 mIU/L; hypogonadism or amenorrhoea; persists after drug review; macroadenoma; pregnancy with a macroprolactinoma
Routine endocrinology	Microprolactinoma; complex macroprolactin result; drug-induced with symptoms where the drug cannot change
Psychiatry	Symptomatic antipsychotic-induced rise
Fertility	Cycles not restored despite normal prolactin (with endocrinology)

10 SPECIAL GROUPS

Pituitary Society 2023 · ESE 2021 · NICE NG257

- **Pregnancy**: plan with endocrinology. Microprolactinoma: the dopamine agonist is usually stopped once pregnancy is confirmed. Macroprolactinoma or tumour near the chiasm: specialist review throughout; new headache or visual change → same day.
- **Infertility**: normalise prolactin first; ovulation usually returns. Measure prolactin in infertility only with an ovulatory disorder, galactorrhoea or a pituitary tumour (NG257).
- **Men and post-menopausal women**: no menstrual clue, so tumours are found later and are more often macroadenomas — always check visual fields.
- **Antipsychotics long term**: sustained rise → hypogonadism and bone loss; assess fracture risk and raise with the psychiatric team.

11 PITFALLS AND NEVER-EVENTS

SfE · SPC · MHRA

- **Never** MRI a single unrested result before repeating it and checking TSH, pregnancy and macroprolactin.
- **Never** stop an antipsychotic yourself; dopamine agonists can worsen psychosis or mania.
- **Never** treat stalk compression with cabergoline, or trust a modest result beside a large mass (hook effect).
- **Never** measure prolactin to monitor a pregnancy; it rises normally.
- Do not send a raised prolactin down a cancer route.

12 REASSURE AND SAY IT LIKE THIS

plain English

Most raised prolactin results are stress, drugs or macroprolactin. Prolactinomas are benign and usually controlled with tablets.

"Your prolactin reading is high, but most of it is a larger, inactive form that attaches to proteins in the blood and can't act on your body. It's a quirk of your blood test, not a disease, so you don't need a scan or treatment unless new symptoms appear."

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Low testosterone in men

CLINIC GUIDE

Interpreting a low testosterone in men 18+. Not for boys or delayed puberty, women, gender-affirming hormone care, or men on androgen deprivation for prostate cancer.

BSSM testosterone deficiency guide **2023**
NICE **NG12** (prostate) · product SPCs
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/testosterone-deficiency.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → how low? → primary or secondary?

! ACT TODAY

BSSM 2023 · NICE NG12

TIER	RESULT OR PICTURE	ACTION
999	Sudden severe headache with visual loss (possible pituitary apoplexy) · vomiting, low BP or collapse in known hypopituitarism (adrenal crisis)	Emergency transfer
Urgent	Testosterone **<5.2 nmol/L** with low or normal LH and FSH, or a **raised prolactin** · headache, visual-field loss or other pituitary-axis problems	Endocrinology ± pituitary MRI **before** any testosterone
Urgent	Low testosterone with anaemia, bone pain and fatigue in an older man	Exclude myeloma, prostate mets first
Suspected cancer pathway	Prostate feels malignant on DRE → refer. PSA above the age threshold with possible prostate symptoms → consider referral: 40–49 **>2.5** · 50–59 **>3.5** · 60–69 **>4.5** · 70–79 **>6.5** µg/L	Suspected cancer pathway referral (NG12); no testosterone until resolved

1 IS IT REAL? TWO GOOD SAMPLES, AND SYMPTOMS

BSSM 2023

TIMING
Two samples on separate days, **7–11 am**, ideally fasting. Afternoon results run lower and cannot diagnose; private one-off results need repeating.

ACUTE ILLNESS
Illness, surgery, crash dieting and heavy alcohol suppress testosterone. Repeat once recovered.

SHBG
Obesity, diabetes and hypothyroidism lower SHBG (total looks low); ageing, liver disease, hyperthyroidism raise it. Borderline → SHBG + calculated free T.

DRUGS
Opioids, glucocorticoids, anabolic steroids (ask directly), and prolactin-raising antipsychotics. Symptoms alone never justify treatment.

2 HOW LOW, AND WHAT IT MEANS

BSSM 2023 action levels, not universal cut-offs · use your laboratory's range

TOTAL TESTOSTERONE (TWO MORNING SAMPLES)	MEANS	ACTION	TIMING
>12 nmol/L	Deficiency unlikely	Look for other causes of the symptoms	Routine
8–12 nmol/L borderline	Possible deficiency; often functional	SHBG + calculated free testosterone; treat functional causes; repeat	Routine
<8 nmol/L + symptoms	Deficiency likely	LH and FSH to classify	Soon
Low + **LH/FSH raised**	**Primary** (testicular) failure	Karyotype if small firm testes; TRT candidate after checks (page 2)	Routine
Low + **LH/FSH low or normal**	**Secondary** (pituitary or hypothalamic)	Prolactin, ferritin + TSAT, other pituitary axes; endocrinology before TRT	Soon
Low on **opioids or after anabolic steroids**	Secondary, often reversible	Drug review; endocrinology if it persists after stopping	Routine
Low + **haematocrit >0.54**	TRT would raise thrombosis risk	Do not start; look for hypoxia, smoking, sleep apnoea	Before any TRT
<5.2 nmol/L + low LH/FSH, or prolactin raised	Possible pituitary tumour	Endocrinology ± pituitary MRI	Urgent

3 SYMPTOMS AND WHO TO TEST

BSSM 2023

Sexual (most specific)	Low desire, erectile dysfunction, fewer morning erections — the three commonest
Physical	Less muscle and body hair, central fat, hot flushes, fragility fracture, anaemia
Mood (least specific)	Low mood, poor concentration, low energy: think depression first
Test	ED or low desire; type 2 diabetes; obesity; long-term opioids, antipsychotics or anticonvulsants

4 FIND THE CAUSE

clue → think → first test

CLUE	THINK	FIRST TEST
Obesity, type 2 diabetes, snoring	**Functional**	HbA1c, STOP-BANG
Long-term opioids or steroids	**Drug-induced**	Drug review
Gym user, small testes, low LH	**Anabolic steroids**	Ask directly
Small firm testes, gynaecomastia	**Klinefelter**	Karyotype
Diabetes, liver disease, arthropathy	**Haemochromatosis**	Ferritin, TSAT
Headache, galactorrhoea, field loss	**Pituitary tumour**	Prolactin; refer
Chemotherapy, mumps, orchidectomy	**Testicular failure**	LH, FSH
Heavy drinking, liver disease	**Alcohol, high SHBG**	LFTs, SHBG
Low mood, tired, poor sleep	**Depression, thyroid, OSA**	PHQ-9, TSH, FBC

5 WORK-UP CHECKLIST

BSSM 2023

- ☐ **Testosterone ×2**, 7–11 am
- ☐ **Prolactin** if LH/FSH not raised
- ☐ HbA1c, lipids, BP · TSH, LFTs, ferritin
- ☐ **LH, FSH** if low
- ☐ **FBC**, haematocrit
- ☐ Sleep apnoea screen
- ☐ **SHBG** (+ albumin) → free T
- ☐ **PSA** after counselling; DRE if indicated
- ☐ **Fertility plans** — ask now

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

BSSM 2023 · product SPCs

CAUSE	FIRST LINE	CHECK AND CONTINUE
Functional	Weight loss, CPAP for sleep apnoea, diabetes control, alcohol within 14 units a week, opioid review, stop anabolic steroids	Reassess symptoms and testosterone once the cause is addressed; response is variable
Primary deficiency *replacement, often long term*	**Testosterone replacement** under the local endocrine/urology protocol: daily gel (fast withdrawal if problems), short-acting IM, or long-acting undecanoate IM (specialist-led). Aim for the mid-normal range	Gel: avoid skin transfer to partners and children. Long-acting injection: observe after each dose (oil microembolism)
Secondary deficiency	Endocrinology first; treat the cause (e.g. prolactinoma with a dopamine agonist may restore testosterone)	TRT only on specialist advice
Wants children now or later	**No testosterone**: it suppresses sperm production. Refer to andrology or reproductive endocrinology	Recovery after stopping is slow and not guaranteed
Erectile dysfunction	PDE5 inhibitor remains first line after the usual checks (nitrates, unstable CVD)	Testosterone is not an ED treatment without confirmed deficiency

7 BEFORE ANY TESTOSTERONE — CONTRAINDICATIONS

BSSM 2023 · SPC

- **Do not start**: locally advanced or metastatic prostate cancer · male breast cancer · wish to preserve or achieve fertility · haematocrit **>0.54** · severe heart failure (NYHA IV).
- **Specialist caution**: untreated severe sleep apnoea · recent major cardiovascular event · thrombophilia or recurrent thrombosis · unexplained prostate nodule or raised PSA · significant LUTS · treated localised prostate cancer.

8 BASELINE RECORD BEFORE STARTING

BSSM 2023

- ☐ Two results with dates
- ☐ FBC, haematocrit
- ☐ CV risk, sleep apnoea
- ☐ Gel-transfer advice
- ☐ LH, FSH ± prolactin
- ☐ PSA; DRE if indicated
- ☐ Fertility discussion
- ☐ Who prescribes and monitors

9 PREPARATIONS AT A GLANCE — REPLACEMENT, PER LOCAL PROTOCOL

product SPCs · BSSM 2023

PREPARATION	ADVANTAGES	WATCH FOR	LEVEL CHECK
Daily gel	Stable levels; quick to stop if problems	Skin transfer: dry, cover, wash hands; irritation; adherence	Timing per the product SPC
Short-acting IM	Can be adjusted or stopped	Peaks and troughs in mood and energy; more erythrocytosis	Trough, before the next dose
Long-acting undecanoate IM	Infrequent dosing; steady levels; specialist-led	Slow to reverse; oil microembolism or anaphylaxis — give in clinic, observe	Trough, before the next dose

10 FOLLOW-UP ON TESTOSTERONE

BSSM 2023 · NG12

- ☐ **3, 6 and 12 months, then yearly**: testosterone (timing per product), FBC, PSA, symptoms, BP, weight
- ☐ **Haematocrit ≥0.54** → withhold or reduce; specialist advice
- ☐ **PSA rise >1.4 ng/mL** in any 12 months, or **>0.4 ng/mL a year** over more than 2 years → urology
- ☐ Prostate feels malignant, or PSA above age threshold → NG12
- ☐ New wish for children → stop and refer
- ☐ No benefit despite good levels → reassess, consider stopping
- ☐ Functional cause fixed → consider a trial off, under specialist guidance
- ☐ Annual review: still indicated? benefit? new contraindications?

11 WHEN AND WHERE TO REFER

BSSM 2023 · NG12

999	Apoplexy; adrenal crisis; collapse or throat swelling after an injection
Same day	Settling cough or breathlessness after an injection; new aggression or mania
Suspected cancer	Malignant-feeling prostate; PSA above age threshold (NG12)
Urgent endocrinology	<5.2 nmol/L with low LH/FSH; raised prolactin; visual or pituitary features
Andrology	Low testosterone and fertility wishes; Klinefelter
Urology	PSA rise on testosterone (box 10)
Endocrinology	Secondary deficiency; primary where no local GP protocol; complex cases

12 PITFALLS AND NEVER-EVENTS

BSSM 2023

- **Never** diagnose on one result, an afternoon sample or symptoms alone.
- **Never** start testosterone with low LH/FSH before endocrine assessment.
- **Never** prescribe before the fertility conversation is had and documented.
- **Never** use testosterone to treat infertility or ED without confirmed deficiency.
- **Never** leave testosterone on repeat without an annual review.
- Do not miss sleep apnoea, depression, opioids or anabolic steroids.

13 FUNCTIONAL CAUSES; SAY IT

BSSM · plain English

Weight loss, treating sleep apnoea, cutting alcohol and reviewing opioids can lift testosterone without lifelong treatment; supplements (zinc, vitamin D) are not a substitute.

"Testosterone is highest first thing, so one afternoon test can't tell us much. I'd like two morning samples, plus a few other hormones that show where any problem starts. If we do treat, it switches off sperm production, so tell me now if you might want children. Weight, sleep and some medicines can lower it too, and fixing those can help."

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Female hormone results

CLINIC GUIDE

Interpreting FSH, LH, oestradiol, progesterone, prolactin and androgens in women 18+. Not for pregnancy, primary amenorrhoea or puberty (paediatrics), fertility-treatment monitoring or gender-affirming care.

NICE **NG23** menopause (updated 2026) · **NG243** · **NG12**
PCOS guideline **2023** · ESHRE POI **2024** · SfE androgen excess **2025**
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/female-hormone-abnormalities.html

PAGE 1 · INTERPRET

is it dangerous? → is it valid? → read the pattern → name the cause

! ACT TODAY

NG243 · SfE apoplexy · NG12

TIER	RESULT OR PICTURE	ACTION
999	**Adrenal crisis**: shock, vomiting, abdominal pain, confusion, hypoglycaemia or low Na^+, especially with known or possible adrenal insufficiency · **Pituitary apoplexy**: sudden severe headache with visual loss, double vision or reduced consciousness	**Hydrocortisone 100 mg** IM/IV now, do not wait for bloods (NG243) · transfer
Urgent	**Rapidly progressive virilisation** (deep voice, clitoromegaly, fast-worsening hirsutism) or testosterone **about 6–7 nmol/L or higher** · raised DHEAS	Urgent endocrinology or gynaecology: possible androgen-secreting tumour
Suspected cancer pathway	Ascites or pelvic or abdominal mass · postmenopausal bleeding aged **55+** not explained by HRT · persistent bloating, early satiety or pelvic pain → follow NG12 (CA-125 and ultrasound)	Suspected cancer pathway referral (NG12). Raised FSH/LH after the menopause is normal — never a tumour marker

1 IS THE RESULT VALID? CHECK BEFORE YOU INTERPRET

NG23 · PCOS 2023 · local laboratory

PREGNANCY FIRST
A pregnancy test before anything else in amenorrhoea. Then ask about breastfeeding and recent delivery.

HORMONES ON BOARD
Combined contraception, high-dose progestogen and HRT make FSH, LH and oestradiol uninterpretable. Do not use FSH to diagnose menopause on them.

TIMING
Progesterone about **7 days before the expected period** (day 21 only in a 28-day cycle). FSH varies: never diagnose POI on one result.

ASSAY AND STRESS
Testosterone: early-morning sample with SHBG, reliable assay. Prolactin rises with stress and breast stimulation; repeat rested, check macroprolactin.

2 READ THE PATTERN

NG23 · ESHRE 2024 · PCOS 2023 · use your laboratory's ranges

RESULT PATTERN	MEANS	ACTION	TIMING
FSH raised, <40, few or no periods	Possible **POI**	Repeat FSH **4–6 weeks** later; raised twice = POI → HRT or CHC	Soon
FSH raised, aged 40–45, symptoms and cycle change	Supports menopause	Manage as perimenopause (NG23)	Routine
Aged **45+** with symptoms	Clinical diagnosis	No FSH, oestradiol or AMH needed	Routine
FSH/LH low or normal + low oestradiol + amenorrhoea	Hypothalamic or pituitary	Weight, exercise, eating, stress; prolactin, TSH; endocrinology if no functional cause	Soon
Testosterone/FAI mildly raised + irregular cycles	**PCOS** likely	Exclude thyroid, prolactin, NCCAH, Cushing's (box 3)	Routine
Testosterone >4 nmol/L	Needs explanation	Refer endocrinology (CKS). **≈5+**: prompt confirmation and investigation, incl. ovarian/adrenal imaging (SfE 2025), esp. if unexpected or immunoassay	Prompt
Testosterone ≈6–7 nmol/L+, virilisation or rapid progression	Possible tumour	Urgent endocrinology; DHEAS if adrenal source suspected	Urgent
17-OHP raised	Non-classical CAH	Endocrinology	Routine
Prolactin raised	Stress, drugs, thyroid, macroprolactin, prolactinoma	See the Raised prolactin guide	By level
Progesterone above / below lab ovulatory threshold	Ovulation likely / anovulation or mistimed	If low, check the timing and repeat before labelling anovulation	Routine

The LH:FSH ratio is not a diagnostic criterion for PCOS. Read FAI against the reporting laboratory's range; a normal testosterone does not exclude clinical hyperandrogenism.

3 NAME THE CAUSE

clue → think → first test

CLUE	THINK	FIRST TEST
Irregular cycles, hirsutism, acne	**PCOS**	Testosterone, SHBG
Flushes, no periods, under 40	**POI**	FSH ×2
Low weight, high exercise, restricted eating	**Hypothalamic**	FSH/LH, oestradiol
Galactorrhoea, headache	**Raised prolactin**	Prolactin
Tired, cold, heavy periods	**Hypothyroidism**	TSH
Striae, bruising, weak thighs	**Cushing's**	Specialist cortisol tests
Early severe hirsutism, family history	**NCCAH**	17-OHP
Weeks of virilisation	**Androgen tumour**	Testosterone, DHEAS

4 WORK-UP CHECKLIST — BY PHENOTYPE

no universal panel

- ☐ **Pregnancy test**
- ☐ **Prolactin**, rested
- ☐ **Testosterone + SHBG**
- ☐ Timed progesterone (fertility)
- ☐ Pelvic USS when indicated
- ☐ **TSH** if indicated
- ☐ **FSH/LH** ± oestradiol
- ☐ DHEAS, 17-OHP if indicated
- ☐ PCOS: BP, BMI, HbA1c, lipids
- ☐ Medicine and contraception list

PCOS in adults: 2 of ovulatory dysfunction, hyperandrogenism, and polycystic ovaries on ultrasound or a raised AMH, after excluding other causes. Adolescents: both of the first two; no ultrasound or AMH. AMH is not used to diagnose POI.

5 USUALLY REASSURING

NG23 · PCOS 2023

- High FSH and LH after the menopause: expected, not a tumour marker.
- A mildly raised prolactin after a stressful sample: often normal on a rested repeat.
- Polycystic-looking ovaries with regular cycles and no hyperandrogenism: not PCOS.
- A low progesterone taken on the wrong day: retime, do not label.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

NG23 · PCOS 2023 · ESHRE 2024 · MHRA

CAUSE	FIRST LINE	CHECK AND CONTINUE
PCOS	Person-centred lifestyle support. **Combined hormonal contraceptive** for cycles and hyperandrogenism if not seeking pregnancy (lowest-VTE-risk choice). Hirsutism: eflornithine cream, laser; co-cyprindiol only after VTE assessment, stop 3–4 cycles after resolution (MHRA)	Infrequent bleeds and no CHC or LNG-IUS → endometrial protection, e.g. medroxyprogesterone 10 mg daily for 10–14 days. Fertility: letrozole via the fertility service
POI *replacement*	**HRT or CHC until at least the average age of menopause** unless contraindicated; younger women often need higher oestrogen doses; progestogen if uterus present	HRT is not contraception; spontaneous pregnancy is possible. DXA at diagnosis
Menopause	HRT for symptoms; oestrogen alone only after hysterectomy. Consider **transdermal** if VTE risk, including BMI **over 30**	Review at 3 months, then at least yearly; no fixed time limit
Hypothalamic amenorrhoea	Restore energy intake, reduce training, eating-disorder services, psychological support	Sex-steroid replacement is a specialist decision; bone assessment if prolonged
Thyroid, prolactin	Treat the cause (Abnormal TFTs and Raised prolactin guides)	Cycles often return once corrected

7 WHEN HRT IS NOT THE WHOLE ANSWER

NG23 · MHRA

HRT UNSUITABLE
Menopause-specific CBT. SSRIs, SNRIs and clonidine are not first line for flushes alone. Fezolinetant: check liver function first (MHRA).

VAGINAL SYMPTOMS
Vaginal oestrogen, alone or with HRT; continue as long as needed; no endometrial monitoring; report bleeding.

LOW DESIRE
Consider testosterone if HRT alone has not helped (NG23); off-label, keep levels in the female range.

SUPPORT
PCOS: no single best diet; avoid weight stigma. FHA: eating-disorder services. POI: the Daisy Network.

8 FOLLOW-UP AND MONITORING

NG23 · PCOS 2023 · ESHRE

- [ ] **Possible POI**: second FSH 4–6 weeks after the first
- [ ] **HRT or POI treatment**: review at 3 months, then at least yearly — symptoms, bleeding, BP, contraception
- [ ] **POI**: DXA at diagnosis; repeat interval by result and risk
- [ ] **PCOS**: BP, BMI, lipids and glycaemic status at diagnosis; glucose or HbA1c every 1–3 years by risk; OGTT if high risk or planning pregnancy
- [ ] **PCOS**: endometrial-protection plan agreed and coded
- [ ] Mood screen in PCOS and POI
- [ ] Unscheduled bleeding on HRT, or PMB → investigate
- [ ] Fezolinetant: liver function before starting and as the SPC/MHRA advise

9 WHEN AND WHERE TO REFER

NG23 · NG12 · NG243

999	Adrenal crisis; pituitary apoplexy
Same day	Severe pelvic pain with suspected torsion
Suspected cancer	Mass or ascites; PMB 55+; abnormal CA-125/ultrasound (NG12)
Urgent endocrinology	Testosterone ≈6–7 nmol/L+, virilisation or rapid progression; high DHEAS; suspected pituitary tumour or Cushing's
Reproductive medicine	Confirmed POI with complex needs; anovulatory infertility
Gynaecology	PCOS with heavy, prolonged or persistent bleeding; ovarian lesion on USS
Endocrinology	Testosterone >4 nmol/L; raised 17-OHP; low gonadotrophins without a functional cause

10 PITFALLS — DO NOT ROUTINELY

NG23 · PCOS 2023

- Use the **LH:FSH ratio** for PCOS, or diagnose PCOS from "polycystic ovaries" alone.
- Take progesterone on day 21 whatever the cycle length.
- Check FSH or oestradiol to adjust HRT, or FSH on combined contraception.
- Diagnose POI from one FSH, or use AMH to diagnose it.
- Use CA-125 as a screening test, or scan the endometrium in asymptomatic PCOS.

11 SPECIAL GROUPS AND NEVER-EVENTS

NG23 · MHRA · ESHRE 2024

- **Contraception**: ovulation in PCOS is unpredictable, and spontaneous pregnancy occurs in POI — HRT is not a contraceptive. Antiandrogens (spironolactone, finasteride) are off-label for hirsutism: exclude pregnancy and ensure reliable contraception; finasteride is specialist-led.
- **On HRT**: suspected VTE → stop and same-day assessment; specialist advice before any restart. Breast cancer history with vaginal symptoms → discuss with the breast or menopause team.
- **Never** leave a woman with confirmed POI without a hormone-replacement discussion at each contact; non-treatment carries the greater risk to bone and heart.
- **Never** attribute rapid virilisation to PCOS, or delay referral to repeat a testosterone when the clinical picture is convincing.

12 SAY IT LIKE THIS

plain English

"Your blood test shows your ovaries have slowed down earlier than usual. We'll repeat it in about six weeks to be sure. If it's confirmed, taking hormones until around the usual age of menopause protects your bones and heart — it replaces what your body should be making, rather than adding extra."

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Anaemia in adults: the low Hb CLINIC GUIDE

Interpreting a low haemoglobin in adults 18+. Not for pregnancy (use maternity thresholds), children, CKD on ESA (renal team), haemoglobinopathy care or active major bleeding (emergency).

BSG IDA guideline 2021 · NICE **NG12** · HTG690
NICE **NG239** (B12, 2024) · NG20 · NICE CKS (iron deficiency)
Reviewed **Oct 2026** · next review **Apr 2027**
Full protocol: gpreasoning.uk/tools/algorithms/anaemia.html

PAGE 1 · ASSESS

is it dangerous? → classify by MCV → find the cause

! RED FLAGS — SORT THEM FIRST

NICE NG12 · HTG690

TIER	FEATURE	ACTION
999	Collapse, shock or major bleeding (haematemesis, melaena) · chest pain or possible ACS, whatever the Hb	Emergency transfer
Same day	Breathless at rest, syncope, angina or heart failure · **rapidly falling Hb** · jaundice or dark urine (haemolysis) · film suggests leukaemia, or pancytopenia with fever or bleeding · **B12 neurology**	Same-day medical or haematology assessment (immediate for suspected leukaemia); treat B12 neurology now
Suspected cancer pathway	**Iron-deficiency anaemia** (any age) or aged **60+ with anaemia** even without iron deficiency → **FIT**: ≥10 µg Hb/g → colorectal pathway · rectal mass → consider referral without FIT · dysphagia, or 55+ with weight loss + upper GI symptoms → upper GI pathway · unexplained lymphadenopathy or splenomegaly → consider lymphoma pathway	NG12 referral; FIT <10 does not exclude cancer if concern persists

Aged 55+ with upper abdominal pain and a low Hb → consider non-urgent direct-access OGD (NG12). Transfusion is decided on symptoms and bleeding, not on the number alone.

1 CONFIRM AND CLASSIFY

use local laboratory ranges

Anaemia: Hb **<130 g/L** in men, **<120 g/L** in non-pregnant women (WHO). Read the MCV with the ferritin, reticulocytes and the rest of the count: mixed deficiency can give a normal MCV.

MCV	COMMONEST CAUSES	NEXT
<80 microcytic	Iron deficiency; thalassaemia trait	Ferritin
80–100 normocytic	Chronic disease, CKD, bleeding, mixed deficiency	Ferritin, CRP, U&E, retics
>100 macrocytic	B12 or folate, alcohol, liver, thyroid, drugs	B12 + folate, LFTs, TFTs

2 FOCUSED HISTORY — WHAT DISCRIMINATES

BSG 2021 · NG239

Where is the blood going?	Heavy periods, melaena, rectal bleeding, haematuria, blood donation
Bowel and upper GI symptoms	Change in habit, dysphagia, dyspepsia, weight loss
Drugs	NSAIDs, aspirin, anticoagulants; PPIs and metformin (B12); methotrexate (folate)
Diet and absorption	Vegan diet, alcohol, coeliac or IBD, gastric or bariatric surgery
Nerves and mind	Numbness, unsteadiness, memory: B12; ask about nitrous oxide
Periods, pregnancies	Heavy menstrual bleeding is the commonest cause before the menopause
Family and heritage	GI cancer, coeliac disease, thalassaemia

3 EXAMINATION FINDINGS THAT CHANGE THE PLAN

examination

Pulse, BP, breathlessness	Compromise → same day or 999
Abdominal or rectal mass	Rectal: consider suspected cancer referral, no FIT needed; abdominal: FIT
Nodes, spleen	Haematological cancer or haemolysis → urgent route
Jaundice	Haemolysis or liver disease
Vibration, proprioception, gait	B12 neuropathy: treat without waiting
PR when indicated	Rectal bleeding, change in bowel habit, suspected melaena
Pallor, koilonychia, glossitis	Support the picture; never rule anaemia in or out

4 KEY DIFFERENTIALS

clue → think → first test

CLUE	THINK	FIRST TEST
Low MCV, low ferritin	**Iron deficiency**	Coeliac serology; FIT; find the loss
Very low MCV, normal ferritin, heritage	**Thalassaemia trait**	Haemoglobinopathy screen (HPLC)
Normocytic, raised CRP or CKD	**Chronic disease**	CRP, eGFR; ferritin with TSAT
High reticulocytes, jaundice	**Haemolysis**	Bilirubin, LDH, haptoglobin, DAT, film
Macrocytic, neuropathy, vegan	**B12 deficiency**	B12, folate; anti-IF antibody
Macrocytic, alcohol	**Alcohol, liver**	LFTs; alcohol history
Other cytopenias, odd film	**Marrow disease**	Film; urgent haematology

A normal MCV does not exclude iron deficiency: mixed iron and B12 or folate deficiency can cancel out. Check ferritin, B12 and folate when the picture does not fit.

5 INVESTIGATIONS CHECKLIST

BSG 2021 · NG239 · NG20

- ☐ **FBC**, film if cytopenias or atypical
- ☐ **Ferritin**, CRP
- ☐ **B12 and folate together**
- ☐ U&E/eGFR, LFTs, TFTs
- ☐ **tTG IgA + total IgA** in IDA
- ☐ **FIT** in IDA, and in 60+ with anaemia
- ☐ Reticulocytes if bleeding or haemolysis
- ☐ Urine dipstick for blood

RESULT	MEANS
Ferritin **<15 µg/L**	Iron deficiency (highly specific)
Ferritin **<45 µg/L**	Consider GI investigation (BSG)
Ferritin normal + inflammation	Deficiency still possible: check TSAT (BSG)
Total B12 **<180 ng/L** · 180–350 · >350	Deficient · indeterminate · unlikely (NG239)

Coeliac serology only while eating gluten. Do not use B12 levels to monitor injections.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary.

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6 MANAGEMENT OF THE COMMONEST CAUSES

BSG 2021 · NICE CKS · NG239 · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Iron deficiency	One standard oral iron tablet **once daily**, e.g. ferrous sulfate 200 mg (65 mg elemental iron); **alternate days** if not tolerated (BSG 2021). IV iron if oral iron fails, is not tolerated or will not be absorbed	Recheck Hb at **2–4 weeks**; continue for **3 months after** Hb normalises. Treat the cause (heavy periods, coeliac) and investigate it
B12 deficiency	Hydroxocobalamin 1 mg IM three times a week for 2 weeks, then every 2–3 months. Neurological features: 1 mg on alternate days until no further improvement, then every 2 months (seek specialist advice)	Review symptoms at about 3 months (1 month in pregnancy). Autoimmune gastritis or ileal resection → lifelong IM. Diet only → oral B12
Folate deficiency	Folic acid 5 mg daily, usually for 4 months — **only once B12 is known to be normal or is being replaced**	Find the cause: diet, alcohol, malabsorption, drugs
Chronic disease, CKD	Treat the underlying condition; iron only if deficiency is shown	CKD anaemia → renal team (NG203)
Thalassaemia trait	**No iron** unless also iron deficient	Explain; partner testing before pregnancy

7 IRON DEFICIENCY — WHO NEEDS GI INVESTIGATION

BSG 2021 · NG12 · HTG690 · NG20

EVERY ADULT WITH IDA
FIT; ≥10 µg Hb/g → colorectal suspected cancer pathway. Coeliac serology (NG20).

MEN, POST-MENOPAUSAL WOMEN
Bidirectional endoscopy (OGD + colonoscopy) unless there is a clear cause — even if FIT is negative.

PRE-MENOPAUSAL WOMEN
Usually menstrual: treat and recheck. Endoscopy if aged 50+, GI symptoms or family history of colorectal cancer.

STILL UNEXPLAINED
Positive coeliac serology → gastroenterology. After negative endoscopy → H. pylori test; specialist advice.

8 SELF-CARE AND THE CONVERSATION

BSG 2021 · NG239

Take iron away from tea, coffee, calcium and antacids · black stools on iron are expected, but tarry, sticky stools with dizziness are not · vegans need a B12 supplement or fortified foods.

"The iron tablets will build you back up, but they don't tell us why you were low. Finding where the iron went is the important part, so the tests I'm arranging matter even when you feel better."

9 IRON TREATMENT PROBLEMS — AND THE FIX

BSG 2021 · SPC · MHRA

Nausea, constipation	Alternate days, a different salt, or take with food (absorbs less)
No Hb rise	Adherence, tea with meals, ongoing loss, coeliac, wrong diagnosis
IV iron	Only where anaphylaxis can be managed; observe afterwards
Ferric carboxymaltose	Check phosphate if repeated or high doses or risk factors (MHRA)

10 SAFETY-NET AND FOLLOW-UP CHECKLIST

BSG 2021 · NICE CKS · NG239

- ☐ **Hb at 2–4 weeks** on iron: no rise → adherence, ongoing loss, malabsorption, wrong diagnosis
- ☐ FIT returned and actioned; unreturned kit chased
- ☐ GI investigation and coeliac result reviewed
- ☐ Iron for **3 months after** Hb normalises; ferritin at the end
- ☐ B12: symptom review at about 3 months; injections booked
- ☐ Recurrent IDA after treatment → investigate again
- ☐ Return the **same day**: breathless, chest pain, faint, black tarry stools, new numbness or weakness
- ☐ Cause recorded in the problem list

11 WHEN AND WHERE TO REFER

NG12 · BSG 2021 · NG239

999	Shock, major bleeding, ACS
Same day	Severe symptoms, falling Hb, haemolysis, suspected leukaemia, B12 neurology
Suspected cancer	FIT ≥10; rectal mass; dysphagia; 55+ weight loss with upper GI symptoms; lymphadenopathy
Urgent GI	Unexplained IDA in men and post-menopausal women, FIT negative
Haematology	Unexplained macrocytosis or normocytic anaemia with other cytopenias; no response to oral iron after review

12 SPECIAL GROUPS AND NEVER-EVENTS

BSG 2021 · NG12 · NG239

- **Pregnancy**: different Hb thresholds — use the maternity pathway. **IBD or bariatric surgery**: IV iron is often preferred (specialist pathway). **On anticoagulants**: IDA is still investigated, not attributed to the drug.
- **Never** give folic acid alone before B12 deficiency is excluded or treated · never treat IDA in a man or post-menopausal woman without investigating the cause · never treat a negative FIT as an all-clear when concern persists.
- **Never** give iron for thalassaemia trait without proven iron deficiency · never delay B12 treatment when there are neurological features · never transfuse, or withhold transfusion, on the Hb number alone.

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Microcytosis: the low MCV CLINIC GUIDE

Interpreting an MCV below the reference range (usually <80 fL) in adults 18+. Not for pregnancy (maternity thresholds), children, known transfusion-dependent thalassaemia or CKD on dialysis (renal team).

BSG IDA 2021 · NICE **NG12** · HTG690 · NG20
BSH iron deficiency 2021 · NHS SCT
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/microcytosis.html

PAGE 1 · INTERPRET

is it dangerous? → iron or not iron? → find the cause

! ACT TODAY — RED FLAGS

NICE NG12 · BSG 2021

TIER	RESULT OR COMBINATION	ACTION
999	Low Hb with collapse, shock or major bleeding (haematemesis, melaena, heavy rectal bleeding)	Emergency transfer
Same day	Low Hb with chest pain, breathlessness at rest, syncope or heart failure · **rapidly falling Hb** · microcytosis with other cytopenias or a film suggesting leukaemia	Same-day medical or haematology assessment; transfusion is a hospital decision on symptoms, not the number alone
Suspected cancer pathway	**Iron-deficiency anaemia** (any age), or aged **60+ with anaemia** even without iron deficiency → **FIT**: ≥10 µg Hb/g → colorectal pathway · rectal mass → consider referral without FIT · dysphagia, or 55+ with weight loss + upper GI symptoms → upper GI pathway	NG12 referral; FIT <10 does not exclude GI pathology if concern persists
Urgent haematology	Marked microcytosis with very low Hb, splenomegaly or jaundice, especially with heritage from a high-prevalence region (thalassaemia intermedia, HbH disease)	Urgent advice; FBC, film, Hb analysis (HPLC)

1 CONFIRM IT IS REAL

BSG 2021

- **Look back first:** a long-standing, stable low MCV with a normal Hb points to a carrier state; a **new or falling** MCV points to iron loss.
- **Ferritin misleads** when there is inflammation, infection, liver disease or malignancy: it rises as an acute-phase protein. Add **CRP and transferrin saturation (TSAT)**; repeat iron studies once an acute illness has settled.
- **Recent iron, IV iron or transfusion** raises ferritin: note the date before interpreting.
- **Serum iron** varies through the day and with meals: do not decide on it alone.
- MCH is often a more reliable marker of iron deficiency than MCV (BSG).
- Anaemia: Hb **<130 g/L** men, **<120 g/L** non-pregnant women (WHO); use local laboratory ranges.

2 RESULT BANDS AND ACTION

BSG 2021 · NHS SCT

RESULT	MEANS	ACTION · TIMING
Ferritin **<15 µg/L**	Absent iron stores (highly specific)	Start iron **now**; find the cause
Ferritin **15–30 µg/L**	Low iron stores likely	Treat; investigate as IDA
Ferritin **30–45 µg/L** with inflammation	Iron deficiency still possible	TSAT; **<45**: consider GI investigation
Normal ferritin, **TSAT <20%**, raised CRP	Chronic disease, often with iron deficiency	Treat the disease; iron trial if deficiency likely
Normal ferritin + CRP, persistent low MCV	Thalassaemia trait likely	Hb analysis (HPLC) routinely
HbA2 ≥3.5%	Beta-thalassaemia carrier	Explain; partner testing if pregnancy possible
Normal HbA2, MCH low, normal ferritin	Possible alpha-thalassaemia trait	Laboratory advice; DNA testing if reproductive risk

Severe iron deficiency can lower HbA2 and mask beta-thalassaemia trait: correct proven iron deficiency, then interpret HbA2.

3 IRON STUDIES PATTERN

BSG 2021 · read together, never one test alone

PATTERN	FERRITIN	TSAT	TIBC / TRANSFERRIN	MCV, FILM	NEXT
Iron deficiency	Low	<20%	High	Low; pencil cells	Treat + find the loss
Chronic disease	Normal or high	<20%	Low or normal	Normal or slightly low	Treat the disease
Mixed	Variable	<20%	Variable	Low, high RDW	Iron trial; ≥10 g/L rise confirms
Sideroblastic	High	High	—	Low; ring sideroblasts in marrow	Film; haematology
Thalassaemia trait	Normal	Normal	Normal	Very low, Hb near normal	HPLC; no iron

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Low ferritin, heavy periods or GI loss	**Iron deficiency**	FIT; coeliac serology; urinalysis
Very low MCV, near-normal Hb, high RBC count, normal ferritin	**Thalassaemia trait**	Hb analysis (HPLC, HbA2)
Raised CRP, chronic inflammatory disease	**Chronic disease ± iron deficiency**	TSAT, CRP, ferritin trend
NSAIDs, aspirin, anticoagulants	**GI blood loss**	FIT; investigate, do not blame the drug
Vegan diet, blood donation, bariatric surgery	**Low intake or absorption**	Still investigate if man or post-menopausal
Diarrhoea, weight loss, family history	**Coeliac disease**	tTG IgA + total IgA (on gluten)
Raised ferritin and TSAT, alcohol, isoniazid, lead	**Sideroblastic anaemia**	Film; haematology; lead level
Other cytopenias, abnormal film	**Marrow disease (MDS)**	Film; urgent haematology

5 WORK-UP CHECKLIST

BSG 2021 · NG20 · NG12

- ☐ **FBC** with MCH, RDW; film if atypical
- ☐ **CRP + TSAT** if inflammation likely
- ☐ **FIT** in IDA, and 60+ with anaemia
- ☐ U&E/eGFR; LFTs if ferritin high
- ☐ **Ferritin** — the key test
- ☐ **tTG IgA + total IgA** in all IDA
- ☐ **Urinalysis** for blood in IDA
- ☐ **HPLC** if ferritin normal + low MCV

Reticulocyte Hb helps where available. Coeliac serology only while eating gluten. Hb analysis with ferritin, not instead of it.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT OF THE COMMONEST CAUSES

BSG 2021 · NHS SCT · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Iron deficiency replacement	One standard oral iron tablet **once daily**, e.g. ferrous sulfate 200 mg (65 mg elemental iron); **alternate days**, another oral preparation or IV iron if not tolerated (BSG 2021). Do not defer iron while awaiting tests unless colonoscopy is imminent	Recheck Hb at **2–4 weeks** (a rise of ≥10 g/L in 2 weeks confirms iron deficiency); continue for about **3 months after** Hb normalises. Treat the cause
Iron deficiency with inflammation	Treat the underlying disease; oral iron first unless ineffective, not tolerated or poorly absorbed	IV iron via local pathway when oral fails or urgent correction is needed
Thalassaemia trait	**No iron** unless iron deficiency is also proven. Explain it is a carrier state, not a disease	Code it in the record; partner testing via NHS SCT screening if pregnancy is planned or ongoing
Coeliac disease	Strict gluten-free diet + iron replacement	Gastroenterology per NG20; recheck FBC and ferritin
Sideroblastic, MDS, HbH disease	Haematology-led	Do not give iron empirically

7 IRON DEFICIENCY — WHO NEEDS GI INVESTIGATION

BSG 2021 · NG12 · HTG690 · NG20

EVERY ADULT WITH IDA
FIT; ≥10 µg Hb/g → colorectal suspected cancer pathway. Coeliac serology and urinalysis.

MEN, POST-MENOPAUSAL WOMEN
Gastroscopy + colonoscopy (CT colonography if unsuitable) unless there is a clear cause — even if FIT is negative.

PRE-MENOPAUSAL WOMEN
Usually menstrual: treat and recheck. Endoscopy if aged 50+, red flags, strong family history or IDA out of proportion.

NEGATIVE ENDOSCOPY
Persistent or recurrent IDA, or poor response → small bowel (capsule) and renal tract investigation.

8 TREAT THE SOURCE TOO

BSG 2021 · NG88 · SPC · MHRA

- **Heavy periods**: manage per NICE NG88 (LNG-IUS, tranexamic acid) alongside iron.
- **NSAIDs**: stop if possible; gastroprotection if essential.
- **IV iron** (local pathway): oral iron ineffective, not tolerated, malabsorbed, or urgent. Observe for at least 30 minutes afterwards (SPC). Ferric carboxymaltose: check phosphate with repeat dosing or risk factors (MHRA).
- Take iron away from tea, coffee, calcium and antacids where practical; adherence comes first.

9 FOLLOW-UP CHECKLIST

BSG 2021 · NHS SCT

- ☐ **Hb at 2–4 weeks** on iron: no rise → adherence, ongoing loss, malabsorption, wrong diagnosis
- ☐ FIT returned and actioned; unreturned kit chased
- ☐ Endoscopy and coeliac results reviewed and recorded
- ☐ Iron for **3 months after** Hb normalises; then ferritin
- ☐ Then FBC periodically (about **every 6 months** at first) to catch recurrence
- ☐ Trait: coded; partner testing offered; HbA2 repeated after iron if iron deficient at testing
- ☐ Declines investigation: informed decision and safety-net recorded
- ☐ Return the **same day**: black tarry stools, vomiting blood, chest pain, faint, worsening breathlessness

10 WHEN AND WHERE TO REFER

NG12 · BSG 2021

999	Shock, major bleeding, chest pain
Same day	Severe symptomatic anaemia, falling Hb, suspected leukaemia
Suspected cancer	FIT ≥10; rectal mass; dysphagia; 55+ weight loss with upper GI symptoms
Urgent GI	IDA in men and post-menopausal women, FIT negative
Haematology	Possible thalassaemia intermedia or HbH disease; sideroblastic or MDS picture; unexplained low MCV after work-up; no response to iron after review
Genetics	Both partners carriers: specialist counselling, urgently in pregnancy

11 SPECIAL GROUPS

BSG 2021 · NHS SCT

- **Pregnancy**: different Hb and ferritin thresholds — use the maternity pathway; carrier found → partner testing now.
- **IBD, coeliac, bariatric surgery**: oral iron often poorly absorbed; IV iron via specialist or local pathway. Previous surgery does not stop the search for other causes.
- **Frail or major comorbidity**: weigh the risks and benefits of endoscopy with the patient; record the decision.
- **Young men with IDA**: investigate as for older adults.

12 PITFALLS AND NEVER-EVENTS

BSG 2021 · NG12

- **Never** put IDA down to diet in a man or post-menopausal woman without GI investigation.
- **Never** treat a negative FIT as an all-clear when concern persists; it does not see the upper GI tract.
- **Never** give iron for a low MCV without checking ferritin: iron does not help thalassaemia trait.
- Do not rely on the Mentzer index alone; use HPLC.
- A normal ferritin with a raised CRP does not exclude iron deficiency.
- Black stools on iron are expected; tarry, sticky stools with dizziness are not.
- A rising Hb on iron does not mean the cause has been found.
- Oesophagitis on gastroscopy does not explain IDA on its own: complete the colonoscopy.
- IDA on anticoagulants or antiplatelets is still investigated.

13 SAY IT LIKE THIS

plain English

"Your red blood cells are smaller than usual. The commonest reason is low iron, and then the important question is where the iron went, so I'm arranging some tests. If your iron is normal, it may be something you were born with that runs in families. That is harmless, but worth knowing before a pregnancy."

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Macrocytosis: the high MCV CLINIC GUIDE

Interpreting an MCV above the reference range (usually >100 fL) in adults 18+. Not for pregnancy (physiological rise; maternity pathway), children or patients under active haematology care.

NICE **NG239** (B12, 2024) · NG12
BSH cobalamin/folate 2014 · SPS
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/macrocytosis.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → B12, folate, alcohol, liver, thyroid, drugs, marrow

! ACT TODAY — RED FLAGS

NG239 · BSH

TIER	RESULT OR COMBINATION	ACTION
999	Macrocytic anaemia with chest pain, syncope, hypoxia, heart failure, confusion or collapse · variceal bleeding	Emergency transfer
Immediate	**Blasts on film** · unexplained **pancytopenia** or rapidly falling counts · severe neutropenia with fever · active bleeding	Phone haematology or acute medicine now
Same day	**Neurological B12 features**: progressive weakness, ataxia, loss of proprioception, sphincter symptoms (subacute combined degeneration), or after nitrous oxide use	Bloods if no delay, then IM hydroxocobalamin **now**; same-day medical or neurology assessment if severe or progressive
Same day	On **methotrexate** with mouth ulcers, sore throat, fever, bruising or falling counts	Withhold methotrexate; urgent FBC, U&E, LFTs; contact the specialist team
Same day	Macrocytosis with jaundice, ascites or confusion (decompensated liver disease)	Same-day hospital assessment

1 CONFIRM IT IS REAL

repeat if unexpected

- **Look back** at previous FBCs: stable, long-standing and isolated, or new and rising?
- **Delayed or stored sample**: red cells swell and the MCV rises. Repeat on a fresh sample if unexpected.
- **Cold agglutinins**: very high MCV with raised MCHC and low red cell count, clumping on the film. Repeat on a sample kept at 37 °C; discuss with haematology.
- **Reticulocytosis** (bleeding, haemolysis, recent treatment) raises the MCV: check reticulocytes.
- **B12 results mislead**: recent injection or supplement; combined oral contraceptive lowers total B12 (use active B12); nitrous oxide gives functional deficiency with a normal B12 (use MMA or homocysteine). Anti-IF antibodies can falsely normalise B12 assays.
- **Pregnancy, smoking** cause a mild physiological rise.

2 RESULT BANDS AND ACTION

NG239 · BSH

RESULT	MEANS	ACTION · TIMING
Isolated, stable mild rise; Hb and counts normal	Usually alcohol, liver, thyroid or drugs	First-line tests; treat cause; routine recheck
Macrocytosis **+ anaemia**	B12 or folate deficiency until proven otherwise	B12, folate, film, retics within days
+ another cytopenia or dysplastic film	Possible MDS or marrow disease	Haematology referral; urgency by counts
Total B12 **<180 ng/L** · active B12 **<25 pmol/L**	Deficiency	Treat; find the cause
Total B12 **180–350 ng/L** · active **25–70 pmol/L**	Indeterminate	MMA if symptomatic; consider treating meanwhile
Total B12 **>350 ng/L** · active **>70 pmol/L**	Deficiency unlikely	Unless strong suspicion or nitrous oxide
Low folate, B12 normal	Folate deficiency	Folic acid; find the cause
MCV **>105 fL on methotrexate**	Drug effect or deficiency	B12, folate, TSH; treat; normal → discuss with specialist (SPS)

There is no single MCV that diagnoses MDS or mandates referral: a marked or rising MCV with anaemia, cytopenias or an abnormal film needs prompt review. Deficiency is not excluded by a normal Hb or MCV (NG239).

3 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Raised GGT, heavy drinking, round macrocytes	**Alcohol** (commonest)	AUDIT-C; LFTs, GGT
Neuropathy, autoimmunity, vegan, metformin, PPI	**B12 deficiency**	B12; anti-IF antibodies
Poor diet, malabsorption, methotrexate, phenytoin	**Folate deficiency**	Serum folate (with B12)
Target cells, stigmata of liver disease	**Liver disease**	LFTs, INR, liver screen
Weight gain, cold intolerance	**Hypothyroidism**	TSH
Hydroxycarbamide, azathioprine, zidovudine	**Drug effect**	Medication review
Jaundice, high reticulocytes	**Haemolysis**	Bilirubin, LDH, haptoglobin, DAT
Older, other cytopenias, dysplastic film	**MDS, marrow disease**	Film; haematology

4 WORK-UP CHECKLIST

NG239 · BSH

- ☐ **Repeat FBC + film** if unexpected
- ☐ **LFTs + GGT**; alcohol history
- ☐ **Reticulocytes**; U&E
- ☐ Anti-IF antibodies if B12 low
- ☐ **B12 and folate together**
- ☐ **TSH**
- ☐ **Full drug review**, incl. nitrous oxide
- ☐ tTG IgA + total IgA if malabsorption

Haemolysis screen if reticulocytes high. Do not request anti-parietal cell antibodies routinely; a negative anti-IF does not exclude autoimmune gastritis.

5 BLOOD FILM DECODED

ask for a film with every unexplained high MCV

FILM FINDING	POINTS TO	NEXT
Oval macrocytes + hypersegmented neutrophils	Megaloblastic: B12 or folate deficiency, antifolate drugs (methotrexate)	B12, folate; drug review — not alcohol alone
Round macrocytes, target cells, acanthocytes	Alcohol, liver disease	LFTs, GGT, INR; alcohol history
Polychromasia	Reticulocytosis: haemolysis, bleeding, recovery after treatment	Reticulocytes, haemolysis screen
Dysplastic neutrophils, abnormal platelets, blasts	MDS, leukaemia	Haematology — immediately if blasts
Red cell agglutination	Cold agglutinins (spurious MCV)	Warm repeat sample; haematology

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT OF THE COMMONEST CAUSES

NG239 · BSH · SPC · SPS

CAUSE	FIRST LINE	CHECK AND CONTINUE
B12: autoimmune gastritis, malabsorption replacement	Hydroxocobalamin **1 mg IM three times a week for 2 weeks**, then **1 mg every 2–3 months** for life	Lifelong IM for autoimmune gastritis, total gastrectomy or terminal ileal resection
B12 with neurological features replacement	Hydroxocobalamin **1 mg IM on alternate days** until no further improvement, then **1 mg every 2 months**; seek specialist advice	Start before results if neurology; same-day assessment if severe or progressive
B12: diet only	Oral cyanocobalamin **50–150 micrograms daily** between meals + dietary advice	Not improving → higher oral dose or IM (NG239)
Folate deficiency replacement	Folic acid **5 mg daily for 4 months — only once B12 is normal or being replaced**	Longer if the cause persists; find the cause
Alcohol	AUDIT-C, brief intervention; alcohol service if dependent; folic acid only if deficient	GGT falls first; MCV takes months. Persists despite abstinence → reassess
Drugs	Methotrexate: folic acid **5 mg once weekly** on a different day. Hydroxycarbamide: macrocytosis expected, do not change the dose for MCV alone	Antiretrovirals: discuss with the HIV team. Hypothyroidism: levothyroxine

7 B12 DEFICIENCY — CHOOSE THE ROUTE BY THE CAUSE

NG239 · SPC

AUTOIMMUNE GASTRITIS
IM hydroxocobalamin for life; also after total gastrectomy or terminal ileal resection. Oral is unreliable without intrinsic factor.

DIET (VEGAN, POOR INTAKE)
Oral cyanocobalamin; fortified foods or a supplement long term. Stop when the cause is addressed and symptoms resolve.

MEDICINES (METFORMIN, PPIS)
Oral or IM by clinical judgement and preference; usually continue the medicine and supplement alongside.

NITROUS OXIDE
Stop exposure; IM hydroxocobalamin without waiting for results; same-day assessment if neurological symptoms.

8 FOLLOW-UP AND MONITORING CHECKLIST

NG239 · SPS

- [ ] Cause found: treat, then **recheck FBC** to confirm the response
- [ ] B12: initial review at about **3 months** (earlier if severe) — symptoms, adherence, cause
- [ ] **Do not use B12 levels** to monitor regular IM injections
- [ ] Poor response → reconsider: iron deficiency, MDS, coeliac, wrong diagnosis
- [ ] Folate: FBC rechecked; continue beyond 4 months if the cause persists
- [ ] Alcohol: GGT and MCV trend as feedback
- [ ] No cause after first and second line → haematology
- [ ] Isolated, stable, no cause: periodic FBC in primary care
- [ ] Return the **same day**: fever, mouth ulcers, bruising or bleeding, new numbness, weakness or unsteadiness

9 WHEN AND WHERE TO REFER

NG239 · NG12 · BSH

999	Chest pain, syncope, hypoxia, collapse with anaemia
Immediate	Blasts; unexplained pancytopenia; neutropenic fever; methotrexate toxicity
Same day	Severe or progressive neurological B12 features; decompensated liver disease
Haematology	Unexplained macrocytic anaemia; any unexplained cytopenia; dysplastic film; rising MCV without a cause; persists after B12/folate correction and abstinence
Gastro-enterology	Autoimmune gastritis with new or worsening dyspepsia, nausea or vomiting (NG12 if cancer suspected); coeliac disease
Neurology	Neuropathy not improving after 3 months of B12; cognitive decline out of proportion
Primary care	Isolated, stable macrocytosis with normal counts: no urgent referral needed

10 SPECIAL GROUPS

NG239 · SPS · BSH

- **On methotrexate**: monitor FBC per the shared-care schedule and watch the MCV trend; folic acid weekly, never daily; acute rise with ulcers or cytopenias = toxicity.
- **Older adults**: unexplained macrocytosis with any cytopenia raises MDS — refer rather than repeat. Follow the haematology plan for transfusion and shared care.
- **Coeliac disease, bariatric surgery**: combined B12, folate and iron deficiency is common — check all three. **Pregnancy**: use active B12 first and the maternity pathway.
- **Autoimmune overlap**: autoimmune gastritis and hypothyroidism often coexist — a high MCV in treated hypothyroidism still needs B12 checked. **Recent B12 injection**: the B12 level is meaningless; read the film and response instead.

11 PITFALLS AND NEVER-EVENTS

NG239 · BSH · SPS

- **Never** give folic acid alone before B12 deficiency is excluded or treated: it can precipitate cord damage.
- **Never** co-prescribe trimethoprim or co-trimoxazole with methotrexate where avoidable: marrow toxicity can be fatal.
- **Never** delay B12 when there are neurological features.
- **Never** stop injections in autoimmune gastritis because symptoms improve.
- Do not blame alcohol without checking B12, folate and the film.
- Ask about nitrous oxide in young adults with neuropathy.

12 SAY IT LIKE THIS

plain English

"Your red blood cells are a bit larger than usual. On its own that isn't harmful, but it's a clue. The commonest reasons are alcohol, low vitamin B12 or folate, the thyroid, the liver or some medicines. These blood tests will tell us which, and most of these causes are easy to put right."

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Normocytic anaemia CLINIC GUIDE

Interpreting a low Hb with a normal MCV (usually 80–100 fL) in adults 18+. Not for pregnancy, children, dialysis or ESA care (renal team) or known sickle cell disease (hospital pathway).

NICE **NG203** · **NG12** · HTG690 · **NG239**
BSG IDA 2021 · BSH
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/normocytic-anaemia.html

PAGE 1 · INTERPRET — is it dangerous? → marrow making too few, or losing too many? → find the cause

! ACT TODAY — RED FLAGS

NICE NG12 · BSH

TIER	RESULT OR COMBINATION	ACTION
999	Anaemia with chest pain, syncope, hypoxia, new confusion, collapse or active major bleeding	Emergency transfer; do not wait to plan transfusion
Immediate	**Anaemia + low platelets + fragmented red cells** (schistocytes), with or without fever, confusion or renal impairment: possible TTP	Hospital and haematology now; do not wait for further tests
Same day	**Rapidly falling Hb** · jaundice or dark urine (haemolysis) · **blasts**, febrile neutropenia or severe cytopenias · sickle cell disease with acute anaemia, fever or chest symptoms	Same-day hospital or haematology assessment
Suspected cancer pathway	Iron-deficiency anaemia, or aged **60+ with anaemia** even without iron deficiency → **FIT** ≥10 µg Hb/g → colorectal pathway · unexplained lymphadenopathy or splenomegaly, especially with fever, night sweats or weight loss → consider haematological cancer referral	NG12 referral; FIT <10 does not exclude cancer if concern persists

1 CONFIRM IT IS REAL

repeat if unexpected

- **Anaemia**: Hb **<130 g/L** men, **<120 g/L** non-pregnant women; use local laboratory ranges. **Look back**: new, falling or long-standing?
- **Repeat the FBC with reticulocytes and a film** if unexpected, before labelling it.
- **Dilution**: pregnancy, fluid overload or IV fluids lower the Hb without true anaemia.
- **Early bleeding and early iron deficiency** are normocytic before the MCV falls.
- **Mixed deficiency** (iron + B12 or folate) cancels out to a normal MCV: a high RDW is a clue. Check ferritin, B12 and folate together.
- **Ferritin rises with inflammation**, infection, liver disease, malignancy: read it with CRP and TSAT.
- **A haemolysed sample** raises LDH and potassium: repeat before calling it haemolysis.

2 RESULT BANDS AND ACTION

NG203 · BSG 2021 · NG12

RESULT	MEANS	ACTION · TIMING
Ferritin **<15 µg/L** (**<30** supports)	Iron deficiency, before microcytosis	Iron now; investigate as IDA
Ferritin normal or high, CRP raised, **TSAT <20%**	Chronic disease, often with iron deficiency	Treat the disease; ferritin **<45**: consider GI investigation
Reticulocytes high	Marrow responding: bleeding or haemolysis	Haemolysis screen; look for blood loss promptly
Retics low, other counts normal	Under-production: chronic disease, CKD, thyroid, early deficiency	Work-up (box 5)
+ another cytopenia or abnormal film	Marrow disease, MDS, myeloma, lymphoma	Urgent haematology; same day if blasts or fever
DAT positive with haemolysis	Autoimmune haemolysis	Urgent haematology discussion
eGFR **>60**	CKD unlikely to be the cause	Investigate other causes
eGFR **30–60**	CKD may contribute	Investigate others; extent by judgement
eGFR **<30**	CKD often the cause	Still consider other causes; renal team

3 HISTORY AND EXAMINATION THAT CHANGE THE PLAN

BSG 2021 · NG12

Where is the blood going?	Heavy periods, melaena, rectal bleeding, haematuria, blood donation	**Jaundice, dark urine**	Haemolysis; liver disease
Chronic illness	RA, IBD, CKD, HIV, TB, cancer: chronic disease likely, but iron deficiency may coexist	**Nodes, spleen, liver**	Lymphoma, leukaemia, haemolysis → urgent route
Drugs	NSAIDs, anticoagulants (loss); chemotherapy, immunosuppressants (marrow); new drugs with haemolysis (G6PD)	**Bone pain or tenderness**	Myeloma, metastases: electrophoresis, calcium
Fever, sweats, weight loss	Infection, lymphoma, malignancy	**Heritage, family history**	Sickle cell, thalassaemia, G6PD deficiency

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
RA, IBD, infection, cancer; raised CRP	**Chronic disease** (common)	CRP, ferritin + TSAT
Reduced eGFR, proteinuria	**CKD**	eGFR; exclude other causes
Heavy periods, NSAIDs, GI symptoms	**Blood loss**, early iron deficiency	Ferritin; FIT; retics
Jaundice, dark urine, high retics	**Haemolysis**	Bilirubin, LDH, haptoglobin, DAT, film
Bone pain, raised calcium, renal impairment	**Myeloma**	Serum protein electrophoresis, free light chains
Other cytopenias, nodes, spleen	**Marrow disease**, lymphoma	Film; haematology
Vegan, malabsorption, neuropathy	**Mixed deficiency**	B12, folate, ferritin
Tiredness, weight gain	**Hypothyroidism**	TSH

5 WORK-UP CHECKLIST

NG203 · NG239 · BSG 2021

- ☐ **FBC + film**, reticulocytes
- ☐ **B12 and folate**
- ☐ Urinalysis (blood, protein)
- ☐ **FIT** if 60+ or iron deficient
- ☐ **Ferritin, CRP**; TSAT if inflamed
- ☐ **U&E/eGFR**, LFTs, bilirubin
- ☐ TSH if clinically indicated
- ☐ Electrophoresis + light chains if suggested

Haemolysis suspected: add LDH, haptoglobin, DAT. Coeliac serology if iron deficient or malabsorption.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 RETICULOCYTES — THE FORK IN THE ROAD

BSH · NG203

HIGH RETICS + JAUNDICE
Haemolysis: LDH, bilirubin, haptoglobin, DAT, film. Schistocytes + low platelets = TTP, immediate.

HIGH RETICS, NO HAEMOLYSIS
Bleeding: GI, menstrual, urinary. FIT; ferritin will fall later.

LOW RETICS, ISOLATED ANAEMIA
Chronic disease, CKD, thyroid, early or mixed deficiency: treat what you find.

LOW RETICS + OTHER CYTOPENIAS
Marrow failure, MDS, leukaemia, myeloma: film and urgent haematology.

7 MANAGEMENT OF THE COMMONEST CAUSES

BSG 2021 · NG203 · NG239 · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Chronic disease	Treat the underlying condition; iron only if coexisting deficiency is shown	No role for iron, B12 or folate alone; monitoring follows the disease
Iron deficiency replacement	One oral iron tablet **once daily**, e.g. ferrous sulfate 200 mg (65 mg elemental iron); **alternate days** or another preparation if not tolerated (BSG 2021)	Hb at **2–4 weeks**; continue **3 months after** Hb normalises. Find the source
B12 deficiency replacement	Hydroxocobalamin 1 mg IM three times a week for 2 weeks, then every 2–3 months; neurological: alternate days until no further improvement, then every 2 months	Review at about 3 months. Lifelong IM for autoimmune gastritis
Folate deficiency	Folic acid 5 mg daily, usually for 4 months — **only once B12 is normal or being replaced**	Find the cause: diet, alcohol, malabsorption, drugs
CKD anaemia	Exclude other causes; iron status from ferritin with TSAT (or Ret-He / hypochromic cells)	Renal team starts ESA or HIF-PHI; Hb aim **100–120 g/L**, not normal
Haemolysis	Haematology-led: do not start steroids unless directed. Cold AIHA: keep warm	G6PD: stop the trigger drug or food; urgent bloods

8 IRON TREATMENT PROBLEMS

BSG 2021 · SPC · MHRA

Nausea, constipation	Alternate days, another preparation, or take with food (absorbs less)
No Hb rise	Adherence, ongoing loss, coeliac, inflammation, wrong diagnosis
IV iron	Local pathway when oral fails, is not absorbed or correction is urgent; observe at least 30 minutes
Ferric carboxymaltose	Check phosphate with repeat dosing or risk factors (MHRA)

9 FOLLOW-UP CHECKLIST

BSG 2021 · NG239 · NG203

- ☐ **Hb at 2–4 weeks** on iron: a rise of ≥10 g/L in 2 weeks confirms iron deficiency
- ☐ No rise → adherence, ongoing loss, coeliac, wrong diagnosis, marrow
- ☐ Iron for **3 months after** Hb normalises
- ☐ B12: review at about **3 months**; do not use B12 levels to monitor injections
- ☐ FIT and haemolysis results returned and actioned
- ☐ Chronic disease: monitoring follows the disease and shared-care plan
- ☐ Declines investigation: informed decision and safety-net recorded
- ☐ Still unexplained after first-line work-up → haematology
- ☐ Return the **same day**: breathless, chest pain, faint, black stools, jaundice or dark urine, new bruising or bleeding

10 WHEN AND WHERE TO REFER

NG12 · NG203 · BSG 2021

999	Chest pain, syncope, hypoxia, confusion, major bleeding
Immediate	Possible TTP; blasts; febrile neutropenia
Same day	Rapidly falling Hb; acute haemolysis; sickle cell complications
Suspected cancer	FIT ≥10; lymphadenopathy or splenomegaly with systemic symptoms
Urgent haematology	Stable pancytopenia; paraprotein; confirmed haemolysis; unexplained anaemia not responding to treatment
Renal	CKD anaemia needing IV iron, ESA or HIF-PHI; NG203 referral criteria
Urgent GI	Iron deficiency in men and post-menopausal women: endoscopy even if FIT negative

11 PITFALLS AND NEVER-EVENTS

NG203 · NG239 · BSG 2021

- **Never** blame CKD from the eGFR alone: investigate other causes first.
- **Never** give folic acid alone before B12 is excluded or treated.
- **Never** give empirical iron that delays assessment of possible marrow disease.
- A normal ferritin with a raised CRP does not exclude iron deficiency.
- A normal MCV does not exclude deficiency: mixed states hide.
- Ignoring a raised reticulocyte count misses bleeding and haemolysis.
- FIT <10 does not end the search when concern persists.
- Do not transfuse, or withhold transfusion, on the Hb number alone.
- Do not start steroids for suspected autoimmune haemolysis unless haematology advises.

12 SPECIAL GROUPS

NG203 · BSH · NG12

- **Pregnancy**: different Hb thresholds — use the maternity pathway. **Sickle cell disease**: acute anaemia, fever, chest or neurological symptoms → same-day hospital, following the patient's own plan.
- **G6PD deficiency**: sudden haemolysis after an infection, drug or fava beans — stop the trigger, urgent bloods, discuss with haematology. **On anticoagulants**: anaemia is still investigated, not blamed on the drug.
- **CKD**: iron deficiency is common and treatable; ESA and HIF-PHI are renal-team decisions. **Older adults**: anaemia at 60+ → FIT, and think of myeloma and MDS.

13 SAY IT LIKE THIS

plain English

"Your blood count is low, but the cells are a normal size, so this is more about why your body isn't making enough, or is losing some, than about a single missing vitamin. These tests will narrow it down, and treating the cause is what brings the count back up."

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Low ferritin: iron deficiency

CLINIC GUIDE

A low serum ferritin, with or without anaemia, in adults 18+. Not for pregnancy, children, dialysis or ESA care (renal team) or IBD under specialist care.

BSG IDA 2021 · NICE **NG12** · HTG690
NG88 · NG20 · NG24 · MHRA
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/low-ferritin.html

PAGE 1 · INTERPRET

is it dangerous? → is it real iron deficiency? → where did the iron go?

! ACT TODAY — RED FLAGS

NICE NG12 · NG24 · BSG 2021

TIER	RESULT OR COMBINATION	ACTION
999	Low ferritin with collapse, haemodynamic instability or major active GI bleeding	Emergency transfer
Same day	Anaemia with chest pain, breathlessness at rest, syncope or heart failure · **rapidly falling Hb**	Same-day hospital assessment; transfusion decided on symptoms, not Hb alone
Suspected cancer pathway	**Iron-deficiency anaemia** → **FIT** ≥10 µg Hb/g → colorectal pathway · rectal or anal mass, unexplained anal ulceration → consider referral without FIT · dysphagia, or 55+ with weight loss + upper GI symptoms → upper GI · unexplained visible haematuria aged **45+** → urology · unexplained haemoptysis aged **40+** → lung	NG12 referral; FIT <10 does not exclude GI pathology in confirmed IDA
Urgent GI	New, unexplained IDA in a **man** (any age) or **post-menopausal woman**	Gastroscopy + colonoscopy, whatever the FIT

Transfusion is a hospital decision (NG24: consider a restrictive threshold of 70 g/L in stable adults, 80 g/L in acute coronary syndrome; single units, then reassess). Iron is still needed afterwards.

1 CONFIRM IT IS REAL

BSG 2021

- **A low ferritin is rarely false**: ferritin is the single most useful test, and a value <15 µg/L is highly specific for iron deficiency.
- **Ferritin is an acute-phase reactant**: a normal ferritin does not exclude iron deficiency when inflammation is present (also raised by infection, liver disease, malignancy). If deficiency is suspected, add **CRP and TSAT**; TSAT <20% supports deficiency.
- **Serum iron** swings with time of day and recent meals or tablets: do not decide on it.
- **Check the Hb, MCV and MCH**: iron deficiency can occur with a normal MCV, early or in mixed deficiency.
- **Repeat** only if the result does not fit the picture; otherwise act on it.
- Anaemia: Hb **<130 g/L** men, **<120 g/L** non-pregnant women; use local laboratory ranges.

2 RESULT BANDS AND ACTION

BSG 2021

RESULT	MEANS	ACTION · TIMING
Ferritin **<15 µg/L**	Absent iron stores	Start iron **now**; find the cause
Ferritin **15–30 µg/L**	Low iron stores	Treat; investigate the cause
Ferritin **30–45 µg/L** + inflammation	Deficiency possible	TSAT; **<45**: consider GI investigation
Low ferritin + **anaemia**	Iron-deficiency anaemia	FIT, coeliac serology, urinalysis; endoscopy in men and post-menopausal women
Low ferritin, **normal Hb**	Non-anaemic iron deficiency	Treat; low threshold to investigate in men, post-menopausal women, GI symptoms or family history
Hb rise **≥10 g/L in 2 weeks** on iron	Confirms iron deficiency	Continue; complete investigation

Pre-menopausal women: GI investigation only with added concern — aged 50+, red flags, strong family history of colorectal cancer, or IDA out of proportion to periods.

3 IRON STUDIES PATTERN

BSG 2021 · read together, never one test alone

PATTERN	FERRITIN	TSAT	TIBC / TRANSFERRIN	CRP	NEXT
Iron deficiency	Low	<20%	High	Normal	Treat + find the loss
Deficiency hidden by inflammation	Normal or borderline	<20%	Normal or low	Raised	Treat the disease; iron trial; consider GI tests
Chronic disease alone	Normal or high	Low	Low or normal	Often raised	Treat the disease; no iron
Thalassaemia trait	Normal	Normal	Normal	Normal	Low MCV persists → HPLC; no iron

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Man or post-menopausal woman	**GI blood loss**, cancer until excluded	FIT; endoscopy
Heavy periods, flooding, clots	**Menstrual loss**	Menstrual history; coeliac serology
Diarrhoea, family history, poor response to iron	**Coeliac disease**	tTG IgA + total IgA (on gluten)
NSAIDs, aspirin, anticoagulants	**GI erosions, ulcers**	Investigate; do not blame the drug
Gastrectomy, bariatric surgery, PPIs	**Malabsorption**	Still look for loss
Haematuria, haemoptysis	**Urinary or lung loss**	Urinalysis; NG12 pathway
Vegan diet, blood donation	**Low intake, donation**	Diagnosis of exclusion in men and post-menopausal women

5 WORK-UP CHECKLIST

BSG 2021 · NG20 · NG12

- ☐ **FBC** with MCV, MCH
- ☐ **tTG IgA + total IgA**
- ☐ **FIT** in IDA
- ☐ B12, folate if mixed picture
- ☐ **CRP**; TSAT if inflamed
- ☐ **Urinalysis** for blood
- ☐ U&E/eGFR
- ☐ HPLC if MCV low out of proportion

Coeliac serology only while eating gluten. Do not defer iron while awaiting tests unless colonoscopy is imminent.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT OF THE COMMONEST CAUSES

BSG 2021 · NG88 · NG20 · SPC · MHRA

CAUSE	FIRST LINE	CHECK AND CONTINUE
Any iron deficiency replacement	One oral iron tablet **once daily**, e.g. ferrous sulfate 200 mg (65 mg elemental iron), ferrous fumarate or gluconate; **alternate days**, another preparation or IV iron if not tolerated. Not twice or three times daily	Hb at **2–4 weeks**; continue for about **3 months after** Hb normalises
Heavy menstrual bleeding	Iron + treat the bleeding per NICE NG88 (LNG-IUS, tranexamic acid, hormonal options)	Gynaecology if treatment fails or a structural cause is suspected
Coeliac disease	Strict gluten-free diet + iron	Gastroenterology per NG20
NSAID-related loss	Stop the NSAID if possible; gastroprotection if essential	Still complete GI investigation
Oral iron fails or will not work	**IV iron** via the local pathway: intolerance, malabsorption (coeliac, IBD, bariatric), ongoing loss, urgent correction	Product-specific dose; observe at least 30 minutes. Ferric carboxymaltose: phosphate with repeat doses or risk factors (MHRA)

7 WHO NEEDS GI INVESTIGATION

BSG 2021 · NG12 · HTG690 · NG20

EVERY ADULT WITH IDA
FIT; ≥10 µg Hb/g → colorectal suspected cancer pathway. Coeliac serology and urinalysis.

MEN, POST-MENOPAUSAL WOMEN
Gastroscopy + colonoscopy together (CT colonography if unsuitable) — even if FIT is negative. Young men too.

PRE-MENOPAUSAL WOMEN
Treat iron and periods; investigate if aged 50+, red flags, strong family history or IDA out of proportion.

NEGATIVE ENDOSCOPY
Recurrent or refractory IDA → small bowel (capsule endoscopy) and renal tract. Long-term iron if the cause is irreversible.

8 GETTING ORAL IRON TO WORK

BSG 2021 · SPC

Nausea, constipation	Alternate days, another preparation, or take with food (absorbs less, but adherence comes first)
Absorption	Empty stomach if tolerated; away from tea, coffee, calcium and antacids where practical. Review whether a PPI is needed
No Hb rise	Adherence, ongoing loss, coeliac, inflammation, mixed deficiency, wrong diagnosis
Duration	Set an end date: about 3 months after Hb normalises — stopping early empties the stores again

9 FOLLOW-UP CHECKLIST

BSG 2021

- ☐ **Hb at 2–4 weeks**: expect a rise of ≥10 g/L in 2 weeks
- ☐ No rise → adherence, ongoing loss, coeliac, mixed deficiency, wrong diagnosis
- ☐ FIT, coeliac and endoscopy results reviewed and recorded
- ☐ Iron for **3 months after** Hb normalises, then stop
- ☐ Ferritin when response is incomplete, it recurs or the cause is unclear
- ☐ FBC periodically (about **every 6 months** at first) to catch recurrence
- ☐ Declines investigation: informed decision recorded
- ☐ Return the **same day**: black tarry stools, vomiting blood, chest pain, faint, worsening breathlessness

10 RECURRENCE AND THE LONG TERM

BSG 2021

- **Recurs after treatment**: is the source still active (periods, NSAIDs, gluten)? Reinvestigate if not explained.
- **Cause irreversible or unknown** after full investigation: long-term iron with a documented, reviewed plan.
- **H. pylori**: test and treat if dyspepsia, or after non-explanatory endoscopy per the GI pathway.

11 WHEN AND WHERE TO REFER

NG12 · BSG 2021 · NG88

999	Collapse, major GI bleeding, instability
Same day	Severe symptomatic or rapidly falling Hb
Suspected cancer	FIT ≥10; rectal mass; dysphagia; 55+ weight loss with upper GI symptoms; haematuria 45+; haemoptysis 40+
Urgent GI	Unexplained IDA in men and post-menopausal women
Gastro-enterology	Positive coeliac serology; recurrent IDA after negative endoscopy
Gynaecology	Heavy bleeding not controlled, or structural cause
Haematology	No response to iron after review, cause not found

12 SPECIAL GROUPS

BSG 2021 · NG24

- **Pregnancy**: different Hb and ferritin thresholds — use the maternity pathway; IV iron only under obstetric care. **IBD, coeliac, bariatric surgery**: oral iron often poorly absorbed; IV iron via specialist or local pathway — previous surgery does not stop the search for other causes.
- **Frail or major comorbidity**: weigh the risks and benefits of endoscopy with the patient and record the decision. **Young men**: investigate as for older adults. **After transfusion**: iron is still needed, and the cause still needs finding.

13 PITFALLS AND NEVER-EVENTS

BSG 2021 · NG12

- **Never** put IDA down to diet in a man or post-menopausal woman.
- **Never** treat a negative FIT as an all-clear in confirmed IDA.
- **Never** arrange only one end: oesophagitis on gastroscopy does not exclude a colonic cancer.
- A rising Hb on iron does not mean the cause is found.
- Black stools on iron are expected; tarry, sticky stools are not.
- IDA on anticoagulants or antiplatelets is still investigated.
- No routine ferritin screening in well vegans or athletes; investigate deficiency as in anyone.

14 SAY IT LIKE THIS

plain English

"Your iron stores are low. The tablets will fill them back up, but they don't tell us why you ran low. Finding where the iron went is the important part, so please do the tests I'm arranging, even once you feel better."

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High ferritin CLINIC GUIDE

Interpreting a raised serum ferritin in adults 18+. Not for children, pregnancy, or patients with transfusional iron overload already under haematology.

BSH ferritin 2018 · haemochromatosis 2018
EASL 2022 · NICE NG50 · NG12
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/high-ferritin.html

PAGE 1 · INTERPRET — is it dangerous? → is it inflammation? → is it iron overload? (the TSAT decides)

! ACT TODAY — RED FLAGS

BSH · NG12

TIER	RESULT OR COMBINATION	ACTION
999	Very high ferritin with haemodynamic instability, hypoxia, confusion, bleeding, organ failure or sepsis	Emergency transfer
Same day	Ferritin **>10,000 µg/L**, or very high with **fever + cytopenias**, splenomegaly, liver dysfunction or coagulopathy: possible HLH or macrophage activation	Same-day haematology or acute medicine discussion, even if stable. Ferritin alone does not diagnose HLH
Same day	High-spiking fever, arthritis, evanescent salmon-pink rash, neutrophilia, very high ferritin: possible Still disease (after excluding infection, cancer, HLH)	Same-day rheumatology or acute medicine discussion
Suspected cancer pathway	Raised ferritin with unexplained lymphadenopathy, splenomegaly, weight loss, night sweats or fever · blasts on film → same-day haematology	NG12 referral by site (lymphoma, leukaemia, myeloma, solid tumour)

1 CONFIRM IT IS REAL

BSH 2018

- **Check CRP with every ferritin**: ferritin is an acute-phase protein. A high ferritin with a high CRP says nothing about iron stores.
- **Recheck when the illness has settled** (CRP normal) before investigating further.
- **Recent iron tablets, IV iron or transfusion** raise ferritin; recent oral iron can also raise TSAT.
- **Alcohol** is often the only cause: ask (AUDIT-C) and look at GGT.
- **TSAT**: a fasting sample is not routinely needed, but **repeat a borderline or unexpected result** before acting on it.
- Do not request HFE genotyping without a TSAT: the genotype alone does not make the diagnosis.

2 RESULT BANDS AND ACTION

BSH 2018 · EASL 2022

RESULT	MEANS	ACTION · TIMING
Raised ferritin + **raised CRP**	Inflammation; stores unknown	Treat the cause; recheck with CRP when settled
Raised ferritin, **TSAT normal**	Alcohol, metabolic, liver, inflammation; iron overload less likely	Treat the cause; recheck; no venesection
Raised ferritin + TSAT **>50% men, >40% women** (persistent)	Possible haemochromatosis	HFE genotype or specialist advice
C282Y homozygous + raised ferritin/TSAT	HFE haemochromatosis	Hepatology or iron-overload service
C282Y/H63D or H63D/H63D	Low penetrance: rarely the cause	Look for other causes; specialist if significant loading
C282Y carrier (heterozygote)	Not at significant risk	No routine monitoring
Ferritin **≥1,000 µg/L**, persistent	Unexplained: risk of liver damage (BSH)	Specialist referral, whatever the HFE result
Ferritin **>10,000 µg/L**	HLH, Still disease, liver injury, sepsis, renal failure, malignancy	**Same-day** discussion

EASL uses a TSAT threshold of 45% in women; follow your local laboratory pathway.

3 HISTORY AND EXAMINATION THAT CHANGE THE PLAN

BSH 2018 · EASL 2022

Alcohol	AUDIT-C; often the only cause and often under-declared	**Liver**	Hepatomegaly, jaundice, spider naevi, ascites → LFTs, ultrasound, specialist
Iron in	Supplements, iron-containing multivitamins, IV iron, transfusions	**Skin, joints**	Bronze or grey pigmentation; 2nd/3rd MCP arthropathy → haemochromatosis
Overload symptoms	Fatigue, joint pain, low libido or erectile dysfunction, diabetes, palpitations	**Nodes, spleen**	Malignancy, HLH, Still disease → urgent route
Family history	Haemochromatosis in a first-degree relative → test even with modest results	**Metabolic**	BMI, waist, BP, HbA1c, lipids: metabolic hyperferritinaemia

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
High CRP, infection, RA, IBD	**Inflammation** (commonest)	CRP; recheck when settled
Drinking, raised GGT	**Alcohol**	AUDIT-C; LFTs, GGT
Obesity, diabetes, fatty liver; TSAT normal	**Metabolic hyperferritinaemia**	HbA1c, lipids, LFTs; liver ultrasound
Raised TSAT, joint pain (2nd/3rd MCPs), fatigue, diabetes, family history	**Haemochromatosis**	Repeat TSAT; HFE genotype
Abnormal LFTs, hepatitis risk	**Liver disease**	LFTs; hepatitis B and C; ultrasound
Many transfusions, iron infusions	**Secondary iron overload**	Iron history; haematology
Reduced eGFR	**Kidney disease**	U&E/eGFR
Nodes, spleen, sweats, cytopenias	**Malignancy**	FBC, film; NG12 route

5 WORK-UP CHECKLIST

BSH 2018

- ☐ **Ferritin + CRP** together
- ☐ **LFTs + GGT**; alcohol history
- ☐ U&E/eGFR
- ☐ Hepatitis B, C if LFTs abnormal
- ☐ **TSAT** (serum iron + transferrin)
- ☐ **FBC**; film if cytopenias
- ☐ HbA1c, lipids, BMI
- ☐ **HFE genotype** if TSAT raised

Liver ultrasound if LFTs abnormal, hepatomegaly or suspected liver disease. MRI liver iron is a specialist test.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 RAISED FERRITIN — THE DECISION IN FOUR BOXES

BSH 2018 · EASL 2022

CRP RAISED
Treat the cause. Recheck ferritin with CRP when well. Do not chase iron overload yet.

CRP NORMAL, TSAT NORMAL
Alcohol, metabolic liver disease, other liver disease, kidney disease. Treat and recheck; refer if persistent and unexplained.

TSAT RAISED
Repeat to confirm; HFE genotype. C282Y homozygous with raised indices → specialist.

FERRITIN ≥1,000 MG/L
Persistent and unexplained: specialist referral regardless of the HFE result, with LFTs, FBC and liver imaging per local pathway.

7 MANAGEMENT OF THE COMMONEST CAUSES

BSH 2018 · EASL 2022 · NG50

CAUSE	FIRST LINE	CHECK AND CONTINUE
Inflammation, infection	Treat the underlying condition. Ferritin here is not an iron store measure	Recheck ferritin + CRP once settled; still raised with normal CRP → TSAT and work-up
Alcohol	AUDIT-C, brief intervention; alcohol service if dependent; abstinence if liver disease	Repeat LFTs, GGT and ferritin after a period of abstinence
Metabolic, fatty liver	Weight loss, activity, alcohol reduction; cardiovascular risk and fibrosis assessment per local pathway	**No venesection** simply to lower ferritin
HFE haemochromatosis specialist-led	**Venesection** by the specialist service: Hb before each session; induction weekly to fortnightly until ferritin is low, then lifelong maintenance	Targets set by the service (EASL 2022: ferritin <50 µg/L in induction, <100 µg/L in maintenance)
Secondary (transfusional) overload	Chelation is a specialist haematology decision	GP: drug monitoring per the specialist plan

8 LIVING WITH HAEMOCHROMATOSIS

BSH 2018 · EASL 2022

- **No iron supplements** or iron-containing multivitamins; avoid high-dose vitamin C supplements. Diet alone does not replace venesection.
- **Alcohol**: keep low; avoid if there is liver disease.
- **Raw oysters and shellfish**: avoid with significant iron overload or liver disease (severe Vibrio infection).
- Some patients can give blood through NHS Blood and Transplant if they meet donor criteria.

9 FOLLOW-UP CHECKLIST

BSH 2018 · EASL · NG50

- ☐ Inflammatory rise: ferritin + CRP rechecked once well
- ☐ Alcohol or metabolic: LFTs and ferritin rechecked after change
- ☐ Haemochromatosis: venesection plan and targets recorded from the specialist letter
- ☐ Cirrhosis: **liver ultrasound every 6 months** for HCC (NG50), ± AFP per hepatology
- ☐ HbA1c, lipids and joints reviewed in haemochromatosis
- ☐ **Adult first-degree relatives** of a C282Y homozygote offered HFE genotype, ferritin and TSAT
- ☐ Return the **same day**: fever with bruising or bleeding, jaundice, confusion, abdominal swelling

10 WHEN AND WHERE TO REFER

BSH 2018 · NG12 · NG50

999	Unstable, septic, organ failure with very high ferritin
Same day	Possible HLH or Still disease; blasts on film
Suspected cancer	Lymphadenopathy, splenomegaly, B symptoms
Hepatology / iron overload	Confirmed or suspected haemochromatosis; persistent ferritin ≥1,000 µg/L; raised ferritin with abnormal LFTs or hepatomegaly; persistent unexplained rise
Haematology	Transfusional overload needing chelation
Rheumatology	Haemochromatosis arthropathy

11 PITFALLS AND NEVER-EVENTS

BSH 2018 · EASL

- **Never** interpret ferritin without a CRP.
- **Never** diagnose haemochromatosis from a genotype alone, or from ferritin alone.
- **Never** venesect for inflammatory or metabolic hyperferritinaemia.
- **Never** diagnose HLH from ferritin alone — but never sit on fever, cytopenias and a very high ferritin.
- Ancestry informs the odds; it is not a reason to withhold testing.
- Do not start HCC surveillance just because ferritin is high or the patient is C282Y homozygous: it is for cirrhosis (or advanced fibrosis on hepatology advice).
- Do not let a normal TSAT reassure you about a ferritin ≥1,000 µg/L.

12 SPECIAL GROUPS AND FAMILIES

BSH 2018 · EASL 2022

- **Family cascade**: siblings of a C282Y homozygote have a 1 in 4 chance of being homozygous — offer adult first-degree relatives HFE genotype, ferritin and TSAT. Avoid routine testing of children unless clinically indicated or advised by genetics.
- **Women before the menopause**: periods protect, so ferritin often rises after the menopause. **Pregnancy**: venesection can usually pause; specialist plan if overload is severe.
- **Transfusion-dependent anaemias** (thalassaemia, sickle cell, MDS): iron overload is managed by haematology with MRI liver and cardiac iron; cardiac symptoms need urgent specialist review.

13 SAY IT LIKE THIS

plain English

"Ferritin is a protein that stores iron, but it also goes up when the body is inflamed, with alcohol or with a fatty liver. Most raised results are not iron overload. One more test, the iron saturation, tells us whether you are actually holding too much iron."

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Low vitamin B12 in adults CLINIC GUIDE

Interpreting a low or borderline B12 in people aged 16+. Not for children under 16, infants of deficient mothers (paediatrics) or inpatient severe megaloblastic anaemia.

NICE **NG239** (2024) · NICE **NG12**
MHRA DSU metformin (Jun 2022) · BSH 2014
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/vitamin-b12-deficiency.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → how low → why

! ACT TODAY — NEUROLOGY AND SEVERE ANAEMIA

NICE NG239 · NG12

TIER	RESULT + PICTURE	ACTION
Same day	**Neurological features** with low or suspected low B12: progressive weakness, unsteadiness, loss of vibration or joint position sense, positive Romberg, sphincter symptoms (subacute combined degeneration) · neurological symptoms after **nitrous oxide** use, even with a normal B12	Urgent specialist advice; give **IM hydroxocobalamin now** — do not wait for MRI or MMA. Delay risks permanent damage
Same day	Severe symptomatic anaemia or **pancytopenia** · acute confusion or psychosis (assess for delirium; do not blame B12 alone) · pregnant or breastfeeding with a baby who is not feeding or developing well	Same-day hospital, paediatric or obstetric assessment
Urgent	Macrocytosis with abnormal film or other cytopenias suggesting marrow disease · new upper GI symptoms in autoimmune gastritis	Urgent haematology advice (immediate if film suggests leukaemia) · NG12 suspected cancer pathway referral where criteria met

1 IS THE LEVEL REAL?

NICE NG239

Normal B12, strong suspicion	Anti-intrinsic factor antibodies can interfere with the assay and give a falsely normal B12: treat on the clinical picture and discuss with the laboratory
Pregnancy	Total B12 falls normally: use **active B12** as the first test
Combined pill	Can lower total B12 without true deficiency: consider active B12
Nitrous oxide	B12 may look normal: test **homocysteine or MMA** first
Injection or supplement	Levels are high afterwards and mean nothing: test before the first dose
Normal Hb and MCV	Do not rule out deficiency: neurology can come first

Indeterminate result with symptoms: send MMA, and consider starting B12 while waiting. No need to repeat a clearly low level before treating.

2 SEVERITY BANDS

NICE NG239

RESULT	MEANS	ACTION AND TIMING
Total B12 **<180 ng/L** Active B12 **<25 pmol/L**	Deficiency confirmed	Find the cause and treat now; same day if neurological features
Total **180–350 ng/L** Active **25–70 pmol/L**	Indeterminate	Symptoms or signs → send **MMA**; consider treating while waiting, especially if rapid deterioration likely, after GI surgery, or pregnant
Total **>350 ng/L** Active **>70 pmol/L**	Deficiency unlikely	Look for another cause; still act on neurology, nitrous oxide or assay interference
MMA above lab range	Supports functional B12 deficiency	Treat as deficient

Bands are NG239 defaults: use your laboratory's thresholds if they differ. Interpret with symptoms, risk factors and the FBC.

3 WHY IS IT LOW?

clue → think → first test

CLUE	THINK	FIRST TEST
No dietary or drug cause, other autoimmune disease, family history	**Autoimmune gastritis** (pernicious anaemia)	Anti-intrinsic factor antibody
Vegan or strict vegetarian diet	**Dietary**	Diet history; still consider AIG if severe
Metformin, long-term PPI	**Medicine-induced**	Dose and duration
Gastrectomy, bariatric surgery, ileal Crohn's or resection	**Malabsorption**	Surgical history; coeliac serology
Young, tingling legs, "balloons"	**Nitrous oxide**	Homocysteine or MMA
Other cytopenias, odd film	**Marrow disease** as well	Film; haematology advice

A negative anti-IF antibody does not exclude autoimmune gastritis: if suspicion remains, consider gastric parietal cell antibody, gastrin or gastroscopy with gastric-body biopsy (NG239).

4 WORK-UP CHECKLIST — SEND AND WHY

NICE NG239

- ☐ **FBC + film**: macrocytosis, hypersegmented neutrophils, other cytopenias
- ☐ **Anti-IF antibody** if autoimmune gastritis suspected
- ☐ **Homocysteine or MMA** if nitrous oxide use
- ☐ **tTG IgA + total IgA** if GI or malabsorptive features
- ☐ **Folate**: often low too; treat B12 first or together
- ☐ **MMA** if indeterminate with symptoms
- ☐ **Ferritin**: mixed deficiency can mask MCV
- ☐ LFTs, U&E; TFTs if clinically indicated

Ask every patient about diet, metformin, PPIs, gastric or bowel surgery, autoimmune disease and nitrous oxide. Examine vibration sense, joint position, gait and Romberg.

5 READING THE RESULT IN FOUR STEPS

NICE NG239

1 · NEUROLOGY?
Check vibration, joint position, gait, Romberg, mood and memory. Any neurological feature → IM B12 today and specialist advice, whatever the level.

2 · WHICH BAND?
<180 ng/L treat · 180–350 with symptoms → MMA and consider treating · >350 look elsewhere unless the picture fits.

3 · WHICH CAUSE?
Diet, medicines, surgery or bowel disease, nitrous oxide; otherwise test anti-IF antibody for autoimmune gastritis.

4 · ANYTHING ELSE?
Read the whole FBC and film: other cytopenias, an abnormal film or a macrocytosis that does not fit → haematology.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 REPLACEMENT BY CAUSE

NICE NG239 · SPC

CAUSE	REPLACEMENT REGIMEN	DURATION
Neurological features (any cause)	Urgent specialist advice. If not immediately available: **hydroxocobalamin 1 mg IM on alternate days until no further improvement**, then **1 mg every 2 months**	Lifelong if the cause persists
Autoimmune gastritis, total gastrectomy, complete terminal ileal resection	**Hydroxocobalamin 1 mg IM three times a week for 2 weeks**, then **1 mg every 2–3 months**	**Lifelong IM** — never stop
Other malabsorption (e.g. coeliac, partial surgery)	Consider IM as above; if oral, cyanocobalamin **at least 1 mg daily**	While the cause persists
Medicine-induced (metformin, PPI)	IM, or oral cyanocobalamin **1 mg daily**, by judgement and preference, while the medicine continues; review whether the medicine is still needed	While on the medicine
Indeterminate + symptoms	Consider starting replacement for the likely cause while the MMA result is awaited	Review when the MMA result is back
Diet-related (vegan)	Oral cyanocobalamin **50–150 micrograms daily**; no response → increase the dose or switch to IM. Advise fortified foods or a supplement	Stop once improved if diet corrected; else continue

Low folate too: start B12 first or at the same time — never folic acid alone. Pregnancy: same doses, review symptoms at 1 month.

7 FOLLOW-UP CHECKLIST

NICE NG239 · MHRA

- ☐ **Review at about 3 months** (1 month in pregnancy; earlier if severe): symptoms, FBC response, adherence
- ☐ **Target**: symptoms resolved or plateaued, Hb and MCV back to normal
- ☐ Not improving → recheck cause and adherence, increase oral dose or **switch to IM**; consider other diagnoses
- ☐ **Do not routinely use B12 levels** to monitor people on regular IM treatment
- ☐ Oral, cause fixed and symptoms resolved → **stop**; cause unknown or ongoing → continue and follow up
- ☐ Lifelong IM: recall system for injections every 2–3 months (every 2 with neurology)
- ☐ Symptoms return before the next injection → reassess dose and frequency
- ☐ Metformin: check B12 if symptoms; consider periodic checks if risk factors (MHRA)

8 WHEN AND WHERE TO REFER

NICE NG239 · NG12

Same day	Suspected SACD, progressive weakness or ataxia, nitrous oxide with neurology, severe anaemia or pancytopenia, acute confusion
Suspected cancer pathway	Upper GI criteria met (NG12), e.g. dysphagia, or 55+ with weight loss and upper GI symptoms
Urgent haematology	Macrocytosis or cytopenias not explained by B12, abnormal film
Gastroenterology	Autoimmune gastritis with new or worsening dyspepsia, nausea or vomiting; uncertain diagnosis needing biopsy
Neurology	Neuropathy not improving after about 3 months of replacement
GP	Most cases: replace, review, keep the schedule going

Routine surveillance gastroscopy is not recommended in autoimmune gastritis (NG239): refer on symptoms.

9 WHO TO TEST, AND WHEN TO RECHECK THE DIAGNOSIS

NICE NG239 · MHRA

SYMPTOMS
Tiredness, sore red tongue, mouth ulcers, pins and needles, numbness, unsteadiness, memory or mood change — with or without anaemia.

RISK GROUPS
Autoimmune disease or a first-degree relative with pernicious anaemia · gastric or bariatric surgery · ileal Crohn's · vegan diet.

MEDICINES AND DRUGS
Metformin (MHRA: test if symptoms) · long-term PPI · recreational nitrous oxide.

POOR RESPONSE
No improvement at review → adherence, wrong route, a second cause (iron, folate, thyroid, marrow) or another neurological diagnosis.

10 SPECIAL GROUPS

NICE NG239 · MHRA

PREGNANCY, BREASTFEEDING
Active B12 as the first test. Same doses; review at 1 month. A breastfed baby of a deficient mother who is not feeding or developing well → same-day paediatric assessment.

NITROUS OXIDE
Ask directly and without judgement. Homocysteine or MMA first. Neurological symptoms → same day, IM B12 now. Stopping the gas is essential.

VEGANS
Everyone needs a reliable B12 source: fortified foods or a supplement. Confirmed deficiency needs a treatment dose, then prevention.

AUTOIMMUNE GASTRITIS
Ask about thyroid and other autoimmune disease; TFTs if indicated. Plan injections around long trips abroad. New dyspepsia → refer.

11 PITFALLS AND NEVER-EVENTS

NICE NG239 · MHRA

- **Never** give folic acid alone before B12 is known to be normal or is being replaced: it can let nerve damage progress.
- **Never** stop injections in autoimmune gastritis because the Hb or B12 is now normal.
- **Never** delay IM B12 for MRI or MMA when there are neurological features.
- A normal B12 does not exclude deficiency after nitrous oxide or with assay interference.
- Do not attribute confusion, macrocytosis or pancytopenia to B12 without excluding other causes.
- Metformin is usually continued while B12 is replaced.

12 SELF-CARE AND THE CONVERSATION

NICE NG239

Vegans: fortified plant milks, cereals or yeast, or a daily supplement · keep every injection appointment, and plan ahead before long trips · come back the **same day** with new tingling, leg weakness, poor balance or falls · come back sooner if symptoms return before the next injection is due · the Pernicious Anaemia Society has patient information.

"Your B12 is low. Your body needs it for healthy blood and nerves. We'll top it up and look for the reason. If the problem is that your stomach can't absorb it, the injections are for life, like insulin in type 1 diabetes — stopping them lets it run down again and can damage the nerves."

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Low folate in adults CLINIC GUIDE

Interpreting a low serum folate in adults 18+, including pregnancy planning. Not for children, inpatient refeeding or specialist haemolytic anaemia and dialysis protocols.

NICE CKS · BSH 2014 · SPS
NICE **NG247** (2025) · **NG239** (2024)
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/low-folate.html

PAGE 1 · INTERPRET — is it dangerous? → is B12 low too? → why is folate low

! ACT TODAY

NG239 · CG100 · NG12

TIER	RESULT + PICTURE	ACTION
999	Low folate with confusion, ataxia or eye-movement problems in alcohol misuse or vomiting (Wernicke) · unstable, septic, bleeding	Emergency transfer; parenteral thiamine if available, without delaying transfer
Same day	**Neurological features** (unsteady gait, loss of proprioception, weakness, cognitive or visual change) — may be **B12** deficiency	Take B12 and folate bloods, start **B12 today**, urgent specialist advice. Never folic acid alone
Same day	Pancytopenia with infection or bleeding · severe symptomatic anaemia · film suggesting leukaemia or MDS · rapidly falling counts	Same-day haematology or hospital assessment
Same day	On **methotrexate** with mouth ulcers, cytopenias, sepsis or rising creatinine	Withhold methotrexate; urgent FBC, LFTs, U&E; specialist the same day

1 IS THE RESULT REAL?

BSH 2014 · CKS

Recent meal or supplement	Serum folate reflects the last few days' intake: a normal level can hide chronic deficiency, and a low one may follow a few poor days
Haemolysed sample	Red cells are rich in folate: haemolysis can falsely raise serum folate. Repeat if the result does not fit
Already on folic acid	The level means little: do not use it to monitor
B12 deficiency	Can distort folate results: always read B12 and folate together
Red-cell folate	Not routinely needed in primary care

A mildly low folate in a well person with a normal FBC: dietary advice and a repeat test is reasonable; treat if the cause persists or there is anaemia.

2 SEVERITY BANDS

use your laboratory's lower limit

RESULT + PICTURE	MEANS	ACTION AND TIMING
Low folate, normal FBC, well	Early or dietary deficiency	Check B12; diet advice; treat if cause persists, risk group or pregnant
Low folate + **MCV >100** or anaemia	Megaloblastic picture	B12 result first, then folic acid; FBC response
Low folate + **low B12**	Combined deficiency	Treat **B12 first or at the same time**
Low folate + other cytopenias or abnormal film	Severe deficiency or marrow disease	Urgent haematology advice; same day if unwell
Low folate + neurological features	Possible B12 neuropathy	Same day: B12 now, specialist advice
Planning pregnancy or first 12 weeks	Neural tube risk	Start folic acid now; do not wait for results

3 READ THE FOLATE WITH THE REST OF THE COUNT

BSH 2014 · CKS

PATTERN	THINK	NEXT
MCV **>100**, hypersegmented neutrophils	Megaloblastic: folate or B12	B12 result before folic acid where feasible
Normal MCV, low folate and low ferritin	Mixed deficiency can cancel out the MCV — think coeliac	tTG IgA; treat both
Macrocytosis, normal folate and B12	Alcohol, liver, thyroid, drugs, MDS	LFTs, GGT, TFTs; film; haematology if unexplained
Low folate, high retics, jaundice	Haemolysis using up folate	Film, LDH, bilirubin; haematology

4 WHY IS IT LOW?

clue → think → first test

CLUE	THINK	FIRST TEST
Poor diet, frailty, eating disorder	**Low intake** (commonest)	Diet history; MUST
Alcohol, raised GGT	**Alcohol**	AUDIT-C; LFTs, GGT
Diarrhoea, weight loss, low iron too	**Coeliac**, small-bowel Crohn's	tTG IgA + total IgA (on gluten)
Methotrexate, phenytoin, carbamazepine, trimethoprim, sulfasalazine	**Drug-related**	Medication review
Jaundice, high retics, known sickle or spherocytosis	**Haemolysis** (high demand)	Film, retics, bilirubin, LDH
Pregnancy, breastfeeding, exfoliative skin disease	**Increased demand**	Context
Other cytopenias, odd film	**Marrow disease**	Film; haematology

5 WORK-UP CHECKLIST

CKS · NG239 · NG20

- ☐ **B12 (total or active)**: before folic acid where feasible
- ☐ **Blood film** if macrocytosis or cytopenias
- ☐ **tTG IgA + total IgA** while eating gluten
- ☐ TFTs if macrocytosis unexplained
- ☐ **FBC + MCV**, reticulocytes
- ☐ **Ferritin**: coeliac and diet cause both
- ☐ **LFTs + GGT**: alcohol, liver
- ☐ U&E; CRP if inflammation suspected

Ask about diet, alcohol, bowel symptoms, medicines, pregnancy plans and haemolytic disease. Examine mouth (glossitis, angular cheilitis), nodes, liver and spleen, and vibration and joint position sense.

6 READING THE RESULT IN FOUR STEPS

CKS · NG239

1 · UNWELL OR NEUROLOGICAL?
Neurology, pancytopenia, sepsis or Wernicke features → same day or 999 first.

2 · WHAT IS THE B12?
Low or unknown with neurology → treat B12 first. Never start folic acid alone in a macrocytic patient with an unknown B12.

3 · WHY?
Diet and alcohol are commonest; always consider coeliac disease and medicines, especially methotrexate.

4 · WHO ELSE NEEDS FOLIC ACID?
Pregnancy planning, high neural tube risk, chronic haemolysis, methotrexate.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 TREATMENT BY CAUSE

NICE CKS · NG247 · SPS · SPC

SITUATION	REPLACEMENT REGIMEN	DURATION AND NOTES
Folate deficiency (diet, alcohol)	**Folic acid 5 mg daily** — once B12 is known to be normal or is being replaced	Usually **4 months**; longer if the cause persists. Treat the cause
Combined B12 + folate	IM hydroxocobalamin first (see B12 guide), with or after folic acid 5 mg daily	Never folic acid alone
Coeliac, malabsorption	Folic acid 5 mg daily, plus iron and B12 as needed; gluten-free diet	Until folate normal and absorption recovered (often >4 months)
Methotrexate	**Folic acid 5 mg once weekly**, on a different day from methotrexate	Throughout methotrexate; any increase is specialist-led
Planning pregnancy, standard risk	**Folic acid 400 micrograms daily**	Before conception until **12 weeks**
Planning pregnancy, high risk	**Folic acid 5 mg daily**: previous or family NTD, type 1 or 2 diabetes, sickle cell or thalassaemia, antiepileptic or HIV medicines	Before conception until **12 weeks**; BMI alone is not an indication
Chronic haemolysis, dialysis	Long-term folic acid per haematology or renal protocol	Specialist-led dose

8 SELF-CARE AND THE CONVERSATION

CKS · NG247

Eat leafy greens, broccoli, pulses and fortified cereals; steam rather than boil · cut down alcohol · anyone who could become pregnant: take folic acid before conception · come back the **same day** with tingling, numbness, unsteadiness, fever or bleeding.

"Your folate — a B vitamin your body needs to make blood cells — is low. We've checked your B12 too, because the two go together and the order of treatment matters. A course of tablets for a few months usually sorts it, but we also need to find out why it dropped."

9 FOLLOW-UP AND MONITORING CHECKLIST

CKS · SPS

- ☐ **B12 result** seen before or alongside the first folic acid prescription
- ☐ **Recheck FBC within about 8 weeks** if there was anaemia or macrocytosis
- ☐ **Target**: Hb and MCV back to normal, symptoms gone
- ☐ **Do not recheck serum folate** while on folic acid
- ☐ No response → B12, iron, haemolysis, alcohol, thyroid, MDS, adherence, malabsorption
- ☐ **Stop** after the course once corrected and the cause is fixed; continue if the cause persists
- ☐ Methotrexate: folic acid on every repeat; FBC, LFTs, U&E per shared-care protocol
- ☐ Coeliac: annual review with FBC, ferritin, B12, folate

10 WHO TO TEST, AND WHEN TO THINK AGAIN

CKS · NG20 · NG247

TEST WHEN
Macrocytosis or anaemia, sore tongue or mouth, unexplained fatigue with poor diet, alcohol misuse, malabsorption.

DO NOT TEST TO DECIDE
pregnancy folic acid: everyone planning pregnancy needs it, whatever the level. No routine folate test at booking.

COELIAC
Low folate with low iron is a classic pattern: test tTG IgA while still eating gluten; refer if positive.

NO RESPONSE
FBC not improving → recheck B12, iron and adherence; consider haemolysis, hypothyroidism, MDS; haematology advice.

11 WHEN AND WHERE TO REFER

CKS · NG12 · NG20

999	Unstable, septic, bleeding, or Wernicke features
Same day	Neurological features; pancytopenia with infection or bleeding; suspected leukaemia; methotrexate toxicity
Urgent haematology	Other cytopenias, abnormal film, no FBC response despite adherence
Gastroenterology	Positive coeliac serology; suspected malabsorption or IBD
Maternity	Low folate in pregnancy with severe anaemia, vomiting or another concern
Dietitian	Severe or multiple deficiencies, eating disorder, bariatric surgery

12 PITFALLS AND NEVER-EVENTS

NG239 · NG247 · MHRA

- **Never** give folic acid alone to a macrocytic or neurologically symptomatic patient before B12 is known: it can mask B12 deficiency while nerve damage progresses.
- **Never** prescribe methotrexate without weekly folic acid.
- **Never** wait for a folate result before starting pregnancy folic acid.
- Folic acid does not remove the teratogenic risk of valproate: follow the MHRA pregnancy prevention programme.
- Do not stop antiepileptics abruptly; plan pregnancy with neurology.
- Severe malnutrition: thiamine before refeeding.

13 SPECIAL GROUPS

NG247 · SPS · MHRA · CG100

PREGNANCY
Low folate alone can be managed in primary care if well: supplement and check B12, diet, coeliac and adherence. Same-day maternity review if severe anaemia, vomiting or another concern.

METHOTREXATE
Check folic acid at every repeat. Toxicity (ulcers, cytopenias, sepsis, renal decline) → withhold, urgent bloods, specialist today. Folinic acid rescue is specialist-led.

ANTIEPILEPTICS
Review folate in those on phenytoin, carbamazepine, phenobarbital or primidone, especially if planning pregnancy. Valproate: pregnancy prevention programme.

ALCOHOL, MALNUTRITION
AUDIT-C and brief intervention. Wernicke features → 999 with parenteral thiamine. Severe malnutrition → thiamine before refeeding; dietitian.

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Raised haematocrit in adults CLINIC GUIDE

Interpreting a high Hb or haematocrit (erythrocytosis) in adults 18+. Not for neonates, children, pregnancy (joint specialist care) or established PV care, which haematology leads.

BSH PV and erythrocytosis guidelines 2019
NICE **TA921** · NG158 · NG209
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/polycythaemia.html

PAGE 1 · INTERPRET

is it dangerous? → is it real and persistent? → primary or secondary

! ACT TODAY

BSH 2019 · NG158

TIER	RESULT + PICTURE	ACTION
999	Raised Hct with stroke or focal neurology, confusion, seizure, chest pain or significant breathlessness (ACS, PE), or severe abdominal pain with ascites (Budd–Chiari)	Emergency pathway
Same day	Hct **≥0.60 (men) / ≥0.56 (women)** with headache, visual disturbance or dizziness (hyperviscosity), thrombosis, bleeding, pregnancy or other major count abnormality · suspected DVT	Same-day specialist discussion; same-day DVT assessment
Prompt	Hct ≥0.60 / ≥0.56 in a well patient	Prompt clinical review and haematology advice; not an automatic emergency
Urgent	JAK2 positive · erythrocytosis with splenomegaly, high platelets or high WCC · unusual-site thrombosis	Urgent haematology referral

1 IS IT REAL AND PERSISTENT?

BSH 2019

Single high result	Repeat a **venous** FBC to confirm persistence (interval per local pathway); review Hb, Hct, WCC, platelets, MCV and film
Dehydration, acute illness	Low plasma volume: repeat when well and hydrated
Diuretics	Relative erythrocytosis: review the need
Smoking, CO exposure	Carboxyhaemoglobin raises Hb and makes pulse oximetry misleading: consider a carboxyhaemoglobin level
Testosterone, anabolic steroids, EPO	Drug-driven: ask directly
Altitude	Hct is higher at altitude

Haematocrit is better than Hb for identifying raised red-cell mass. Persistent erythrocytosis should be investigated, not repeatedly re-tested.

2 SEVERITY BANDS (VENOUS HCT)

BSH 2019

RESULT	MEANS	ACTION AND TIMING
Single raised Hct	Possible erythrocytosis	Repeat venous FBC after correcting reversible factors
Persistent Hct **>0.52 men / >0.48 women**	Erythrocytosis: investigate (not a diagnosis of PV)	**JAK2 V617F + serum EPO** per local pathway; look for secondary causes
Hct **≥0.60 men / ≥0.56 women**	True erythrocytosis can be assumed; higher viscosity and thrombosis risk	Prompt clinical assessment; same day if symptoms, thrombosis or pregnancy
+ high platelets, WCC or splenomegaly	Suggests myeloproliferative neoplasm	Urgent haematology

JAK2 AND EPO	MEANS
JAK2 positive	Clonal MPN likely (present in about 95% of PV): urgent haematology
JAK2 negative, **low EPO**	Still possible PV: haematology (exon 12 is specialist)
JAK2 negative, raised EPO	Supports secondary cause; find it

3 EXAMINATION THAT CHANGES THE PLAN

BSH 2019

Spleen	Palpable → favours PV or another MPN: urgent haematology	**Cyanosis, clubbing, crackles**	Hypoxic lung or heart disease
Plethora, scratch marks	Supportive of PV, never diagnostic	**BP, renal bruit, palpable kidneys**	Renal cause; vascular risk
Calf swelling, focal signs	Thrombosis: same day or 999	**Hepatomegaly, ascites**	Budd–Chiari: emergency

4 PRIMARY OR SECONDARY?

clue → think → first test

CLUE	THINK	FIRST TEST
Itch after bathing, splenomegaly, high platelets or WCC, gout, erythromelalgia	**Polycythaemia vera**	JAK2 V617F + EPO
COPD, cyanosis, low SpO_2	**Hypoxic lung disease**	SpO_2, spirometry
Snoring, apnoeas, sleepiness	**Sleep apnoea**	Sleep pathway
Heavy smoker	**Smoking, CO**	Carboxyhaemoglobin
Testosterone or steroid use	**Drug-induced**	Drug history
Kidneys palpable, haematuria, transplant	**Renal** (cysts, RCC)	Urinalysis; renal ultrasound
Obese, hypertensive, on diuretics	**Relative**	Repeat after addressing factors
Young or lifelong, family history	**Congenital**	Haematology advice

5 WORK-UP CHECKLIST

BSH 2019

- ☐ **Repeat venous FBC** + film
- ☐ U&E, LFTs, calcium
- ☐ **SpO_2** at rest; spirometry if COPD suspected
- ☐ Urinalysis; renal ultrasound if indicated
- ☐ **JAK2 V617F** + **serum EPO**
- ☐ **Ferritin**: iron deficiency can mask PV
- ☐ **Carboxyhaemoglobin** in smokers or CO exposure
- ☐ Sleep assessment if OSA suspected

Normal daytime SpO_2 does not exclude nocturnal hypoxia. No routine CT, AFP or marrow in primary care.

6 READING THE RESULT IN FOUR STEPS

BSH 2019

1 · SYMPTOMS OR CLOT?
Headache, visual change, chest pain, breathlessness, focal neurology, leg swelling → 999 or same day before anything else.

2 · REAL AND PERSISTENT?
Repeat venous FBC after correcting dehydration, diuretics and smoking. Above 0.52 / 0.48 on repeat → investigate.

3 · CLONAL?
JAK2 with EPO. Positive JAK2, or high platelets, WCC or a spleen → urgent haematology.

4 · WHICH SECONDARY CAUSE?
Lungs, sleep, smoking, testosterone, kidneys. Unexplained, young or lifelong → haematology advice.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 MANAGEMENT BY CAUSE

BSH 2019 · NICE TA921 · NG209

CAUSE	FIRST LINE	WHO LEADS, AND THE TARGET
Polycythaemia vera (confirmed)	Venesection to **Hct <0.45** (men and women); **low-dose aspirin** unless contraindicated; cytoreduction (e.g. hydroxycarbamide, interferon) for high-risk disease (age, previous thrombosis) or poor control; ruxolitinib if hydroxycarbamide fails or is not tolerated (TA921)	**Haematology-led.** GP prescribes hydroxycarbamide only under formal shared care. Do not start aspirin in suspected PV before specialist advice
Secondary (hypoxia, OSA, drugs, renal)	Treat the cause: COPD care, sleep pathway and CPAP, stop smoking, review testosterone or steroids, renal or cardiac referral	Primary care if asymptomatic with a known cause. **No routine venesection**; no aspirin for the Hct alone. The PV target does not apply
Relative	Address hydration, diuretics, smoking, alcohol, weight; control BP and cardiovascular risk	Primary care; no venesection. Persistent and unexplained → JAK2/EPO or haematology advice
Smoking-related	Behavioural support plus varenicline, cytisinicline, combination NRT or a nicotine vape (NG209)	Quitting may lower the Hct

8 CARDIOVASCULAR RISK — ALWAYS PART OF THE PLAN

BSH 2019 · NG209

SMOKING
The biggest modifiable driver in both PV and secondary erythrocytosis. Offer support and medication at every contact.

BP AND LIPIDS
Treat per NICE. QRISK does not include polycythaemia, so it may underestimate risk in PV.

WEIGHT AND SLEEP
Weight loss can improve OSA and relative erythrocytosis; refer suspected OSA to the sleep pathway.

DIURETICS AND FLUIDS
Review diuretics; avoid dehydration, but no fixed fluid volume in heart failure, CKD or fluid restriction.

9 FOLLOW-UP AND MONITORING CHECKLIST

BSH 2019

- [ ] **Repeat Hct** after correcting dehydration, diuretics, smoking
- [ ] **JAK2 and EPO** results actioned; positive → urgent referral
- [ ] **PV target**: Hct **<0.45** — set and monitored by haematology
- [ ] Rising Hct, platelets or WCC on treatment → inform haematology
- [ ] Hydroxycarbamide: FBC and hold thresholds per shared-care protocol only
- [ ] Low ferritin after venesection is expected: **no iron** in PV without haematology advice
- [ ] Annual: thrombosis, bleeding, spleen, itch, BP, lipids, smoking
- [ ] Secondary: recheck Hct once the cause is treated; refer if rising or symptomatic

10 WHEN AND WHERE TO REFER

BSH 2019

999	Stroke, ACS, PE with instability, confusion or seizure, Budd–Chiari picture
Same-day haematology	Hyperviscosity symptoms; Hct ≥0.60/≥0.56 with symptoms, thrombosis or pregnancy; suspected leukaemia or transformation; pregnancy with PV
Urgent haematology	JAK2 positive · JAK2 negative with low EPO · persistent unexplained erythrocytosis · with splenomegaly, high platelets or WCC · with thrombosis · young or lifelong · B symptoms
Advice and guidance	Borderline persistent Hct with normal EPO and negative JAK2; family history
Respiratory, renal	Hypoxic lung disease or OSA; polycystic kidneys, transplant, renal mass

11 SPECIAL SITUATIONS IN KNOWN OR SUSPECTED PV

BSH 2019 · NG158

PREGNANCY
Known or suspected PV, or planning pregnancy → early joint haematology and obstetric care. Do not start or stop hydroxycarbamide or ruxolitinib without specialist advice.

SURGERY, TRAVEL
Tell the surgical team and haematology. Long journeys: keep mobile and hydrated; drug prophylaxis only after individual risk assessment.

CLOT ON TREATMENT
Stroke, MI, PE or DVT despite treatment → emergency care, then haematology review of control and escalation.

TRANSFORMATION
Growing spleen, falling Hb or platelets, tear-drop cells or blasts on the film → urgent haematology (same day if blasts).

12 PITFALLS AND NEVER-EVENTS

BSH 2019

- **Never** diagnose PV from a raised Hct alone, or exclude it on a negative JAK2.
- **Never** venesect secondary erythrocytosis in primary care, or aim for a normal Hct in hypoxic lung disease.
- **Never** give iron in PV without haematology advice.
- Do not label persistent unexplained erythrocytosis "relative" without EPO and JAK2.
- Normal SpO_2 in a smoker does not exclude carboxyhaemoglobin or nocturnal hypoxia.
- Do not adjust cytoreduction outside the agreed shared-care pathway.

13 SELF-CARE AND THE CONVERSATION

BSH 2019 · NG209

Stop smoking · avoid dehydration, especially in heat, illness and travel · keep moving on long journeys · control BP and weight · avoid high altitude if possible.

"Your blood is thicker than it should be — there are more red cells than normal. Often that's from something like smoking, a lung or sleep problem, or being a bit dry, and we'll check for those. Sometimes the bone marrow makes too many on its own, so I'm also sending a blood test for that. Thick blood can clot, so tell us straight away about headaches, sight changes, chest pain or a swollen leg."

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Leucocytosis: the high white cell count

CLINIC GUIDE

A raised total white cell count in adults 18+. Not for children, pregnancy (counts rise normally), or known leukaemia, lymphoma or MPN under haematology (follow the specialist plan).

NICE **NG12** (suspected cancer, updated 2026)
BSH eosinophilia 2017 · BSH CLL 2025 · iwCLL 2018
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/leucocytosis.html

PAGE 1 · INTERPRET

is it dangerous? → which cell is high? → reactive or clonal?

! ACT TODAY — RED FLAGS FIRST

NICE NG12 · local sepsis pathway

TIER	FEATURE	ACTION
999 / emergency	Shock or sepsis, reduced consciousness · high WCC with headache, confusion, visual change, breathlessness, chest pain or priapism (**suspected leucostasis**) · active significant bleeding · suspected acute leukaemia and unstable	Emergency assessment; transport set by the clinical state
Same day	**Blasts** or an abnormal immature-cell population on the film, at any count · WCC **≥100 × 10⁹/L** (hyperleukocytosis) · high WCC with new anaemia or low platelets · marked splenomegaly with abnormal counts · leucoerythroblastic film · unexplained bruising, petechiae or bleeding	Same-day haematology discussion; WCC ≥100 in a stable patient usually needs hospital assessment, not automatic 999
NG12 routes	Unexplained pallor, fatigue, fever, recurrent infection, generalised lymphadenopathy, bruising, bleeding, petechiae or hepatosplenomegaly → consider a **very urgent FBC within 48 hours** · unexplained nodes or spleen → consider lymphoma suspected cancer pathway referral	Under 25 with petechiae or hepatosplenomegaly → **immediate** specialist assessment

Urgency comes from the differential, the film, the other counts, the trend and how the patient is — not the total WCC alone.

1 CONFIRM IT IS REAL

repeat when the cause settles

- **Read the whole FBC**: absolute differential, Hb, platelets, previous results and the laboratory comment or flags.
- **Transient rises**: infection, exercise, acute stress, surgery or trauma, steroids, smoking. A single mild rise in a well patient with a clear cause is usually reactive.
- **Artefact**: platelet clumps or nucleated red cells can inflate an automated count; a **blood film** settles it.
- **Repeat** once the suspected reactive cause has resolved. Sooner, or discuss with haematology, if rising or with cytopenias, organomegaly, abnormal cells or systemic symptoms.

2 SEVERITY BANDS BY CELL TYPE

use local ranges · BSH

RESULT (×10⁹/L)	MEANS	ACTION · TIMING
Neutrophils **>7.5**, clear cause, well	Reactive neutrophilia	Treat cause; repeat once resolved
Lymphocytes **>4**	Viral, stress, CLL	Film; CLL needs **≥5** clonal B cells for **3 months** on flow
Eosinophils **0.5–1.5**, well	Mild: allergy, drugs, parasites	History, travel, drugs; may not need extensive tests
Eosinophils **≥1.5** persistent	Hypereosinophilia	**Urgent** haematology advice; same day if organ involvement
Monocytes **>1** persistent	Chronic infection, CMML	Film, infection screen; haematology if cytopenias
Basophils **>0.1** persistent	CML, PV, MF	Haematology if with neutrophilia, left shift or spleen
Any lineage, persistent, unexplained	Possible clonal disease	**Urgent** haematology via local pathway
WCC **≥100** or blasts	Leukaemia until proven otherwise	**Same-day** haematology

3 EXAMINE

nodes · spleen · liver · skin

Temperature, pulse, BP	Fever >38 °C → infection, lymphoma; low BP, fast pulse → sepsis pathway
Lymph nodes	Generalised, hard, fixed or painless → lymphoma, CLL; tender, mobile → reactive
Spleen, liver	Massive spleen → CML, myelofibrosis; moderate → CLL, lymphoma, EBV
Skin	Petechiae, bruising → leukaemia; vasculitic rash; erythema nodosum (TB, sarcoid, IBD)

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Fever, focal infection	**Infection** (neutrophilia)	CRP; CXR or urine culture if indicated
On prednisolone, beta-agonist, G-CSF	**Drug effect**	Medicine review; repeat
Smoker, obesity, mild stable rise	**Lifestyle neutrophilia**	Film if persistent
Joint, bowel or vasculitic disease	**Inflammation**	CRP, ESR
Sore throat, nodes, young adult	**EBV, CMV** (lymphocytosis)	Film; serology
Older, well, smudge cells, nodes	**CLL**	Flow cytometry via local pathway
Asthma, eczema, new drug, travel	**Allergy, drugs, parasites**	Stool microscopy, Strongyloides serology
Asthma + sinus disease + neuropathy	**EGPA**	ANCA (may be negative); urine dip
Weight loss, cough, murmur	**TB, endocarditis** (monocytosis)	CXR; blood cultures
Big spleen, basophils, left shift	**CML, MPN**	Film; haematology (not BCR-ABL1 alone)
Blasts, cytopenias, bruising	**Acute leukaemia**	Same-day haematology

5 WORK-UP CHECKLIST

target to the lineage

- ☐ **Repeat FBC** + differential
- ☐ **CRP** and/or ESR
- ☐ LDH if haematological concern
- ☐ **Eosinophilia**: B12, then stool, Strongyloides, IgE, ANCA, tryptase as directed
- ☐ **Lymphocytosis**: EBV/CMV if suggested; immunoglobulins if infections
- ☐ **Blood film** where indicated
- ☐ U&E, LFTs, bone profile
- ☐ CXR, urine culture if indicated
- ☐ **Monocytosis**: HIV test; cultures if endocarditis
- ☐ Examine nodes, spleen, liver, skin

FILM SHOWS	MEANS
Blasts	Acute leukaemia → same-day haematology
Smudge cells	CLL → flow cytometry via local pathway
Atypical lymphocytes	EBV or CMV most often
Left shift, basophilia	CML → haematology
Leucoerythroblastic	Marrow infiltration or fibrosis → same day

No urate for the count alone, no JAK2 or BCR-ABL1 without haematology input, no bone marrow in primary care.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

BSH 2017 · BSH CLL 2025 · Green Book

CAUSE	FIRST LINE	CHECK AND CONTINUE
Infection	Treat a clinically diagnosed infection per the current antimicrobial guideline and local formulary. A raised WCC alone is **not** a reason for antibiotics	Repeat FBC after recovery if the count was unexplained or markedly abnormal
Steroids, other drugs	Review the medicine. Do not taper steroids just to correct the WCC; never stop needed steroids abruptly	Interpret the count with symptoms, CRP and the differential
Smoking	Stop-smoking support with varenicline or NRT and the local service — for health, not a guaranteed WCC response	Repeat only if other features change
Eosinophilia (allergy, drugs, parasites)	Stop a causative drug where possible. Antiparasitic treatment depends on the parasite, exposure and patient — follow specialist advice	Before steroids with relevant travel: Strongyloides test or specialist advice (risk of hyperinfection)
Monocytosis	Look for chronic infection (TB, endocarditis) and autoimmune disease; offer an HIV test	Haematology if persistent, progressive or with cytopenias, spleen or dysplasia
CLL (watch and wait)	No treatment while asymptomatic; follow the haematology monitoring plan. Treat infections promptly. Flu and pneumococcal vaccines; avoid live vaccines if immunosuppressed	Contact haematology for rapid rise, new cytopenia, growing nodes or spleen, B symptoms
CML, MPN, other	Specialist-led. GP role: act on the monitoring plan, spot infection, bleeding, clots and side effects	Aspirin in ET or PV is a haematology decision

7 LIVING WITH CLL OR AN MPN

UKHSA Green Book · NHS

INFECTIONS
Annual flu and pneumococcal vaccines (Green Book chapter 25). No live vaccines if immunosuppressed. Treat infections early.

ALCOHOL, SMOKING
No more than 14 units a week; none with liver involvement or on chemotherapy. Offer stop-smoking support.

SKIN
On immunosuppression: SPF 30 or more with 4-star UVA protection, avoid midday sun, report new skin lesions.

ACTIVITY
About 150 minutes a week of moderate exercise helps fatigue. Avoid contact sports with low platelets or a big spleen.

8 FOLLOW-UP AND MONITORING

local laboratory pathway

- ☐ **Repeat FBC** once the reactive cause has resolved: target back within the laboratory range
- ☐ Persistent and unexplained → film and urgent haematology via the local pathway
- ☐ Ask each time about fevers, sweats, weight loss, nodes, abdominal swelling
- ☐ Rising count, new cytopenia or blasts → same-day haematology
- ☐ Diagnosis and plan in the problem list
- ☐ Normal on repeat → **stop**; no further routine testing
- ☐ Watch the **trend**: a steady rise matters even when each value looks modest
- ☐ Known CLL, CML or MPN: follow the specialist-written schedule
- ☐ **Re-refer**: new B symptoms or organomegaly, count rising despite treatment, new cytopenias

9 WHEN AND WHERE TO REFER

NG12 · BSH · local laboratory pathway

EMERGENCY
Sepsis or shock, suspected leucostasis, reduced consciousness, significant bleeding, priapism, unstable suspected acute leukaemia.

SAME-DAY HAEMATOLOGY
Blasts, WCC ≥100, symptomatic or fast-rising count, new anaemia or low platelets, marked spleen, leucoerythroblastic film, eosinophilia with organ damage.

SUSPECTED CANCER / URGENT
NG12 symptoms: very urgent FBC; nodes or spleen → lymphoma pathway. Persistent unexplained count of any lineage, or eosinophils ≥1.5 → urgent haematology.

PRIMARY CARE
Mild rise, clear reactive cause, well patient, reassuring differential and film; settles once the cause is treated.

Other specialties: infectious diseases (persistent fever, travel, TB, HIV) · respiratory (eosinophilia with asthma) · rheumatology (suspected vasculitis, EGPA).

10 PITFALLS AND NEVER-EVENTS

BSH · NG12

- **Never** judge urgency on the total WCC alone: blasts or cytopenias at a modest count matter more.
- **Never** give antibiotics for a raised WCC without an infection.
- **Never** start steroids for unexplained eosinophilia with relevant travel without thinking of Strongyloides.
- **Never** call it CLL on the count alone: it needs clonal B cells confirmed by flow cytometry.
- **Never** request BCR-ABL1 or JAK2 from primary care on the count alone — discuss with haematology.
- Neutrophilia does not prove bacterial infection; a normal CRP does not exclude it.
- Eosinophilia with asthma, sinus disease or neuropathy may be EGPA even if ANCA is negative.
- No endocarditis prophylaxis just because of leucocytosis, CLL or a big spleen.

11 SAFETY-NET

tell the patient

999 or emergency: sudden severe headache, confusion, vision loss, chest pain or breathlessness, a painful erection that will not go down, heavy bleeding, fever with rigors or feeling very unwell. **Same day:** new bruising or tiny red spots, new lumps in the neck, armpit or groin, a swollen tummy, drenching sweats or weight loss.

12 SAY IT LIKE THIS

plain English

"Your white cells, the cells that fight infection, are a little high. Most of the time that is the body responding to an infection, a medicine or smoking, and it settles. I'll repeat the test when you're better; if it stays high without a reason we'll ask a blood specialist."

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Leucopenia: the low white cell count

CLINIC GUIDE

A low white cell or neutrophil count in adults 18+. Not for children, or for counts expected during chemotherapy (follow the acute oncology plan). Low lymphocytes alone: see the lymphopenia guide.

NICE **CG151** neutropenic sepsis · **NG12** (2026)
BSH neutropenia bands · NG239 · SPCs
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/leucopenia.html

PAGE 1 · INTERPRET

is it sepsis? → how low is the ANC? → isolated or pancytopenia?

! ACT TODAY — RED FLAGS FIRST

NICE CG151 · NG12 · SPC

TIER	FEATURE	ACTION
Immediate / 999	Unwell on **anticancer treatment** (now or recent), febrile or not · any low count with fever, rigors or sepsis features · shock, confusion, breathlessness · severe mucositis, cannot swallow or drooling	Immediate hospital assessment; contact acute oncology. Do not wait for the count. Neutropenic sepsis = neutrophils **≤0.5 × 10⁹/L** with temperature **>38 °C** or other sepsis signs (CG151)
Same day	New or falling ANC **<0.5**, even if afebrile · new pancytopenia, dysplasia or rapidly falling counts · purpura or bleeding · **suspected drug-induced agranulocytosis** (clozapine, carbimazole, sulfasalazine, DMARDs) with sore throat, fever or very low ANC	Same-day haematology or acute medical discussion. Carbimazole: **stop now**. Clozapine: follow the monitoring service and mental health team
Immediate specialist	Blasts, or a film or count reported as acute leukaemia	Immediate haematology referral

Fever can be absent in neutropenic or steroid-treated patients, and CRP can be normal despite serious infection. A stable known chronic neutropenia goes through the local haematology route, not 999.

1 CONFIRM IT IS REAL

the ANC is the number that matters

- **Look at the differential**: a low WCC may be low neutrophils, low lymphocytes or both. ANC = WCC × neutrophil %.
- **Transient**: viral infections (flu, EBV, CMV) commonly lower the count; repeat once recovered.
- **Artefact**: delayed or old samples and laboratory flags — repeat and ask for a **blood film** if it does not fit.
- **Trend** beats a single value: a falling ANC matters more than a stable low one.
- **Duffy-null-associated (benign ethnic) neutropenia**: well, stable, isolated, normal Hb and platelets, normal film, no recurrent infection. Never diagnose it from ethnicity alone.

2 SEVERITY BANDS — THE ANC

BSH bands · ×10⁹/L

ANC	MEANS	ACTION · TIMING
<1.5	Neutropenia	Repeat with differential; review drugs
1.0–1.5 mild	Low infection risk; Duffy-null, viral, drug	Stable, isolated, well: repeat and investigate selectively; routine haematology if persistent and unexplained
0.5–1.0 moderate	Autoimmune, drugs, HIV, TB	**Urgent** haematology advice if persistent; timing by trend
<0.5 severe	High infection risk	New or unexplained: **same-day** discussion even without fever
Any ANC + fever or unwell	Possible neutropenic sepsis	**Immediate** hospital assessment
Low WCC + low Hb + low platelets	Pancytopenia: marrow failure or infiltration	**Urgent** haematology even if mild; same day with red flags

3 EXAMINE

infection · organs · autoimmune

Temperature, pulse, BP	Fever, fast pulse, low BP → sepsis pathway; fever may be absent
Mouth, throat	Ulcers, gingivitis, mucositis → agranulocytosis; thrush
Nodes, spleen	HIV, SLE, lymphoma; Felty (RA + spleen), portal hypertension
Skin, joints	Petechiae (marrow failure); malar rash (SLE); synovitis (RA)
Chest	Pneumonia may have few signs without neutrophils

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Recent viral illness, improving	**Post-viral**	Repeat FBC when recovered
Well, stable, isolated, African or Middle Eastern ancestry	**Duffy-null-associated**	Repeat counts; film; haematology if <1.0 or falling
Clozapine, carbimazole, sulfasalazine, azathioprine, methotrexate	**Drug-induced**	Urgent FBC; drug-specific pathway
Also valproate, captopril, beta-lactams	**Drug, by timing**	Medicine review; film
Macrocytosis, glossitis, neuropathy	**B12 or folate**	B12, folate
Rash, joint pain, mouth ulcers	**SLE, RA, Felty**	ANA, RF, anti-dsDNA if suggested
Risk factors or other cytopenias	**HIV**, hepatitis	HIV test; hepatitis B, C if indicated
Night sweats, weight loss, cough	**TB**	CXR; specialist advice
Pancytopenia, bruising, fatigue	**Aplastic anaemia, MDS, leukaemia**	Film; urgent haematology
Recent chemotherapy or radiotherapy	**Treatment effect**	Acute oncology plan
Big spleen, alcohol, liver disease	**Hypersplenism**	LFTs; ultrasound

5 WORK-UP CHECKLIST

target to the pattern

- ☐ **Repeat FBC** + differential, trend
- ☐ **Medicine review**, including OTC
- ☐ **B12 and folate**, esp. macrocytosis
- ☐ ANA, RF, anti-dsDNA only if suggested
- ☐ **Pancytopenia**: retics, LDH, haptoglobin, TFTs
- ☐ **Blood film**: dysplasia, blasts
- ☐ U&E, LFTs; CRP if indicated
- ☐ **HIV test** when appropriate
- ☐ Immunoglobulins if recurrent infection
- ☐ EBV/CMV only in a compatible illness

No bone marrow and no G-CSF from primary care. Never delay transfer or referral to "complete investigations" when the ANC is <0.5 or there is pancytopenia.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

CG151 · NG239 · SPC · Green Book

CAUSE	FIRST LINE	CHECK AND CONTINUE
Suspected neutropenic sepsis	Immediate transfer; empirical antibiotics are given at once **in hospital**. Do not give oral antibiotics and send home, and do not wait for the ANC	Shocked or deteriorating: sepsis pathway while arranging 999
Drug-induced	Withhold the likely drug where clinically safe; urgent FBC. **Carbimazole**: stop immediately. **Clozapine**: the brand monitoring service (CPMS, DMS or ZTAS: WBC and neutrophils, green/amber/red; not EMA/US ANC-only thresholds) and the mental health team decide — do not stop it alone in a stable patient. **DMARDs**: withhold, contact the shared-care team	Repeat FBC as agreed with the specialist team; still falling → same-day haematology. Rechallenge is a specialist decision
B12 or folate deficiency replacement	**Replacement:** hydroxocobalamin 1 mg IM three times a week for 2 weeks, then every 2–3 months (neurological features: alternate days until no further improvement). Folic acid 5 mg daily — **never alone** until B12 deficiency is excluded or treated	Repeat FBC to confirm the counts recover
Duffy-null-associated	No treatment. Explain it is a normal variant; infection risk is generally not increased. Give written information	Record it to avoid repeat work-ups; haematology if <1.0 persistently, falling or new abnormalities
Post-viral	Reassure; safety-net	Repeat FBC after recovery; no further routine tests once normal
Felty, hypersplenism	Manage the RA or liver disease with the specialist team; Felty treatment is specialist-led	Rheumatology or hepatology plus haematology advice
Autoimmune, marrow, HIV	Specialist-led: steroids, immunosuppression, rituximab, G-CSF, antibiotic or antifungal prophylaxis are not started in primary care	No prophylaxis just because the ANC is <1.0 or <0.5

7 LIVING WITH A LOW NEUTROPHIL COUNT

proportionate advice · Green Book

HANDS AND FOOD
Hand washing is the single most useful step. Normal diet with ordinary safe food handling; no "neutropenic diet" for stable mild neutropenia.

PEOPLE AND PETS
Normal social contact; avoid obvious infections such as chickenpox where practical. Healthy pets and gardening are fine with gloves and hand washing.

MOUTH
Brush twice daily with a soft brush. Keep routine dental care; dental work during profound or treatment-related neutropenia follows specialist advice.

VACCINES
Annual flu and pneumococcal (Green Book chapter 25). Live vaccines depend on immunosuppression, not the ANC alone.

8 FOLLOW-UP AND MONITORING

local haematology advice

- ☐ **Drug-induced**: repeat FBC at the interval agreed with the specialist team; the count should be recovering
- ☐ **Post-viral**: repeat once recovered; target back within the laboratory range, then **stop**
- ☐ **Possible Duffy-null**: repeat counts to show the ANC is stable and isolated with a normal film; then reassure
- ☐ Chronic idiopathic neutropenia: follow the haematology plan
- ☐ Re-refer: ANC falling despite stopping the drug, new Hb or platelet drop, recurrent significant infections
- ☐ Watch the **trend**: a falling ANC over months suggests marrow or autoimmune disease even if each value is "mild"
- ☐ Recurrent significant infections at an acceptable count → immunoglobulins and haematology review
- ☐ Diagnosis and plan in the problem list

9 SAFETY-NET

tell the patient

Immediately (999 if confused, breathless or collapsing): fever, shivers or feeling suddenly unwell — especially on cancer treatment (call the acute oncology line). **Same day:** sore throat, mouth ulcers, cough, burning urine, diarrhoea, tummy pain, new bruising or bleeding.

10 WHEN AND WHERE TO REFER

CG151 · NG12 · local haematology pathway

IMMEDIATE
Suspected neutropenic sepsis; any low count with fever, sepsis or deterioration; blasts or a film reported as acute leukaemia.

SAME-DAY HAEMATOLOGY
New or falling ANC <0.5 (afebrile); new pancytopenia, dysplasia or rapidly falling counts; suspected drug agranulocytosis.

URGENT HAEMATOLOGY
Persistent ANC 0.5–1.0 · persistent unexplained low WCC · abnormal film · neutropenia with spleen, nodes, recurrent infection or B symptoms · mild stable pancytopenia.

ROUTINE OR PRIMARY CARE
Routine: persistent ANC 1.0–1.5 not explained by Duffy-null. Primary care: stable mild isolated, recovering post-viral, confirmed Duffy-null.

Infectious diseases or HIV team: every person with HIV and a cytopenia; suspected TB. Recurrent infections are judged by severity, site, unusual organisms and hospital treatment.

11 PITFALLS AND NEVER-EVENTS

CG151 · SPC

- **Never** manage suspected neutropenic sepsis with oral antibiotics in the community or wait for the count.
- **Never** rely on fever: it may be absent on steroids or with neutropenia.
- **Never** label Duffy-null-associated neutropenia on ethnicity alone, or when there are other cytopenias.
- **Never** give folic acid alone before B12 deficiency is excluded or treated.
- **Never** stop clozapine on your own in a stable patient — contact the monitoring service and mental health team today.
- Carbimazole users should know to stop it and get an urgent FBC for sore throat, mouth ulcers or fever.
- A neutropenic patient may have pneumonia or sepsis with few signs and a normal CRP.
- A stable chronic ANC of 0.8 in a well patient is not the same as a new falling 0.8 with fever.

12 SAY IT LIKE THIS

plain English

"Your neutrophils, the white cells that fight bacteria, are lower than usual. Often that's after a virus, a medicine or just how your body is made. I'll repeat it, but if you get a fever, shivers or feel suddenly very unwell, don't wait — get seen straight away."

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Lymphocytosis: the high lymphocyte count

CLINIC GUIDE

A raised absolute lymphocyte count in adults 18+. Not for children (use age-specific ranges and the paediatric pathway), or known CLL or lymphoma under active haematology treatment.

NICE **NG12** (suspected cancer, updated 2026)
BSH CLL guidelines · iwCLL 2018 · NICE CKS
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/lymphocytosis.html

PAGE 1 · INTERPRET

blasts or cytopenias? → reactive or persistent? → clonal or not?

! ACT TODAY — RED FLAGS FIRST

NICE NG12 · BSH

TIER	FEATURE	ACTION
Immediate	**Blasts**, or a count or film reported as acute leukaemia · severe cytopenias with deterioration, severe infection or bleeding · glandular fever with stridor, breathing difficulty or unable to swallow saliva · sudden left upper abdominal or shoulder-tip pain, dizziness or collapse (splenic rupture)	Immediate haematology discussion and hospital assessment; 999 for airway compromise or suspected splenic rupture
Urgent / suspected cancer	Rapidly rising count · new anaemia, low platelets or neutropenia · unexplained lymphadenopathy or splenomegaly (consider the lymphoma **suspected cancer pathway referral**, especially with fever, night sweats, weight loss, itch or breathlessness) · abnormal lymphocytes with a mediastinal mass	Urgent haematology or NG12 route; same day if systemically unwell
NG12 FBC	Pallor, persistent fatigue, unexplained fever, recurrent infection, generalised nodes, bruising, bleeding, petechiae, hepatosplenomegaly	Consider a **very urgent FBC within 48 hours**. Under 25: **offer** it; petechiae or hepatosplenomegaly → immediate specialist assessment

No single lymphocyte count is an emergency threshold on its own: serious disease can occur at modest counts, and a well patient with a high, stable CLL count may not need same-day referral.

1 CONFIRM IT IS REAL

absolute count, not percentage

- **Use the absolute count**: a high lymphocyte % with low neutrophils is a relative rise, not lymphocytosis.
- **Look back**: CLL is often visible on old FBCs for years.
- **Timing**: during or just after a viral illness the rise is usually reactive.
- **Repeat** after recovery, commonly at **4–6 weeks**, to confirm it has settled.
- **Film** if persistent, rising, or with cytopenias, nodes or spleen. Smear cells support CLL but are not diagnostic; diagnosis is by **flow cytometry**.

2 SEVERITY BANDS

clonal B cells ×10⁹/L · iwCLL · BSH

RESULT	MEANS	ACTION · TIMING
Raised with acute viral illness, well	Reactive	Repeat 4–6 weeks after recovery
Persistent after recovery, unexplained	Needs a cause	Film, examine, HIV if indicated; flow via local pathway
Clonal B **<0.5**, nothing else	Low-count MBL	Follow the flow report
Clonal B **0.5 to <5**, no nodes, spleen or cytopenia	High-count MBL; CLL needing treatment **~1–2%** a year	Periodic FBC and review per the haematology letter
Clonal B **≥5** for **≥3 months**	**CLL**	Haematology plan; most watch and wait
Any count + cytopenias, nodes, spleen, B symptoms, fast rise	CLL needing review, lymphoma	**Urgent** haematology or NG12
Blasts	Acute leukaemia	**Immediate**

CLL and MBL thresholds apply to the **clonal B-cell** count on flow cytometry, not the total lymphocyte count.

3 EXAMINE

nodes · spleen · other cell lines

Lymph nodes	Neck, axillae, groins: count the areas (Binet); tender → reactive
Spleen, liver	CLL, lymphoma, EBV; a big spleen in glandular fever → no contact sport
Throat, skin	Tonsillar swelling, airway; bruising, petechiae, pallor
Rest of the FBC	Hb, neutrophils, platelets: any new fall raises urgency

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Sore throat, fever, nodes, young adult	**EBV** (glandular fever)	Film (atypical lymphocytes); EBV serology if needed; LFTs
Similar illness, Monospot negative	**CMV**	CMV serology if it changes management
Paroxysmal cough, whoop, vomiting	**Pertussis** (notifiable)	PCR or serology by duration
Flu-like illness, rash, risk	**HIV seroconversion**	4th-generation HIV test
Smoker, mild stable rise	**Smoking**	Exclude other causes
Splenectomy	**Post-splenectomy**	None if mild and stable
Older, well, smear cells, nodes	**CLL or MBL**	Flow cytometry via local pathway
Cleft nuclei, nodes, spleen	**Lymphoma** in the blood	Flow cytometry; haematology
Neutropenia, spleen, RA picture	**LGL leukaemia**	Film; specialist assessment
Fatigue, low BP, pigmentation	**Addison's disease**	9am cortisol, U&E
Weight loss, tremor, palpitations	**Hyperthyroidism**	TFTs

5 WORK-UP CHECKLIST

persistent or unexplained

- ☐ **Repeat FBC** + differential
- ☐ **Previous counts** reviewed
- ☐ CRP/ESR in context
- ☐ EBV, CMV, pertussis only if the illness fits
- ☐ Examine nodes, liver, spleen, skin
- ☐ **Blood film**
- ☐ LFTs, U&E
- ☐ **HIV test** where indicated
- ☐ **Flow cytometry** via local pathway
- ☐ Immunoglobulins if infections

FILM SHOWS	SUGGESTS
Atypical (reactive) lymphocytes	Viral, often EBV — not specific
Small mature cells ± smear cells	CLL → flow cytometry
Cleft or irregular nuclei	Lymphoma → flow cytometry
Large granular lymphocytes	LGL expansion → specialist
Blasts	Acute leukaemia → immediate

EBV: interpret VCA IgM/IgG and EBNA, not IgM alone; Monospot can be falsely negative early. CT staging and bone marrow are arranged by haematology.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

NICE CKS · BSH CLL · UKHSA Green Book

CAUSE	FIRST LINE	CHECK AND CONTINUE
Glandular fever (EBV)	Paracetamol or an NSAID, fluids, salt-water gargles. No antivirals (aciclovir does not help). **No routine steroids** — specialist-led only for airway obstruction, severe ITP or haemolysis, or neurological complications. Avoid **amoxicillin/ampicillin**: if an antibiotic is needed, phenoxymethylpenicillin or clarithromycin	No contact sport or heavy lifting for at least **4 weeks** (CKS). Repeat FBC (and LFTs if abnormal) after recovery. A rash on amoxicillin during EBV is **not** penicillin allergy
Pertussis, other viral	Manage the infection per its own pathway; pertussis is notifiable to UKHSA	Repeat FBC after recovery
Smoking	Exclude a malignant cause; then stop-smoking support	The count often settles over months after stopping
MBL	No treatment. Explain it is common and usually stays stable	FBC and review at the frequency in the haematology letter
CLL (watch and wait)	**No treatment for the count alone**; treatment is haematology-led for active disease (marrow failure, symptomatic nodes or spleen, rapid rise, B symptoms, steroid-resistant AIHA or ITP)	GP monitoring only with a **written haematology plan**: Hb, platelets, lymphocyte trend, re-referral triggers
CLL infections	Treat infections promptly. Inactivated flu, COVID-19, Shingrix and pneumococcal vaccines per the Green Book; **no live vaccines** unless haematology advises	Recurrent or severe infections → immunoglobulins; replacement only after specialist assessment

7 LIVING WITH CLL, RECOVERING FROM GLANDULAR FEVER

BSH CLL · NICE CKS

INFECTIONS IN CLL
Same-day advice for fever over 38 °C, worsening cough or spreading skin infection; on treatment, use the haematology contact number.

SKIN IN CLL
Higher skin cancer risk: daily high-SPF sunscreen, avoid peak sun, report new, bleeding or non-healing lesions.

MIND
"Watch and wait" is the evidence-based treatment. Patient support: CLL Support and Lymphoma Action; NHS Talking Therapies for anxiety or low mood.

AFTER GLANDULAR FEVER
Gradual return to work, study and exercise, guided by symptoms. Avoid alcohol while unwell or if LFTs are raised. Persistent fatigue → reassess.

8 CLL: BINET STAGE

haematology stages

STAGE	FINDINGS
A	Hb ≥100 g/L, platelets ≥100 × 10^9/L, fewer than 3 node areas
B	Same counts, 3 or more node areas
C	Hb <100 g/L or platelets <100 × 10^9/L, any nodes

Stage alone does not start treatment; active-disease criteria do.

9 FOLLOW-UP AND MONITORING

written plan where diagnosed

- [ ] **Reactive**: repeat FBC after recovery, commonly **4–6 weeks**; target back within the laboratory range, then **stop**
- [ ] Not settled: film, previous counts, examine, HIV if indicated, flow per the local pathway
- [ ] **MBL**: FBC at the frequency in the haematology letter
- [ ] **CLL**: FBC per the plan; falling Hb or platelets, rapid rise, B symptoms, new nodes or spleen → haematology
- [ ] CLL: vaccines reviewed; immunoglobulins known; skin checks and sun protection
- [ ] Mood after a CLL diagnosis: offer support and NHS Talking Therapies
- [ ] Glandular fever: contact-sport advice given and recorded
- [ ] Cause, repeat result and plan recorded in the notes

10 WHEN AND WHERE TO REFER

NG12

Immediate	Blasts or suspected acute leukaemia; severe cytopenia with deterioration, severe infection or bleeding
Suspected cancer	Unexplained nodes or spleen (lymphoma pathway); NG12 symptoms → very urgent FBC
Urgent haematology	New or progressive anaemia, low platelets or neutropenia; rapid rise; recurrent severe infection; bruising; growing nodes or spleen
Advice or routine	Persistent unexplained lymphocytosis after reactive causes; abnormal flow; new MBL or CLL needing a plan
Primary care	Reactive with a clear cause, settling; mild stable post-splenectomy; known MBL with a plan

11 PITFALLS AND NEVER-EVENTS

BSH · CKS

- **Never** diagnose CLL from the total count or smear cells: it needs flow cytometry.
- **Never** give amoxicillin in glandular fever, or code the rash as penicillin allergy.
- **Never** give live vaccines in CLL without haematology advice.
- CLL: falling Hb with raised bilirubin and LDH and a positive DAT is **AIHA**; falling platelets with bruising may be **ITP** — urgent haematology, even if the CLL is stable.
- Mild stable lymphocytosis after splenectomy needs no action; a rising count does.
- Routine spleen ultrasound is not needed after glandular fever; seek advice for athletes or a persisting spleen.
- Persistent or severe fatigue after glandular fever: reassess for other diagnoses and low mood.
- CLL and COVID-19: check interactions with CLL drugs before nirmatrelvir/ritonavir; use the NHS antiviral pathway.

12 SAFETY-NET

tell the patient

999: breathing difficulty, cannot swallow saliva, sudden left-sided tummy or shoulder-tip pain, collapse. **Same day:** fever over 38 °C, worsening cough or breathlessness, spreading skin infection, new bruising or bleeding. **Within a week:** new lumps, drenching sweats, weight loss.

13 SAY IT LIKE THIS

plain English

"Your lymphocytes, one type of white cell, are raised. After a virus that's very common and settles. If it stays up we'll do a special test on the blood. If that shows a slow-growing condition like CLL, most people just need regular checks — watching it carefully is the right treatment, not doing nothing."

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Lymphopenia: the low lymphocyte count

CLINIC GUIDE

A low absolute lymphocyte count in adults 18+. Not for children or infants (age-specific ranges; suspected SCID → paediatric immunology), or people under active HIV, transplant or chemotherapy care (follow that team's plan).

NICE **NG60** HIV testing · **NG12** (2026) · NG243
BHIVA testing and standards · Green Book
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/lymphopenia.html

PAGE 1 · INTERPRET

is it dangerous? → is it transient? → why is it low?

! ACT TODAY — RED FLAGS FIRST

NG12 · BHIVA · local pathways

TIER	FEATURE	ACTION
Same-day hospital	Fever or sepsis with severe lymphopenia · suspected opportunistic infection (PCP, CMV, disseminated fungal, toxoplasmosis, TB) · severe respiratory, neurological or systemic illness · very unwell with fever, cytopenias, organomegaly, hepatitis or very high ferritin (**suspected HLH**) · pancytopenia, blasts or major film abnormality	Immediate hospital assessment; offer an HIV test. Do not start HLH treatment in primary care
Urgent / suspected cancer	Lymphopenia with fever, night sweats, weight loss and nodes or a mediastinal mass · progressive unexplained lymphopenia with other cytopenias, abnormal film, nodes or spleen	Suspected cancer pathway referral (lymphoma, NG12) — do not wait for LDH or CT
HIV service	New positive HIV test · suspected seroconversion · known HIV with low CD4	HIV specialist review **within 14 days** (BHIVA); same day or within 24 hours if unwell, pregnant, suspected seroconversion or AIDS-defining illness

In confirmed HIV, CD4 <200 cells/mm³ is severe immunosuppression; a low total lymphocyte count does not tell you the CD4 count, and CD4 <200 without HIV is not AIDS.

1 CONFIRM IT IS REAL

repeat unless unwell

- **Absolute count**, read with neutrophils, Hb, platelets, the film and previous FBCs: new, persistent or progressive?
- **Transient causes**: acute viral illness (COVID-19, flu, EBV, CMV), sepsis, recent steroids, physiological stress. These usually recover; confirm with a repeat.
- **Repeat** FBC + differential at about **4–8 weeks**, or sooner by severity — unless the patient is unwell or there are features of immunodeficiency, malignancy or marrow disease.
- **Steroids** lower the measured count by redistribution; that alone does not mean clinically important T-cell deficiency.

2 SEVERITY BANDS

practical descriptors · ×10⁹/L

LYMPHOCYTES	MEANS	ACTION · TIMING
1.0–1.5 mild	Usually reactive: viral, stress, steroids	Repeat after the trigger settles; no immediate immunology work-up
0.5–1.0 moderate	More likely significant if persistent	Offer HIV test, drug review, film; persistent, progressive or other cytopenias → haematology or immunology
<0.5 severe	Opportunistic infection risk	HIV test promptly; film; subsets with specialist advice; prompt immunology or haematology discussion
Any + fever, sepsis or opportunistic infection	Clinical immunodeficiency	**Same-day** hospital
Falling on serial counts	Progressive	**Urgent** haematology

Practical descriptors, not validated national severity categories; local laboratory ranges take precedence. What matters more: **persistence and trend**; recurrent or opportunistic infection; other cytopenias, abnormal film, nodes or spleen; drugs, HIV risk, systemic disease, immunosuppression. No PCP prophylaxis on the total count alone.

3 EXAMINE

nodes · spleen · mouth · skin

Lymph nodes	Generalised → HIV, lymphoma, EBV; regional → reactive, TB, cancer
Spleen	EBV, HIV, lymphoma, leishmaniasis
Mouth	Oral candidiasis → T-cell deficiency
Skin	Kaposi sarcoma, skin lymphoma, lupus rash
Chest, sats	Breathless or hypoxic with few signs → think PCP; same day

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Recent viral illness or sepsis	**Infection** (commonest)	Repeat FBC after recovery
Prednisolone, methotrexate, azathioprine, mycophenolate, rituximab, natalizumab, radiotherapy	**Drugs** (very common)	Medicine review; shared-care bloods
Thrush, recurrent or unusual infection, weight loss, risk	**HIV**	4th-generation HIV test
Cough, night sweats, travel	**TB**	CXR; specialist advice
Rash, joint pain, mouth ulcers	**SLE**, RA, Sjögren's	ANA; dsDNA, C3/C4 if positive
Cough, nodes, erythema nodosum	**Sarcoidosis**	CXR
Sweats, weight loss, nodes	**Lymphoma**	NG12 route; LDH
Recurrent chest or sinus infections	**CVID**, antibody deficiency	IgG, IgA, IgM
Weight loss, poor intake, malabsorption	**Malnutrition**, coeliac	Albumin; coeliac serology
Advanced kidney disease	**Uraemia**	U&E, eGFR

Zinc deficiency only in the right context (malnutrition, malabsorption, bariatric surgery, chronic diarrhoea).

5 WORK-UP CHECKLIST

guided by history and examination

- ☐ **Repeat FBC** + differential
- ☐ **Blood film** if persistent or marked
- ☐ **HIV test**: offer and recommend
- ☐ **Medicine review**
- ☐ Immunoglobulins if infections
- ☐ U&E, LFTs; CRP, ESR in context
- ☐ ANA if SLE features
- ☐ CXR if TB, lymphoma or sarcoid
- ☐ EBV, CMV only if the illness fits
- ☐ CD4 subsets: specialist advice only

RESULT	MEANS
HIV positive	HIV service; ART for all, whatever the CD4
Low IgG, IgA or IgM + infections	Clinical immunology (possible CVID)
ANA positive + features	dsDNA, C3/C4; rheumatology
Film: blasts, other cytopenias	Same-day haematology

Unexplained low white cells or platelets are an HIV indicator condition: test without waiting for risk factors (NG60, BHIVA). No routine CD4 for mild isolated lymphopenia.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

BHIVA · NG243 · BSR · Green Book

CAUSE	FIRST LINE	CHECK AND CONTINUE
Post-viral, sepsis	Reassure; safety-net	Repeat FBC at 4–8 weeks; if not normal: film, HIV test, targeted tests
Corticosteroids	Review indication and dose with the prescriber; **never stop abruptly**. Steroid Emergency Card for prednisolone **5 mg a day or more for 4 weeks or longer**; sick-day rules (NG243)	Judge infection risk on the whole regimen, not the count; PCP prophylaxis only per specialist or local protocol
DMARDs, biologics	Follow the drug's shared-care protocol. **Do not stop** for isolated mild lymphopenia	Withhold and contact the specialist for significant or progressive cytopenia, neutropenia, serious infection or a protocol threshold
HIV	Refer to the HIV service: ART is recommended for everyone with HIV, whatever the CD4; prophylaxis is started by the HIV team	CD4 and viral load per the HIV service; annual bloods and sexual health review in shared care
SLE	Rheumatology: hydroxychloroquine for all; immunosuppression for active disease	FBC trend; urinalysis and BP for nephritis
CVID	Immunoglobulin replacement (IV or subcutaneous) is started and dosed by clinical immunology	Treat chest infections promptly — do not wait; vaccines per immunology advice
Oral thrush	Topical nystatin or miconazole for mild disease; systemic fluconazole for extensive or recurrent disease, at the oral-candidiasis dose after checking interactions and LFTs	Recurrent → look for HIV, diabetes, inhaled steroids, dentures

7 LIVING WITH A WEAK IMMUNE SYSTEM

proportionate advice · Green Book

FOOD AND HANDS
Wash hands with soap before meals and after exposures. Avoid raw meat, raw eggs and unpasteurised dairy; avoid cat litter in severe T-cell deficiency.

TRAVEL
Travel health review before going abroad: live travel vaccines need specialist assessment; malaria prophylaxis and bite avoidance; food safety.

PREGNANCY
HIV: continue ART throughout. CVID: replacement continues. DMARDs: check safety with the prescriber before conception.

MIND AND SUPPORT
NHS Talking Therapies; address HIV stigma openly. Support: Terrence Higgins Trust, aidsmap, UKPIN. CVID: carry a letter for emergencies.

8 FOLLOW-UP AND MONITORING

local pathways

- [ ] **Transient**: FBC at 4–8 weeks after the illness or steroid course; target back within the laboratory range, then **stop**
- [ ] Not normalised: film, HIV test, targeted tests by history
- [ ] Persistent unexplained severe lymphopenia → haematology or immunology within a week
- [ ] On a DMARD: falling trend → discuss with rheumatology
- [ ] HIV with CD4 falling below 200 → HIV service for prophylaxis and ART review
- [ ] Steroid card and sick-day rules given and recorded
- [ ] Vaccines reviewed against Green Book chapter 6; annual inactivated flu, COVID-19 and pneumococcal (chapter 25)
- [ ] Recurrent oral thrush → HIV test, diabetes check, inhaler technique; oral medicine if it keeps recurring

9 WHEN AND WHERE TO REFER

NG12 · BHIVA · local haematology and immunology pathways

SAME-DAY HOSPITAL
Fever or sepsis with severe lymphopenia; suspected opportunistic infection or HLH; pancytopenia or blasts.

HIV SERVICE
New diagnosis: seen within 14 days; same day or within 24 hours if unwell, pregnant, seroconverting or AIDS-defining illness.

HAEMATOLOGY
Suspected lymphoma (NG12); progressive or persistent unexplained lymphopenia with other cytopenias, odd film, nodes, spleen or B symptoms.

IMMUNOLOGY, RHEUMATOLOGY
Recurrent or unusual infections, low immunoglobulins, severe persistent lymphopenia without HIV · active SLE · cytopenia on a DMARD.

10 PITFALLS AND NEVER-EVENTS

Green Book · BHIVA · UKHSA

- **Never** start PCP, antifungal or herpes prophylaxis on the lymphocyte count alone.
- **Never** decide on live vaccines from the count: check Green Book chapter 6 and ask the specialist. Inactivated vaccines (flu, COVID-19, pneumococcal) are safe; Shingrix (two doses) where indicated.
- **Never** stop steroids abruptly or a DMARD just because the count is below range.
- **Never** skip the HIV test because the patient "is not at risk".
- Non-immune and immunosuppressed after a chickenpox or shingles exposure → same-day advice; post-exposure prophylaxis is oral aciclovir or valaciclovir (UKHSA).
- Oral thrush: a single 150 mg fluconazole dose is the vaginal regimen, not the oral one.

11 SAFETY-NET

tell the patient

Same day or 999: fever with shivers, breathlessness, a new cough, confusion, severe headache, or feeling suddenly very unwell. **Soon:** mouth or throat thrush, recurring chest or sinus infections, night sweats, weight loss, new lumps. On long-term steroids: follow your sick-day rules.

12 SAY IT LIKE THIS

plain English

"Your lymphocytes, one kind of white cell, are a bit low. That's very common after a virus or with steroid tablets, and it usually comes back up. I'll repeat the test, and I recommend an HIV test as a routine part of checking — we offer it to everyone with this result."

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Eosinophilia CLINIC GUIDE

A raised eosinophil count in adults 18+. Not for children, or known HES, EGPA or eosinophilic leukaemia under specialist care.

BSH eosinophilia guideline (Butt et al., 2017)
BIA eosinophilia in travellers and migrants (2024)
UKHSA migrant health guide · NICE **NG12** · MHRA
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/eosinophilia.html

PAGE 1 · INTERPRET

is there organ damage? → is it real? → how high, how long? → find the cause

! ACT TODAY — RED FLAGS

BSH 2017 · NG12

TIER	PICTURE WITH EOSINOPHILIA (ANY COUNT)	ACTION
999	Haemodynamic instability, severe chest pain, syncope, acute heart failure or respiratory compromise	Emergency transfer
Same day	**Heart**: chest pain, breathlessness, palpitations, thromboembolism → ECG, troponin, echo · **Nerves**: new neuropathy, mononeuritis multiplex, focal deficit, confusion, seizures · **Lungs**: hypoxia, infiltrates, haemoptysis · **DRESS**: new drug + fever + rash or facial swelling + organ involvement · **Strongyloides** risk + immunosuppressed + sepsis · blasts on the film	Same-day hospital; stop a suspected culprit drug. Do not just send an outpatient ANCA
Suspected cancer pathway	Unexplained lymphadenopathy or splenomegaly, especially with fever, night sweats, weight loss or itch (lymphoma)	Consider NG12 referral; urgent haematology advice

The count does not predict organ damage: 2.0 × 10⁹/L with cardiac or neurological involvement is more urgent than a well patient at 8.0. Organ damage → urgent assessment without waiting for duration criteria.

1 CONFIRM IT IS REAL

BSH 2017

Repeat	An unexpected result: repeat FBC with differential **and a blood film** before further tests
Film	Confirms the count and morphology; excludes blasts
Previous counts	Long-standing, stable, mild = usually atopy
Drug timeline	Anything started in the previous 8 weeks, including OTC NSAIDs and herbal products
Steroids	Lower the count and can hide the cause — and are dangerous in undiagnosed Strongyloides

Hypereosinophilia = ≥1.5 × 10⁹/L on two occasions, usually at least 4 weeks apart. Persistent ≥1.5 needs investigation (BSH).

2 SEVERITY BANDS

descriptive only · ×10⁹/L

COUNT	MEANS	ACTION · TIMING
0.5–1.5 mild, well	Atopy, drugs, parasites	History, drug review; repeat guided by context
Mild + travel or migration	Helminth, especially Strongyloides	Strongyloides serology; stool via microbiology
1.5–5.0 moderate	Helminths, EGPA, drugs, lymphoma, Addison's	Initial tests now; repeat to confirm persistence
Persistent **≥1.5**, unexplained	Possible clonal disease or HES	Urgent haematology advice; ECG, troponin, echo
>5.0 severe	HES, eosinophilic leukaemia, T-cell lymphoma, heavy parasite load	Urgent haematology; cardiac assessment
Any count + organ damage	Eosinophil-mediated injury	Same day

At any count, an abnormal film, blasts, cytopenias, splenomegaly or constitutional symptoms → urgent haematology advice. Repeat testing is guided by the count, persistence and symptoms — not automatically.

3 FOCUSED HISTORY — WHAT DISCRIMINATES

BSH 2017 · BIA 2024

WHERE HAVE YOU LIVED?
Country of birth, migration, travel, barefoot soil or freshwater contact — even decades ago.

WHAT ARE YOU TAKING?
New prescription, OTC NSAID, herbal or supplement; match the drug timeline to the count.

ALLERGY HISTORY
Asthma (especially new or difficult), eczema, rhinitis, nasal polyps, food allergy.

ORGAN SYMPTOMS
Chest pain, breathlessness, numbness, rash, diarrhoea, weight loss, sweats, nodes.

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Asthma, eczema, hay fever	**Atopy** (commonest)	Clinical; IgE if unclear
New drug: allopurinol, antiepileptic, antibiotic, NSAID	**Drug reaction** or DRESS	Stop if safe; LFTs, U&E
Born or lived in the tropics, barefoot soil	**Strongyloides**	Strongyloides IgG
Freshwater exposure in Africa	**Schistosomiasis**	Serology; urine, stool
Dog contact, pica	**Toxocara**	Toxocara serology
Late asthma, polyps, neuropathy, purpura	**EGPA**	ANCA (often negative), urine, CXR
Nodes, sweats, weight loss, itch	**Hodgkin or T-cell lymphoma**	Film, LDH; NG12 route
Splenomegaly, cytopenias, odd film	**Clonal: HES, CEL**	Haematology: FIP1L1-PDGFRA
Cough, wheeze, fleeting CXR shadows	**Löffler syndrome** (larval migration)	CXR; stool may be negative early
Tired, low BP, low sodium	**Addison's**	9am cortisol

5 WORK-UP CHECKLIST

BSH initial tests · targeted extras

- ☐ **Repeat FBC + film**
- ☐ **Bone profile**, LDH
- ☐ **Vitamin B12** (raised in clonal disease)
- ☐ **Strongyloides IgG** if exposure
- ☐ ANCA only if EGPA suspected
- ☐ **U&E**, eGFR, LFTs
- ☐ **ESR and/or CRP**
- ☐ Medication and travel review
- ☐ Urine dip; CXR if chest symptoms
- ☐ ECG, troponin, echo if persistent ≥1.5 or cardiac symptoms

Stool microscopy has low sensitivity for Strongyloides — serology is the key test. Bone marrow and molecular tests are haematology-led.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

BSH 2017 · BIA 2024 · MHRA · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Atopy	Optimise treatment: inhaled steroid for asthma, intranasal steroid for rhinitis, non-sedating antihistamine, emollients and topical steroid for eczema	The count may stay up despite good control; persistent or rising → look for another cause
Drug-induced	Stop the suspected drug if clinically safe; specialist advice if essential. Document the reaction; Yellow Card	Recheck FBC after stopping; not settling → other cause. **Never rechallenge** after DRESS, SJS/TEN, hepatitis or nephritis
Strongyloides	Ivermectin, regimen per tropical medicine or local protocol. Immunosuppressed or needs immunosuppression → discuss with ID/tropical medicine urgently	Follow with the eosinophil count, not serology (stays positive for months)
Other helminths	Treat per microbiology or tropical medicine advice, guided by exposure and serology	Recheck FBC at about 3 months: should normalise
EGPA, HES, clonal	**Specialist-led**: urgent rheumatology (EGPA) or haematology (HES, FIP1L1-PDGFRA is imatinib-sensitive)	Shared monitoring per specialist plan; steroid safety checks

7 BEFORE ANY IMMUNOSUPPRESSION — STRONGYLOIDES FIRST

BIA 2024 · UKHSA migrant health guide

WHO
Born in, or lived in, an endemic area (South and South-East Asia, sub-Saharan Africa, South America, parts of southern Europe). Infection can persist for decades.

WHEN
Before high-dose or prolonged steroids, rituximab, anti-TNF and other immunosuppressants — whatever the eosinophil count.

TEST
Strongyloides IgG serology. Positive → ID/tropical medicine and treat **before** immunosuppression starts.

CAN'T WAIT?
Empirical treatment only on specialist advice, when urgent immunosuppression cannot wait or testing is unavailable.

Steroids in undiagnosed Strongyloides can trigger hyperinfection with Gram-negative sepsis, which is often fatal.

8 ADVICE AND THE CONVERSATION

plain English

- Travel: wear shoes on soil and sand, safe water and food, avoid freshwater swimming in endemic areas
- Pets: regular worming; wash hands after soil or pet contact
- Report rash or fever after starting a new medicine

"Eosinophils are a type of white blood cell that go up with allergies, some medicines and certain infections picked up abroad. Yours are a little high. We'll look at your medicines and travel, repeat the test, and check for a common hidden worm infection before any steroid treatment."

9 DRESS — RECOGNISE AND ACT

MHRA · BSH

Drug started usually **2–8 weeks** before · fever · widespread rash or facial swelling · eosinophilia or atypical lymphocytes · lymphadenopathy · liver, kidney, lung or heart involvement.

Act: stop the culprit, same-day hospital or dermatology; steroids are specialist-led. Record as a severe allergy; never rechallenge. Common culprits: allopurinol, carbamazepine, lamotrigine, sulfonamides, vancomycin, abacavir. MHRA: HLA-B*15:02 testing before carbamazepine in Han Chinese or Thai ancestry.

10 FOLLOW-UP AND MONITORING

BSH 2017 · BIA 2024

- ☐ Recheck after each intervention: drug stopped, parasite treated, atopy optimised
- ☐ After parasite treatment: FBC at about **3 months**
- ☐ Persistent eosinophilia → ID or haematology review
- ☐ Atopy: no routine annual FBC for a past mild rise
- ☐ Drug reaction recorded as an allergy
- ☐ Return the **same day**: chest pain, breathlessness, fainting, new numbness or weakness, rash with fever

11 WHEN AND WHERE TO REFER

BSH 2017 · NG12

999	Unstable, acute heart failure, respiratory compromise
Same day	Cardiac, neurological or lung involvement; DRESS; Strongyloides sepsis
Suspected cancer	Unexplained lymphadenopathy or splenomegaly (NG12)
Urgent haematology	Persistent unexplained ≥1.5; abnormal film, cytopenias, splenomegaly, constitutional symptoms; rising count
Same-week rheumatology	Suspected EGPA — ANCA positivity not required
ID / tropical medicine	Helminth suspected or confirmed; positive pre-immunosuppression screen
GP manages	Well, mild, clear atopy, normal examination; uncomplicated drug cause

12 PITFALLS AND NEVER-EVENTS

BSH · BIA · MHRA

- **Never** start steroids in an at-risk patient without thinking Strongyloides.
- **Never** use a negative ANCA to exclude EGPA.
- Do not grade urgency by the count alone.
- Do not blame atopy for a rising or persistent count.
- Allopurinol + rash + fever = DRESS until proved otherwise.
- Steroids lower the count and can mask the cause.
- Negative stool tests do not exclude Strongyloides.

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Thrombocytosis: the high platelet count

CLINIC GUIDE

Platelets above the laboratory range (usually >450 × 10^9/L) in adults 18+. Not for children, pregnancy, or known ET, PV or myelofibrosis already under haematology (follow the specialist plan).

NICE **NG12** (suspected cancer, updated 2026)
BSH thrombocytosis 2010 · BSH mutation-negative thrombocytosis 2025
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/thrombocytosis.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → reactive or clonal?

! ACT TODAY — RED FLAGS FIRST

NICE NG12 · NG128 · NG158 · BSH

TIER	FEATURE	ACTION
999	Stroke or ongoing neurological deficit · acute coronary syndrome · suspected PE · splanchnic or cerebral venous thrombosis · digital ischaemia (cold, pale or blue digit) · major bleeding · sudden vision loss	Emergency pathway, whatever the count
Same day	Platelets **≥1000 × 10^9/L** with bleeding, thrombosis, neurological or visual symptoms · bleeding with very high platelets (acquired von Willebrand) · new or rapidly enlarging spleen · film suggesting leukaemia, CML or myelofibrosis · new or severe erythromelalgia · suspected DVT (NG158)	Same-day assessment and haematology discussion. Resolved TIA → specialist assessment **within 24 hours** (NG128)
Cancer routes (NG12)	Aged **40+** with thrombocytosis → consider urgent direct-access **chest X-ray** · women **55+** with thrombocytosis and visible haematuria or unexplained vaginal discharge → consider urgent direct-access **ultrasound** · **55+** with thrombocytosis and nausea, vomiting, weight loss, reflux, dyspepsia or upper abdominal pain → consider non-urgent direct-access **OGD**	Dysphagia, or 55+ with weight loss and upper GI symptoms → suspected cancer pathway referral

A count of 1000 or more without symptoms and with a clear reactive cause needs prompt review and specialist advice, not automatic admission. The count correlates poorly with thrombotic risk (BSH).

1 CONFIRM IT IS REAL AND PERSISTENT

BSH 2025

- **Look back** at previous FBCs: a single high count does not establish persistence.
- **Spurious high count**: red-cell fragments or very small red cells can be counted as platelets by the analyser. Ask for a **blood film** if the count does not fit.
- **Transient causes**: recent surgery, trauma, bleeding or infection. Treat the cause, then repeat the FBC.
- **Repeat** at a clinically judged interval — sooner with symptoms, a rising count, thrombosis, bleeding or an abnormal film.
- **Persistent** = raised for at least **3 months** (BSH 2025); longer observation may suit mild or variable counts.

2 SEVERITY BANDS

count + context → action · BSH · NG12

RESULT	MEANS	ACTION · TIMING
>450, single, clear cause	Usually reactive	Treat the cause; repeat FBC once settled
>450, aged 40+	NG12 lung route	Consider urgent CXR now
>450 for **3 months**, unexplained	Possible MPN (ET, PV, CML, MF)	Film, JAK2 per local pathway; **urgent haematology**
Any raised count + spleen, thrombosis, B symptoms or odd film	Clonal until proven otherwise	**Urgent** haematology; same day if acute
≥1000, well, clear cause	Bleeding risk (acquired VWS)	Prompt review; haematology advice
≥1000 + symptoms	Bleeding or thrombosis risk	**Same-day** haematology

No count separates reactive from clonal: reactive counts can exceed 1000, and ET can sit just above 450. Do not wait for 600 or 700 before testing.

3 EXAMINE

spleen · extremities · nodes

Spleen, liver	Splenomegaly → MPN or CML; usually absent in reactive
Hands and feet	Erythromelalgia, digital ischaemia → ET or PV
Face, palms	Plethora → PV (use the polycythaemia pathway)
Nodes, chest, abdomen	Lymphoma, lung or GI cancer, infection
Neurology, legs	Focal deficit, visual field loss (stroke, TIA); calf swelling (DVT)
Abdomen	Ascites with a big spleen → Budd–Chiari (hepatic vein clot)

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Fever, cough, UTI symptoms	**Infection**	CRP; CXR, urine culture
Joint, bowel or vasculitic disease	**Inflammation** (RA, IBD)	CRP, ESR
Low MCV, heavy periods, GI loss	**Iron deficiency**, bleeding	Ferritin ± TSAT; find the loss
Smoker, weight loss, age 40+	**Solid cancer** (lung, GI, endometrial)	NG12 routes above
Recent operation or trauma	**Post-operative**	Repeat FBC when recovered
Splenectomy, coeliac, sickle cell	**Hyposplenism**	Film (Howell–Jolly bodies)
Itch after bathing, plethora, high Hb	**PV**	Hb, haematocrit; JAK2
High WCC, basophils, big spleen	**CML**	Film; BCR-ABL1 via haematology
Persistent, unexplained, any age	**ET**	Film; JAK2 per local pathway

Iron deficiency can coexist with ET; normal CRP does not exclude a reactive cause or an MPN.

5 WORK-UP CHECKLIST

BSH · NG12

- ☐ **Repeat FBC** and look back
- ☐ **Ferritin** ± transferrin saturation
- ☐ U&E, LFTs: baseline
- ☐ Urine dip; culture if UTI
- ☐ FIT or OGD if GI symptoms or IDA
- ☐ **Blood film**: persistent or odd
- ☐ **CRP, ESR**: reactive clue
- ☐ **CXR** if aged 40+ (NG12)
- ☐ **JAK2 V617F** if persistent and unexplained
- ☐ LDH, urate if MPN suspected

RESULT	MEANS
Film: left shift, basophilia	CML → urgent haematology
Film: tear-drop cells, leucoerythroblastic	Myelofibrosis → urgent haematology
JAK2 positive	Supports an MPN (not diagnostic of ET) → refer
JAK2 negative, still persistent	Still refer: some ET is triple-negative

JAK2-negative does not exclude ET: CALR, MPL and bone marrow are specialist-directed. No marrow in primary care.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

BSH · BSG 2021 · NICE CKS · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Infection, inflammation	Treat the cause. Antibiotics only for documented or strongly suspected bacterial infection	Repeat FBC once settled; refer if still raised and unexplained
Iron deficiency replacement	**Replacement:** one oral iron tablet **once daily**, e.g. ferrous sulfate 200 mg; **alternate days** if not tolerated (BSG 2021). Investigate the source of loss	Recheck FBC and iron studies after treatment — platelet normalisation is variable. Do not delay iron to "see if platelets settle"
After surgery, trauma or bleeding	No treatment for the count; treat any bleeding cause and iron deficiency	Repeat FBC after recovery; refer if still raised and unexplained
Suspected cancer	NG12 route: CXR, pelvic ultrasound, OGD or suspected cancer pathway referral	Track the result; a normal test does not end the search if still unexplained
Suspected or confirmed ET	**Haematology-led.** Aspirin only if advised (caution with platelets >1000, acquired VWS, bleeding or anticoagulation). Cytoreduction (hydroxycarbamide; interferon in younger patients or pregnancy planning; anagrelide second line) is started by haematology	GP prescribing only under a written shared-care protocol; follow its FBC, renal and liver monitoring
PV, CML, myelofibrosis	Refer; treatment is specialist-led (PV: use the polycythaemia pathway)	Cardiovascular risk managed through standard pathways

Reactive thrombocytosis: no aspirin for the count itself, no cytoreduction and no NSAIDs because the platelets are raised.

7 ET OR PV — CUT THE CLOT RISK

BSH · NICE CVD pathways

STOP SMOKING
Smoking adds to arterial clot risk. Offer behavioural support with drug treatment or a nicotine vape, and the stop-smoking service (NG209).

BP, LIPIDS, DIABETES
Manage through the usual NICE pathways (NG136, NG238, NG28). Thrombocytosis does not change the usual targets.

FLUIDS AND MOVING
Avoid dehydration, especially in heat, illness or flying. Keep mobile on long journeys; travel prophylaxis is decided on VTE risk, not the count.

ALCOHOL, SURGERY
No more than 14 units a week, spread out. Tell surgeons and anaesthetists: the peri-operative plan is agreed with haematology.

8 WHEN AND WHERE TO REFER

NG12 · BSH 2025 · NG128 · NG158

999
Stroke, ACS, suspected PE, splanchnic or cerebral vein thrombosis, digital ischaemia, major bleeding, collapse.

SAME-DAY HAEMATOLOGY
Symptomatic count ≥1000, bleeding with extreme count, new or enlarging spleen, film suggesting leukaemia, CML or myelofibrosis, recent thrombosis.

URGENT HAEMATOLOGY
Persistent unexplained count at any age (even JAK2-negative or below 700) · positive JAK2, CALR, MPL or BCR-ABL1 · spleen, B symptoms, high WCC or Hb · unusual-site clot.

PRIMARY CARE
Clear reactive cause settling with treatment. Refer if it becomes unexplained, disproportionate, progressive or other counts change.

Other specialties for the reactive cause: respiratory (mass on chest X-ray), gastroenterology (IBD, upper GI symptoms), rheumatology (RA, vasculitis). Use the local haematology referral pathway and include previous counts, film, ferritin, CRP and any JAK2 result.

9 FOLLOW-UP AND MONITORING

BSH · shared care

- [ ] **Repeat FBC** once the presumed cause is treated: target back within the laboratory range
- [ ] Iron deficiency: FBC and iron studies after replacement
- [ ] Still raised and unexplained at **3 months** → JAK2 per local pathway or urgent haematology
- [ ] NG12 test results (CXR, ultrasound, OGD) reviewed and actioned
- [ ] On cytoreduction: bloods at the intervals in the shared-care protocol; never stop or change the dose without specialist advice
- [ ] ET or PV: annual review of clots, bleeding, symptoms and cardiovascular risk
- [ ] Very high or rapidly rising count on monitoring → **same-day** haematology discussion
- [ ] **Re-refer**: count rising or a clot on treatment, intolerance of treatment, pregnancy planned
- [ ] Discharge from monitoring once the count is normal and the cause resolved
- [ ] Diagnosis and plan recorded in the problem list

10 PITFALLS AND NEVER-EVENTS

BSH · NG12

- **Never** start aspirin or cytoreduction for suspected ET in primary care, or advise self-starting aspirin.
- **Never** wait for a count of 600 or 700 before testing or referring persistent unexplained thrombocytosis.
- **Never** forget the NG12 routes: a raised count at 40+ is a reason to consider a chest X-ray.
- **Never** take normal CRP or ESR as proof of ET, or a negative JAK2 as proof it is reactive.
- Thrombosis at an **unusual site** (portal, splenic, hepatic or cerebral vein) may be the first sign of an MPN.
- Known ET with a **falling** count and a growing spleen may be progression to myelofibrosis: tell haematology.
- Ferritin can be normal or high in inflammation: check transferrin saturation before ruling out iron deficiency.
- Pregnancy with ET or PV → early haematology and obstetric referral; do not stop cytoreduction or start aspirin or LMWH yourself.

11 SAFETY-NET

tell the patient

999: face, arm or speech symptoms (even if they settle), chest pain, sudden breathlessness, sudden vision loss, a cold or blue finger or toe, heavy bleeding, severe abdominal pain with swelling, confusion or a fit.
Same day: one-sided leg swelling, burning red hands or feet, unusual bleeding or bruising, severe headache.

12 SAY IT LIKE THIS

plain English

"Your platelets, the cells that help blood clot, are a bit high. Most often that is the body reacting to an infection, inflammation or low iron, and it settles once we treat the cause. I'll recheck the blood test, and if it stays high without a reason we'll ask a blood specialist to look at it."

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Low platelets in adults CLINIC GUIDE

Interpreting a platelet count below 150 × 10⁹/L in adults 18+. Not for children and young people (paediatric pathway), patients on chemotherapy (oncology hotline) or inpatient HIT management.

BSH TTP 2023 · BSH HIT 2023 · ITP consensus 2019
NICE **NG24** · **NG12** · NG240 · Green Book ch. 7
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/thrombocytopenia.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → how low, how fast → why → work it up

! ACT TODAY

BSH 2023 · NG24 · NG240 · NG12

TIER	RESULT + PICTURE	ACTION
999	Any low count with **major bleeding**, severe headache, confusion, focal signs or seizure (intracranial bleed, TTP) · **non-blanching rash with fever** or unwell (meningococcal, sepsis, DIC) · bleeding from several sites	Emergency transfer
999	Low platelets + **schistocytes**, raised LDH, low haptoglobin (**TTP**/TMA) · pregnancy with hypertension, headache, visual change or epigastric pain (HELLP)	Immediate TTP or obstetric emergency care; do not wait for ADAMTS13
Same day	Platelets **<20 × 10⁹/L** or falling rapidly, even without bleeding (especially <10) · blasts or pancytopenia · mucosal bleeding or widespread petechiae	Same-day haematology assessment
Same day	Platelet fall on **heparin** (suspected **HIT**), or after quinine, antibiotics or a new drug	Stop all heparin; 4Ts score; haematology today; non-heparin anticoagulant per specialist

1 IS IT REAL?

repeat FBC + film

Unexpected low count, well, no bleeding	EDTA can make platelets **clump** (pseudothrombocytopenia): repeat FBC with a film; if clumps, send a **citrate** sample as the laboratory advises
Clotted or difficult sample	Falsely low: repeat
Recent viral illness	Transient falls are common: repeat once recovered
Compare	Previous counts: new, falling or long-standing and stable?

Assume artefact only in a well patient without bleeding, other cytopenias or film abnormalities. Never delay an urgent referral to exclude clumping.

2 SEVERITY BANDS

platelets × 10⁹/L

COUNT	MEANS	ACTION AND TIMING
100–150	Mild	Isolated, stable, well: review drugs, alcohol, liver, infection; repeat FBC and film; safety-net
50–100	Moderate	Investigate; discuss or refer per local haematology pathway, especially if new, falling or symptomatic
<50	Severe	Unexplained → same-day haematology discussion
<20	Very severe	Same-day haematology assessment, even if no bleeding
<10	Critical	Same-day assessment; specialist decides on treatment
Any count + bleeding, schistocytes, blasts	Count does not set the risk	999 or same day as above

The rate of fall, other cell lines, clotting, anticoagulants and planned procedures matter as much as the number.

3 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Isolated, well, large platelets on film	**ITP** (exclusion)	HIV, hepatitis B/C; film
New drug, heparin	**Drug-induced, HIT**	Drug history; 4Ts
Alcohol, spleen, stigmata of liver disease	**Liver disease, hypersplenism**	LFTs, albumin, PT; ultrasound
Recent viral illness	**Post-viral**	Repeat when recovered
Late pregnancy, mild, isolated	**Gestational**	BP, urine, LFTs
Rash, joints, mouth ulcers	**SLE**, antiphospholipid	ANA
Other cytopenias, blasts, nodes	**Marrow disease**, lymphoma	Film; same-day haematology
Fever, hypotension, oozing	**Sepsis, DIC**	999; clotting, fibrinogen

4 WORK-UP CHECKLIST

send and why

- ☐ **Repeat FBC + film**: clumps, schistocytes, blasts
- ☐ **LFTs**, albumin: liver disease
- ☐ **HIV, hepatitis B and C** if ITP suspected
- ☐ ANA if features of SLE
- ☐ **Clotting, fibrinogen** if bleeding, liver disease or DIC possible
- ☐ U&E; **LDH**, reticulocytes if haemolysis possible
- ☐ B12, folate
- ☐ Immunoglobulins if suspected ITP

Examine skin and mouth for petechiae and purpura, nodes, liver and spleen, BP and fundi if severe. No routine antiplatelet antibody tests; no marrow in primary care.

5 WHAT THE FILM TELLS YOU

laboratory film review

FILM FINDING	THINK	FILM FINDING	THINK
Platelet clumps	Pseudothrombocytopenia: citrate sample	**Schistocytes**	TTP, HUS or DIC: emergency
Large platelets, otherwise normal	Consistent with ITP	**Blasts or dysplasia**	Leukaemia or MDS: same day

6 READING THE RESULT IN FOUR STEPS

BSH · NG24 · NG12

1 · BLEEDING OR UNWELL?
Major bleeding, neurology, fever with a rash, schistocytes or pregnancy symptoms → 999.

2 · REAL?
Repeat FBC with a film; clumps → citrate sample. Never let this delay an urgent referral.

3 · HOW LOW, HOW FAST?
<20 or falling fast → same day · unexplained <50 → same-day discussion · <100 → discuss or refer.

4 · ISOLATED OR NOT?
Other cytopenias, nodes or spleen point away from ITP towards marrow, liver or lymphoma.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 MANAGEMENT BY CAUSE

BSH 2023 · ITP consensus · NG24 · Green Book

CAUSE	FIRST LINE	WHO LEADS, AND WHAT NEXT
ITP	Observe or treat is a **haematology decision**: corticosteroids or IVIg (IVIg when a fast rise is needed); later TPO receptor agonists, rituximab or splenectomy	Haematology-led. GP: avoid NSAIDs and aspirin, manage bleeding emergencies, prescribe only under shared care. **No routine prophylactic platelet transfusion**
Suspected HIT	**Stop all heparin**, including flushes; 4Ts score; non-heparin anticoagulant (e.g. argatroban, fondaparinux, a DOAC in selected patients) per specialist	Same-day haematology. Do not wait for lab confirmation if probability is intermediate or high. HIT is prothrombotic: never just stop and leave
Other drug-induced	Stop the likely culprit if safe; consider alternatives for essential drugs	Repeat FBC per advice; still falling → same day. Record the reaction; Yellow Card
Liver disease, alcohol	Reduce or stop alcohol; assess cirrhosis and portal hypertension	Hepatology per liver pathway
Splenectomy planned	Vaccinate at least **2 weeks** before: pneumococcal, MenACWY, MenB; annual influenza	Green Book chapters 7 and 25; lifelong infection advice

Platelet transfusion (NICE NG24): prophylaxis below 10 is not routine in ITP, HIT, TTP or chronic marrow failure; procedures usually need above 50, and above 100 for brain or posterior eye surgery — specialist decision.

8 SELF-CARE AND THE CONVERSATION

BSH · NG24

Use paracetamol, not ibuprofen or aspirin, unless haematology advises · electric razor, soft toothbrush · tell your dentist · nosebleed: lean forward and pinch the soft part for 10–15 minutes · do not stop blood thinners without advice.

"Platelets are the tiny cells that help your blood clot. Yours are lower than normal. Often that's from a virus, a medicine or the immune system, and it can settle by itself, but we need to find the reason. Any bleeding that won't stop, a bad headache, or a rash with a temperature — call 999."

9 FOLLOW-UP CHECKLIST

local pathway · haematology plan

- ☐ **Repeat FBC + film** per local pathway; plot the **trend**
- ☐ **Mild, stable, isolated**: recheck; refer if persistent, falling or <100
- ☐ **Post-viral**: should normalise; persistent → discuss
- ☐ **Drug stopped**: count should rise; still falling → same day
- ☐ **Known ITP**: FBC at intervals set by haematology; falling or bleeding → re-refer
- ☐ Steroids: glucose, mood; bone protection on long courses
- ☐ Splenectomy: vaccines up to date; standby antibiotics as advised
- ☐ Before any procedure, agree a platelet threshold with haematology

10 WHEN AND WHERE TO REFER

NG12 · local pathway

999	Major bleeding, suspected intracranial bleed, TTP, DIC, HIT with thrombosis, HELLP, non-blanching rash with fever
Same-day haematology	Platelets <20 or falling fast; significant unexplained <50; schistocytes or blasts; pancytopenia; suspected HIT or drug-induced immune thrombocytopenia
Discuss or refer	<100, especially if new, falling or symptomatic; splenomegaly; abnormal film
Suspected cancer pathway	Unexplained lymphadenopathy or splenomegaly (NG12)
Routine haematology	Suspected or known ITP; persistent unexplained thrombocytopenia
Obstetrics	New, falling or moderate thrombocytopenia in pregnancy

11 PITFALLS AND NEVER-EVENTS

BSH 2023 · NG24

- **Never** continue any heparin, including flushes, when HIT is suspected.
- **Never** wait for the full pentad or ADAMTS13 to act on suspected TTP.
- **Never** stop anticoagulants or antiplatelets without discussing the clot risk.
- Do not call it ITP without a film and a search for secondary causes.
- Low platelets alone in pregnancy is not HELLP — but check BP, urine and LFTs.
- Splenomegaly argues against simple ITP.

12 SPECIAL GROUPS

BSH 2023 · NG24 · NG133

PREGNANCY
Mild, isolated, late and well: likely gestational. New, falling or moderate → obstetric and haematology assessment. Hypertension or symptoms → emergency.

ON ANTICOAGULANTS
Do not stop automatically: discuss the same day, weighing bleeding against clot risk (valves, recent VTE, AF).

BEFORE PROCEDURES
Dental work, injections, surgery: agree a platelet threshold with haematology; usually above 50 for invasive procedures (NG24).

HEAVY PERIODS
Tranexamic acid or hormonal options with haematology advice; avoid tranexamic acid in TTP, HIT or DIC. Iron if anaemic.

13 WHO NEEDS AN FBC FAST

NICE NG12

ADULTS: VERY URGENT FBC
Within **48 hours** for unexplained bruising, bleeding or petechiae, pallor, persistent fatigue, fever, recurrent infection, generalised lymphadenopathy or hepatosplenomegaly.

COUNT SAYS LEUKAEMIA
FBC or film reported as acute leukaemia → immediate specialist assessment (standard practice, not an NG12 criterion).

CHILDREN AND YOUNG PEOPLE
Under 25 with unexplained petechiae or hepatosplenomegaly → immediate specialist assessment.

NON-BLANCHING RASH
With fever or unwell, at any age → 999 (meningococcal disease, NG240).

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Pancytopenia in adults

CLINIC GUIDE

Low Hb, low white cells and low platelets together, in adults 18+. Not for children and young people (paediatric pathway), patients on chemotherapy (use the oncology hotline) or established aplastic anaemia or MDS care.

NICE **NG12** · CG151 · **NG239** (2024)
BSH aplastic anaemia 2024 · MHRA
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/pancytopenia.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → marrow, nutrition, drugs or spleen

! ACT TODAY — NEVER A BENIGN FINDING

NG12 · CG151 · BSH

TIER	RESULT + PICTURE	ACTION
999	Neutrophils **≤0.5 × 10⁹/L** with fever, rigors or feeling unwell (neutropenic sepsis) · major bleeding	Emergency admission for immediate IV antibiotics; do not wait for cultures
Same day	**Blasts** or a film or count reported as **acute leukaemia** · platelets **<20 × 10⁹/L** or any bleeding, petechiae, gum bleeding · rapidly falling counts	Immediate (same-day) haematology assessment
Same day	On **methotrexate, carbimazole, clozapine** or another marrow-toxic drug	Stop the drug (clozapine: per the monitoring service), urgent FBC; same-day specialist advice
Suspected cancer pathway	Pancytopenia with unexplained **lymphadenopathy or splenomegaly**, especially with night sweats, fever or weight loss	Consider NG12 lymphoma referral, plus urgent haematology for the count

1 IS IT REAL?

repeat within days if well

Low platelets only on analyser	EDTA clumping gives false thrombocytopenia: check the film or repeat in a citrate tube
Clotted or delayed sample	Can lower platelets and distort the count: repeat
Recent viral illness	Transient falls are common: repeat FBC and film once well
Low neutrophils, Black or Middle Eastern heritage	Constitutionally lower neutrophils are common, but explain only an isolated mild neutropenia, not pancytopenia
Alcohol, liver disease	Direct marrow toxicity and splenic pooling: still investigate

A blood film reviewed by the laboratory is mandatory: the analyser cannot see blasts, dysplasia or hypersegmented neutrophils. Refer in parallel — do not wait for every result.

2 SEVERITY BANDS

per cell line · NG12 · CG151 · BSH

RESULT	MEANS	ACTION AND TIMING
Neutrophils **≤0.5** + fever	Neutropenic sepsis until proved otherwise	**999**
Platelets **<20**, or any bleeding	Spontaneous bleeding risk	Same-day haematology
Blasts on film	Acute leukaemia	Same-day haematology
Two or more lines falling on serial FBCs	Progressive marrow problem	Urgent haematology
Unexplained stable pancytopenia, normal film	Cause not yet found	Urgent haematology referral while first-line bloods run
Mild, macrocytic, low B12 or folate, no blasts or nodes	Megaloblastic pancytopenia	Replace; recheck FBC; refer if not recovering

Neutrophils and platelets × 10⁹/L. Severe aplastic anaemia is defined by haematology (Camitta criteria) — any neutrophils <0.5 or platelets <20 needs same-day discussion.

3 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
High MCV, hypersegmented neutrophils, neuropathy	**B12 or folate**	B12, folate, anti-IF antibody
Blasts, fast onset, bone pain	**Acute leukaemia**	Film; same-day haematology
Older, macrocytic, dysplasia	**MDS**	Film; urgent haematology
Nodes, spleen, B symptoms	**Lymphoma, CLL, myeloma**	LDH; NG12 pathway
New drug	**Drug-induced**	Drug history; stop it
Hypocellular picture, young, low retics	**Aplastic anaemia**	Retics; urgent haematology
Big spleen, cirrhosis	**Hypersplenism**	LFTs; ultrasound
Viral illness, risk factors, rash, joints	**HIV, hepatitis, EBV, SLE**	Serology; ANA

4 WORK-UP CHECKLIST

send in parallel with referral

- ☐ **FBC + differential**, reticulocytes
- ☐ **B12 + folate**; ferritin
- ☐ U&E, calcium, TFTs
- ☐ **HIV**, hepatitis B and C, EBV, CMV
- ☐ Blood cultures if febrile (hospital)
- ☐ **Blood film** (laboratory review)
- ☐ LFTs, bilirubin, **LDH**, urate
- ☐ CRP, ESR
- ☐ DAT if haemolysis possible
- ☐ Coagulation if bleeding

Ask about onset, bleeding, infections, B symptoms, drugs, alcohol and travel. Examine for pallor, petechiae, nodes, liver, spleen, mouth ulcers and jaundice. Marrow biopsy is a hospital test.

5 WHAT THE FILM TELLS YOU

laboratory film review

FILM FINDING	THINK	FILM FINDING	THINK
Blasts	Acute leukaemia: same day	**Tear-drop cells**	Myelofibrosis, marrow infiltration
Hypersegmented neutrophils	B12 or folate deficiency	**Smear cells**	CLL
Dysplastic cells	MDS	**Spherocytes**	Haemolysis
Schistocytes	TTP or DIC: same day	**Parasites**	Malaria (travel): same day

6 READING THE RESULT IN FOUR STEPS

NG12 · CG151

1 · SEPTIC OR BLEEDING?
Fever with low neutrophils → 999. Bleeding or platelets <20 → same day.

2 · WHAT DOES THE FILM SHOW?
Blasts or schistocytes → same day. Dysplasia → urgent haematology.

3 · SIMPLE CAUSE?
B12, folate, drugs, alcohol, viral illness. Only a clear cause with a reassuring film stays GP-led.

4 · NODES OR SPLEEN?
Examine. Unexplained lymphadenopathy or splenomegaly → suspected cancer pathway referral.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 MANAGEMENT OF THE COMMONEST CAUSES

NG239 · CKS · BSH · MHRA · SPC

CAUSE	FIRST LINE	THEN
B12 deficiency	**Hydroxocobalamin 1 mg IM three times a week for 2 weeks**, then every 2–3 months; neurological features: **1 mg on alternate days until no further improvement**, then every 2 months, with specialist advice	Recheck FBC; all three lines should recover. No recovery → haematology
Folate deficiency	**Folic acid 5 mg daily**, usually for 4 months — only once B12 is normal or being replaced	Find the cause: diet, alcohol, coeliac
Drug-induced	**Stop the drug now.** Carbimazole: same-day advice; do not switch to propylthiouracil. Clozapine: act today with the monitoring service and mental health team (a red result means stop). Methotrexate: folinic acid rescue is specialist-led	Serial FBC with haematology; record the reaction; Yellow Card
Alcohol	Stop alcohol; replace B12 and folate if low; thiamine where indicated	Recheck FBC; refer if not recovering or spleen enlarged
Marrow failure, leukaemia, MDS	Haematology-led: transfusion, growth factors, immunosuppression, chemotherapy or transplant	GP: sepsis safety-net, vaccines, shared-care drugs

8 FOLLOW-UP AND MONITORING CHECKLIST

BSH · CG151

- ☐ **Film result** read and actioned the same day
- ☐ **Referral sent in parallel**; result chased; patient knows the plan
- ☐ B12 or folate cause: **recheck FBC**; target all three lines normal; no recovery → refer
- ☐ Drug stopped: serial FBC until recovery; allergy or adverse reaction recorded
- ☐ **Neutropenic patients**: written safety-net and 24-hour haematology number recorded
- ☐ Under haematology: shared-care drug monitoring per protocol
- ☐ **No live vaccines** when immunocompromised; inactivated vaccines are fine
- ☐ Mood, fatigue and fertility questions addressed

9 WHEN AND WHERE TO REFER

NG12 · CG151

999	Neutropenic sepsis; major bleeding; confusion or focal neurology with low platelets
Same-day haematology	Blasts or count suggesting acute leukaemia; platelets <20 or bleeding; rapidly falling counts; drug-induced with neutropenia
Suspected cancer pathway	Unexplained lymphadenopathy or splenomegaly (NG12)
Urgent haematology	Any unexplained pancytopenia · dysplasia or unexplained macrocytosis on film · suspected MDS · progressive fall · splenomegaly without malignancy features
GP-led	Clear B12 or folate cause, normal film apart from megaloblastic change, no nodes or spleen — with a recheck

10 WHO NEEDS AN FBC — AND WHO NEEDS IT FAST

NICE NG12

ADULTS: VERY URGENT FBC
Within **48 hours** for pallor, persistent fatigue, unexplained fever, persistent or recurrent infection, generalised lymphadenopathy, unexplained bruising, bleeding or petechiae, hepatosplenomegaly.

COUNT SAYS LEUKAEMIA
FBC or film reported as acute leukaemia, at any age → immediate specialist assessment (standard practice, not an NG12 criterion).

CHILDREN AND YOUNG PEOPLE
Under 25 with unexplained petechiae or hepatosplenomegaly → immediate specialist assessment (paediatric or young adult pathway).

LYMPHOMA
Adults with unexplained lymphadenopathy or splenomegaly → consider a suspected cancer pathway referral.

11 SPECIAL GROUPS

CG151 · MHRA · SPC

ON CHEMOTHERAPY
Expected low counts: any fever or feeling unwell → the 24-hour oncology or haematology hotline or 999, not GP telephone advice alone.

ANTITHYROID DRUGS
Counsel on carbimazole to report sore throat, mouth ulcers or fever: urgent FBC, stop if neutropenic.

CLOZAPINE
Mandatory monitoring via the brand service (CPMS, DMS or ZTAS; WBC and neutrophils, not ANC-only); low counts → act today on its instructions (red → stop).

TRAVEL
Fever after malaria-area travel → same-day malaria testing. Jaundice, big liver and spleen after travel → think leishmaniasis; specialist advice.

12 PITFALLS AND NEVER-EVENTS

NG12 · NG239 · CG151

- **Never** treat a febrile neutropenic patient with oral antibiotics at home.
- **Never** attribute pancytopenia to B12 or folate without a film, or give folic acid alone.
- **Never** continue carbimazole or methotrexate, or leave a clozapine count unactioned, while the cause is unclear.
- Do not wait for all first-line bloods before referring unexplained pancytopenia.
- A "mild" stable pancytopenia in an older person can still be MDS.
- Avoid NSAIDs and aspirin while platelets are low; plan dental extractions with haematology.

13 SELF-CARE AND THE CONVERSATION

CG151 · BSH

Fever, shivering or feeling very unwell → **999 or straight to hospital** · new bruises, spots, bleeding gums or breathlessness → contact us the **same day** · avoid NSAIDs, aspirin and contact sports · good food hygiene and handwashing.

"All three types of blood cell are lower than they should be. That can come from something simple, like a vitamin shortage, but it can also mean the bone marrow — the body's blood factory — isn't working properly, so I'm asking the blood specialists to look at it quickly. Until we know, a temperature or feeling shivery means hospital straight away."

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Abnormal liver blood tests CLINIC GUIDE

Abnormal ALT, AST, ALP, GGT, bilirubin or albumin in adults 18+. Not for children, pregnancy or post-partum (maternity pathway), or liver disease already under specialist care.

BSG liver blood tests 2017 · NICE **NG49**
NICE **NG50** · **NG12** · CG165 · NG238
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/abnormal-lfts.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → which pattern? → find the cause

! ACT TODAY — RED FLAGS

NICE NG12 · BSG 2017

TIER	RESULT OR PICTURE	ACTION
999	Confusion or drowsiness (encephalopathy) · haematemesis or melaena · shock or severe sepsis · suspected acute liver failure: acute liver injury, **INR typically ≥1.5** and new encephalopathy, without known cirrhosis · paracetamol overdose, including staggered	Emergency transfer; paracetamol → immediate hospital assessment
Same day	Jaundice + fever + RUQ pain (cholangitis) · new ascites · marked or rapidly rising transaminases with jaundice, raised INR, hypoglycaemia or renal impairment · suspected severe drug-induced liver injury	Same-day hospital assessment; stop a suspected non-essential drug
Suspected cancer pathway	**Jaundice aged 40+** → pancreatic cancer referral · aged **60+** with weight loss and diarrhoea, back or abdominal pain, nausea, vomiting, constipation or new diabetes → urgent direct-access CT (ultrasound if CT unavailable) · liver mass on imaging	NG12 referral; not a 999 transfer unless other emergency features

The size of the ALT rise does not predict how serious the disease is (BSG 2017): bilirubin, INR, albumin, glucose, renal function and the patient decide urgency. Isolated raised INR: think warfarin, vitamin K deficiency or malabsorption first.

1 CONFIRM IT IS REAL — AND LIVER

BSG 2017

Haemolysed sample	Raises AST (and potassium, LDH) but not ALT: repeat if AST rises alone
Muscle	Exercise, injury, myositis raise AST more than ALT: check **CK**
Bone ALP	Raised ALP with normal GGT: bone, vitamin D deficiency, Paget's, metastases
Fasting, illness	Raise unconjugated bilirubin in Gilbert's
Previous results	Compare first: a long-standing stable result changes the pace

Do not simply repeat. Most abnormal results are still abnormal a month later. Repeat in **4–6 weeks** only after a clear transient cause (stopped drug or alcohol, acute illness) — and send the aetiology screen alongside.

2 SEVERITY BANDS — WHAT THE PATTERN MEANS

BSG 2017 · NG238

RESULT	MEANS	ACTION · TIMING
Synthetic failure: INR up, albumin down, low platelets	Advanced disease or acute failure	Same day if acute; else urgent hepatology
Jaundice, conjugated bilirubin up	Obstruction or hepatic disease	USS urgently; 40+ suspected cancer pathway
Cholestatic: ALP + GGT up	Biliary obstruction, PBC, PSC, drugs	USS + AMA; ducts dilated → urgent
Hepatocellular: ALT/AST up, well	MASLD, alcohol, drugs, viral, autoimmune	Full aetiology screen now, whatever the level
On a statin: transaminases **<3× ULN**	Not a reason to stop	Continue; screen if persistent
Transaminases **≥3× ULN** on a statin	Possible drug effect	Withhold, investigate, recheck
Isolated ALP up, GGT normal	Probably bone	Bone profile, vitamin D
Isolated GGT up	Alcohol, MASLD, enzyme inducers	History and risk review; do not dismiss
Isolated unconjugated bilirubin, others normal	Gilbert's once haemolysis excluded	FBC, retics, LDH, haptoglobin; reassure

3 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
BMI, diabetes, BP, lipids	**MASLD** (commonest)	USS; FIB-4, then local pathway
Men >50, women >35 units/week	**Alcohol-related**	AUDIT; elastography (NG50)
New drug, herbal, bodybuilding product in 6 months	**Drug-induced**	Stop if safe; trend
Injecting, transfusion, born abroad, sexual risk	**Hepatitis B or C**	HBsAg; HCV antibody → RNA
Other autoimmune disease, female	**Autoimmune hepatitis, PBC**	ANA, SMA, AMA, immunoglobulins
Cholestasis with IBD	**PSC**	USS; specialist MRCP
Diabetes, arthralgia, family history	**Haemochromatosis**	Ferritin + TSAT (>45%)
Under 40, neuropsychiatric signs	**Wilson's**	Caeruloplasmin
Pain, pale stools, dark urine	**Gallstones, obstruction**	USS
All tests up, unwell or septic	**Sepsis, drugs, infiltration**	FBC, CRP, USS
Weight loss, sweats, nodes	**Malignancy, lymphoma**	USS; urgent route

AST:ALT >2 supports advanced alcohol-related disease but never diagnoses it alone.

4 WORK-UP CHECKLIST — THE AETIOLOGY SCREEN

BSG 2017 · NG49

- ☐ **Liver ultrasound**
- ☐ **HCV antibody**, reflex RNA
- ☐ **AMA, ANA, SMA**
- ☐ FBC: low platelets = portal hypertension
- ☐ HbA1c, lipids: metabolic risk
- ☐ **HBsAg** (hepatitis B)
- ☐ **Ferritin + TSAT** together
- ☐ **Immunoglobulins**
- ☐ INR, albumin, U&E
- ☐ Caeruloplasmin if under 40 (Wilson's)

Screen whatever the level or duration of the abnormality (BSG). MASLD: **FIB-4 <1.3** = low risk of advanced fibrosis (aged under 65); ≥1.3 → second-line test via the local pathway. NICE NG49: **ELF ≥10.51** = advanced fibrosis → refer.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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5 MANAGEMENT BY THE COMMONEST CAUSES

NG49 · NG50 · CG165 · NG238 · CG115

CAUSE	FIRST LINE	CHECK AND CONTINUE
MASLD	Gradual, sustained weight loss: around **7–10%** improves liver fat and inflammation; 150 min/week activity; treat diabetes, BP and lipids. No liver-specific drug in primary care	Stage fibrosis (FIB-4 then local pathway). ELF <10.51 → retest every **3 years** (NG49)
Alcohol-related	Abstinence goal; brief intervention; alcohol service if dependent. Plan withdrawal — no abrupt stop in dependence. Thiamine where indicated	Elastography for men >50, women >35 units/week (NG50). Relapse prevention via the alcohol service
Drug-induced	Stop the suspected drug where safe; record all supplements and timing	Recheck in 4–6 weeks; jaundice or raised INR → same day
Statin, transaminases raised	Do not stop for MASLD or a rise **<3× ULN** (NG238). ≥3× ULN → withhold and investigate	Restart cautiously once resolved and explained
Hepatitis B or C	HBsAg positive → refer all adults (CG165). HCV antibody → RNA; positive → treatment pathway (curable)	Antivirals are specialist-chosen. Offer HIV test
Autoimmune, PBC	Refer: autoimmune hepatitis is treatable; PBC (cholestatic pattern + AMA) is treated with ursodeoxycholic acid by the specialist	A positive antibody alone is read with the pattern, IgG and imaging
Haemochromatosis	Ferritin + TSAT >45% → confirm overload and genotype; refer. Avoid iron supplements	Venesection targets set by the specialist
Gilbert's syndrome	**No treatment**; explain it is harmless	Discharge once haemolysis excluded

6 MASLD — STAGING THE FIBROSIS

BSG 2017 · NICE NG49 · local pathway

1 · CONFIRM MASLD
Fatty liver on USS or metabolic risk, alcohol within lower-risk limits, negative aetiology screen.

2 · FIB-4 FIRST
(age × AST) ÷ (platelets × √ALT). **<1.3** = low risk (under 65): lifestyle, recheck per local pathway.

3 · FIB-4 ≥1.3
Second-line test: ELF or elastography via the local pathway. Do not mix thresholds from different pathways.

4 · ELF ≥10.51
Advanced fibrosis → hepatology. ELF <10.51 → retest every 3 years (NG49).

A low FIB-4 makes advanced fibrosis unlikely but does not exclude it; FIB-4 is less reliable in younger adults, over 65 and during acute illness.

7 ADVICE AND THE CONVERSATION

NG49 · NG50

- Alcohol: no more than **14 units/week**, spread over 3 or more days; none at all with alcohol-related disease or cirrhosis
- Gradual weight loss, 150 min/week of activity, stop smoking
- Stop unnecessary supplements and herbal remedies
- Hepatitis A and B vaccination if chronic liver disease or at risk (Green Book)

"Your liver test is a little off. On its own that doesn't tell us much, so rather than just repeating it I'm checking the common causes — fat in the liver, alcohol, medicines and viruses — and whether there is any scarring. Most people with a result like this can be looked after here."

8 FOLLOW-UP AND MONITORING

BSG 2017 · NG49 · NG50

- ☐ **4–6 weeks** after stopping a drug or alcohol, or after a transient illness
- ☐ MASLD: weight, metabolic and alcohol review at **3–6 months**
- ☐ Low-risk MASLD: reassess fibrosis per local pathway (ELF every 3 years)
- ☐ Cirrhosis: **6-monthly ultrasound** ± AFP for liver cancer (NG50)
- ☐ Statin: transaminases within **3 months** of starting and at **12 months**, then only if indicated
- ☐ Hepatitis B: liver cancer surveillance as set by the specialist (CG165)
- ☐ Discharge only when the cause is found or serious disease and advanced fibrosis are excluded
- ☐ Return the **same day**: yellow eyes, fever with pain, swelling tummy, bruising, confusion

9 WHEN AND WHERE TO REFER

BSG 2017 · NG12 · NG50

999	Encephalopathy, GI bleed, shock, acute liver failure, severe paracetamol poisoning
Same day	Cholangitis, new ascites, rising tests with jaundice or raised INR, severe DILI
Suspected cancer	Jaundice 40+; 60+ weight loss + qualifying symptom (CT); liver mass
Urgent hepatology	HBsAg or HCV RNA positive, suspected autoimmune hepatitis, cirrhosis, iron overload, suspected Wilson's
Routine	Advanced fibrosis (ELF ≥10.51 or high-risk local test), suspected PBC or PSC, persistent abnormality after a negative screen
Keep in practice	Low-risk MASLD, explained isolated GGT, Gilbert's

HBsAg positive: with the referral send HBeAg/anti-HBe, HBV DNA, anti-HCV and anti-HDV where appropriate, and offer an HIV test (CG165).

10 PITFALLS AND NEVER-EVENTS

BSG 2017 · NG238 · NG12

- **Never** just repeat an unexplained result without the aetiology screen.
- **Never** stop a statin for transaminases below 3× ULN.
- Normal LFTs do **not** exclude cirrhosis, hepatitis B or C.
- Do not label Gilbert's from one bilirubin or without excluding haemolysis.
- Jaundice at 40+ is a suspected cancer pathway referral, not a wait-and-see.
- Ask about herbal, over-the-counter and bodybuilding products.
- Do not wait for an ALT threshold: the extent of the rise does not grade the disease.
- Cirrhosis from any cause: advise complete abstinence from alcohol (NG50).

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Jaundice and raised bilirubin CLINIC GUIDE

New jaundice or a raised bilirubin in adults 18+. Not for neonates or children, or known liver disease under specialist care. Pregnancy: same-day obstetric assessment.

NICE **NG12** · **NG85** · **BSG** liver blood tests 2017
NICE **NG49** · **NG50** · CG165 · SPC
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/jaundice.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → which type? → find the cause

! ACT TODAY — RED FLAGS

NICE NG12 · BSG 2017

TIER	RESULT OR PICTURE	ACTION
999	Jaundice with confusion, drowsiness or asterixis (acute liver failure) · hypoglycaemia · shock or poor perfusion · active GI bleeding · cholangitis with hypotension or confusion · unwell after overdose	Emergency transfer
Same day	Fever or rigors + RUQ pain + jaundice (**cholangitis**) · raised INR, bruising, renal impairment · rapidly deepening jaundice · suspected alcoholic hepatitis · any suspected paracetamol overdose, including staggered · jaundice in **pregnancy**	Same-day hospital; paracetamol → ED now, do not wait for bloods
Suspected cancer pathway	**Jaundice aged 40+**, with or without weight loss → pancreatic cancer referral · aged **60+** with weight loss and diarrhoea, back or abdominal pain, nausea, vomiting, constipation or new diabetes → urgent direct-access CT (ultrasound if CT unavailable)	NG12 referral; do not wait for imaging — a normal ultrasound does not exclude pancreatic cancer

Pale stools, dark urine and itch = obstructive picture → urgent ultrasound and HPB route. A palpable, non-tender gallbladder with jaundice raises concern for malignant obstruction.

1 CONFIRM IT IS REAL

examination · laboratory

Look at the sclera	In natural light. Yellow skin with white sclera = carotenaemia, not jaundice
Fractionate	Ask the laboratory for conjugated and unconjugated bilirubin
Haemolysed sample	Can distort bilirubin and AST: repeat if the picture does not fit
Fasting, illness	Raise unconjugated bilirubin in Gilbert's — repeat when well
Raised INR	Check for warfarin, vitamin K deficiency, malabsorption before calling it liver failure

Visible jaundice is never just watched: send the first-line bloods the same day. Isolated mildly raised bilirubin with otherwise normal tests can be repeated once, when well.

2 SEVERITY BANDS — TYPE AND URGENCY

BSG 2017 · NG12

RESULT	MEANS	ACTION · TIMING
Bilirubin up + INR up, confusion or low glucose	Acute liver failure	999
Conjugated, ALP + GGT up (cholestatic)	Obstruction: stones, pancreas, cholangiocarcinoma; PBC, PSC, drugs	Urgent USS; 40+ suspected cancer pathway
ALT/AST predominant (hepatitic), INR normal, well	Viral, alcohol, drug, autoimmune hepatitis	Same-week screen; specialist input
Hepatitic + raised INR or unwell	Severe hepatitis	Same day
Unconjugated, anaemia, high retics	Haemolysis	Film, LDH, haptoglobin, DAT; haematology
Unconjugated only, all else normal	Gilbert's syndrome	Exclude haemolysis; reassure

For suspected drug injury the R-ratio (ALT/ULN ÷ ALP/ULN) classifies: ≥5 hepatocellular, 2–5 mixed, ≤2 cholestatic.

3 EXAMINATION FINDINGS THAT CHANGE THE PLAN

3-minute examination

FEVER, PULSE, BP
Fever with jaundice = cholangitis until proved otherwise → same-day hospital; unstable → 999.

ASTERIXIS, CONFUSION
Hepatic encephalopathy → acute liver failure → 999.

GALLBLADDER, MASS, NODES
Non-tender palpable gallbladder, epigastric mass or Virchow's node → malignancy → suspected cancer referral.

CHRONIC LIVER SIGNS
Spider naevi, palmar erythema, splenomegaly, ascites → cirrhosis → hepatology; new ascites same day.

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Colicky RUQ pain, fatty food	**Gallstones in the duct**	USS
Painless, weight loss, 40+	**Pancreatic or biliary cancer**	Suspected cancer referral
New drug, antibiotic, herbal	**Drug-induced**	Stop if safe; trend
Heavy drinking, tender liver	**Alcoholic hepatitis**	Same-day specialist
Travel, shellfish, contacts	**Hepatitis A or E**	Hep A IgM, hep E
Injecting, sexual risk, born abroad	**Hepatitis B or C**	HBsAg; HCV antibody → RNA
Other autoimmune disease, arthralgia	**Autoimmune hepatitis**	ANA, SMA, IgG
Itch, fatigue, cholestatic, female	**PBC**	AMA
Cholestasis with IBD	**PSC**	USS; specialist MRCP
Anaemia, dark urine, no pale stools	**Haemolysis**	Film, retics, LDH, DAT
Young, after fasting or illness	**Gilbert's**	Fractionated bilirubin

5 WORK-UP CHECKLIST

BSG 2017 · NG12

- ☐ **Bilirubin, fractionated**
- ☐ **INR**, albumin
- ☐ U&E, glucose
- ☐ **USS**: cholestatic pattern
- ☐ Hep A IgM, hep E if hepatitic
- ☐ Ferritin + TSAT
- ☐ **ALT, AST, ALP, GGT**
- ☐ **FBC** + platelets
- ☐ CRP if infection possible
- ☐ **HBsAg, HCV antibody**
- ☐ AMA, ANA, SMA, immunoglobulins
- ☐ Film, retics, LDH, haptoglobin

Do not send CA 19-9 without imaging evidence of malignancy: false positives are common, especially with obstruction.

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6 MANAGEMENT BY THE COMMONEST CAUSES

NG12 · NG85 · CG165 · NG50 · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Obstruction (stones, mass)	Urgent USS and HPB route; cholangitis → hospital. Suspected pancreatic cancer: pancreatic-protocol CT before biliary drainage, unless drainage is urgent (NG85)	40+ → suspected cancer pathway, whatever the ultrasound shows
Alcoholic hepatitis	Same-day specialist assessment. Steroids are specialist-only. Thiamine if at risk; no abrupt stop in dependence without assisted withdrawal	Alcohol service; complete abstinence long term
Drug-induced	Stop a suspected non-essential drug; specialist advice for essential ones. Report on a Yellow Card	Recheck LFTs and INR; worsening → urgent specialist. Do not rechallenge
Viral hepatitis	Hep A or E: supportive, avoid alcohol, notify. HBsAg positive → refer (CG165). HCV RNA positive → treatment pathway	HBV: vaccinate household contacts. HCV: specialist antivirals, screen for HBV first
Cholestatic itch	**Colestyramine 4–8 g daily** (SPC); give other medicines well apart, as the SPC directs. Rifampicin only on specialist advice	Constipation, fat-soluble vitamin malabsorption
PBC	Hepatology; ursodeoxycholic acid is usually specialist-initiated	Second-line treatment is specialist-led
Gilbert's syndrome	**No treatment** and no routine monitoring once confirmed	Explain; record the diagnosis

7 IMAGING AND THE ROUTE

NG12 · NG85 · BSG 2017

CHOLESTATIC PATTERN
Ultrasound first: dilated ducts, stones, mass. Urgency by sepsis, pain and speed of rise.

SEPTIC OR IN PAIN
Suspected cholangitis → hospital the same day, not an outpatient scan.

USS NORMAL, STILL CHOLESTATIC
MRCP via the specialist for stones, strictures or PSC; AMA for PBC.

SUSPECTED PANCREATIC CANCER
Refer at 40+ without waiting; pancreatic-protocol CT before any biliary stent (NG85).

8 FOLLOW-UP AND MONITORING

BSG 2017 · NG49 · NG50

- [ ] Referral booked and the result of every test actioned
- [ ] Stable, selected patients: LFTs + FBC at **1–2 weeks** — never instead of referral
- [ ] Drug-induced: LFTs falling after the drug is stopped
- [ ] Not improving at **4 weeks** → escalate
- [ ] Falling bilirubin does not exclude cancer or obstruction
- [ ] Cirrhosis: **6-monthly ultrasound** ± AFP (NG50)
- [ ] Gilbert's confirmed: no further tests
- [ ] Return the **same day**: deeper yellow, fever, unable to eat or drink, bruising, swelling tummy; **999** for confusion or vomiting blood

9 WHEN AND WHERE TO REFER

NG12 · BSG 2017 · CG165

999	Encephalopathy, hypoglycaemia, shock, GI bleed, septic cholangitis, unwell after overdose
Same day	Coagulopathy, renal impairment, fever, significant pain, rapidly rising bilirubin, alcoholic hepatitis, paracetamol overdose, pregnancy
Suspected cancer	Jaundice 40+; 60+ weight loss + qualifying symptom (CT); duct dilatation or mass on imaging
Urgent hepatology	Cholestatic LFTs with normal USS (PBC, PSC); HBsAg or HCV RNA positive; autoimmune hepatitis supported by pattern and IgG
Routine	Stable chronic viral hepatitis, haemochromatosis, stable PBC, persistent unexplained derangement
Keep in practice	Confirmed Gilbert's syndrome

No single bilirubin or INR value decides the tier: the whole picture, the trend and the cause do.

10 PITFALLS AND NEVER-EVENTS

NG12 · BSG 2017

- **Never** let a normal ultrasound stop a suspected cancer referral at 40+.
- **Never** treat suspected cholangitis as an outpatient ultrasound.
- Do not label Gilbert's with abnormal ALT, ALP, GGT, albumin or INR.
- Do not delay acetylcysteine for staggered paracetamol overdose.
- Painless jaundice is more worrying, not less.
- Conjugated bilirubin is not the same as obstruction: hepatitis raises it too.
- Ask about herbal remedies, supplements and recent antibiotics.
- A raised INR is not automatically liver failure: check anticoagulants and vitamin K first.
- **Pregnancy**: jaundice is never routine — same-day obstetric assessment (cholestasis, HELLP, acute fatty liver).
- Do not start rifampicin for itch without specialist advice (liver toxicity, interactions).

11 ADVICE AND THE CONVERSATION

NG50 · Green Book

- No alcohol until the cause is known; complete abstinence in alcohol-related disease or cirrhosis
- Hepatitis A and B vaccines for chronic liver disease (Green Book)
- Avoid NSAIDs and sedatives in significant liver disease; tailor the paracetamol dose
- Itch: cool skin, emollients; colestyramine if cholestatic

"The yellow colour means a substance called bilirubin is building up. Sometimes that's harmless, but it can mean a blockage or inflammation in the liver, so I want to find the cause quickly. I'm arranging blood tests and a scan, and a fast-track hospital referral so nothing is missed."

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Raised creatine kinase (CK) CLINIC GUIDE

A raised CK in adults 18+, including on a statin. Not for children, suspected acute coronary syndrome (use troponin) or known neuromuscular disease under specialist care.

NICE **NG238** (2023) · SPC
NHS England statin intolerance pathway
MHRA Drug Safety Update (2012)
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/raised-creatine-kinase.html

PAGE 1 · INTERPRET — is it rhabdomyolysis? → is it real? → how many times the ULN? → find the cause

! ACT TODAY — RED FLAGS

NG238 · NHS England pathway

TIER	RESULT OR PICTURE	ACTION
999	Shock, collapse or confusion · severe hyperkalaemia or ECG changes · tense, swollen, very painful limb after crush, prolonged lie or injecting (**compartment syndrome**) · chest pain that sounds cardiac (troponin, not CK)	Emergency transfer; compartment syndrome → emergency surgical assessment, no waiting for tests
Same day	**Any CK** with AKI, oliguria, hyperkalaemia, dark urine, severe pain or weakness, systemic illness or unable to keep fluids down · CK **≥10× ULN** · weakness with breathing or swallowing difficulty	Same-day medical assessment or admission; stop causative drugs
Urgent specialist	Objective progressive proximal weakness with a high CK that persists after stopping a statin (immune-mediated necrotising myopathy) · dermatomyositis rash	Urgent rheumatology or neurology

Rhabdomyolysis = CK >10× ULN with renal impairment, or >50× ULN (NHS England pathway). Weakness, dark urine, AKI or hyperkalaemia move a patient up a tier whatever the number. CK peaks 1–3 days after injury, so an early value may still be rising.

1 CONFIRM IT IS REAL

repeat at rest

Use multiples of ULN	Ranges differ between laboratories, and are higher in men and people of African ancestry
Exercise	The commonest cause: unaccustomed, eccentric or extreme exercise
Injections, falls	IM injections, trauma, seizures, surgery
Macro-CK	Persistently raised, well, normal examination → ask the laboratory

Well, no weakness, normal renal function and clear urine: avoid strenuous exercise, IM injections and heavy alcohol for about **7 days**, then repeat. Never delay assessment for suspected rhabdomyolysis.

2 SEVERITY BANDS — MULTIPLES OF ULN

NHS England pathway · NG238

CK	MEANS	ACTION · TIMING
Any + red flags	Possible rhabdomyolysis	Same day; 999 if unstable
>50×	Rhabdomyolysis	Same-day hospital; stop statin
10–50×	Significant muscle injury	Same-day assessment; U&E, K^+, urine
4–10×, well	Secondary cause likely: drug, endocrine, exertion	Repeat; renal function, urine, targeted tests
<4×, well	Usually exercise or incidental	Rest, repeat in about 7 days
Persistent + weakness	Inflammatory or hereditary myopathy	Rheumatology or neurology
Persistent, mild, well	Idiopathic, macro-CK, ethnicity	Observe or non-urgent advice; document

3 FOCUSED HISTORY AND EXAMINATION — WHAT DISCRIMINATES

examine strength, not just pain

PAIN OR WEAKNESS?
Ache after exertion or on a statin is usually benign. True proximal weakness (rising from a chair, stairs, arms up) points to myopathy.

DRUGS AND SUBSTANCES
Statin or fibrate dates and dose changes, interacting drugs, antipsychotics, colchicine, hydroxychloroquine, retinoids; alcohol, cocaine, supplements.

PRECIPITANTS
New or eccentric exercise, trauma, seizure, surgery, long lie, heat, recent viral illness.

SIGNS
Heliotrope rash, Gottron papules, wasting, calf hypertrophy, thyroid signs; dark urine; family history of muscle disease.

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Gym, run, spin class	**Exertion**	Repeat at rest
Statin or fibrate, recent dose change or interacting drug	**Statin myopathy**	Statin intolerance pathway
Crush, long lie, seizure, heat	**Rhabdomyolysis**	U&E, K^+, urine dip
Alcohol, cocaine, MDMA	**Toxic myopathy**	U&E, LFTs
Weight gain, cold, slow reflexes	**Hypothyroidism**	TSH
Proximal weakness, rash, dysphagia	**Inflammatory myopathy**	Rheumatology; myositis antibodies
Breathless, arthritis, mechanic's hands	**Antisynthetase syndrome**	Rheumatology
Cramps and dark urine with exercise, "second wind"	**McArdle disease**	Neurology, metabolic
Young, calf hypertrophy, family history	**Muscular dystrophy**	Neurology
Recent viral illness, calf pain	**Viral myositis**	Recheck when recovered

5 WORK-UP CHECKLIST

by history, not a routine package

- ☐ **Repeat CK** at rest
- ☐ **Urine dipstick**
- ☐ Calcium, phosphate, magnesium
- ☐ ECG if potassium abnormal or unwell
- ☐ **U&E/eGFR, potassium**
- ☐ **TSH**
- ☐ Glucose, LFTs
- ☐ Drug and supplement review

Dipstick positive for blood with few or no red cells supports myoglobinuria but does not prove it — act on the picture, do not wait for microscopy. Add vitamin D, inflammatory or autoimmune tests only when indicated.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT BY THE COMMONEST CAUSES

NG238 · NHS England pathway · MHRA · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Exertion, well	Rest from hard exercise, hydrate; return gradually once CK normalises	Repeat CK after about 7 days; recurrent exertional rhabdomyolysis → specialist
Statin, muscle symptoms	Check CK, renal function, TSH. Follow the statin intolerance pathway below — stop only where it says	Rechallenge, lower dose or different statin; most people can stay on one
Drug interaction	Simvastatin with clarithromycin, erythromycin, some azoles or HIV protease inhibitors; dose limits with amlodipine, diltiazem, verapamil, amiodarone (MHRA 2012). Fibrates, fusidic acid, ciclosporin, colchicine add risk	Check the SPC before adding any interacting drug
Hypothyroidism	Treat; CK often normalises without myopathy work-up	Recheck CK when euthyroid
Alcohol, drugs	Reduce alcohol, thiamine where at risk; stop cocaine, MDMA, stimulant supplements	Recheck after abstinence
Rhabdomyolysis	Hospital: IV fluids, potassium and renal monitoring	After discharge: renal recovery, culprit recorded

7 ON A STATIN — THE NATIONAL CK BANDS

NHS England statin intolerance pathway · NG238

<4× ULN, TOLERABLE
Statin cause less likely. Look for other causes; continue, or reassess and restart at a lower dose or alternative statin.

4–10× ULN OR INTOLERABLE
Stop the statin for 4–6 weeks. Rechallenge once symptom-free: lower dose or different statin.

10–50× ULN
Stop the statin. Check renal function; renal impairment → treat as rhabdomyolysis. Specialist advice if CK does not settle.

>50× ULN
Rhabdomyolysis: stop the statin, same-day hospital. Do not wait for urine myoglobin.

Before starting (NG238): persistent unexplained muscle pain → measure CK; >5× ULN → re-measure after 7 days; still >5× → do not start; raised but <5× → start at a lower dose. **Do not** measure CK routinely on a statin without symptoms. Several statins not tolerated → lipid specialist; consider non-statin options (NG238).

8 ADVICE AND THE CONVERSATION

plain English

- Build up new or intense exercise gradually; drink enough; avoid training in extreme heat or when unwell
- Some bodybuilding and weight-loss supplements contain stimulants — stop them while CK is raised
- On a statin: in blinded trials most muscle aches were as common on placebo (Herrett et al., BMJ 2021) — a reason to rechallenge, not to give up

"Your muscle enzyme is raised, which is very common after exercise. Please rest from hard exercise for a week and we'll repeat it. If your urine goes dark like cola, you pass much less urine, or your muscles become very painful or weak, get seen the same day."

9 FOLLOW-UP AND MONITORING

NG238 · NHS England pathway

- [ ] Recheck CK about **7 days** after rest if well
- [ ] Track the trend, not one value
- [ ] Still raised → targeted secondary tests ± referral
- [ ] Statin stopped: rechallenge once symptom-free
- [ ] Weakness worsening off the statin → urgent specialist
- [ ] After rhabdomyolysis: renal function recovered, culprit recorded
- [ ] Hypothyroid: CK rechecked once euthyroid
- [ ] CK recorded as a multiple of ULN with the recheck date
- [ ] Return the **same day**: dark urine, passing much less urine, severe pain or weakness; **999** for collapse, confusion or a tense, swollen limb

10 WHEN AND WHERE TO REFER

NG238 · NHS England pathway

999	Shock, severe hyperkalaemia, ECG changes, compartment syndrome, confusion
Same day	Rhabdomyolysis, AKI, dark urine, CK ≥10× ULN; breathing or swallowing weakness
Urgent rheumatology	Progressive proximal weakness; dermatomyositis rash; CK not falling after stopping a statin
Neurology / metabolic	Young, family history, exertional symptoms, recurrent rhabdomyolysis
Lipid clinic	Several statins not tolerated
GP manages	Exertional, drug or thyroid cause in a well patient

Refer with serial CK as multiples of ULN with dates, statin dates, renal and urine results, and a clear statement on weakness.

11 PITFALLS AND NEVER-EVENTS

NG238 · MHRA

- **Never** wait for CK or imaging in suspected compartment syndrome.
- **Never** stop a statin for good on a mildly raised CK alone.
- Do not check CK routinely on a statin without symptoms.
- Do not use fixed absolute cut-offs: use multiples of ULN.
- Pain alone is not weakness: examine proximal strength.
- Check interactions before prescribing a macrolide with simvastatin.
- CK also rises in heart muscle injury: chest pain needs troponin.
- A high CK that does not fall after stopping a statin is not simple myalgia.

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Raised CRP or ESR CLINIC GUIDE

Raised CRP, ESR or plasma viscosity in adults 18+. Not for children, pregnancy, or monitoring an established rheumatic disease (use its own pathway).

NICE **NG253** · **NG12** · NG237 · NG250
BSR GCA 2020 · PMR 2010 · TA518
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/high-crp-esr.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → how high, and which pattern? → find the cause

! ACT TODAY — RED FLAGS

NICE NG253 · BSR 2020 · NG12

TIER	PICTURE	ACTION
999	Suspected sepsis with **NEWS2 ≥7**, or 5–6 with one parameter scoring 3 or added concern · shock or altered consciousness · back pain with new leg weakness, saddle numbness or bladder change	Emergency transfer, pre-alert. NEWS2 5–6 → urgent senior review
Same day	Aged **50+** with new headache, scalp tenderness, jaw claudication or visual symptoms (**GCA**) · **visual loss** or diplopia with suspected GCA · organ-threatening vasculitis (rapidly worsening renal function, active urine sediment, lung haemorrhage) · back pain + fever or IV drug use (spinal infection) · hypercalcaemia, AKI or cord compression with possible myeloma	GCA: **start prednisolone now**, do not wait for results; visual symptoms → emergency eye or acute assessment today
Suspected cancer pathway	Results suggesting myeloma · unexplained lymphadenopathy or splenomegaly (with fever, sweats, weight loss, itch) → lymphoma · other cancer symptoms → site-specific NG12 route	An isolated raised ESR is not a referral criterion

Do not use CRP to rule sepsis in or out: serious infection can occur with a modest or initially normal CRP. Decide on observations, NEWS2 and the patient.

1 CONFIRM IT IS REAL

laboratory · timing

Timing	CRP rises within hours and falls fast; ESR rises and falls over days to weeks — a lagging ESR after recovery is expected
ESR raised by	Age, female sex, anaemia, renal disease, pregnancy, paraprotein
ESR lowered by	Polycythaemia
CRP raised by	Obesity, smoking, hard exercise in the previous 48 hours, recent surgery or trauma
Masked by	Steroids and tocilizumab suppress CRP (and fever) — assess infection clinically

Mild, unexplained rise in a well patient: **repeat in 4–6 weeks** with a clinical review — not a cascade of tests, and not endless rechecks without examining the patient. Use your laboratory's ESR range for age and sex.

2 SEVERITY BANDS

supportive only — no value diagnoses

RESULT	MEANS	ACTION · TIMING
CRP **<5** mg/L	Normal; does not exclude early infection	Clinical judgement
CRP **5–20**	Obesity, smoking, minor infection, early arthritis	Treat cause; recheck 4–6 weeks if unexplained
CRP **20–100**	Significant inflammation or infection	Examine, find the source today
CRP **>100**	Bacterial or severe viral infection, GCA, vasculitis, pancreatitis, cancer	Careful same-day assessment with NEWS2 — not automatic antibiotics
High ESR, CRP normal or mild	Myeloma, anaemia, renal disease, SLE, age	FBC, U&E, calcium; SPEP + FLC if plausible
Both markedly high	GCA/PMR, TB, endocarditis, lymphoma, vasculitis	Targeted work-up; GCA same day

A rising trend matters more than one value: a CRP that keeps climbing over months needs more thought than a stable mild rise over years.

3 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Fever, focal symptoms	**Infection**	Source: MSU, CXR, cultures
50+, headache, jaw claudication	**GCA**	Start steroid; fast-track USS/biopsy
50+, shoulder/hip girdle pain and stiffness	**PMR**	Exclude GCA, RA, myositis, cancer
60+, back pain, fracture, high ESR	**Myeloma**	FBC, calcium, SPEP + FLC
Nodes, sweats, weight loss	**Lymphoma, cancer**	FBC, film, LDH; CXR
Synovitis, morning stiffness	**Inflammatory arthritis**	RF, anti-CCP
Rash, mouth ulcers, serositis	**SLE, CTD**	ANA, dsDNA, ENA
Kidney, lung, ENT, purpura	**ANCA vasculitis**	Urine dip, ANCA, U&E
Diarrhoea, blood, weight loss	**IBD**	Faecal calprotectin
Murmur, fever, IV drug use	**Endocarditis**	Blood cultures; same day
Overweight, smoker, well	**Metabolic, low grade**	Lifestyle; recheck

4 WORK-UP CHECKLIST

targeted, not scattergun

- ☐ **FBC** + film
- ☐ **U&E**, LFTs, **calcium**
- ☐ Urine dipstick + ACR
- ☐ **SPEP + serum FLC** if myeloma plausible
- ☐ CXR if chest, nodes or weight loss
- ☐ RF + anti-CCP if synovitis
- ☐ ANA, ANCA only with features
- ☐ Blood cultures before antibiotics if septic
- ☐ CK, TSH if girdle symptoms
- ☐ LDH if nodes or sweats

Myeloma (NG12): aged 60+ with persistent bone or back pain or unexplained fracture → FBC, calcium, ESR or plasma viscosity, SPEP + serum free light chains (urine Bence-Jones if FLC unavailable).

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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5 MANAGEMENT BY THE COMMONEST CAUSES

BSR 2020 · BSR/BHPR 2010 · NG253 · NG250 · TA518

CAUSE	FIRST LINE	CHECK AND CONTINUE
Bacterial infection	Treat the source on clinical grounds and NEWS2, not on CRP alone. Blood cultures before antibiotics if possible	Not improving → reassess diagnosis, source control (abscess), antibiotic choice. Pneumonia: CRP at 3–4 days only if concern about failure (NG250)
GCA (suspected)	**Prednisolone 40–60 mg** daily now, same-day fast-track referral (BSR). **Visual loss**: IV methylprednisolone **0.5–1 g** daily for up to 3 days, or **60 mg** oral prednisolone if IV is not immediately available	Temporal artery ultrasound or biopsy urgently — do not delay steroid. Taper is rheumatology-guided; tocilizumab only for relapsing or refractory GCA (TA518, specialist)
PMR	**Prednisolone 15 mg** daily: 15 mg for 3 weeks → 12.5 mg for 3 weeks → 10 mg for 4–6 weeks → then reduce by 1 mg every 4–8 weeks (BSR/BHPR, individualise)	A prompt response supports but does not prove PMR. Ask about GCA symptoms at every review
Metabolic, low grade	Weight loss, stop smoking, 150 min/week activity, alcohol within 14 units/week, dental review	Do not chase occult cancer in a well patient with a stable mild rise
Steroid safety	Individual plan: BP, weight, HbA1c, fracture risk and bone protection; gastric protection by GI risk	Steroid card; sick-day advice; do not stop abruptly

6 SUSPECTED GCA — THE SAME-DAY SEQUENCE

BSR 2020 · local fast-track pathway

1 · RECOGNISE
Aged 50+: new headache, scalp tenderness, jaw or tongue claudication, visual symptoms, PMR symptoms. Markers can be modest or rarely normal.

2 · TREAT NOW
Prednisolone **40–60 mg** today. Visual loss: IV methylprednisolone **0.5–1 g** up to 3 days, or **60 mg** oral if IV not immediately available.

3 · REFER TODAY
GCA fast-track for ultrasound or biopsy, ideally soon after steroid starts — never delay treatment for it.

4 · PROTECT
Fracture-risk assessment and bone protection, GI risk, glucose and BP; steroid card.

7 FOLLOW-UP AND MONITORING

BSR · SPS · NG250

- [ ] Mild unexplained rise: recheck in **4–6 weeks with** a clinical review
- [ ] Still raised: FBC + film, renal, liver, calcium, urine, targeted tests
- [ ] Treated infection: judge on the patient; CRP trend supports only
- [ ] PMR/GCA: symptoms with ESR/CRP during the taper
- [ ] Flare = symptoms return; never raise steroid on markers alone
- [ ] PMR on treatment beyond 2 years → review diagnosis, refer
- [ ] Long-term steroid: BP, weight, HbA1c, fracture risk reviewed (SPS)
- [ ] Plan and reasoning documented
- [ ] Return the **same day**: new vision change, headache or jaw pain; fever with back pain; **999** if very unwell

8 WHEN AND WHERE TO REFER

NG253 · BSR · NG12

999	High-risk sepsis, shock, cord compression or cauda equina
Same day	GCA with visual symptoms (emergency eye/acute); organ-threatening vasculitis (renal, rheumatology); spinal infection; severe hypercalcaemia or AKI with possible myeloma; film suggesting leukaemia
GCA fast-track	Every suspected GCA, same day — steroid already started
Suspected cancer	Myeloma results; lymphoma features; site-specific NG12 symptoms
Urgent rheumatology	Suspected CTD or vasculitis without organ threat; atypical PMR
GP manages	Identified, treated cause; metabolic rise; typical PMR responding

9 PITFALLS AND NEVER-EVENTS

BSR · NG253 · NG12

- **Never** wait for ESR/CRP before starting steroid in suspected GCA.
- **Never** diagnose or exclude sepsis on CRP.
- Normal markers do not exclude GCA, PMR or serious infection.
- On steroids or tocilizumab, CRP and fever may stay low despite infection.
- PMR not responding to 15 mg → rethink the diagnosis.
- High ESR with normal CRP in an older patient: think myeloma.
- SLE flares and active IBD can have a normal CRP.
- A positive ANCA or ANA alone does not make the diagnosis.
- Procalcitonin is not a routine primary-care test.

10 ADVICE AND THE CONVERSATION

plain English

- Avoid hard exercise for 48 hours before a repeat test
- On steroids: carry the steroid card; never stop suddenly
- Weight loss, stopping smoking and dental care lower a low-grade CRP

"This test shows there is some inflammation in your body, but it doesn't tell us where or why. It can be as simple as a recent infection or extra weight. I'll examine you and do a few targeted tests rather than lots of random ones, and recheck it in a few weeks."

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Abnormal immunoglobulins CLINIC GUIDE

Adults 18+ with a paraprotein, abnormal serum free light chains (FLC), polyclonal rise or low immunoglobulins. Not for children, known myeloma under active treatment (haematology-led) or established CVID care (immunology-led).

NICE **NG12** (2026) · **NG35** · **NG234** (2023)
BSH MGUS 2023 · IMWG 2014 · Myeloma UK
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/abnormal-immunoglobulin.html

PAGE 1 · INTERPRET

is it dangerous? → is it real? → which band? → what is driving it?

! ACT TODAY — RED FLAGS

NG35 · NG234 · NG12 · IMWG 2014

TIER	RESULT OR COMBINATION	ACTION
999 / emergency	Paraprotein + **leg weakness, sensory loss, saddle anaesthesia, bladder or bowel dysfunction** (suspected MSCC) · AKI or rapidly falling eGFR · severe or symptomatic hypercalcaemia · sepsis on myeloma treatment · pathological fracture with deformity or instability	MSCC: contact MSCC coordinator now; whole-spine MRI **within 24 h**; dexamethasone **16 mg** oral as soon as possible (NG234). Do not delay transfer for imaging
Same day	**IgM paraprotein + hyperviscosity** symptoms: visual disturbance, headache, mucosal bleeding, confusion · severe spinal pain without neurology	Discuss with haematology or acute oncology now; spinal pain: MSCC advice within 24 h
Suspected cancer pathway	Paraprotein or abnormal FLC + attributable **CRAB**: calcium **>2.75 mmol/L** (or >0.25 above ULN) · creatinine **>177 µmol/L** or CrCl <40 mL/min · Hb **<100 g/L** (or >20 below LLN) · lytic lesion · or FLC ratio **≥100** with involved FLC ≥100 mg/L	Haematology suspected cancer pathway referral (NG12: results suggest myeloma)

Paraprotein size alone never diagnoses or excludes myeloma. Severe hypercalcaemia, AKI or sepsis: follow the local acute pathway.

1 CONFIRM IT IS REAL

BSH 2023 · local laboratory

FLC ratio, CKD	Both chains rise as eGFR falls: use the lab's **renal reference range** before calling it abnormal
Infection, inflammation	Raise FLC and globulin polyclonally; repeat when well if no band and no CRAB
Small or faint band	Needs **immunofixation** to type it; read the lab comment
Low IgG	Confirm it persists when well; check for protein loss (urine PCR, albumin), steroids, rituximab, CLL
Trends	Compare on the **same laboratory platform**: assays differ
Normal SPEP	Does not exclude light-chain myeloma: add serum FLC if suspicion persists (NG35)

2 SEVERITY BANDS

Myeloma UK · BSH 2023

RESULT	MEANS	ACTION · TIMING
Band + CRAB, or FLC ratio ≥100	Possible active myeloma	**Suspected cancer pathway**
New IgG band **>15 g/L**; IgA or IgM **>10 g/L**; any **IgD or IgE**; markedly abnormal FLC	Higher risk of myeloma or lymphoma	**Suspected cancer pathway** (Myeloma UK)
Below these, well, normal FBC, Ca, eGFR	Probable MGUS	Routine haematology or advice and guidance to stage risk
Known MGUS rise **>25% and ≥5 g/L**	Possible progression	Re-refer within about 1 week
Polyclonal rise, no band	Reactive	Find the cause
Low IgG + recurrent infection	Antibody deficiency	Clinical immunology
IgA **<0.07 g/L**, IgG and IgM normal	Selective IgA deficiency	IgG-based coeliac test; transfusion alert

3 WHEN TO SEND A MYELOMA SCREEN

NG12 · NG35 · BSH 2023

NICE NG12	Aged **60+** with persistent bone pain, especially back pain, or unexplained fracture → offer FBC, calcium, ESR or plasma viscosity, protein electrophoresis and **serum FLC** (urine BJP if FLC unavailable)
Also consider	Unexplained anaemia, renal impairment or hypercalcaemia · raised ESR or plasma viscosity · recurrent bacterial infections · raised total protein or globulin
Think amyloid	Nephrotic syndrome, heart failure without a cause, neuropathy, macroglossia, periorbital purpura → FLC + SPEP + immunofixation

4 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Bone pain, anaemia, high Ca, renal	**Myeloma**	FLC, immunofixation
Incidental small band, well	**MGUS**	FBC, Ca, eGFR, FLC
IgM band, nodes, blurred vision	**Waldenström**	Plasma viscosity, FBC
Nephrotic, HF, neuropathy, macroglossia	**AL amyloid**	FLC, urine PCR
Polyclonal, liver or HIV risk	**Chronic infection, liver, autoimmune**	LFT, HIV, hepatitis, ANA
Low IgG, CLL or rituximab	**Secondary deficiency**	FBC, film, drug history
Recurrent chest infections	**CVID**	Ig levels; immunology

5 WORK-UP CHECKLIST

NG12 · NG35 · BSH 2023

- ☐ **SPEP + immunofixation**: class, light chain
- ☐ **Serum FLC** κ, λ and ratio
- ☐ **FBC**, film if cytopenias
- ☐ **Adjusted calcium**, albumin
- ☐ **Creatinine, eGFR**
- ☐ Quantitative IgG, IgA, IgM
- ☐ ESR or plasma viscosity
- ☐ Urine BJP only if FLC unavailable
- ☐ Urine PCR if low IgG or oedema
- ☐ **tTG IgA + total IgA** together

Imaging is arranged by haematology: whole-body MRI first-line, low-dose CT if unsuitable; no isotope bone scan (NG35).

6 MGUS: PUTTING RISK IN PROPORTION

Mayo model · BSH 2023

Most incidental bands are MGUS. Progression to myeloma or a related disorder is about **1% a year**; most people never progress.

RISK FACTOR (MAYO)	WHY IT MATTERS
Paraprotein **≥15 g/L**	Larger clone
Non-IgG (IgA, IgM)	IgM progresses to lymphoma or Waldenström
Abnormal FLC ratio	Clone making excess light chain

Original Mayo data: 20-year progression about 5% with none of these, about 58% with all three. Haematology sets the risk group and the follow-up plan.

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 INCIDENTAL PARAPROTEIN — THE FOUR STEPS

NG35 · BSH 2023 · Myeloma UK

1 · TYPE IT
Immunofixation (class, κ or λ) and quantify in g/L; serum FLC and ratio, renal range if CKD.

2 · LOOK FOR CRAB
FBC, adjusted calcium, creatinine; ask about bone or back pain, infections, weight loss, neuropathy.

3 · CHOOSE THE TIER
CRAB, FLC ratio ≥100, IgG >15, IgA/IgM >10, IgD/IgE → suspected cancer pathway. Otherwise routine.

4 · FOLLOW THE LETTER
Haematology stages risk; GP monitors only if delegated, with clear re-referral triggers.

8 MANAGEMENT OF THE COMMONEST CAUSES

BSH 2023 · NG35 · NG20 · Green Book

CAUSE	GP ACTION	CONTINUE AND CHECK
MGUS	Haematology referral or advice and guidance for characterisation and risk stratification (isotype, size, FLC ratio). No treatment needed	GP monitoring **only if the haematology letter delegates it**, with tests, interval and re-referral triggers. Typically bloods at about **6 months**, then per plan
Suspected myeloma	Suspected cancer pathway; stop NSAIDs; encourage fluids	**Do not start** bisphosphonate or denosumab: bone treatment is haematology-led, after dental review (NG35)
IgM paraprotein	Haematology referral; ask about hyperviscosity, lymph nodes, sweats, weight loss	Same-day discussion if visual disturbance, bleeding or confusion; plasmapheresis is a specialist decision, not set by an IgM level
Polyclonal rise	Treat the cause: liver disease, chronic infection, autoimmune disease, sarcoid	No haematology referral if no band
Antibody deficiency	Clinical immunology; treat infections promptly; no live vaccines	Immunoglobulin replacement is specialist-initiated; no universal IgG target
Selective IgA deficiency	Coeliac: use **tTG IgG or DGP IgG** (NG20)	Record a transfusion alert with any reaction history, not a drug allergy (BSH 2023)

9 FOLLOW-UP AND MONITORING CHECKLIST

BSH 2023 · Myeloma UK

- [ ] Haematology plan on file: risk group, tests, interval
- [ ] Recheck: **paraprotein, FBC, calcium, creatinine**, FLC if in plan
- [ ] Same laboratory platform; record the trend
- [ ] Re-refer: rise **>25% and ≥5 g/L**, new anaemia, renal decline, high Ca, bone pain
- [ ] Also re-refer: neuropathy, nephrotic syndrome, HF, macroglossia, purpura (MGCS)
- [ ] Stop routine monitoring only on haematology advice or when no longer in the patient's interest
- [ ] Flu, pneumococcal and Shingrix per Green Book if immunosuppressed
- [ ] Safety-net: back pain with leg weakness or bladder change → **999**

10 SAFETY-NET FOR THE PATIENT

BSH 2023 · Myeloma UK

- Contact us **promptly**, not at the next test: new bone or back pain, thirst or passing more urine, unusual tiredness, repeated infections, foamy urine or swelling.
- **999 or A&E**: back pain with leg weakness, numbness, or loss of bladder or bowel control.
- Myeloma UK has plain-English MGUS information.

11 WHEN AND WHERE TO REFER

NG12 · NG234 · Myeloma UK

Emergency	Suspected MSCC, AKI, severe hypercalcaemia, symptomatic hyperviscosity, sepsis on treatment
Suspected cancer	Band or FLC + CRAB · FLC ratio ≥100 · lytic lesion · new IgG >15 g/L · IgA or IgM >10 g/L · IgD or IgE · any band with clinical suspicion
Within 1 week	Known MGUS: rise >25% and ≥5 g/L, new anaemia or renal decline
Routine	New small band, well, normal bloods: MGUS staging
Immunology	Persistent low IgG with recurrent or severe bacterial infections, bronchiectasis

12 SHARED CARE AFTER A MYELOMA DIAGNOSIS

NG35 · MHRA · Green Book

- **Lenalidomide**: VTE prophylaxis as the haematology protocol sets; Pregnancy Prevention Programme run by the haematology centre.
- **Infection**: treat promptly; no live vaccines on active treatment; prophylaxis as haematology advises.
- **Steroids**: check glucose and BP. **Bone treatment**: renal function before each zoledronic acid dose; dental care, ONJ awareness.
- Kidneys: hydration; avoid NSAIDs. Early advance care planning; Myeloma UK and Macmillan support.

13 PITFALLS AND NEVER-EVENTS

NG35 · BSH 2023 · NG20

- **Never** reassure on a normal SPEP alone when myeloma is suspected: add serum FLC. **Never** diagnose myeloma from the paraprotein size or an isolated abnormal FLC ratio.
- **Never** read an FLC ratio in CKD against the standard range · **never** trust a negative tTG IgA without a total IgA · **never** delay life-saving transfusion for IgA-deficient components.
- **Never** start bisphosphonate or denosumab for suspected myeloma in primary care · **never** lose a delegated MGUS patient from recall · avoid NSAIDs and iodinated contrast when renal function is at risk.
- **Never** assume every IgA-deficient patient needs IgA-deficient blood (only after a previous severe reaction, with transfusion medicine) · **never** label selective IgA deficiency as CVID · no universal IgG target in CVID.

14 SAY IT LIKE THIS

MGUS

"The blood test found a protein made by one family of immune cells. In most people this stays harmless for life, and it is not cancer. We check it with blood tests so that we would spot the small number of people where it starts to change."

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Raised tumour markers CLINIC GUIDE

Adults 18+ with a raised PSA, CA125, CEA, AFP, hCG or other tumour marker in primary care. Not for children, population screening programmes or specialist surveillance decisions (follow the written oncology plan).

NICE **NG12** (updated 2026) · HTG690 (FIT)
NICE CG122 · NG151 · NG50 · CG165 · NG131
PCRMP · RCOG GTG 38 · NG110
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway:
gpreasoning.uk/tools/algorithms/abnormal-cancer-markers.html

PAGE 1 · INTERPRET

symptoms first → is it real? → which pathway? → what else raises it?

! ACT TODAY — RED FLAGS

NG12 · RCOG GTG 38 · HTG690

TIER	RESULT OR COMBINATION	ACTION
999 / same day	**Positive pregnancy test + bleeding** or pain (ectopic, molar pregnancy) · collapse or heavy bleeding · abnormal film, blasts, severe cytopenias or acutely unwell	Assess stability; early pregnancy unit or gynaecology; same-day haematology discussion
Suspected cancer pathway	Prostate feels malignant on DRE, any PSA · **ascites or pelvic/abdominal mass** (no CA125 needed) · CA125 at or above the NG12 age threshold **and** ultrasound suggests ovarian cancer · **FIT ≥10 µg Hb/g**, or rectal or anal mass or anal ulceration	Suspected cancer pathway referral (28-day Faster Diagnosis Standard)
Consider suspected cancer	PSA above the NG12 age threshold in a man with possible symptoms · unexplained lymphadenopathy or splenomegaly	Discuss preferences and comorbidities, then refer
Urgent specialist	New liver lesion, suspicious ultrasound or **progressive AFP rise** in chronic liver disease · persistent hCG after a pregnancy event	Hepatology via local liver pathway · GTD centre

Isolated CEA, AFP, CA19-9 or LDH values are not referral criteria. Markers rarely diagnose cancer alone: symptoms, examination, FIT and imaging drive the decision.

1 CONFIRM IT IS REAL

NG12 · local laboratory

MARKER	BENIGN OR SPURIOUS RISES
PSA	UTI or prostatitis (do not test while active) · recent ejaculation, vigorous exercise · retention, catheter, instrumentation · take blood before DRE. **5-alpha-reductase inhibitors lower PSA**
CA125	Menstruation, pregnancy, endometriosis, fibroids, PID · ascites of any cause, liver disease, heart failure
CEA	**Smoking**, COPD, IBD, liver disease, pancreatitis
AFP	Pregnancy, active hepatitis, liver regeneration
CA19-9	Biliary obstruction, cholangitis, pancreatitis; absent in Lewis-negative people
Chromogranin A	PPIs, renal impairment

Repeat PSA only when infection or a pre-test factor may explain it, once resolved (local laboratory advice). Use the same laboratory for trends.

2 RESULT BANDS

NG12 (2026) · PCRMP

RESULT	MEANS	ACTION · TIMING
PSA above age threshold: **40–49 >2.5 · 50–59 >3.5 · 60–69 >4.5 · 70–79 >6.5** µg/L (ng/mL)	Raised for age (symptomatic men)	**Consider** suspected cancer pathway referral; <40 or 80+: clinical judgement
CA125 at or above age threshold (NG12 2026): 40–49 ≥35 · 50–59 ≥31 · 60–69 ≥24 · 70–79 ≥25 · 80+ ≥31 IU/mL	Possible ovarian cancer	**Urgent direct-access USS** abdomen and pelvis
CA125 raised, aged **39 or under**	Often benign	Not used alone: consider urgent USS if symptoms persist
FIT **≥10 µg Hb/g**	Possible colorectal cancer	**Suspected cancer pathway**
Isolated raised CEA, CA19-9 or LDH	Non-specific	Not a referral criterion: assess symptoms, smoking, liver
AFP rising in cirrhosis or hepatitis B	Possible HCC	**Urgent** hepatology discussion
hCG persistent after a pregnancy	Possible GTD	GTD centre via local service

The old single CA125 cut-off of 35 IU/mL now applies only at 40–49: lower thresholds apply from age 50.

3 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
LUTS, hard nodular prostate	**Prostate cancer**	PSA + DRE
Smooth enlarged prostate, LUTS	**BPH**	PSA vs age threshold
Fever, perineal pain, dysuria	**Prostatitis**	MSU; treat, then repeat
Bloating, early satiety, 50+	**Ovarian cancer**	CA125, USS
Cyclical pelvic pain, under 40	**Endometriosis**	USS; gynaecology
Smoker, no bowel symptoms	**Smoking-related CEA**	Stop retesting; FIT if symptoms

4 WORK-UP CHECKLIST

NG12 · NG131 · CG122

- ☐ **Why was it sent?** Symptoms, history
- ☐ **Examination**: abdomen, pelvis, DRE
- ☐ **Urine dip ± MSU** before PSA
- ☐ **Repeat PSA** only if confounded
- ☐ **Urgent USS** abdomen and pelvis if CA125 at or above threshold
- ☐ **FIT** for bowel symptoms
- ☐ **LFTs** with CEA, AFP or CA19-9
- ☐ **Pregnancy test** with hCG
- ☐ FBC, film if haematological
- ☐ Family history; BRCA, Lynch

5 WHICH MARKER, WHEN

NG12 · NG50 · RCOG

Use, with a pathway	**PSA**: possible symptoms, or informed request · **CA125**: persistent symptoms, aged 40+ · **FIT**: bowel symptoms · **AFP**: hepatology surveillance · **hCG**: pregnancy, GTD
Specialist only	CEA, CA19-9, CA15-3, chromogranin A, NSE, CYFRA 21-1, S100, calcitonin, thyroglobulin (with anti-Tg antibodies) · isolated LDH
Never	A "tumour marker screen" for vague symptoms or reassurance

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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6 MANAGEMENT OF THE COMMONEST SITUATIONS

NG12 · NG110 · NG151 · NG50 · CG165

SITUATION	FIRST LINE	CHECK AND CONTINUE
Raised PSA, possible prostatitis	MSU first; antibiotic for **14 days**, then review (NG110)	Repeat PSA once settled; still above the age threshold → suspected cancer pathway referral
Raised PSA, LUTS awaiting urology	Tamsulosin **400 micrograms once daily** for bothersome LUTS	Note any 5-ARI in the referral; MRI and biopsy are urology decisions (NG131)
CA125 raised, USS normal	Look for other causes: endometriosis, fibroids, liver disease, heart failure	Safety-net; new ascites, mass or worse symptoms → suspected cancer pathway
Incidental raised CEA	Ask about bowel symptoms; FIT if present. Smoking, liver disease, IBD	No referral on CEA alone; do not keep retesting without a plan
Known colorectal cancer	CEA at least **every 6 months** for the first **3 years** per the oncology plan (NG151)	Rising trend → contact the colorectal team; no universal percentage rule
Cirrhosis, hepatitis B	HCC surveillance: **6-monthly ultrasound** ± AFP via hepatology (NG50, CG165)	Progressive AFP rise or new lesion → urgent hepatology

7 SUSPECTED OVARIAN CANCER — THE NG12 ROUTE

NG12 (2026) · CG122

EXAMINE FIRST
Ascites or a pelvic or abdominal mass → suspected cancer pathway now; do not wait for CA125.

AGED 40+
Persistent or frequent symptoms → CA125. At or above the NG12 (2026) age threshold → urgent direct-access USS.

AGED 39 OR UNDER
Do not use CA125 alone; consider urgent USS of abdomen and pelvis for persistent symptoms.

USS RESULT
Suggests cancer → suspected cancer pathway. Normal → other causes, safety-net, review date.

8 FOLLOW-UP AND DOCUMENTATION CHECKLIST

NG12 · specialist plans

- ☐ Result, units and range recorded with the **reason for testing**
- ☐ Benign causes and confounders considered
- ☐ Action, destination, urgency and **who reviews** the result
- ☐ Referral seen and outcome received; system flag for review
- ☐ Planned review within a time frame agreed with the person (NG12)
- ☐ Surveillance follows the **written specialist plan**; tests happen
- ☐ Stop repeat markers that have no indication or plan
- ☐ Safety-net: return if symptoms persist, change or worsen

9 WHEN AND WHERE TO REFER

NG12 · CG122 · RCOG

Same day	Pregnancy with bleeding or pain · abnormal film or blasts · acutely unwell
Suspected cancer	Malignant prostate · PSA above age threshold with symptoms (consider) · ascites or mass · CA125 + suggestive USS · FIT ≥10, rectal or anal mass
Urgent specialist	Hepatology: rising AFP, new liver lesion · GTD centre: persistent hCG
Routine	Gynaecology: persistent symptoms, normal USS, likely benign cause
Secondary care only	RMI I after USS; refer if ≥250 (CG122)

10 PSA IN MEN WITHOUT SYMPTOMS

PCRMP · UK NSC

- No national PSA screening programme. Offer an **informed discussion**: false positives, overdiagnosis, MRI or biopsy may follow.
- A man **aged 50+** who still wants a test after counselling may have one.
- Weigh age, ethnicity, family history, **BRCA2**, comorbidity and life expectancy. Known BRCA2 or strong family history → specialist or clinical genetics advice.

11 MARKERS IN KNOWN CANCER — FOLLOW THE WRITTEN PLAN

NG131 · NG151 · OV05 · specialist plans

CANCER	MARKER AND PLAN	GP ROLE
Prostate	PSA per urology or oncology plan; after radical prostatectomy usually undetectable or very low	Any rise outside the plan → discuss with urology
Colorectal	CEA at least every 6 months, plus CT, for the first 3 years (NG151)	Check tests happen; rising CEA → colorectal team
Ovarian	CA125 per gynae-oncology plan; early retreatment on CA125 alone gave no benefit	Record and pass on; avoid alarm
Thyroid	Thyroglobulin only with anti-Tg antibodies (antibodies falsely lower Tg)	Specialist follow-up only

12 PITFALLS AND NEVER-EVENTS

NG12 · CG122 · NG131 · OV05

- **Never** use tumour markers to screen people without symptoms (except informed PSA) · **never** refer, or reassure, on an isolated CEA, CA19-9, AFP or LDH.
- **Never** wait for CA125 when there is ascites or a mass · **never** let a normal CA125 or negative FIT override persistent symptoms · **never** use CA125 alone under 40.
- **Never** test PSA during active UTI or prostatitis · do not arrange a bone scan just because PSA is above 10 · PSA velocity, density and free-to-total ratio are secondary-care tools.
- After ovarian cancer, a CA125 rise alone does not trigger retreatment (MRC OV05): pass it to oncology without alarm.

13 SAY IT LIKE THIS

raised marker

"This blood test can be raised for lots of reasons, and most of them are not cancer. On its own it can't tell us what is going on, so I'd like to arrange the right scan or referral to find out properly, and I'll make sure we follow it up."

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Raised PSA CLINIC GUIDE

Men and people with a prostate aged 18+ with a raised or rising PSA. Not for men under active surveillance or on ADT (follow the urology or oncology plan), or acute retention and sepsis (emergency pathways).

NICE **NG12** (updated 2026) · **NG131** (2019)
PCRMP (2016) · GIRFT urology (2024)
NICE NG110 · CG97 · NG234 · MHRA (2024)
Reviewed **Oct 2026** · next review **Apr 2027**
Full pathway: gpreasoning.uk/tools/algorithms/high-psa.html

PAGE 1 · INTERPRET — is he unwell? → is it real? → which threshold? → symptomatic or asymptomatic?

! ACT TODAY — RED FLAGS

NG234 · NG110 · NG12

TIER	RESULT OR COMBINATION	ACTION
999 / emergency	Raised PSA or known prostate cancer + back pain with **limb weakness, sensory change, saddle anaesthesia, bladder or bowel dysfunction** (suspected MSCC)	MSCC coordinator now; whole-spine MRI **within 24 h**; dexamethasone **16 mg** oral as soon as possible (NG234)
Same day	**Acute urinary retention** · fever, rigors, perineal pain, tender prostate with sepsis, retention, inability to take oral treatment or no improvement in **48 h**	Same-day catheterisation · hospital for prostatitis with sepsis or abscess; avoid prostatic massage
Urgent	Very high PSA with **bone pain**, weight loss or systemic features	Urgent urology or oncology contact; staging and ADT are specialist-led
Suspected cancer pathway	Prostate feels malignant on DRE, **any PSA**	Refer: suspected cancer pathway referral

A normal neurological examination does not exclude early MSCC. Do not start ADT or anti-androgens in primary care.

1 CONFIRM IT IS REAL

PCRMP · GIRFT 2024 · SPC

Ejaculation, vigorous exercise	Avoid for **48 h** before the test (cycling included)
UTI or prostatitis	Do not test while active; MSU; repeat once settled, not within **6 weeks** of infection
Biopsy, catheter, cystoscopy, retention	Raise PSA: interpret with caution; biopsy: wait 6 weeks
5-ARI (finasteride, dutasteride)	Roughly **halves** PSA after 6 months; a confirmed rise from the lowest on-treatment value needs assessment
Lab and timing	Same laboratory for trends; take blood before DRE where possible

Do not repeat a raised PSA routinely before referral unless a probable transient cause exists (GIRFT 2024). A DRE is not needed before referral.

2 THRESHOLDS AND ACTION

NG12 2026 · PCRMP

AGE	SYMPTOMATIC: PSA ABOVE (NG12)	ACTION
<40	Clinical judgement	**Consider** suspected cancer pathway referral, discussing his preferences and comorbidities
40–49	**>2.5** µg/L (ng/mL)	
50–59	**>3.5**	
60–69	**>4.5**	
70–79	**>6.5**	
80+	Clinical judgement	

OTHER RESULT	ACTION · TIMING
No symptoms, aged 50–69, PSA **≥3.0** (PCRMP)	Urology referral per local pathway
Below threshold, but suspicious DRE, Black ancestry, strong family history or BRCA2	Lower threshold for urology assessment
Rising after prostatectomy or radiotherapy	Contact the treating team
Within range, symptoms	Explain; safety-net; reassess if they persist

3 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Hard, nodular, asymmetric gland	**Prostate cancer**	Refer: MRI first
Smooth enlarged gland, LUTS	**BPH**	IPSS; age threshold
Fever, perineal pain, dysuria	**Acute prostatitis**	MSU, then treat
Dysuria, positive dip	**UTI**	MSU; repeat later
Recent catheter, retention	**Instrumentation**	Interpret with caution
Bone pain, weight loss	**Metastatic cancer**	Urgent urology
Visible haematuria	**Bladder or prostate cancer**	Urology (NG12)
On a 5-ARI, rising from lowest value	**Possible cancer**	Assess; refer

4 WORK-UP CHECKLIST

NG12 · NG131 · CG97

- ☐ **Symptoms**: LUTS, ED, haematuria
- ☐ **DRE** if trained and appropriate
- ☐ **5-ARI** or prostate surgery
- ☐ Ethnicity: Black ancestry
- ☐ Bladder palpation if retention
- ☐ IPSS for LUTS
- ☐ **Urine dip ± MSU**
- ☐ Previous PSA values
- ☐ Family history, **BRCA2**
- ☐ Bone pain, weight loss
- ☐ Lower-limb neurology if back pain
- ☐ Comorbidity, life expectancy

5 MEN WITHOUT SYMPTOMS

PCRMP · UK NSC · GIRFT

- No national PSA screening programme (UK NSC). Offer a **balanced discussion**: false positives, overdiagnosis, MRI or biopsy may follow, a normal result does not exclude cancer.
- A man **aged 50+** who still wants a test after counselling may have it (PCRMP).
- **Higher risk**: Black men, strong family history, known **BRCA2** variant: discuss testing earlier; follow clinical genetics advice.
- Do not arrange BRCA testing just because of one high PSA.

6 WHAT TO PUT IN THE REFERRAL

NG12 · GIRFT 2024

- ☐ PSA value, date, earlier values
- ☐ Family history, ethnicity, BRCA2
- ☐ Symptoms; IPSS
- ☐ Comorbidity, preferences discussed
- ☐ DRE finding, if done
- ☐ Bone pain, weight loss
- ☐ 5-ARI or prostate surgery
- ☐ Anticoagulants (for biopsy)

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7 MANAGEMENT OF THE COMMONEST CAUSES

NG110 · MHRA 2024 · CG97 · NG131 · SPC

CAUSE	FIRST LINE	CHECK AND CONTINUE
Acute bacterial prostatitis (stable)	MSU before antibiotics. **Trimethoprim 200 mg twice daily for 14 days**, then review (NG110). Fluoroquinolones only when other antibiotics are inappropriate (MHRA Jan 2024)	Review at 14 days: stop, or continue a further 14 days if needed. Admit if septic or not improving in 48 h. Repeat PSA once settled
BPH with bothersome LUTS	Alpha-blocker, e.g. **tamsulosin 400 micrograms MR once daily**; warn about postural dizziness and floppy iris syndrome	First review at **4–6 weeks** (CG97, QS45), then regular review of symptoms and side effects
Enlarged prostate, high progression risk	Add a 5-ARI (finasteride **5 mg** daily) if prostate estimated **>30 g** or PSA **>1.4** (CG97)	Record a baseline PSA first; then judge against the lowest on-treatment value. Warn about sexual side effects and mood change
Raised PSA, symptomatic	Suspected cancer pathway referral above the age threshold; MRI before biopsy in urology (NG131)	State PSA, date, DRE, 5-ARI use and family history in the referral
Known cancer, PSA rising	Contact the treating urology or oncology team	Do not label recurrence or arrange PSMA PET-CT yourself
Asymptomatic request	Informed discussion (PCRMP); a man **aged 50+** may then have a test	Prostate Cancer UK decision aid; no national screening programme

8 SELF-CARE AND SUPPORT

PCRMP · NG131

- Before a repeat test: no ejaculation or vigorous exercise, including cycling, for 48 h.
- Unable to pass urine, fever with shivering, or new back pain → contact us the same day.
- Prostate Cancer UK: decision aid and specialist nurses. Regular exercise helps during and after treatment.

9 TREATMENT SIDE EFFECTS — GP ROLE

NG131 · specialist plan

- **Hot flushes on ADT**: medroxyprogesterone **20 mg daily**, initially for 10 weeks (NG131); cyproterone if ineffective.
- **Bone**: assess fracture risk at ADT start; no bisphosphonate solely to prevent osteoporosis (NG131).
- **Metabolic**: weight, BP, HbA1c, lipids; exercise.
- **ED after treatment**: PDE5 inhibitors on the NHS; specialist nurse or psychosexual support.
- ADT itself is specialist-initiated; shared care only by formal agreement.

10 FOLLOW-UP AND MONITORING CHECKLIST

NG131 · CG97 · NG12

- ☐ Referral accepted and appointment received; outcome reviewed
- ☐ Discharged after low-risk MRI or biopsy: **written PSA re-referral threshold** and interval on file
- ☐ After radical treatment: PSA no earlier than **6 weeks**, at least **6-monthly for 2 years**, then at least **yearly** (NG131)
- ☐ Active surveillance: follow the urology protocol
- ☐ On a 5-ARI: compare with the lowest on-treatment value
- ☐ Below threshold: reassess if symptoms persist or progress
- ☐ Safety-net: back pain with leg weakness or bladder change → **999**; unable to pass urine → same day

11 WHEN AND WHERE TO REFER

NG12 · NG234 · GIRFT

Emergency	Suspected MSCC · acute retention · prostatitis with sepsis or abscess
Suspected cancer	Malignant-feeling prostate, any PSA · symptomatic and above the NG12 age threshold (consider) · very high PSA with bone pain
Treating team	PSA rise after radical treatment; any rise outside an active surveillance plan
Routine urology	LUTS failing medical treatment · recurrent UTI · persistent residual urine
Genetics	Known BRCA2 or strong family history: follow clinical genetics advice

12 AFTER REFERRAL — WHAT HAPPENS NEXT

NG131 · GIRFT 2024

REFERRAL
Suspected cancer pathway; 28-day Faster Diagnosis Standard. Fit men usually go straight to MRI.

MRI FIRST
Multiparametric MRI before any biopsy (NG131); a low-risk MRI may avoid biopsy.

BIOPSY IF SUSPICIOUS
Targeted biopsy guided by the MRI; risk group set by urology.

OUTCOME
Discharge with a written PSA plan, active surveillance, or radical treatment after shared decision.

13 PITFALLS AND NEVER-EVENTS

NG12 · NG131 · NG234

- **Never** test during active UTI or prostatitis · **never** let a normal DRE or normal PSA override a malignant-feeling prostate or symptoms.
- **Never** forget 5-ARI halving · **never** use PSA velocity, density or free-to-total ratio as primary-care referral rules.
- **Never** start ADT or anti-androgen flare cover in primary care · do not order a bone scan just because PSA is above 10.
- **Never** lose track of a suspected cancer referral: confirm the appointment and the outcome · **never** arrange genetic testing on one high PSA alone.

14 SAY IT LIKE THIS

raised PSA

"PSA is a protein the prostate makes. It goes up with a larger prostate, infection and some cancers, so a raised result doesn't mean you have cancer. The next step is usually an MRI scan through the hospital team, which tells us much more."

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Allergy test results CLINIC GUIDE

Adults and children with a positive specific IgE or skin prick test, raised total IgE or tryptase. Not for acute anaphylaxis care beyond first treatment, patch testing (dermatology) or immunotherapy decisions (specialist).

NICE **NG258** anaphylaxis (2026, replaced CG134)
NICE **CG116** · CG183 · HTG413 · RCUK 2021
BSACI · MHRA · Reviewed **Oct 2026**
next review **Apr 2027** · Full pathway:
gpreasoning.uk/tools/algorithms/allergy-test-abnormalities.html

PAGE 1 · INTERPRET — was there a reaction? → sensitised or allergic? → how risky? → who needs adrenaline?

! ACT TODAY — RED FLAGS

RCUK 2021 · NG258 · BSACI

TIER	SITUATION	ACTION
999	Sudden airway, breathing or circulation problem, usually with skin or mucosal change (skin may be absent) · tongue or laryngeal swelling	IM adrenaline 1 mg/mL, thigh: **500 micrograms** adult and child >12 · **300** age 6–12 · **150** age 6 months–6 years · **100–150** under 6 months. Repeat after **5 min**. Lie flat
Same day	Reaction compatible with anaphylaxis, **no adrenaline auto-injector** (AAI) · systemic reaction to a bee or wasp sting	Prescribe **two** in-date AAIs by brand, train, written plan; refer to specialist allergy service (NG258)
Urgent allergy	Food reaction with airway symptoms · persistently raised baseline tryptase with severe or venom anaphylaxis · recurrent angioedema without urticaria (low C4)	Urgent allergy, immunology or paediatric allergy referral

Antihistamines and steroids never delay adrenaline. No AAI is needed after a reaction to an easily avoidable drug. A positive test without a compatible history is never a reason for an AAI.

1 CONFIRM IT IS REAL

NG258 · CG116 · BSACI

Sensitisation ≠ allergy	A positive sIgE or SPT only means IgE is present; read it against the history
Broad panels	Generate false positives and needless exclusion diets: test only what the history suggests
Total IgE	Non-specific: atopy, parasites, rarely IgE myeloma
Tryptase timing (16+)	Sample **as soon as possible** after treatment starts; second ideally **1–2 h**, no later than **4 h** from onset; baseline after recovery
Normal tryptase	Does not exclude anaphylaxis, especially food or respiratory reactions
Ranges	Assay- and laboratory-dependent; SPT by trained staff with resuscitation facilities

2 RESULT BANDS

BSACI · NG258 · HTG413

RESULT + HISTORY	MEANS	ACTION
Positive + **compatible immediate reaction**	Probable IgE allergy	Avoid; refer if systemic or significant
Positive, **food eaten without symptoms**	Sensitisation only	Keep eating it; no AAI
Positive, unclear history	Uncertain	Specialist; food challenge only there
PR-10 component: **Ara h 8, Cor a 1**	Pollen cross-reactivity	Usually mild oral symptoms; specialist advice
Storage protein: **Ara h 2, Cor a 9/14**	Primary nut sensitisation	Higher risk: allergy clinic
Raised total IgE alone	Non-specific	No action on its own
Baseline tryptase persistently raised (>20 µg/L is a minor criterion)	Possible mast-cell disease	Allergy or haematology
Eosinophils **>0.5 × 10⁹/L**	Atopy, parasites, drugs	Context; repeat

3 CAUSES

clue → think → first test

CLUE	THINK	FIRST TEST
Itchy mouth with raw apple, hay fever	**Pollen-food syndrome**	History; no panel
Urticaria, vomiting minutes after food	**IgE food allergy**	Targeted sIgE
Seasonal sneezing, May–July	**Grass pollen**	Treat; sIgE if unclear
Year-round rhinitis, worse in bed	**House dust mite**	HDM sIgE
Swelling, no itch or wheals	**Hereditary angioedema**	C4, C1 inhibitor
Childhood rash on amoxicillin	**Likely not penicillin allergy**	Structured history

4 WORK-UP CHECKLIST

CG116 · NG258 · CG183

- ☐ **Allergy-focused history**
- ☐ Exact trigger, cooked vs raw
- ☐ Cofactors: exercise, alcohol, NSAIDs
- ☐ **Targeted sIgE** only
- ☐ **C4** if angioedema alone
- ☐ **Onset**: minutes vs hours
- ☐ Tolerated since?
- ☐ **Asthma control**
- ☐ FBC, eosinophils
- ☐ Drugs: **beta-blocker**, ACE inhibitor

No serum IgG, Vega, kinesiology or hair analysis (CG116). No serum IgE to diagnose drug allergy outside specialist care (CG183).

5 COMPONENT RESULTS — READ WITH SPECIALIST ADVICE

HTG413 · BSACI

PATTERN	COMPONENTS	WHAT IT SUGGESTS
Higher risk	Ara h 2 (peanut) · Cor a 9/14 (hazelnut) · Tri a 14 (wheat) · Ses i 1 (sesame) · casein Bos d 8 · ovomucoid Gal d 1	Primary sensitisation; heat-stable, so baked forms may not be tolerated
Moderate	Milk whey Bos d 5 · egg ovalbumin Gal d 2	Heat-labile: baked forms often tolerated
Lower	PR-10: Ara h 8 · Cor a 1 · Mal d 1 (apple)	Birch-pollen cross-reactivity; usually oral symptoms only

Components refine probability; they do not predict severity or decide who needs an AAI.

6 WHO NEEDS AN AAI?

NG258 · BSACI

Yes	Reaction compatible with anaphylaxis to food, venom, latex or unknown trigger: two AAIs as an interim measure until specialist review
Usually no	Reaction to an easily avoidable drug · positive test alone · large local sting reaction · uncomplicated pollen-food syndrome
Ask allergy	Food allergy with asthma, peanut or tree nut, cofactor reactions, raised baseline tryptase

Decision support, not a substitute for clinical judgement. Check doses in the SPC and your local formulary. Use your laboratory's reference ranges.

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7 MANAGEMENT OF THE COMMONEST SITUATIONS

NG258 · MHRA · BSACI · CG183 · SPC

SITUATION	FIRST LINE	CHECK AND CONTINUE
Anaphylaxis history	**Two** in-date AAIs by brand; dose by weight, age and product licence; brand-specific training; written BSACI action plan	Use the second after **5 minutes** if no better; call 999. Refer to allergy (NG258). Recheck child doses as weight rises
Hay fever (pollen sIgE positive)	Intranasal steroid, e.g. fluticasone propionate 2 sprays each nostril **once daily**, + cetirizine **10 mg** once daily	Start before the season; immunotherapy only via specialist if uncontrolled
House dust mite	Antihistamine + intranasal steroid first	Avoidance measures have variable benefit; SLIT is specialist-initiated (TA1045)
Pollen-food syndrome	Explain; cooked forms often tolerated; avoid only foods that cause symptoms	Reassess with specialist if systemic symptoms, nuts, or cofactors
Penicillin allergy label	Structured history: drug, timing, symptoms, treatment, severity	Severe immediate or severe delayed reaction → avoid, refer. Low risk → governed de-labelling pathway
Child, food allergy	Confirm with history and targeted tests; dietitian if milk or several foods are excluded	Paediatric allergy if anaphylaxis, severe eczema with poor growth

8 AFTER SUSPECTED ANAPHYLAXIS — THE GP'S JOB

NG258 · MHRA · BSACI

RECORD
Time of onset, features, treatment given, suspected trigger and cofactors.

TRYPTASE
Check timed samples were taken in hospital; baseline after recovery via the specialist pathway.

EQUIP
Two AAIs by brand, technique taught, written action plan, school or carer plan.

REFER
Specialist allergy service for every suspected anaphylaxis (NG258); venom → immunotherapy assessment.

9 URTICARIA AND ANGIOEDEMA

BSACI · TA339 · SPC

- **Chronic spontaneous urticaria**: wheals recurring over more than 6 weeks; allergy tests and routine bloods rarely help.
- Regular second-generation antihistamine; if needed, off-label up-dosing to **four times** the usual dose. Omalizumab is specialist-initiated (TA339).
- **Angioedema without wheals**: stop ACE inhibitors; check C4 and C1 inhibitor.
- Wheals lasting over 24–48 h, bruising or pain → consider urticarial vasculitis.

10 EVERYDAY SAFETY

FSA · Anaphylaxis UK

- The **14 major allergens** must be declared on pre-packed food; Natasha's Law covers food pre-packed for direct sale.
- Ask staff about loose or freshly prepared food; carry an allergy card.
- Anaphylaxis UK and Allergy UK: training, school and peer support.

11 ANNUAL REVIEW AND MONITORING CHECKLIST

NG258 · MHRA · BSACI

- [ ] **Two** in-date AAIs physically carried; expiry dates
- [ ] Dose right for current weight (children)
- [ ] Brand-specific technique shown; retrain if brand changes
- [ ] Written action plan up to date; school plan for children
- [ ] **Asthma control**: poor control raises the risk of fatal food anaphylaxis
- [ ] New reactions, cofactors, new drugs (**beta-blockers**)
- [ ] Referral made and specialist plan received
- [ ] De-labelled allergy recorded as resolved; patient told in writing

12 WHEN AND WHERE TO REFER

NG258 · BSACI · CG116

999	Anaphylaxis; tongue or laryngeal swelling
Urgent allergy	Any anaphylaxis not yet referred · venom systemic reaction (immunotherapy assessment) · significant food reaction · raised baseline tryptase · angioedema without urticaria
Paediatric allergy	Food anaphylaxis · severe eczema, multiple foods, poor growth · suspected eosinophilic oesophagitis
Specialist drug allergy	High-risk drug reaction histories; clinically important labels
GP-managed	Rhinitis, sensitisation without reaction, uncomplicated pollen-food syndrome

13 PITFALLS AND NEVER-EVENTS

CG116 · CG183 · MHRA

- **Never** prescribe an AAI or a food-exclusion diet for a positive test alone · **never** order broad IgE panels.
- **Never** diagnose drug allergy with serum IgE in primary care · never de-label after a severe reaction.
- Beta-blockers blunt adrenaline: tell the allergy team. Montelukast: neuropsychiatric warning (MHRA).
- **Never** use serum IgG, Vega, kinesiology or hair tests · never use components alone to predict severity.

14 SAY IT LIKE THIS

positive test, no reaction

"The test shows your immune system recognises this food, but you've been eating it without any trouble, and that tells us more than the blood test does. You don't need to stop eating it."

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